19XR
Hermetic Centrifugal Liquid Chillers
50/60 Hz
With HFC-134a

Start-Up, Operation, and Maintenance Instructions

SAFETY CONSIDERATIONS

Centrifugal liquid chillers are designed to provide safe and reliable service when operated within design specifications. When operating this equipment, use good judgment and safety precautions to avoid damage to equipment and property or injury to personnel.

Be sure you understand and follow the procedures and safety precautions contained in the chiller instructions as well as those listed in this guide.

⚠ DANGER

DO NOT VENT refrigerant relief valves within a building. Outlet from rupture disc or relief valve must be vented outdoors in accordance with the latest edition of ANSI/ASHRAE 15 (American National Standards Institute/American Society of Heating, Refrigeration, and Air Conditioning Engineers). The accumulation of refrigerant in an enclosed space can displace oxygen and cause asphyxiation.

PROVIDE adequate ventilation in accordance with ANSI/ASHRAE 15, especially for enclosed and low overhead spaces. Inhalation of high concentrations of vapor is harmful and may cause heart irregularities, unconsciousness, or death. Misuse can be fatal. Vapor is heavier than air and reduces the amount of oxygen available for breathing. Product causes eye and skin irritation. Decomposition products are hazardous.

DO NOT USE OXYGEN to purge lines or to pressurize a chiller for any purpose. Oxygen gas reacts violently with oil, grease, and other common substances.

NEVER EXCEED specified test pressures, VERIFY the allowable test pressure by checking the instruction literature and the design pressures on the equipment nameplate.

DO NOT USE air for leak testing. Use only refrigerant or dry nitrogen.

DO NOT VALVE OFF any safety device.

BE SURE that all pressure relief devices are properly installed and functioning before operating any chiller.

RISK OF INJURY OR DEATH by electrocution. High voltage is present on motor leads even though the motor is not running when a solid-state or inside-delta mechanical starter is used. Open the power supply disconnect before touching motor leads or terminals.

⚠ WARNING

DO NOT WELD OR FLAMECUT any refrigerant line or vessel until all refrigerant *(liquid and vapor)* has been removed from chiller. Traces of vapor should be displaced with dry air or nitrogen and the work area should be well ventilated. *Refrigerant in contact with an open flame produces toxic gases.*

DO NOT USE eyebolts or eyebolt holes to rig chiller sections or the entire assembly.

DO NOT work on high-voltage equipment unless you are a qualified electrician.

DO NOT WORK ON electrical components, including control panels, switches, starters, or oil heater until you are sure ALL POWER IS OFF and no residual voltage can leak from capacitors or solid-state components.

LOCK OPEN AND TAG electrical circuits during servicing. IF WORK IS INTERRUPTED, confirm that all circuits are deenergized before resuming work.

AVOID SPILLING liquid refrigerant on skin or getting it into the eyes. USE SAFETY GOGGLES. Wash any spills from the skin with soap and water. If liquid refrigerant enters the eyes, IMMEDIATELY FLUSH EYES with water and consult a physician.

NEVER APPLY an open flame or live steam to a refrigerant cylinder. Dangerous over pressure can result. When it is necessary to heat refrigerant, use only warm (110 F [43 C]) water.

DO NOT REUSE disposable (nonreturnable) cylinders or attempt to refill them. It is DANGEROUS AND ILLEGAL. When cylinder is emptied, evacuate remaining gas pressure, loosen the collar and unscrew and discard the valve stem. DO NOT INCINERATE.

CHECK THE REFRIGERANT TYPE before adding refrigerant to the chiller. The introduction of the wrong refrigerant can cause damage or malfunction to this chiller.

Operation of this equipment with refrigerants other than those cited herein should comply with ANSI/ASHRAE15 (latest edition). Contact Carrier for further information on use of this chiller with other refrigerants.

DO NOT ATTEMPT TO REMOVE fittings, covers, etc., while chiller is under pressure or while chiller is running. Be sure pressure is at 0 psig (0 kPa) before breaking any refrigerant connection.

CAREFULLY INSPECT all relief devices, rupture discs, and other relief devices AT LEAST ONCE A YEAR. If chiller operates in a corrosive atmosphere, inspect the devices at more frequent intervals.

DO NOT ATTEMPT TO REPAIR OR RECONDITION any relief device when corrosion or build-up of foreign material (rust, dirt, scale, etc.) is found within the valve body or mechanism. Replace the device.

DO NOT install relief devices in series or backwards.

USE CARE when working near or in line with a compressed spring. Sudden release of the spring can cause it and objects in its path to act as projectiles.

⚠ CAUTION

DO NOT STEP on refrigerant lines. Broken lines can whip about and release refrigerant, causing personal injury.

DO NOT climb over a chiller. Use platform, catwalk, or staging. Follow safe practices when using ladders.

USE MECHANICAL EQUIPMENT (crane, hoist, etc.) to lift or move inspection covers or other heavy components. Even if components are light, use mechanical equipment when there is a risk of slipping or losing your balance.

BE AWARE that certain automatic start arrangements CAN ENGAGE THE STARTER, TOWER FAN, OR PUMPS. Open the disconnect *ahead of* the starter, tower fans, or pumps.

USE only repair or replacement parts that meet the code requirements of the original equipment.

DO NOT VENT OR DRAIN waterboxes containing industrial brines, liquid, gases, or semisolids without the permission of your process control group.

DO NOT LOOSEN waterbox cover bolts until the waterbox has been completely drained.

DOUBLE-CHECK that coupling nut wrenches, dial indicators, or other items have been removed before rotating any shafts.

DO NOT LOOSEN a packing gland nut before checking that the nut has a positive thread engagement.

PERIODICALLY INSPECT all valves, fittings, and piping for corrosion, rust, leaks, or damage.

PROVIDE A DRAIN connection in the vent line near each pressure relief device to prevent a build-up of condensate or rain water.

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INTRODUCTION

Prior to initial start-up of the 19XR unit, those involved in the start-up, operation, and maintenance should be thoroughly familiar with these instructions and other necessary job data. This book is outlined so that you may become familiar with the control system before performing start-up procedures. Procedures in this manual are arranged in the sequence required for proper chiller start-up and operation.

⚠ WARNING

This unit uses a microprocessor control system. Do not short or jumper between terminations on circuit boards or modules; control or board failure may result.

Be aware of electrostatic discharge (static electricity) when handling or making contact with circuit boards or module connections. Always touch a chassis (grounded) part to dissipate body electrostatic charge before working inside control center.

Use extreme care when handling tools near boards and when connecting or disconnecting terminal plugs. Circuit boards can easily be damaged. Always hold boards by the edges and avoid touching components and connections.

This equipment uses, and can radiate, radio frequency energy. If not installed and used in accordance with the instruction manual, it may cause interference to radio communications. It has been tested and found to comply with the limits for a Class A computing device pursuant to Subpart J of Part 15 of FCC Rules, which are designed to provide reasonable protection against such interference when operated in a commercial environment. Operation of this equipment in a residential area is likely to cause interference, in which case the user, at his own expense, will be required to take whatever measures may be required to correct the interference.

Always store and transport replacement or defective boards in anti-static shipping bag.

ABBREVIATIONS AND EXPLANATIONS

Frequently used abbreviations in this manual include:

CCN	—	Carrier Comfort Network
CCW	—	Counterclockwise
CW	—	Clockwise
ECDW	—	Entering Condenser Water
ECW	—	Entering Chilled Water
EMS	—	Energy Management System
HGBP	—	Hot Gas Bypass
I/O	—	Input/Output
LCD	—	Liquid Crystal Display
LCDW	—	Leaving Condenser Water
LCW	—	Leaving Chilled Water
LED	—	Light-Emitting Diode
LID	—	Local Interface Device
OLTA	—	Overload Trip Amps
PIC	—	Product Integrated Control
PSIO	—	Processor Sensor Input/Output Module
RLA	—	Rated Load Amps
SCR	—	Silicon Control Rectifier
SI	—	International System of Units
SMM	—	Starter Management Module
TXV	—	Thermostatic Expansion Valve

Words printed in all capital letters or in italics may be viewed on the LID (e.g., LOCAL, CCN, ALARM, etc.).

Words printed in *both* all capital letters and italics can also be viewed on the LID and are parameters (e.g., *CONTROL MODE*, *COMPRESSOR START RELAY*, *ICE BUILD OPTION*, etc.) with associated values (e.g., modes, temperatures, percentages, pressures, on, off, etc.).

Words printed in all capital letters and in a box represent softkeys on the LID control panel (e.g., $\boxed{\text{ENTER}}$, $\boxed{\text{EXIT}}$, $\boxed{\text{INCREASE}}$, $\boxed{\text{QUIT}}$, etc.).

Factory-installed additional components are referred to as options in this manual; factory-supplied but field-installed additional components are referred to as accessories.

The PSIO software version number of your 19XR unit will be located on the PSIO module.

CHILLER FAMILIARIZATION (Fig. 1 and 2)

Chiller Information Plate — The information plate is located on the right side of the chiller control center panel.

System Components — The components include the cooler and condenser heat exchangers in separate vessels, motor-compressor, lubrication package, control center, and motor starter. All connections from pressure vessels have external threads to enable each component to be pressure tested with a threaded pipe cap during factory assembly.

Cooler — This vessel (also known as the evaporator) is located underneath the compressor. The cooler is maintained at lower temperature/pressure so that evaporating refrigerant can remove heat from water flowing through its internal tubes.

Condenser — The condenser operates at a higher temperature/pressure than the cooler and has water flowing through its internal tubes in order to remove heat from the refrigerant.

Motor-Compressor — This component maintains system temperature and pressure differences and moves the heat-carrying refrigerant from the cooler to the condenser.

Control Center — The control center is the user interface for controlling the chiller. It regulates the chiller's capacity as required to maintain proper leaving chilled water temperature. The control center:

- registers cooler, condenser, and lubricating system pressures
- shows chiller operating condition and alarm shutdown conditions
- records the total chiller operating hours
- sequences chiller start, stop, and recycle under microprocessor control
- provides access to other CCN (Carrier Comfort Network) devices

19XR 52 51 473 CQ H 62 –

Description
19XR — High Efficiency Hermetic Centrifugal Liquid Chiller

Cooler Size
10-12 (Frame 1, Length:10 ft)
15-17 (Frame 1, Length:12 ft)
20-22 (Frame 2, Length:10 ft)
30-32 (Frame 3, Length:12 ft)
35-37 (Frame 3, Length:14 ft)
40-42 (Frame 4, Length:12 ft)
45-47 (Frame 4, Length:14 ft)
50-52 (Frame 5, Length:12 ft)
55-57 (Frame 5, Length:14 ft)
60-62 (Frame 6, Length:12 ft)
65-67 (Frame 6, Length:14 ft)
70-72 (Frame 7, Length:14 ft)
75-77 (Frame 7, Length:16 ft)
80-82 (Frame 8, Length:14 ft)
85-87 (Frame 8, Length:16 ft)

Condenser Size
10-12 (Frame 1, Length:10 ft)
15-17 (Frame 1, Length:12 ft)
20-22 (Frame 2, Length:10 ft)
30-32 (Frame 3, Length:12 ft)
35-37 (Frame 3, Length:14 ft)
40-42 (Frame 4, Length:12 ft)
45-47 (Frame 4, Length:14 ft)
50-52 (Frame 5, Length:12 ft)
55-57 (Frame 5, Length:14 ft)
60-62 (Frame 6, Length:12 ft)
65-67 (Frame 6, Length:14 ft)
70-72 (Frame 7, Length:14 ft)
75-77 (Frame 7, Length:16 ft)
80-82 (Frame 8, Length:14 ft)
85-87 (Frame 8, Length:16 ft)

Compressor Code
First Digit Indicates Compressor Frame Size

Special Order Indicator
– — Standard
S — Special Order

Motor Voltage Code

Code	Volts-Phase-Hertz
60	200-3-60
61	230-3-60
62	380-3-60
63	416-3-60
64	460-3-60
65	575-3-60
66	2400-3-60
67	3300-3-60
68	4160-3-60
69	6900-3-60
50	230-3-50
51	346-3-50
52	400-3-50
53	3000-3-50
54	3300-3-50
55	6300-3-50

Motor Efficiency Code
H — High Efficiency
S — Standard Efficiency

Motor Code

BD	CB	DB	EH
BE	CC	DC	EJ
BF	CD	DD	EK
BG	CE	DE	EL
BH	CL	DF	EM
	CM	DG	EN
	CN	DH	EP
	CP	DJ	
	CQ		

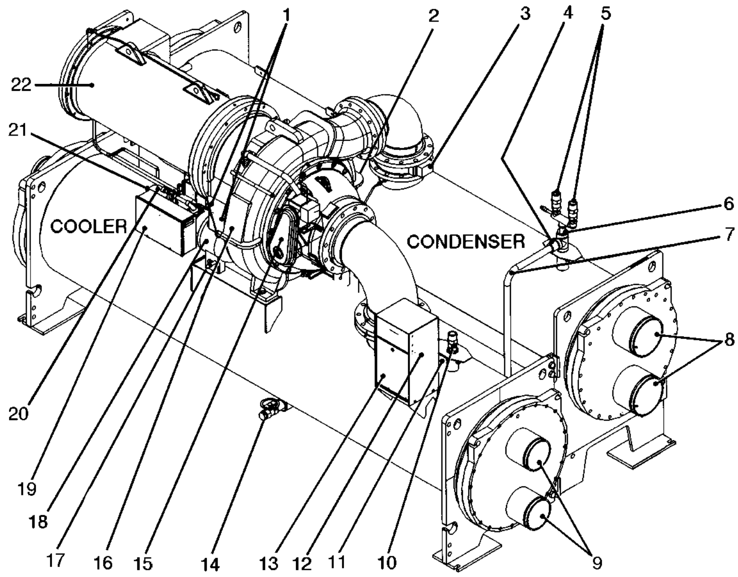
MODEL NUMBER NOMENCLATURE

94 20 J 43758

Year of Manufacture — 94
Week of Year — 20
Place of Manufacture — J
Unique Number — 43758

SERIAL NUMBER BREAKDOWN

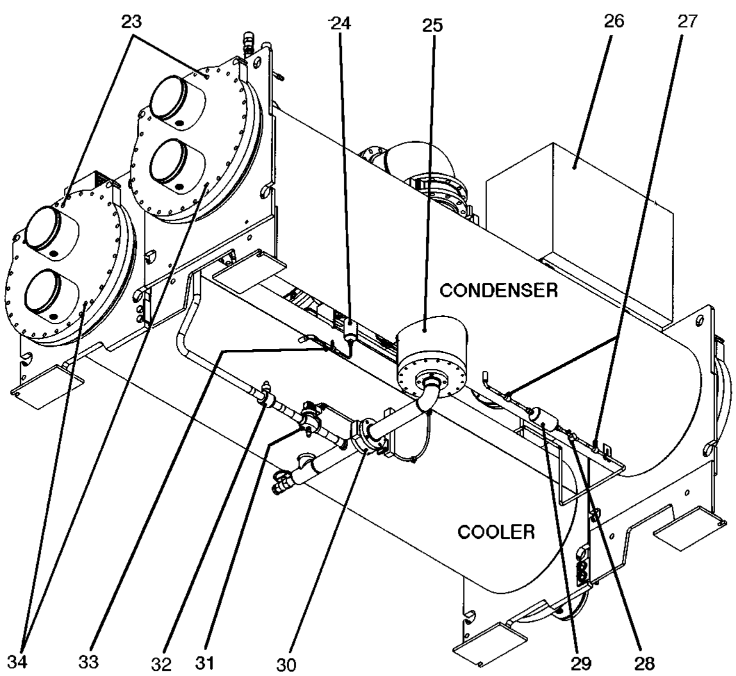
Fig. 1 — 19XR Identification



FRONT TOP VIEW

LEGEND

1 — Oil Level Sight Glass
2 — Diffuser Actuator (Hidden/19XR5 Only)
3 — Discharge Isolation Valve
4 — Condenser Pumpout Connection
5 — Condenser Safety Relief Valves
6 — Three-Way Condenser Relief Valve
7 — Hot Gas Bypass Line
8 — Condenser Waterbox Nozzles
9 — Cooler Waterbox Nozzles
10 — Cooler Safety Relief Valves
11 — Cooler Pumpout Connection
12 — Machine Identification Nameplate
13 — Control Panel
14 — Refrigerant Charging Valve
15 — Guide Vane Actuator
16 — Compressor/Transmission
17 — Oil Drain/Charging Valve
18 — Oil Pump
19 — Auxiliary Power Panel
20 — Oil Filter Isolation Valve
21 — Oil Filter
22 — Motor



BOTTOM REAR VIEW

23 — Waterbox Vents
24 — Oil Reclaim Filter
25 — Float Chamber
26 — Unit-Mounted Starter
27 — Refrigerant Filter/Drier Isolation Valves
28 — Sight Glass/Moisture Indicator
29 — Refrigerant Filter/Drier
30 — Cooler Liquid Line Isolation Valve
31 — Hot Gas Bypass Valve (Option)
32 — Hot Gas Bypass Isolation Valve (Option)
33 — Oil Reclaim Filter Isolation Valve
34 — Waterbox Vents

Fig. 2 — Typical 19XR Components

Factory-Mounted Starter (Optional) — The starter allows for the proper start and disconnect of electrical energy for the compressor-motor, oil pump, oil heater, and control panels.

Storage Vessel (Optional) — There are 2 sizes of storage vessels available. The vessels have double relief valves, a magnetically-coupled dial-type refrigerant level gage, a one-inch FPT drain valve, and a ½-in. male flare vapor connection for the pumpout unit.

NOTE: If a storage vessel is not used at the jobsite, factory-installed isolation valves on the chiller may be used to isolate the chiller charge in either the cooler or condenser. An optional pumpout system is used to transfer refrigerant from vessel to vessel.

REFRIGERATION CYCLE

The compressor continuously draws refrigerant vapor from the cooler at a rate set by the amount of guide vane opening. As the compressor suction reduces the pressure in the cooler, the remaining refrigerant boils at a fairly low temperature (typically 38 to 42 F [3 to 6 C]). The energy required for boiling is obtained from the water flowing through the cooler tubes. With heat energy removed, the water becomes cold enough to use in an air conditioning circuit or for process liquid cooling.

After taking heat from the water, the refrigerant vapor is compressed. Compression adds still more heat energy, and the refrigerant is quite warm (typically 98 to 102 F [37 to 40 C]) when it is discharged from the compressor into the condenser.

Relatively cool (typically 65 to 90 F [18 to 32 C]) water flowing into the condenser tubes removes heat from the refrigerant and the vapor condenses to liquid.

The liquid refrigerant passes through orifices into the FLASC (Flash Subcooler) chamber (Fig. 3). Since the FLASC chamber is at a lower pressure, part of the liquid refrigerant flashes to vapor, thereby cooling the remaining liquid. The FLASC vapor is recondensed on the tubes which are cooled by entering condenser water. The liquid drains into a float chamber between the FLASC chamber and cooler. Here a float valve forms a liquid seal to keep FLASC chamber vapor from entering the cooler. When liquid refrigerant passes through the valve, some of it flashes to vapor in the reduced pressure on the cooler side. In flashing, it removes heat from the remaining liquid. The refrigerant is now at a temperature and pressure at which the cycle began.

MOTOR AND LUBRICATING OIL COOLING CYCLE

The motor and the lubricating oil are cooled by liquid refrigerant taken from the bottom of the condenser vessel (Fig. 3). Refrigerant flow is maintained by the pressure differential that exists due to compressor operation. After the refrigerant flows past an isolation valve, an in-line filter, and a sight glass/moisture indicator, the flow is split between the motor cooling and oil cooling systems.

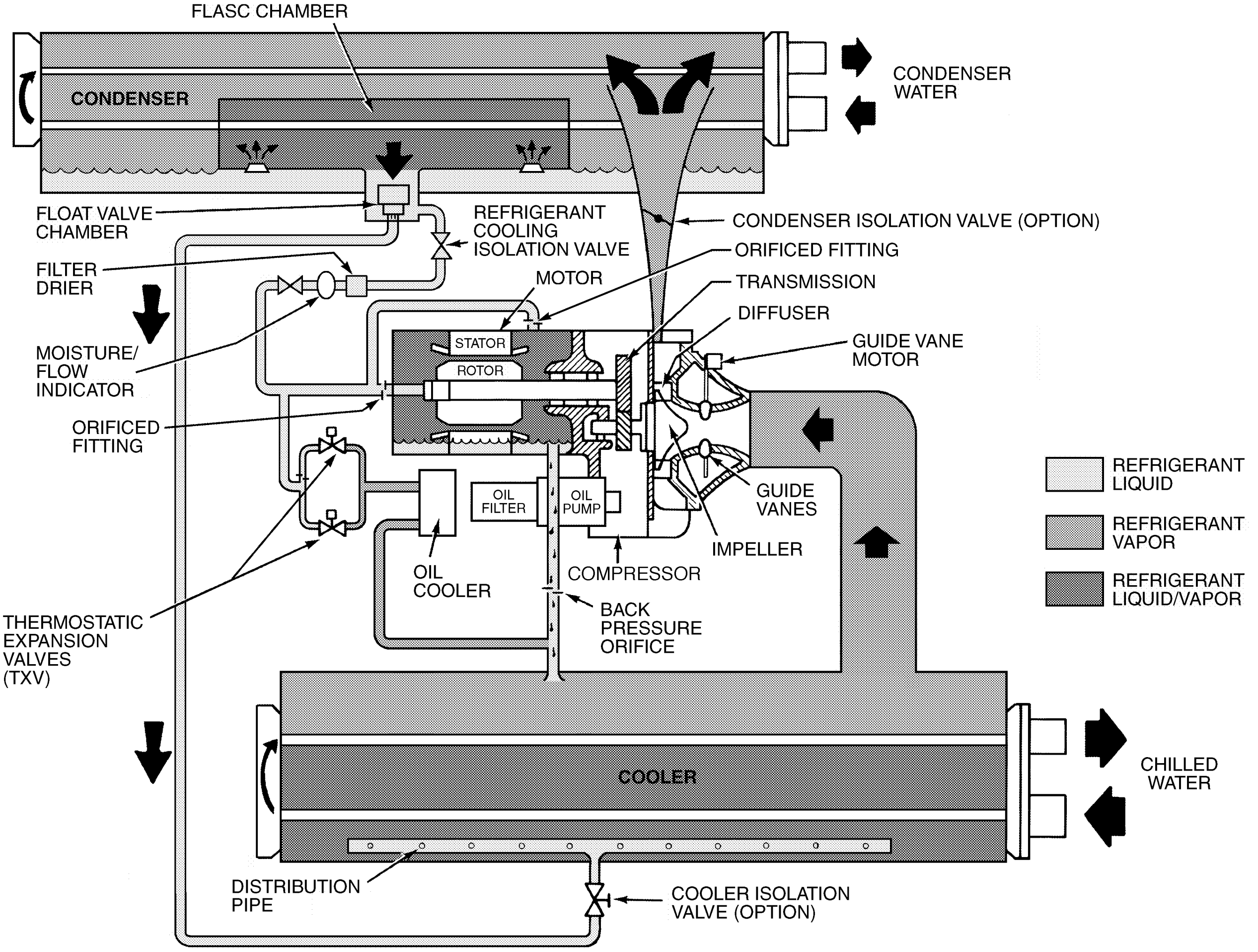


Fig. 3 — Refrigerant Motor Cooling and Oil Cooling Cycles

Flow to the motor cooling system passes through an orifice and into the motor. Once past the orifice, the refrigerant is directed over the motor by a spray nozzle. The refrigerant collects in the bottom of the motor casing and is then drained back into the cooler through the motor refrigerant drain line. A back pressure valve or an orifice in this line maintains a higher pressure in the motor shell than in the cooler/oil sump. The motor is protected by a temperature sensor imbedded in the stator windings. An increase in motor winding temperature past the motor override set point overrides the temperature capacity control to hold, and if the motor temperature rises 10° F (5.5° C) above this set point, closes the inlet guide vanes. If the temperature rises above the safety limit, the compressor shuts down.

Refrigerant that flows to the oil cooling system is regulated by thermostatic expansion valves (TXVs). The TXVs regulate flow into the oil/refrigerant plate and frame-type heat exchanger (the oil cooler in Fig. 3). The expansion valve bulbs control oil temperature to the bearings. The refrigerant leaving the oil cooler heat exchanger then returns to the chiller cooler.

LUBRICATION CYCLE

Summary — The oil pump, oil filter, and oil cooler make up a package located partially in the transmission casting of the compressor-motor assembly. The oil is pumped into a filter assembly to remove foreign particles and is then forced into an oil cooler heat exchanger where the oil is cooled to proper operational temperatures. After the oil cooler, part of the flow is directed to the gears and the high speed shaft bearings; the remaining flow is directed to the motor shaft bearings. Oil drains into the transmission oil sump to complete the cycle (Fig. 4).

Details — Oil is charged into the lubrication system through a hand valve. Two sight glasses in the oil reservoir permit oil level observation. Normal oil level is between the middle of the upper sight glass and the top of the lower sight glass when the compressor is shut down. The oil level should be visible in at least one of the 2 sight glasses during operation. Oil sump temperature is displayed on the LID (Local Interface Device) default screen. During compressor operation, the oil sump temperature ranges between 125 to 150 F (52 to 66 C).

The oil pump suction is fed from the oil reservoir. An oil pressure relief valve maintains 18 to 25 psid (124 to 172 kPad) differential pressure in the system at the pump discharge. This differential pressure can be read directly from the LID default screen. The oil pump discharges oil to the oil filter assembly. This filter can be closed to permit removal of the filter without draining the entire oil system (see Maintenance sections, pages 61 to 66, for details). The oil is then piped to the oil cooler heat exchanger. The oil cooler uses refrigerant from the condenser as the coolant. The refrigerant cools the oil to a temperature between 120 and 140 F (49 to 60 C).

As the oil leaves the oil cooler, it passes the oil pressure transducer and the thermal bulb for the refrigerant expansion valve on the oil cooler. The oil is then divided. Part of the oil flows to the thrust bearing, forward pinion bearing, and gear spray. The rest of the oil lubricates the motor shaft bearings and the rear pinion bearing. The oil temperature is measured in the bearing housing as it leaves the thrust and forward journal bearings. The oil then drains into the oil reservoir at the base of the compressor. The PIC (Product Integrated Control) measures the temperature of the oil in the sump and maintains the temperature during shutdown (see Oil Sump Temperature Control section, page 31). This temperature is read on the LID default screen.

During the chiller start-up, the PIC energizes the oil pump and provides 45 seconds of prelubrication to the bearings after pressure is verified before starting the compressor. During shutdown, the oil pump will run for 60 seconds to post-lubricate after the compressor shuts down. The oil pump can also be energized for testing purposes during a Control Test.

Ramp loading can slow the rate of guide vane opening to minimize oil foaming at start-up. If the guide vanes open quickly, the sudden drop in suction pressure can cause any refrigerant in the oil to flash. The resulting oil foam cannot be pumped efficiently; therefore, oil pressure falls off and lubrication is poor. If oil pressure falls below 15 psid (103 kPad) differential, the PIC will shut down the compressor.

If the controls are subject to a power failure that lasts more than 3 hours, the oil pump will be energized periodically when the power is restored. This helps to eliminate refrigerant that has migrated to the oil sump during the power failure. The controls energize the pump for 60 seconds every 30 minutes until the chiller is started.

Oil Reclaim System — The oil reclaim system returns oil lost from the compressor housing back to the oil reservoir by recovering the oil from 2 areas on the chiller. The guide vane housing is the primary area of recovery. Oil is also recovered by skimming it from the operating refrigerant level in the cooler vessel.

PRIMARY OIL RECOVERY MODE — Oil is normally recovered through the guide vane housing on the chiller. This is possible because oil is normally entrained with refrigerant in the chiller. As the compressor pulls the refrigerant up from the cooler into the guide vane housing to be compressed, the oil normally drops out at this point and falls to the bottom of the guide vane housing where it accumulates. Using discharge gas pressure to power an eductor, the oil is drawn from the housing and is discharged into the oil reservoir.

SECONDARY OIL RECOVERY METHOD — The secondary method of oil recovery is significant under light load conditions, when the refrigerant going up to the compressor suction does not have enough velocity in which to bring oil along. Under these conditions, oil collects in a greater concentration at the top level of the refrigerant in the cooler. This oil and refrigerant mixture is skimmed from the side of the cooler and is then drawn up to the guide vane housing. There is a filter in this line. Because the guide vane housing pressure is much lower than the cooler pressure, the refrigerant boils off, leaving the oil behind to be collected by the primary oil recovery method.

STARTING EQUIPMENT

The 19XR requires a motor starter to operate the centrifugal hermetic compressor motor, the oil pump, and various auxiliary equipment. The starter is the main field wiring interface for the contractor.

See Carrier Specification Z-375 for specific starter requirements. All starters must meet these specifications in order to properly start and satisfy mechanical safety requirements. Starters may be supplied as separate, free-standing units or may be mounted directly on the chiller (unit mounted) for low-voltage units only.

Three separate circuit breakers are inside the starter. Circuit breaker CB1 is the compressor motor circuit breaker. The disconnect switch on the starter front cover is connected to this breaker. Circuit breaker CB1 supplies power to the compressor motor.

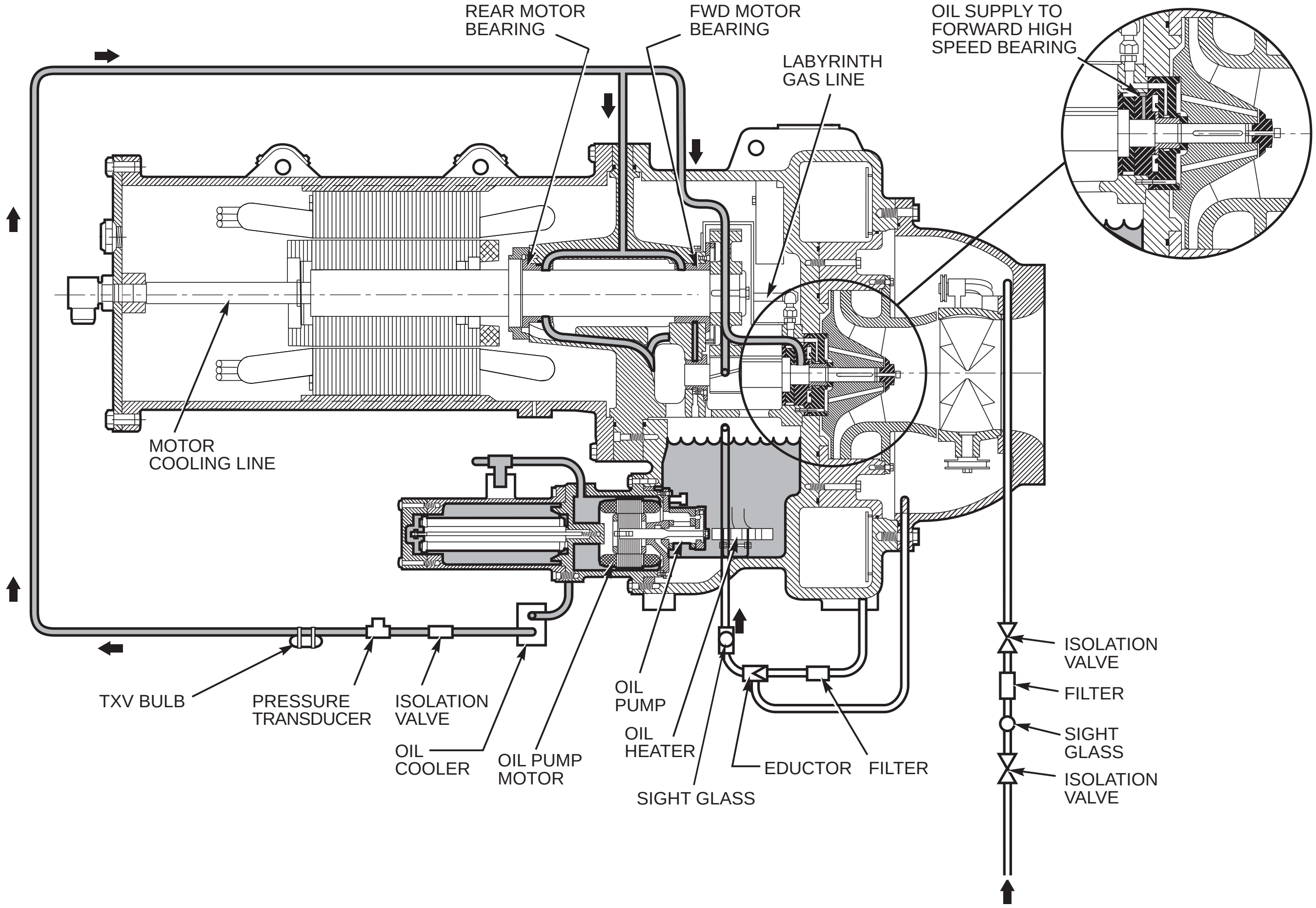


Fig. 4 — Lubrication System

⚠ WARNING

The main circuit breaker (CB1) on the front of the starter disconnects the main motor current only. Power is still energized for the other circuits. Two more circuit breakers inside the starter must be turned off to disconnect power to the oil pump, PIC controls, and oil heater.

Circuit breaker CB2 supplies power to the control center, oil heater, and portions of the starter controls.

Circuit breaker CB3 supplies power to the oil pump. Both CB2 and CB3 are wired in parallel with CB1 so that power is supplied to them if the CB1 disconnect is open.

All starters must include a Carrier control module called the Starter Management Module (SMM). This module controls and monitors all aspects of the starter. See the Controls section on page 10 for additional SMM information. All starter replacement parts are supplied by the starter manufacturer.

Unit-Mounted Solid-State Starter (Optional) — The 19XR chiller may be equipped with a solid-state, reduced-voltage starter (Fig. 5 and 6). This starter's primary function is to provide on-off control of the compressor motor. This type of starter reduces the peak starting torque, reduces the motor inrush current, and decreases mechanical shock. This capability is summed up by the phrase ″soft starting.'' The solid-state starter's manufacturer name is located inside the starter access door.

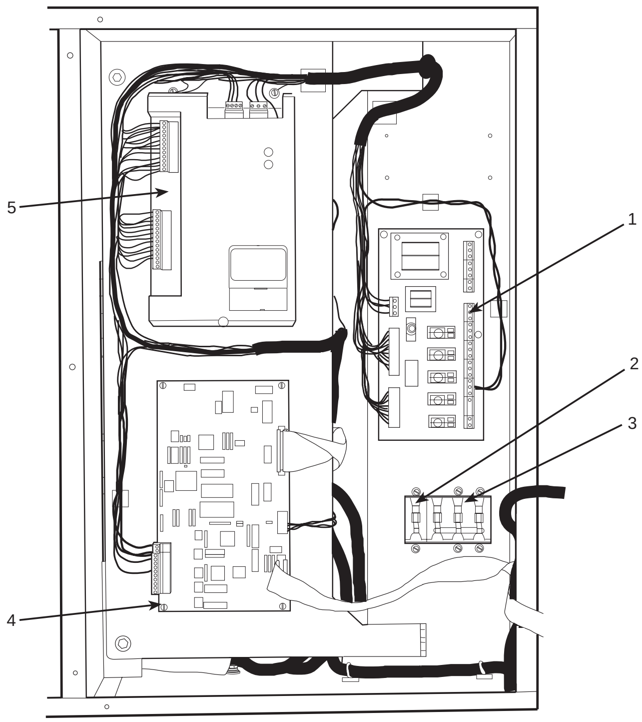
A solid-state, reduced-voltage starter operates by reducing the starting voltage. The starting torque of a motor at full voltage is typically 125% to 175% of the running torque. When the voltage and the current are reduced at start-up, the starting torque is reduced as well. The object is to reduce the starting voltage to just the voltage necessary to develop the torque required to get the motor moving. The voltage is reduced by silicon controlled rectifiers (SCRs). The voltage and current are then ramped up in a desired period of time. Once full voltage is reached, a bypass contactor is energized to bypass the SCRs.

⚠ WARNING

When voltage is supplied to the solid-state circuitry, the heat sinks in the starter are at line voltage. Do not touch the heat sinks while voltage is present or serious injury will result.

There is a display on the front of the Benshaw, Inc., solid-state starters that is useful for troubleshooting and starter check-out. The display indicates:

- voltage to the SCRs
- SCR control voltage
- power indication
- proper phasing for rotation
- start circuit energized
- over-temperature
- ground fault
- current unbalance
- run state



LEGEND
1 — Relay Card
2 — Circuit Breaker 2 (CB2): Machine Control and Heater Power
3 — Circuit Breaker 3 (CB3): Oil Pump Power
4 — Central Processing Unit (CPU) Card
5 — Starter Management Module (SMM)

Fig. 5 — Solid-State Starter Box, Internal View

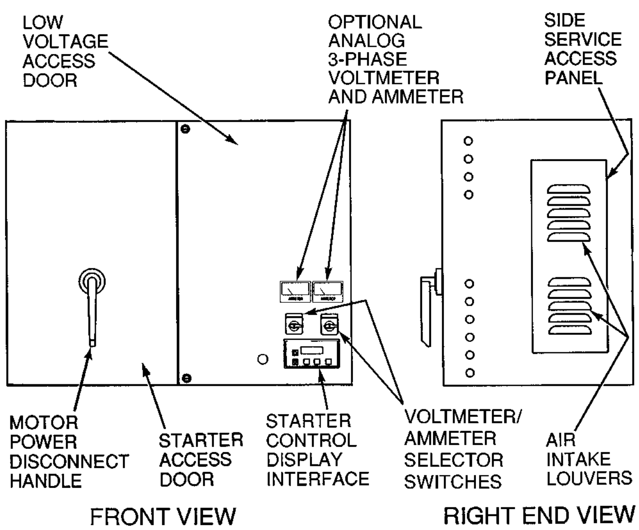


Fig. 6 — Typical Starter External View (Solid-State Starter Shown)

Unit-Mounted Wye-Delta Starter (Optional) — The 19XR chiller may be equipped with a wye-delta starter mounted on the unit. This starter is intended for use with low-voltage motors (under 600 v). It reduces the starting current inrush by connecting each phase of the motor windings into a wye configuration. This occurs during the starting period when the motor is accelerating up to speed. After a time delay and once the motor is up to speed, the starter automatically connects the phase windings into a delta configuration.

CONTROLS

Definitions

ANALOG SIGNAL — *An analog signal* varies in proportion to the monitored source. It quantifies values between operating limits. (Example: A temperature sensor is an analog device because its resistance changes in proportion to the temperature, generating many values.)

DISCRETE SIGNAL — *A discrete signal* is a 2-position representation of the value of a monitored source. (Example: A switch produces a discrete signal indicating whether a value is above or below a set point or boundary by generating an on/off, high/low, or open/closed signal.)

VOLATILE MEMORY — *Volatile memory* is memory incapable of being sustained if power is lost and subsequently restored.

⚠ CAUTION

The memory of the PSIO module is volatile. If the battery in a module is removed or damaged, all programming will be lost.

General — The 19XR hermetic centrifugal liquid chiller contains a microprocessor-based control center that monitors and controls all operations of the chiller (see Fig. 7). The microprocessor control system matches the cooling capacity of the chiller to the cooling load while providing state-of-the-art chiller protection. The system controls cooling load within the set point plus the deadband by sensing the leaving chilled water or brine temperature and regulating the inlet guide vane via a mechanically-linked actuator motor. The guide vane is a variable flow pre-whirl assembly that controls the refrigeration effect in the cooler by regulating the amount of refrigerant vapor flow into the compressor. An increase in guide vane opening increases capacity. A decrease in guide vane opening decreases capacity. The microprocessor-based control center protects the chiller by monitoring the digital and analog inputs and executing capacity overrides or safety shutdowns, if required.

PIC System Components — The chiller control system is called the PIC (Product Integrated Control). See Table 1. The PIC controls the operation of the chiller by monitoring all operating conditions. The PIC can diagnose a problem and let the operator know what the problem is and what to check. It promptly positions the guide vanes to maintain leaving chilled water temperature. It can interface with auxiliary equipment such as pumps and cooling tower fans to turn them on when required. It continually checks all safeties to prevent any unsafe operating condition. It also regulates the oil heater while the compressor is off and regulates the hot gas bypass valve, if installed.

The PIC can interface with the Carrier Comfort Network (CCN) if desired. It can communicate with other PIC-equipped chillers and other CCN devices.

The PIC consists of 3 modules housed inside 3 major components. The component names and corresponding control voltages are listed below (also see Table 1):

- control center
  — all extra low-voltage wiring (24 v or less)
- power panel
  — 230 or 115 v control voltage (per job requirement)
  — up to 600 v for oil pump power
- starter cabinet
  — chiller power wiring (per job requirement)

Table 1 — Major PIC Components and Panel Locations*

PIC COMPONENT	PANEL LOCATION
Processor/Sensor Input/Output Module (PSIO)	Control Center
Starter Management Module (SMM)	Starter Cabinet
Local Interface Device (LID)	Control Center
Chiller Control Module (CCM)	Control Center
6-Pack Relay Board	Control Center
8-Input Modules (Optional)	Control Center
Differential Oil Pressure/Power Supply Module	Control Center
Oil Heater Contactor (1C)	Power Panel
Oil Pump Contactor (2C)	Power Panel
Hot Gas Bypass Relay (3C) (Optional)	Power Panel
Control Transformers (T1, T2, T3, T4)	Power Panel
Control and Oil Heater Voltage Selector (S1)	Power Panel
Temperature Sensors	See Fig. 7
Pressure Transducers	See Fig. 7

*See Fig. 5 and Fig. 7-11.

PROCESSOR/SENSOR INPUT/OUTPUT MODULE (PSIO) — The PSIO is the brain of the PIC. This module contains all the operating software needed to control the chiller. The 19XR uses 4 pressure transducers and 8 thermistors to sense pressures and temperatures (Fig. 8 and 9). These are connected to the PSIO module. The PSIO also provides outputs to the guide vane actuator, oil pump, oil heater, hot gas bypass (optional), and alarm contact. The PSIO communicates with the LID, the SMM, and the optional 8-input modules for user interface and starter management.

STARTER MANAGEMENT MODULE (SMM) — This module is located in the starter cabinet. This module initiates PSIO commands for starter functions such as start/stop of the compressor, condenser, chilled water pumps, tower fan, spare alarm contacts, and the shunt trip. The SMM monitors starter inputs such as flow switches, line voltage, remote start contact, spare safety, condenser high pressure, oil pump interlock, motor current signal, starter 1M and run contacts, and kW transducer input (optional). The SMM contains logic capable of safely shutting down the chiller if communications with the PSIO are lost.

LOCAL INTERFACE DEVICE (LID) — The LID is mounted to the control center (Fig. 10 and 11) and allows the operator to interface with the PSIO or other CCN devices. It is the input center for all local chiller set points, schedules, set-up functions, and options. The LID has a STOP button, an alarm light, 4 buttons for logic inputs, and a backlit display. The backlight will automatically turn off after 15 minutes of non-use. The functions of the 4 buttons or ‘‘softkeys’’ are menu driven and are shown on the display directly above the softkeys.

The viewing angle on the LID display can be adjusted for optimum viewing. Access the adjustment through the hole on the right side of the control panel next to the LID.

CHILLER CONTROL MODULE (CCM) — This module located in the control panel receives analog signals from the impeller discharge monitor transducer as an indication of correct diffuser ring positioning. The signals are converted to digital form and routed to the PSIO for processing.

6-PACK RELAY BOARD — This device is a cluster of 6 pilot relays located in the control center. It is energized by the PSIO for the oil pump, oil heater, alarm, optional hot gas bypass relay, and motor cooling solenoid.

8-INPUT MODULES — These modules, which have 8 spare inputs each, are optional, and there can be 2 per chiller. One module is factory-installed in the control center panel when ordered. The 8-input modules are used whenever chilled water reset, demand reset, or reading a spare sensor is required. The sensors and/or 4 to 20 mA signals are field-installed.

The spare temperature sensors must have the same temperature/resistance curve as the other temperature sensors on this unit. These sensors are 5,000 ohm at 75 F (25 C).

OIL HEATER CONTACTOR (1C) — This contactor is located in the power panel (Fig. 10 and 11) and operates the heater at either 115 or 230 v. It is controlled by the PIC to maintain oil temperature during chiller shutdown.

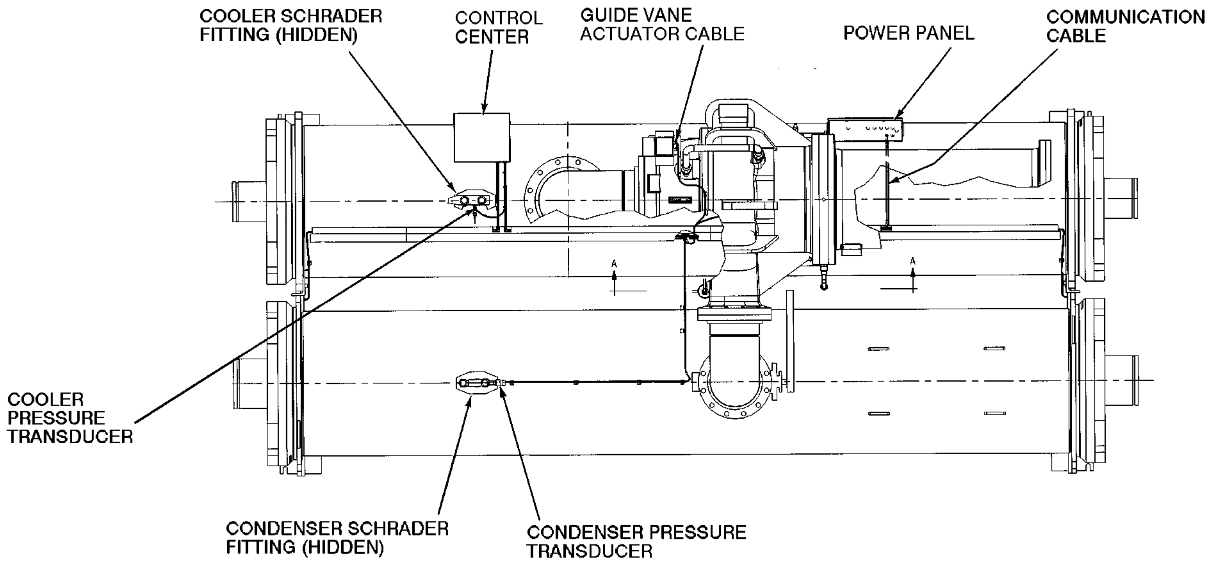
OIL PUMP CONTACTOR (2C) — This contactor is located in the power panel. It operates all 200 to 575-v oil pumps. The PIC energizes the contactor to turn on the oil pump as necessary.

HOT GAS BYPASS CONTACTOR RELAY (3C) (Optional) — This relay, located in the power panel, controls the opening of the hot gas bypass valve. The PIC energizes the relay during low load, high lift conditions.

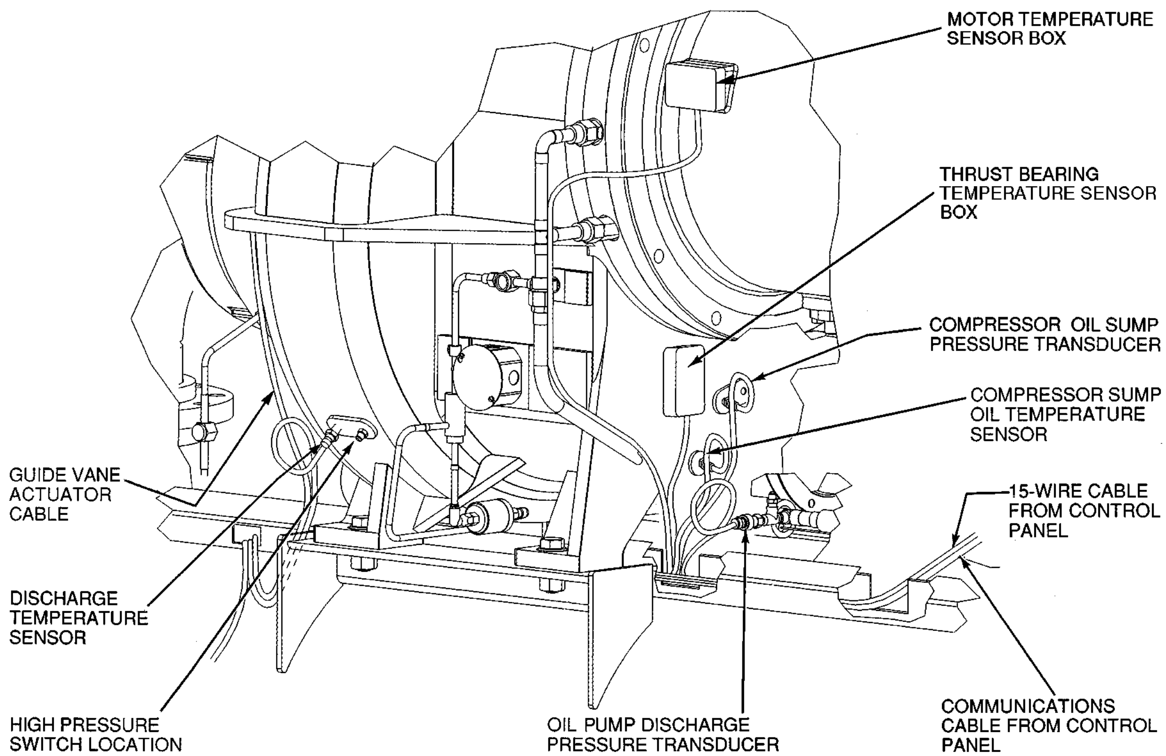
CONTROL TRANSFORMERS (T1, T2, T3 [compressor frame 5 only], T4) — These transformers convert incoming control voltage to either 21 vac power for the PSIO and options modules, or 24 vac power for the 3 power panel contactor relays. The transformers are located in the power panel.

OIL DIFFERENTIAL PRESSURE/POWER SUPPLY MODULE — This module, located in the control center, provides 5 vdc power for the transducers and LID backlight. This module outputs the difference between two pressure transducer input signals. The module subtracts oil supply pressure from transmission sump pressure and outputs the difference as an oil differential pressure signal to the PSIO. The PSIO converts this signal to differential oil pressure. To calibrate this reading, refer to the Troubleshooting, Checking Transducers section.

CONTROL AND OIL HEATER VOLTAGE SELECTOR (S1) — This voltage selector allows you to use either the 115 v or 230 v control power coming into the power panel. The switch is set to the voltage used at the jobsite.



TOP VIEW



VIEW A − A (COMPRESSOR DETAIL)

Fig. 7 — 19XR Controls and Sensor Locations

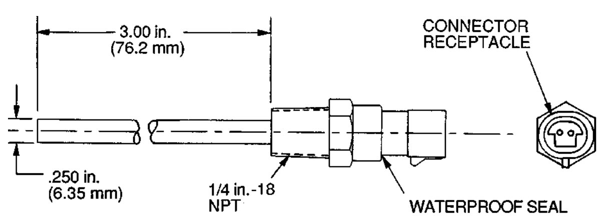


Fig. 8 — Control Sensors (Temperature)

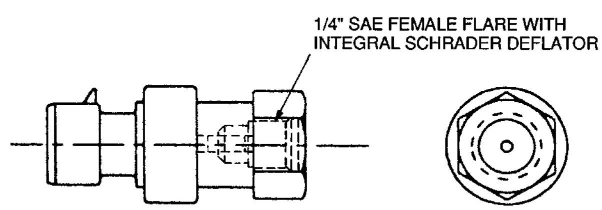
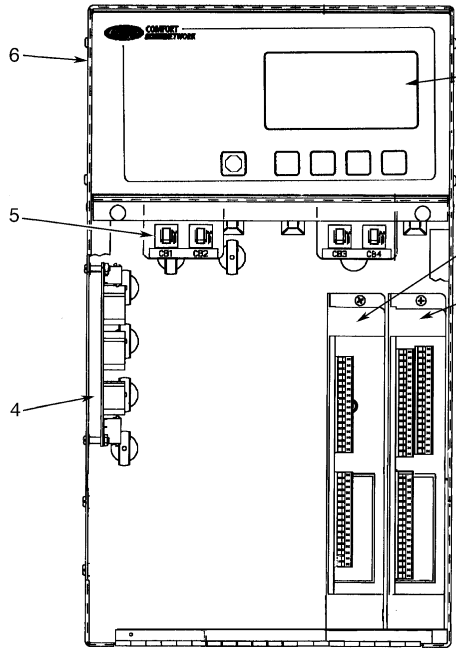
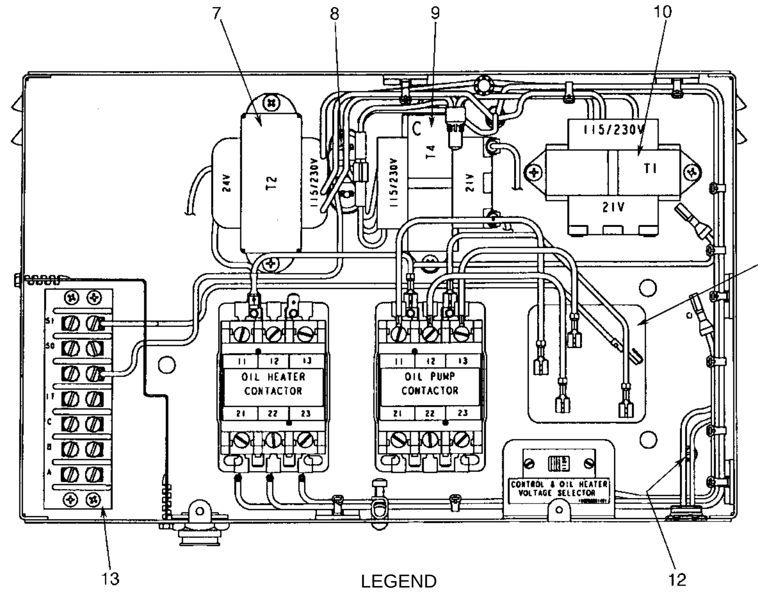


Fig. 9 — Control Sensors (Pressure Transducer, Typical)

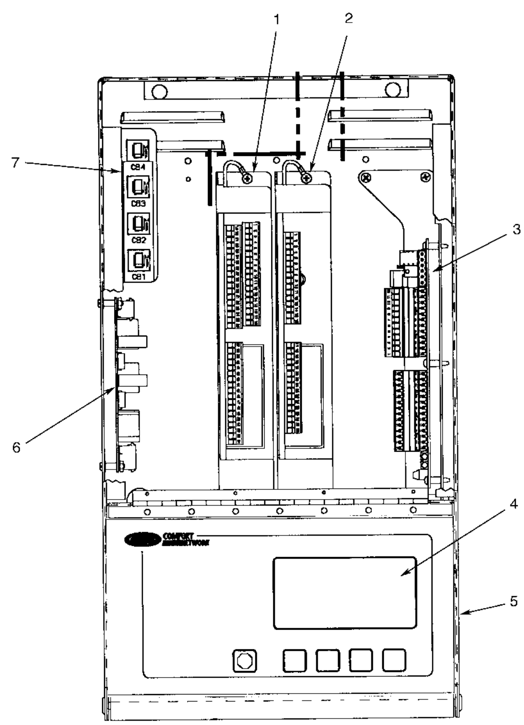


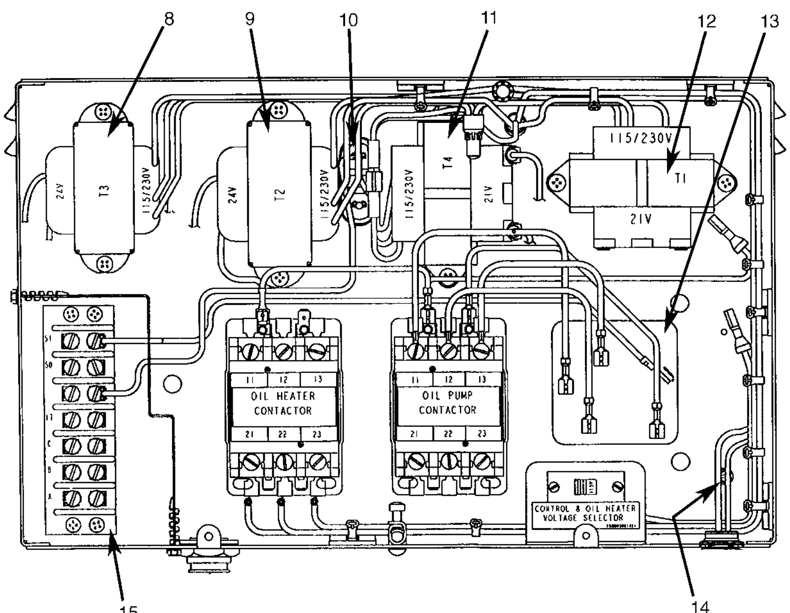


LEGEND

1 — LID (Local Interface Device)
2 — Optional 8-Input Module (One of 2 Available)
3 — PSIO (Processor Sensor Input/Output Module)
4 — 6-Pack Relay Board
5 — Circuit Breakers (4)
6 — Oil Differential Pressure/Power Supply Module (Hidden — Behind LID)
7 — T2 — 24 VAC Power Transformer for Hot Gas Bypass Relay, Oil Pump Relay, Oil Heater Relay, Motor Cooling Solenoid, Oil Reclaim Solenoid
8 — Oil Pressure Switch
9 — T4 — 24 VAC, Optional 8-Input Module Transformer
10 — T1 — 24 VAC, Control Center Transformer
11 — 3C Hot Gas Bypass Relay (Optional) Location
12 — Guide Vane Actuator Motor Wiring
13 — Factory Terminal Connections

Fig. 10 — Control and Power Panels with Options (Compressor Frames 2-4)





LEGEND

1 — PSIO (Processor Sensor Input/Output Module)
2 — Optional 8-Input Module (One of 2 Available)
3 — CCM — Chiller Control Module
4 — LID (Local Interface Device)
5 — Oil Differential Pressure/Power Supply Module (Hidden — Behind LID)
6 — 6-Pack Relay Board
7 — Circuit Breakers (4)
8 — T3 — 24 VAC Chiller Control Module Transformer
9 — T2 — 24 VAC Power Transformer for Hot Gas Bypass Relay, Oil Pump Relay, Oil Heater Relay, Motor Cooling Solenoid, Oil Reclaim Solenoid
10 — Oil Pressure Switch
11 — T4 — 24 VAC, Optional 8-Input Module Transformer
12 — T1 — 24 VAC, Control Center Transformer
13 — 3C Hot Gas Bypass Relay (Optional) Location
14 — Guide Vane and Diffuser Actuator Motor Wiring
15 — Factory Terminal Connections

Fig. 11 — Control and Power Panels with Options (Compressor Frame 5)

LID Operation and Menus (Fig. 12-18)

GENERAL

- The LID display automatically reverts to the default screen after 15 minutes if no softkey activity takes place and if the chiller is not in the pumpdown mode (Fig. 12).
- If a screen other than the default screen is displayed on the LID, the name of that screen is in the upper right corner (Fig. 13).
- The LID may be set to display either English or SI units. Use the LID configuration screen (accessed from the Service menu) to change the units. See the Service Operation section, page 38.
- Local Operation — The PIC can be placed in local operating mode by pressing the LOCAL softkey. The PIC then accepts commands from the LID only and uses the Local Time Schedule to determine chiller start and stop times.
- CCN Operation — The PIC can be placed in the CCN operating mode by pressing the CCN softkey. The PIC then accepts modifications from any CCN interface or module (with the proper authority), as well as from the LID. The PIC uses the CCN time schedule to determine start and stop times.

ALARMS AND ALERTS — An alarm shuts down the compressor. An alert does not shut down the compressor, but it notifies the operator that an unusual condition has occurred. An alarm (*) or alert (!) is indicated on the STATUS screens on the far right field of the LID display screen.

Alarms are indicated when the control center alarm light (!) flashes. The primary alarm message is displayed on the default screen. An additional, secondary message and troubleshooting information are sent to the ALARM HISTORY table.

When an alarm is detected, the LID default screen will freeze (stop updating) at the time of alarm. The freeze enables the operator to view the chiller conditions at the time of alarm. The STATUS tables will show the updated information. Once all alarms have been cleared (by pressing the RESET softkey), the default LID screen will return to normal operation.

LID MENU ITEMS — To perform any of the operations described below, the PIC must be powered up and have successfully completed its self test. The self test takes place automatically, after power-up.

Press the MENU softkey to view the list of menu structures: STATUS , SCHEDULE , SETPOINT , and SERVICE .

- The STATUS menu allows viewing and limited calibration or modification of control points and sensors, relays and contacts, and the options board.
- The SCHEDULE menu allows viewing and modification of the local and CCN time schedules.
- The SETPOINT menu allows set point adjustments, such as the entering chilled water and leaving chilled water set points.

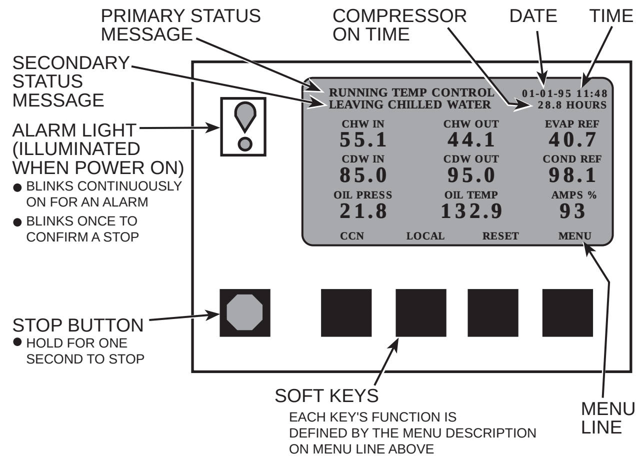


Fig. 12 — LID Default Screen

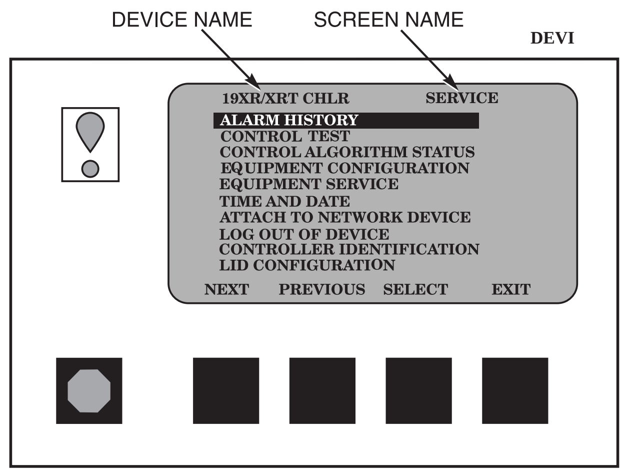


Fig. 13 — LID Service Screen

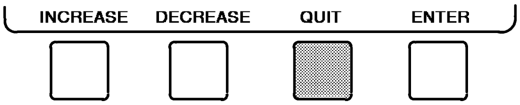
- The SERVICE menu can be used to view or modify information on the Alarm History, Control Test, Control Algorithm Status, Equipment Configuration, Equipment Service, Time and Date, Attach to Network Device, Log Out of Network Device, Controller Identification, and LID Configuration screens.

For more information on the menu structures, refer to Fig. 15.

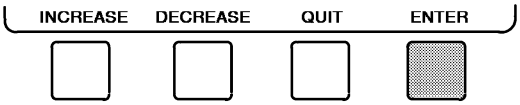
Press the softkey that corresponds to the menu structure to be viewed : STATUS , SCHEDULE , SETPOINT , or SERVICE . To view or change parameters within any of these menu structures, use the NEXT and PREVIOUS softkeys to scroll down to the desired item or table. Use the SELECT softkey to select that item. The softkey choices that then appear depend on the selected table or menu. The softkey choices and their functions are described below.

BASIC LID OPERATIONS (Using the Softkeys) — To perform any of the operations described below, the PIC must be powered up and have successfully completed its self test.

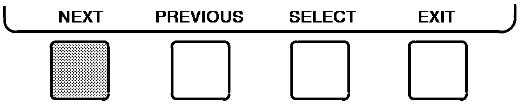
- Press QUIT to leave the selected decision or field without saving any changes.



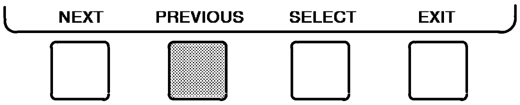
- Press ENTER to leave the selected decision or field and save changes.



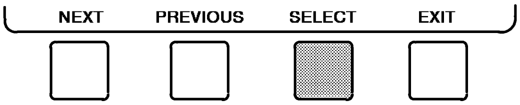
- Press NEXT to scroll the cursor bar down in order to highlight a point or to view more points below the current screen.



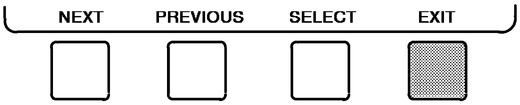
- Press PREVIOUS to scroll the cursor bar up in order to highlight a point or to view points above the current screen.



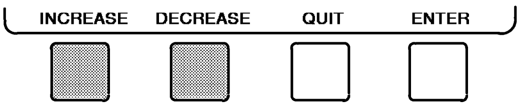
- Press SELECT to view the next screen level (highlighted with the cursor bar), or to override (if allowable) the highlighted point value.



- Press EXIT to return to the previous screen level.

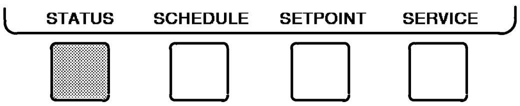


- Press INCREASE or DECREASE to change the highlighted point value.

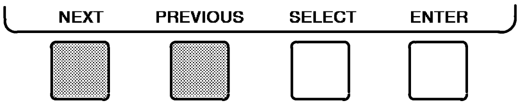


TO VIEW POINT STATUS (Fig. 14) — The point status tables show the actual value of all of the temperatures, pressures, relays, and actuators sensed and controlled by the PIC.

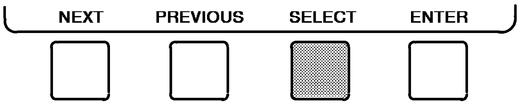
1. On the menu screen, press STATUS to view the list of point status tables.



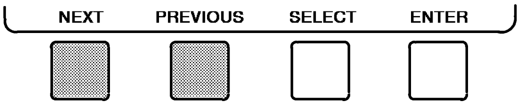
2. Press NEXT or PREVIOUS to highlight the desired status table. The list of tables is:
   - STATUS01 — Status of control points and sensors
   - STATUS02 — Status of relays and contacts
   - STATUS03 — Status of both optional 8-input modules and sensors



3. Press SELECT to view the desired point status table.



4. On the point status table, press NEXT or PREVIOUS until the desired point is displayed on the screen.



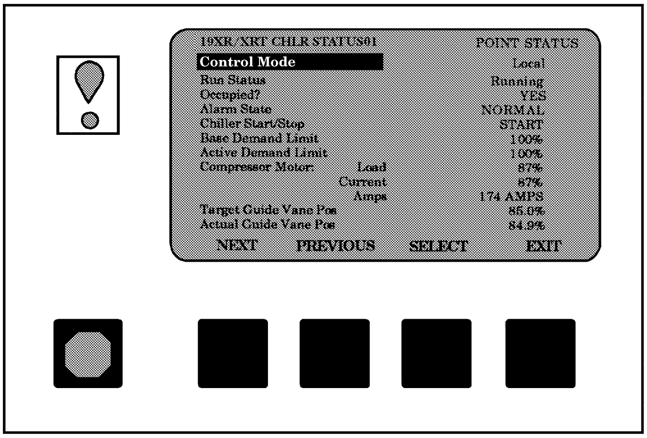
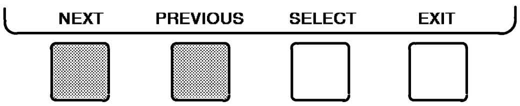


Fig. 14 — Example of Point Status Screen (STATUS01)

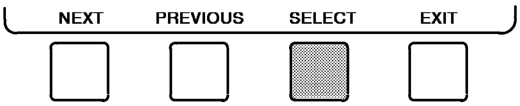
OVERRIDE OPERATIONS

<u>To Override a Value or Status</u>

1. From any point status screen, press NEXT or PREVIOUS to highlight the desired value.



2. Press SELECT to select the highlighted value. Then:



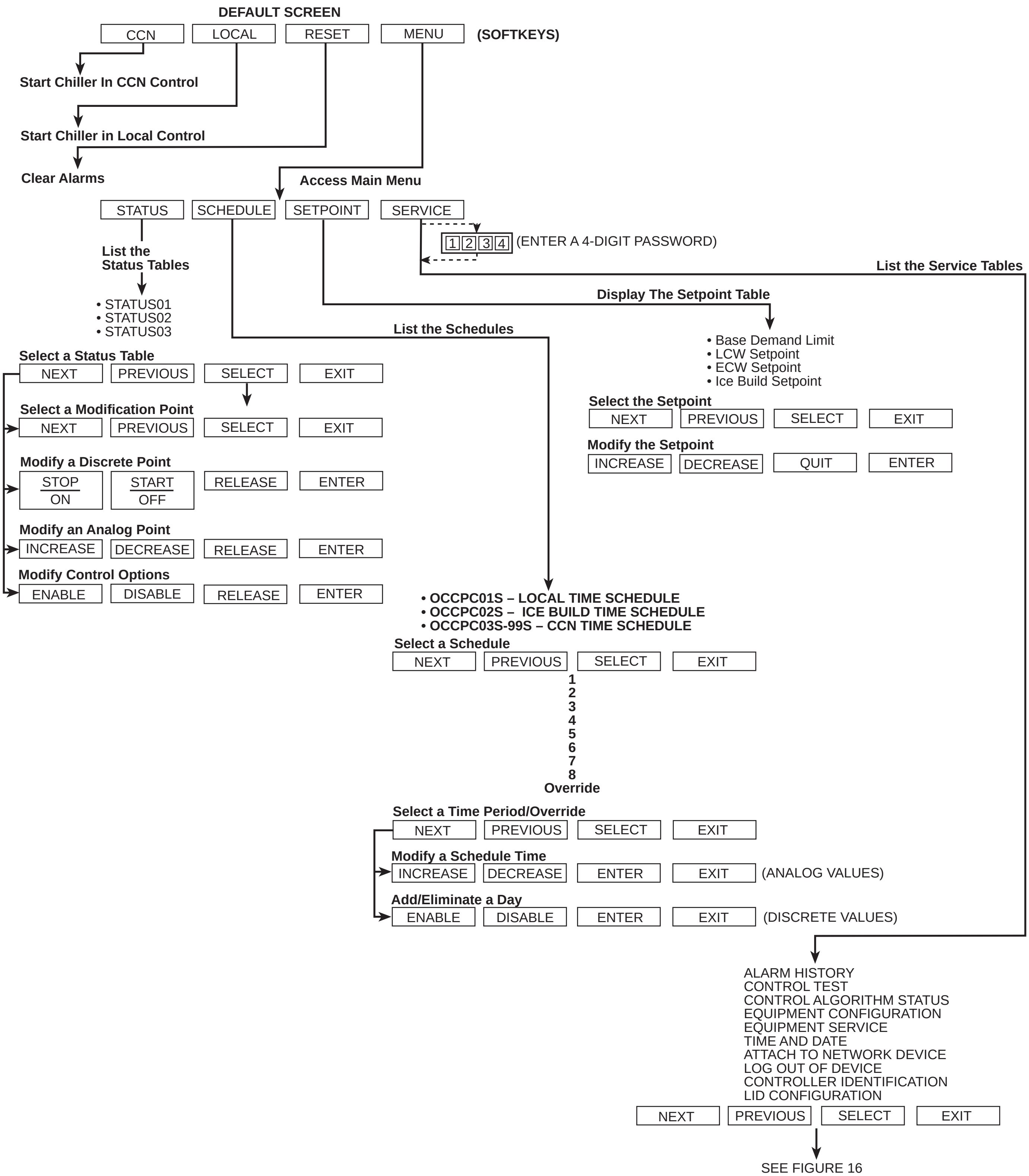


Fig. 15 — 19XR LID Menu Structure

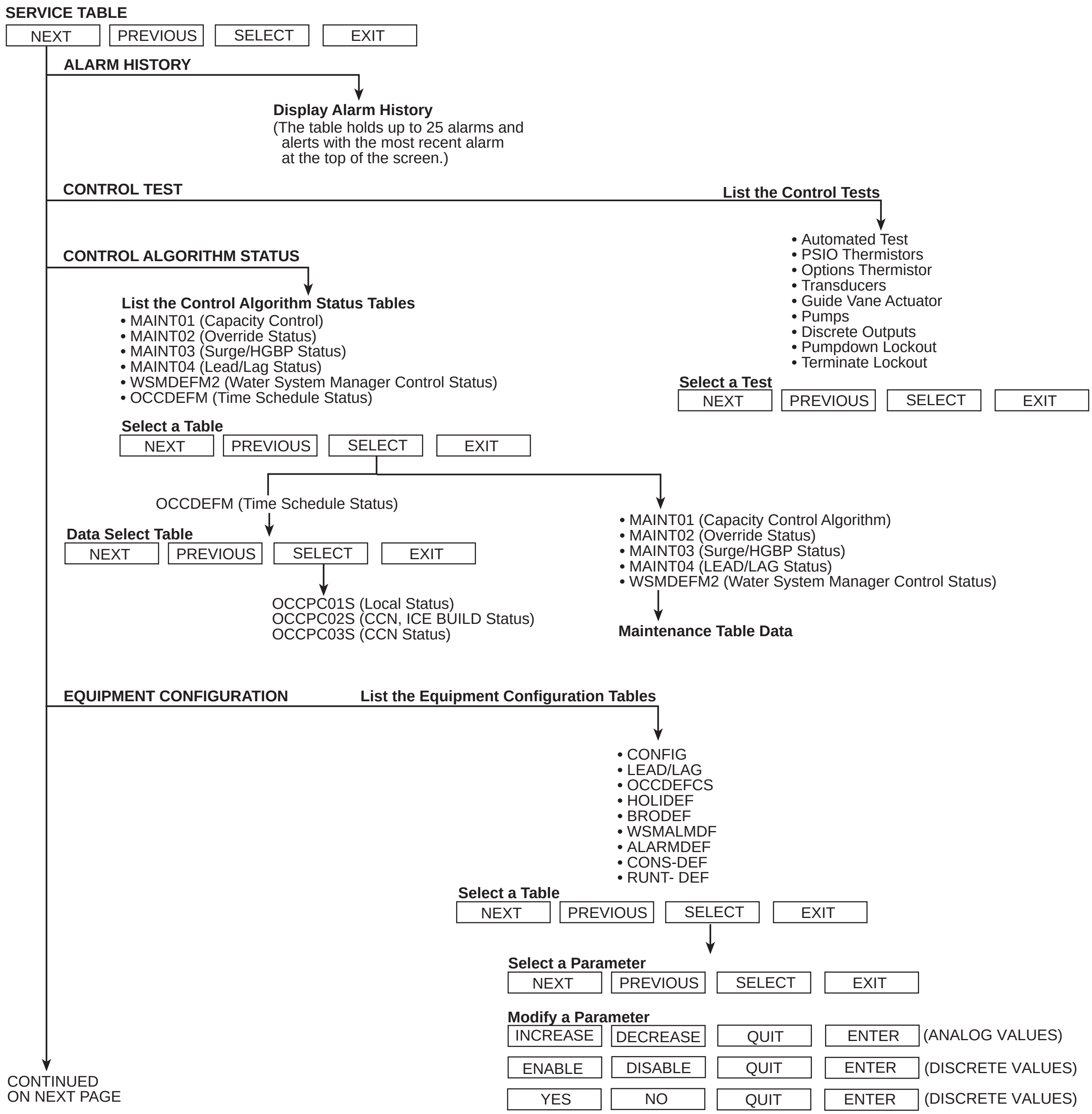
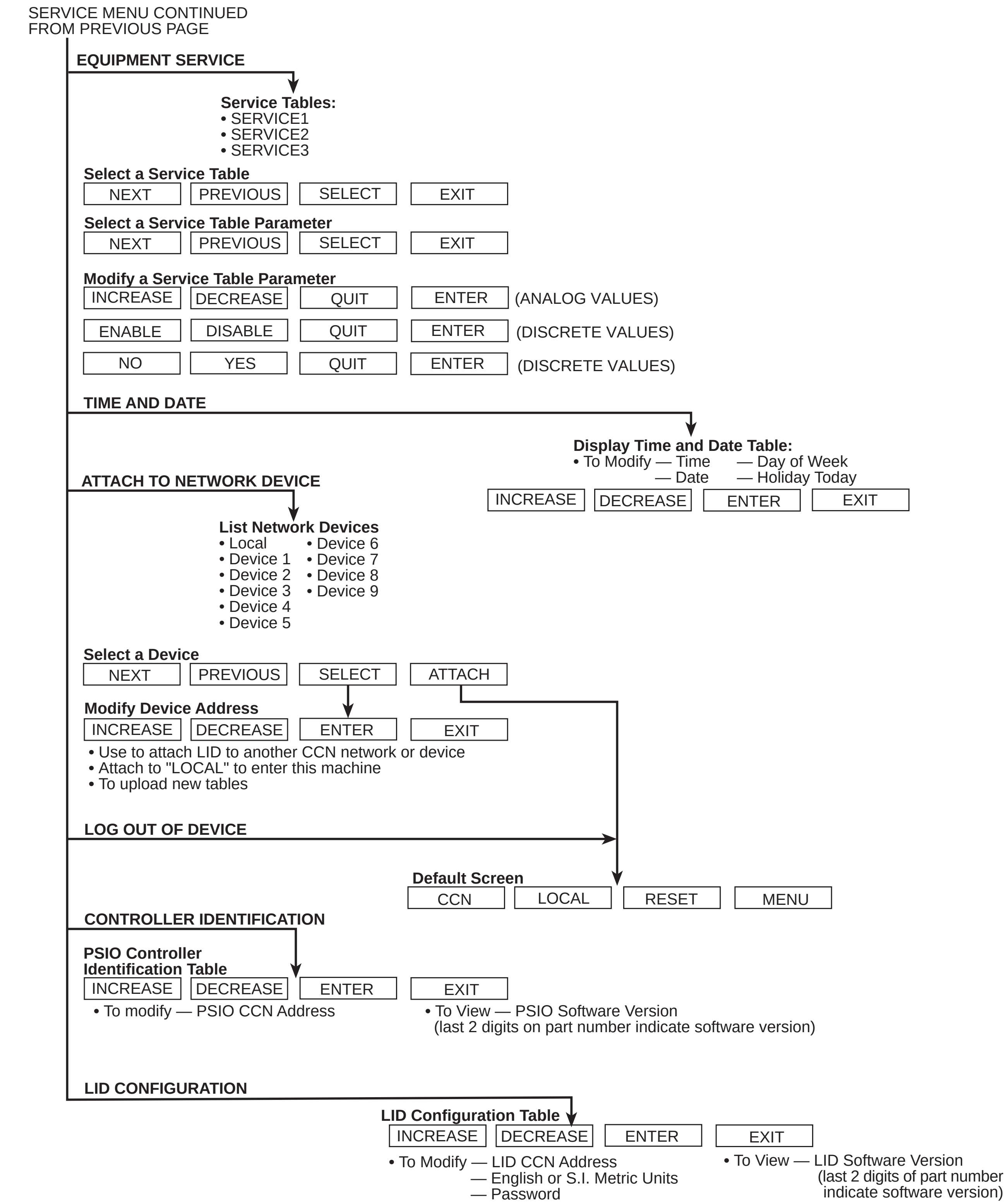


Fig. 16 — 19XR Service Menu Structure

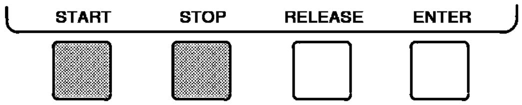


LEGEND

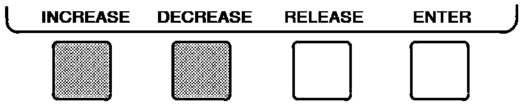
CCN — Carrier Comfort Network
LID — Local Interface Device

Fig. 16 — 19XR Service Menu Structure (cont)

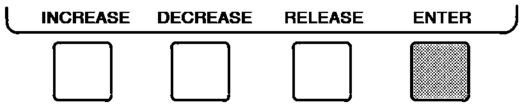
For Discrete Points — Press START or STOP to select the desired state.



For Analog Points — Press INCREASE or DECREASE to select the desired value.



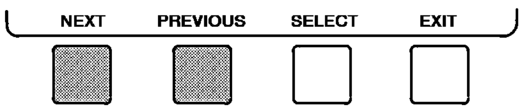
4. Press ENTER to register the new value.



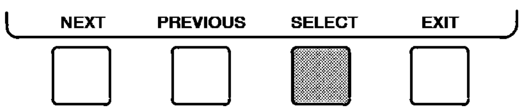
NOTE: When overriding or changing metric values, it is necessary to hold down the softkey for a few seconds in order to see a value change, especially on kilopascal values.

To Remove an Override

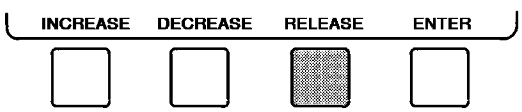
1. On the point status table press NEXT or PREVIOUS to highlight the desired value.



2. Press SELECT to access the highlighted value.



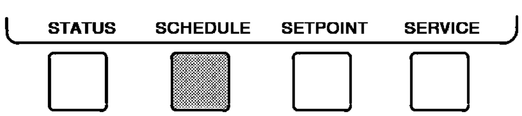
3. Press RELEASE to remove the override and return the point to the PIC's automatic control.



Override Indication — An override value is indicated by "SUPVSR," "SERVC," or "BEST" flashing next to the point value on the STATUS table.

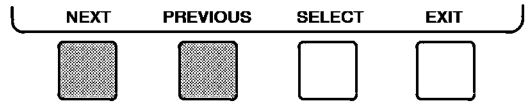
TIME SCHEDULE OPERATION (Fig. 17)

1. On the Menu screen, press SCHEDULE.

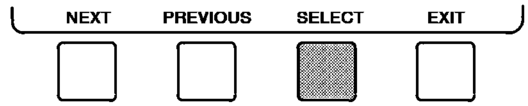


2. Press NEXT or PREVIOUS to highlight the desired schedule.

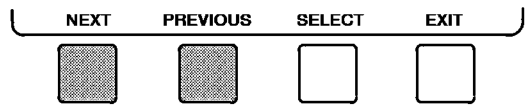
   OCCPC01S — LOCAL Time Schedule
   OCCPC02S — ICE BUILD Time Schedule
   OCCPC03-99S — CCN Time Schedule (Actual number is defined in CONFIG screen.)



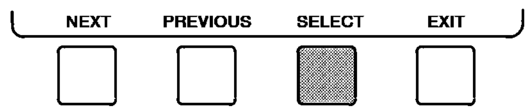
3. Press SELECT to view the desired time schedule.



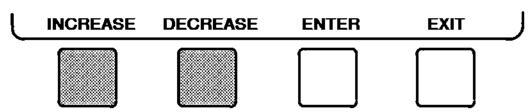
4. Press NEXT or PREVIOUS to highlight the desired period or override to change.



5. Press SELECT to access the highlighted period or override.



6. a. Press INCREASE or DECREASE to change the time values. Override values are in one-hour increments, up to 4 hours.



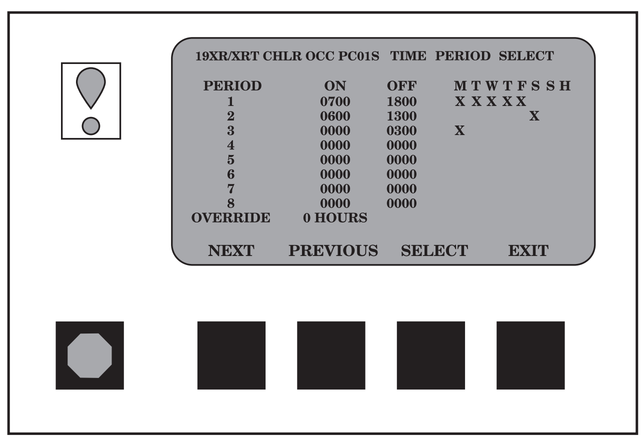
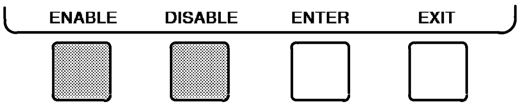
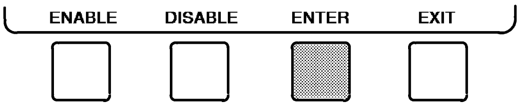


Fig. 17 — Example of Time Schedule Operation Screen

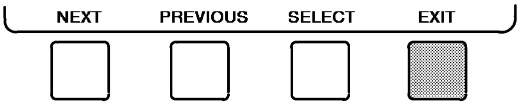
b. Press ENABLE to select days in the day-of-week fields. Press DISABLE to eliminate days from the period.



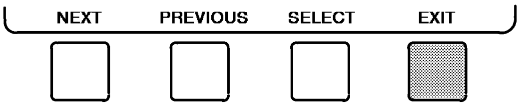
7. Press ENTER to register the values and to move horizontally (left to right) within a period.



8. Press EXIT to leave the period or override.



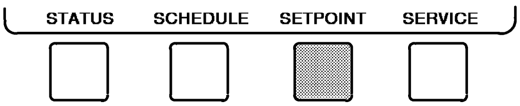
9. Either return to Step 4 to select another period or override, or press EXIT again to leave the current time schedule screen and save the changes.



10. The Holiday Designation (HOLIDEF table) may be found in the Service Operation section, page 38. The month, day, and duration for the holiday must be assigned. The Broadcast function in the BRODEF table also must be enabled for holiday periods to function.

TO VIEW AND CHANGE SET POINTS (Fig. 18)

1. To view the SETPOINT table, from the MENU screen press SETPOINT .



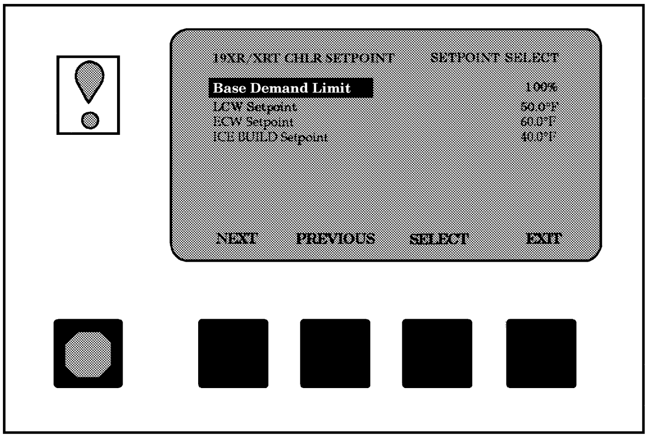
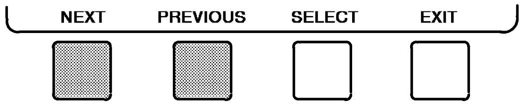


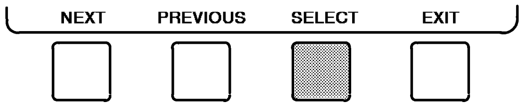
Fig. 18 — Example of Set Point Screen

2. There are 4 set points on this screen: BASE DEMAND LIMIT, LCW SETPOINT (leaving chilled water set point), ECW SETPOINT (entering chilled water set point), and ICE BUILD SETPOINT. Only one of the chilled water set points can be active at one time. The set point that is active is determined from the SERVICE menu. See the Service Operation section, page 38. The ice build (ICE BUILD) function is also activated and configured from the SERVICE menu.

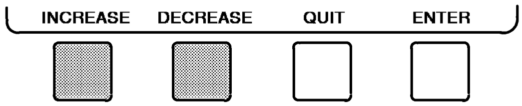
3. Press NEXT or PREVIOUS to highlight the desired set point entry.



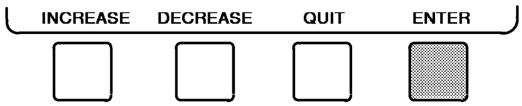
4. Press SELECT to modify the highlighted set point.



5. Press INCREASE or DECREASE to change the selected set point value.



6. Press ENTER to save the changes and return to the previous screen.



SERVICE OPERATION — To view the menu-driven programs available for Service Operation, see Service Operation section, page 38. For examples of LID display screens, see Table 2.

Table 2 — LID Display Data

IMPORTANT: The following notes apply to all Table 2 examples.

1. Only 12 lines of information appear on the LID screen at any one time. Press the NEXT or PREVIOUS softkey to highlight a point or to view items below or above the current screen. If you have a chiller with a backlit LID, press the NEXT softkey twice to page forward; press the PREVIOUS softkey twice to page back.
2. To access the information shown in Examples 5 through 13, enter your 4-digit password after pressing the SERVICE softkey. If no softkeys are pressed for 15 minutes, the LID automatically logs off (to prevent unrestricted access to PIC controls) and reverts to the default screen. If this happens, you must reenter your password to access the tables shown in Examples 5 through 13.
3. Terms in the Description column of these tables are listed as they appear on the LID screen.
4. The LID may be configured in English or Metric (SI) units using the LID CONFIGURATION screen. See the Service Operation section, page 38, for instructions on making this change.
5. The items in the Reference Point Name column *do not appear on the LID screen*. They are data or variable names used in CCN or Building Supervisor (BS) software. They are listed in these tables as a convenience to the operator if it is necessary to cross reference CCN/BS documentation or use CCN/BS programs. For more information, see the 19XR CCN literature.
6. Reference Point Names shown in these tables in all capital letters can be read by CCN and BS software. Of these capitalized names, those preceded by a dagger can also be changed (that is, written to) by the CCN, BS, and the LID. Capitalized Reference Point Names preceded by two daggers can be changed only from the LID. Reference Point Names in lower case type can be viewed by CCN or BS only by viewing the whole table.
7. Alarms and Alerts: An asterisk *in the far right field of a LID status screen* indicates that the chiller is in an alarm state; an exclamation point in the far right field of the LID screen indicates an alert state. The asterisk (or exclamation point) indicates that the value on that line has exceeded (or is approaching) a limit. For more information on alarms and alerts, see the Alarms and Alerts section, page 14.

LEGEND

CCN	—	Carrier Comfort Network
CHW	—	Chilled Water
CHWR	—	Chilled Water Return
CHWS	—	Chilled Water Supply
ECW	—	Entering Chilled Water
HGBP	—	Hot Gas Bypass
LCW	—	Leaving Chilled water
mA	—	Milliamps
P	—	Pressure
T	—	Temperature

EXAMPLE 1 — STATUS01 DISPLAY SCREEN

To access this display from the **LID default** screen:

1. Press **MENU** .
2. Press **STATUS** (**STATUS01** will be highlighted).
3. Press **SELECT** .

DESCRIPTION	RANGE	UNITS	REFERENCE POINT NAME (ALARM HISTORY)
Control Mode	Reset, Off, Local, CCN		MODE
Run Status	Timeout, Recycle, Prestart, Startup, Ramping, Running, Demand, Override, Shutdown, Abnormal, Pumpdown, Tripout		STATUS
Occupied ?	No/Yes		OCC
Alarm State	Normal/Alarm		ALM
†Chiller Start/Stop	Stop/Start		CHIL_S_S
Base Demand Limit	40-100	%	DLM
†Active Demand Limit	40-100	%	DEM_LIM
Compressor Motor Load	0-9999	%	CA_L
Current	0-9999	%	CA_P
††Amps	0-9999	AMPS	CA_A
†Target Guide Vane Pos	0-100	%	GV_TRG
Actual Guide Vane Pos	0-100	%	GV_ACT
Diffuser Actuator Pos	100-0	%	DIFF_ ACT
Water/Brine: Setpoint	10-120 (–12.2-48.9)	DEG F (DEG C)	SP
†Control Point	10-120 (–12.2-48.9)	DEG F (DEG C)	LCW_STPT
Entering Chilled Water	–40-245 (–40-118)	DEG F (DEG C)	ECW
Leaving Chilled Water	–40-245 (–40-118)	DEG F (DEG C)	LCW
Entering Condenser Water	–40-245 (–40-118)	DEG F (DEG C)	ECDW
Leaving Condenser Water	–40-245 (–40-118)	DEG F (DEG C)	LCDW
Evaporator Refrig Temp	–40-245 (–40-118)	DEG F (DEG C)	ERT
†Evaporator Pressure	–6.7-420 (–46-2896)	PSI (kPa)	ERP
Condenser Refrig Temp	–40-245 (–40-118)	DEG F (DEG C)	CRT
†Condenser Pressure	–6.7-420 (–46-2896)	PSI (kPa)	CRP
Discharge Temperature	–40-245 (–40-118)	DEG F (DEG C)	CMPD
Bearing Temperature	–40-245 (–40-118)	DEG F (DEG C)	MTRB
Motor Winding Temp	–40-245 (–40-118)	DEG F (DEG C)	MTRW
Oil Sump Temperature	–40-245 (–40-118)	DEG F (DEG C)	OILT
†Oil Pressure	–6.7-420 (–46-2896)	PSID (kPad)	OILPD
Line Voltage: ††Percent	0-999	%	V_P
Actual	0-99999	VOLTS	V_A
†Remote Contacts Input	Off/On		REMCON
Total Compressor Starts	0-65535		c_starts
Starts in 12 Hours	0-8		STARTS
Compressor Ontime	0-500000.0	HOURS	c_hrs
†Service Ontime	0-32767	HOURS	S_HRS
†Compressor Motor kW	0-9999	kW	CKW

NOTE: All values can be read by a CCN device. Descriptions shown with (†) support write operations for BEST programming language, datatransfer, and overrides. Descriptions shown with (††) can be adjusted from the LID.

Table 2 — LID Display Data (cont)

EXAMPLE 2 — STATUS02 DISPLAY SCREEN

To access this display from the **LID default** screen:

1. Press **MENU** .
2. Press **STATUS** .
3. Scroll down to highlight **STATUS02**.
4. Press **SELECT** .

DESCRIPTION	POINT TYPE		UNITS	REFERENCE POINT NAME (ALARM HISTORY)
	INPUT	OUTPUT		
Hot Gas Bypass Relay		X	OFF/ON	HGBR
†Chilled Water Pump		X	OFF/ON	CHWP
Chilled Water Flow	X		NO/YES	EVFL
†Condenser Water Pump		X	OFF/ON	CDP
Condenser Water Flow	X		NO/YES	CDFL
Compressor Start Relay		X	OFF/ON	CMPR
Compressor Start Contact	X		OPEN/CLOSED	1CR_AUX
Compressor Run Contact	X		OPEN/CLOSED	RUN_AUX
Starter Fault Contact	X		OPEN/CLOSED	STR_FLT
Pressure Trip Contact	X		OPEN/CLOSED	PRS_TRIP
Single Cycle Dropout	X		NORMAL/ALARM	V1_CYCLE
Oil Pump Relay		X	OFF/ON	OILR
Oil Heater Relay		X	OFF/ON	OILH
†Tower Fan Relay		X	OFF/ON	TFR
Compr. Shunt Trip Relay		X	OFF/ON	TRIPR
Alarm Relay		X	NORMAL/ALARM	ALM
Spare Prot Limit Input	X		ALARM/NORMAL	SPR_PL

NOTE: All values can be read from a CCN terminal. Descriptions shown with (†) support write operations from the LID only.

EXAMPLE 3 — STATUS03 DISPLAY SCREEN

To access this display from the **LID default** screen:

1. Press **MENU** .
2. Press **STATUS**.
3. Scroll down to highlight **STATUS03**.
4. Press **SELECT** .

DESCRIPTION	RANGE	UNITS	REFERENCE POINT NAME (ALARM HISTORY)
OPTIONS BOARD 1			
Demand Limit 4-20 mA	4-20	mA	DEM_OPT
Temp Reset 4-20 mA	4-20	mA	RES_OPT
Common CHWS Sensor	–40-245 (–40-118)	DEG F (DEG C)	CHWS
Common CHWR Sensor	–40-245 (–40-118)	DEG F (DEG C)	CHWR
Remote Reset Sensor	–40-245 (–40-118)	DEG F (DEG C)	R_RESET
Temp Sensor — Spare 1	–40-245 (–40-118)	DEG F (DEG C)	SPARE1
Temp Sensor — Spare 2	–40-245 (–40-118)	DEG F (DEG C)	SPARE2
Temp Sensor — Spare 3	–40-245 (–40-118)	DEG F (DEG C)	SPARE3
OPTIONS BOARD 2			
4-20 mA — Spare 1	4-20	mA	SPARE1_M
4-20 mA — Spare 2	4-20	mA	SPARE2_M
Temp Sensor — Spare 4	–40-245 (–40-118)	DEG F (DEG C)	SPARE4
Temp Sensor — Spare 5	–40-245 (–40-118)	DEG F (DEG C)	SPARE5
Temp Sensor — Spare 6	–40-245 (–40-118)	DEG F (DEG C)	SPARE6
Temp Sensor — Spare 7	–40-245 (–40-118)	DEG F (DEG C)	SPARE7
Temp Sensor — Spare 8	–40-245 (–40-118)	DEG F (DEG C)	SPARE8
Temp Sensor — Spare 9	–40-245 (–40-118)	DEG F (DEG C)	SPARE9

NOTE: All values shall be variables available for read operation to a CCN network. All descriptions on this screen support write operations for BEST programming language, data-transfer, and overriding.

EXAMPLE 4 — SETPOINT DISPLAY SCREEN

To access this display from the **LID default** screen:

1. Press **MENU** .
2. Press **SETPOINT** .

DESCRIPTION	CONFIGURABLE RANGE	UNITS	REFERENCE POINT NAME	DEFAULT VALUE
Base Demand Limit	40-100	%	DLM	100
LCW Setpoint	20-120 (–6.6-48.9)	DEG F (DEG C)	lcw_sp	50.0 (10.0)
ECW Setpoint	20-120 (–6.6-48.9)	DEG F (DEG C)	ecw_sp	60.0 (15.6)
ICE BUILD Setpoint	20- 60 (–6.6-15.6)	DEG F (DEG C)	ice_sp	40.0 (4.4)

Table 2 — LID Display Data (cont)

EXAMPLE 5 — CONFIGURATION (CONFIG) DISPLAY SCREEN

To access this display from the **LID default** screen:

1. Press **MENU** .
2. Press **SERVICE** .
3. Scroll down to highlight **EQUIPMENT CONFIGURATION**.
4. Press **SELECT** .
5. Scroll down to highlight **CONFIG**.
6. Press **SELECT** .

DESCRIPTION	CONFIGURABLE RANGE	UNITS	REFERENCE POINT NAME	DEFAULT VALUE
RESET TYPE 1				
Degrees Reset at 20 mA	–30-30 (–17-17)	DEG F (DEG C)	deg_20ma	10Δ(6Δ)
RESET TYPE 2				
Remote Temp (No Reset)	–40-245 (–40-118)	DEG F (DEG C)	res_rt1	85 (29)
Remote Temp (Full Reset)	–40-245 (–40-118)	DEG F (DEG C)	res_rt2	65 (18)
Degrees Reset	–30-30 (–17-17)	DEG F (DEG C)	res_rt	10Δ(6Δ)
RESET TYPE 3				
CHW Delta T (No Reset)	0-15 (0-8)	DEG F (DEG C)	restd_1	10Δ(6Δ)
CHW Delta T (Full Reset)	0-15 (0-8)	DEG F (DEG C)	restd_2	0Δ(0Δ)
Degrees Reset	–30-30 (–17-17)	DEG F (DEG C)	deg_chw	5Δ(3Δ)
Select/Enable Reset Type	0-3		res_sel	0
ECW CONTROL OPTION	DISABLE/ENABLE		ecw_opt	DISABLE
Demand Limit at 20 mA	40-100	%	dem_20ma	40
20mA Demand Limit Option	DISABLE/ENABLE		dem_sel	DISABLE
Auto Restart Option	DISABLE/ENABLE		astart	DISABLE
Remote Contacts Option	DISABLE/ENABLE		r_contact	DISABLE
Temp Pulldown Deg/Min	2-10 (1.1-5.6)	ΔF (ΔC)	tmp_ramp	3 (1.7)
Load Pulldown %/Min	5-20		kw_ramp	10
Select Ramp Type: Temp = 0, Load = 1	0/1		ramp_opt	1
Loadshed Group Number	0-99		ldsgrp	0
Loadshed Demand Delta	0-60	%	ldsdelta	20
Maximum Loadshed Time	0-120	MIN	maxldstm	60
CCN Occupancy Config:				
Schedule Number	3-99		occpcxxe	3
Broadcast Option	DISABLE/ENABLE		occbrcst	DISABLE
ICE BUILD Option	DISABLE/ENABLE		ibopt	DISABLE
ICE BUILD TERMINATION 0 =Temp, 1 =Contacts, 2 =Both	0-2		ibterm	0
ICE BUILD Recycle Option	DISABLE/ENABLE		ibrecyc	DISABLE

NOTE: Δ = Difference or change.

EXAMPLE 6 — LEAD/LAG CONFIGURATION DISPLAY SCREEN

To access this display from the **LID default** screen:

1. Press **MENU** .
2. Press **SERVICE** .
3. Scroll down to highlight **EQUIPMENT CONFIGURATION**.
4. Press **SELECT** .
5. Scroll down to highlight **LEAD/LAG**.
6. Press **SELECT** .

LEAD/LAG CONFIGURATION SCREEN

DESCRIPTION	CONFIGURABLE RANGE	UNITS	REFERENCE POINT NAME	DEFAULT VALUE
LEAD/LAG SELECT DISABLE =0, LEAD =1, LAG =2, STANDBY =3	0-3		leadlag	0
Load Balance Option	DISABLE/ENABLE		loadbal	DISABLE
Common Sensor Option	DISABLE/ENABLE		commsens	DISABLE
LAG Percent Capacity	25-75	%	lag_per	50
LAG Address	1-236		lag_add	92
LAG START Timer	2-60	MIN	lagstart	10
LAG STOP Timer	2-60	MIN	lagstop	10
PRESTART FAULT Timer	2-30	MIN	preflt	5
STANDBY Chiller Option	DISABLE/ENABLE		stndopt	DISABLE
STANDBY Percent Capacity	25-75	%	stnd_per	50
STANDBY Address	1-236		stnd_add	93

Table 2 — LID Display Data (cont)

EXAMPLE 7 — SERVICE1 DISPLAY SCREEN

To access this display from the **LID default** screen:

1. Press **MENU** .
2. Press **SERVICE** .
3. Scroll down to highlight **EQUIPMENT SERVICE**.
4. Press **SELECT** .
5. Scroll down to highlight **SERVICE1**.
6. Press **SELECT** .

DESCRIPTION	CONFIGURABLE RANGE	UNITS	REFERENCE POINT NAME	DEFAULT VALUE
Motor Temp Override	150-200 (66-93)	DEG F (DEG C)	mt_over	200 (93)
Cond Press Override	90-200 (620-1241)	PSI (kPa)	cp_ over	125 (862)
Refrig Override Delta T	2-5 (1-3)	DEG F (DEG C)	ref_over	3Δ (1.6Δ)
Chilled Medium	Water/Brine		medium	WATER
Brine Refrig Trippoint	0-40 (–17.7-4)	DEG F (DEG C)	br_trip	33 (1)
Compr Discharge Alert	125-200 (52-93)	DEG F (DEG C)	cd_alert	200 (93)
Bearing Temp Alert	165-185 (74-85)	DEG F (DEG C)	tb_alert	175 (79)
Water Flow Verify Time	0.5-5	MIN	wflow_t	5
Oil Press Verify Time	15-300	SEC	oilpr_t	15
Water/Brine Deadband	0.5-2.0 (0.3-1.1)	DEG F (DEG C)	cw_db	1.0 (0.6)
Recycle Restart Delta T	2.0-10.0 (1.1-5.6)	DEG F (DEG C)	rcyc_dt	5 (2.8)
Recycle Shutdown Delta T	0.5-4.0 (0.27-2.2)	ΔDEG F (Δ DEG C)	rcycsdt	1.0 (0.6)
Surge Limit/HGBP Option	0/1		srg_hgbp	0
Select: Surge=0, HGBP=1				
Surge/HGBP Delta T1	0.5-15 (0.3-8.3)	DEG F (DEG C)	hgb_dt1	1.5 (0.8)
Surge/HGBP Delta P1	30-170 (207-1172)	PSI (kPa)	hgb_dp1	50 (345)
Min. Load Points (T1/P1)				
Surge/HGBP Delta T2	0.5-15 (0.3-8.3)	DEG F (DEG C)	hgb_dt2	10 (5.6)
Surge/HGBP Delta P2	30-170 (207-1172)	PSI (kPa)	hgb_dp2	85 (586)
Full Load Points (T2/P2)				
Surge/HGBP Deadband	1-3 (0.6-1.6)	DEG F (DEG C)	hgb_dp	1 (0.6)
Surge Delta Percent Amps	5-20†	%	surge_a	10†
Surge Time Period	5-7†	MIN	surge_t	6†
Demand Limit Source	0/1		dem_src	0
Select: Amps=0, Load=1				
Amps Correction Factor	1-8		corfact	3
Motor Rated Load Amps	1-9999	AMPS	a_fs	200
Motor Rated Line Voltage	1-99999	VOLTS	v_fs	460
Meter Rated Line kW	1-9999	kW	kw_fs	600
Line Frequency	0/1	HZ	freq	0
Select: 0=60 Hz, 1=50 Hz				
Compr Starter Type	REDUCE/FULL		starter	REDUCE
Condenser Freeze Point	–20-35 (–28.9-1.7)	DEG F (DEG C)	cdfreeze	34 (1)
Soft Stop Amps Threshold	40-100	%	softstop	100

NOTE: Δ = Difference or change.

†Version 3 software. These values are recommended for Versions 1 and 2 software also.

Table 2 — LID Display Data (cont)

EXAMPLE 8 — SERVICE2 DISPLAY SCREEN

To access this display from the **LID default** screen:

1. Press **MENU** .
2. Press **SERVICE** .
3. Scroll down to highlight **EQUIPMENT SERVICE**.
4. Press **SELECT** .
5. Scroll down to highlight **SERVICE2**.
6. Press **SELECT** .

DESCRIPTION	CONFIGURABLE RANGE	UNITS	REFERENCE POINT NAME	DEFAULT VALUE
OPTIONS BOARD 1				
20 mA POWER CONFIGURATION Ext = 0, Int = 1				
RESET 20 mA Power Source	0, 1		res_20 ma	0
DEMAND 20 mA Power Source	0, 1		dem_20 ma	0
SPARE ALERT ENABLE Disable = 0, High = 1, Low = 2 Temp = Alert Threshold				
CHWS Temp Enable	0-2		chws_en	0
CHWS Temp Alert	–40-245 (–40-118)	DEG F (DEG C)	chws_al	245 (118)
CHWR Temp Enable	0-2		chwr_en	0
CHWR Temp Alert	–40-245 (–40-118)	DEG F (DEG C)	chwr_al	245 (118)
Reset Temp Enable	0-2		rres_en	0
Reset Temp Alert	–40-245 (–40-118)	DEG F (DEG C)	rres_al	245 (118)
Spare Temp 1 Enable	0-2		spr1_en	0
Spare Temp 1 Alert	–40-245 (–40-118)	DEG F (DEG C)	spr1_al	245 (118)
Spare Temp 2 Enable	0-2		spr2_en	0
Spare Temp 2 Alert	–40-245 (–40-118)	DEG F (DEG C)	spr2_al	245 (118)
Spare Temp 3 Enable	0-2		spr3_en	0
Spare Temp 3 Alert	–40-245 (–40-118)	DEG F (DEG C)	spr3_al	245 (118)
OPTIONS BOARD 2				
20 mA POWER CONFIGURATION External = 0, Internal = 1				
SPARE 1 20 mA Power Source	0, 1		sp1_20 ma	0
SPARE 2 20 mA Power Source	0, 1		sp2_20 ma	0
SPARE ALERT ENABLE Disable = 0, High = 1, Low = 2 Temp = Alert Threshold				
Spare Temp 4 Enable	0-2		spr4_en	0
Spare Temp 4 Alert	–40-245 (–40-118)	DEG F (DEG C)	spr4_al	245 (118)
Spare Temp 5 Enable	0-2		spr5_en	0
Spare Temp 5 Alert	–40-245 (–40-118)	DEG F (DEG C)	spr5_al	245 (118)
Spare Temp 6 Enable	0-2		spr6_en	0
Spare Temp 6 Alert	–40-245 (–40-118)	DEG F (DEG C)	spr6_al	245 (118)
Spare Temp 7 Enable	0-2		spr7_en	0
Spare Temp 7 Alert	–40-245 (–40-118)	DEG F (DEG C)	spr7_al	245 (118)
Spare Temp 8 Enable	0-2		spr8_en	0
Spare Temp 8 Alert	–40-245 (–0-118)	DEG F (DEG C)	spr8_al	245 (118)
Spare Temp 9 Enable	0-2		spr9_en	0
Spare Temp 9 Alert	–40-245 (–40-118)	DEG F (DEG C)	spr9_al	245 (118)

NOTE: This screen provides the means to generate alert messages based on exceeding the ‘‘Temp Alert’’ threshold for each point listed. If the ‘‘Enable’’ is set to 1, a value above the ‘‘Temp Alert’’ threshold generates an alert message. If the ‘‘Enable’’ is set to 2, a value below the ‘‘Temp Alert’’ threshold generates an alert message. If the ‘‘Enable’’ is set to 0, alert generation is disabled.

Table 2 — LID Display Data (cont)

EXAMPLE 9 — SERVICE3 DISPLAY SCREEN

To access this display from the **LID default** screen:

1. Press **MENU** .
2. Press **SERVICE** .
3. Scroll down to highlight **EQUIPMENT SERVICE**.
4. Press **SELECT** .
5. Scroll down to highlight **SERVICE3**.

DESCRIPTION	CONFIGURABLE RANGE	UNITS	REFERENCE POINT NAME	DEFAULT VALUE
Proportional Inc Band	2-10		gv_inc	6.5
Proportional Dec Band	2-10		gv_de	6.0
Proportional ECW Gain	1-3		gv_ecw	2.0
Guide Vane Travel Limit	30-100	%	gv_lim	50
Guide Vane Full Span mA	15-22	mA	gv_ma	18
Diffuser Control	Disable/Enable		diff_ctl	Dsable
Guide Vane 25% Load Pt.	0-78	%	gv_25	25
Diffuser 25% Load Pt.	0-100	%	df_25	100
Guide Vane 50% Load Pt.	0-78	%	gv_50	25
Diffuser 50% Load Pt.	0-100	%	df_50	100
Guide Vane 75% Load Pt.	0-78	%	gv_75	25
Diffuser 75% Load Pt.	0-100	%	df_75	100
Diffuser Full Span mA	15-22	mA	diff_mA	18

EXAMPLE 10 — MAINTENANCE (MAINT01) DISPLAY SCREEN

To access this display from the **LID default** screen:

1. Press **MENU** .
2. Press **SERVICE** .
3. Scroll down to highlight **ALGORITHM STATUS**.
4. Press **SELECT** .
5. Scroll down to highlight **MAINT01**.

DESCRIPTION	RANGE/STATUS	UNITS	REFERENCE POINT NAME
CAPACITY CONTROL			
Control Point	10-120 (–12.2-48.9)	DEG F (DEG C)	ctrlpt
Leaving Chilled Water	–40-245 (–40-118)	DEG F (DEG C)	LCW
Entering Chilled Water	–40-245 (–40-118)	DEG F (DEG C)	ECW
Control Point Error	–99-99 (–55-55)	DEG F (DEG C)	cperr
ECW Delta T	–99-99 (–55-55)	DEG F (DEG C)	ecwdt
ECW Reset	–99-99 (–55-55)	DEG F (DEG C)	ecwres
LCW Reset	–99-99 (–55-55)	DEG F (DEG C)	lcwres
Total Error + Resets	–99-99 (–55-55)	DEG F (DEG C)	error
Guide Vane Delta	–2-2	%	gvd
Target Guide Vane Pos	0-100	%	GV_TRG
Actual Guide Vane Pos	0-100	%	GV_ACT
Actual Diffuser Pos	0-100	%	DIFF_ACT
Proportional Inc Band	2-10		gv_inc
Proportional Dec Band	2-10		gv_dec
Proportional ECW Gain	1-3		gv_ecw
Water/Brine Deadband	0.5-2 (0.3-1.1)	DEG F (DEG C)	cwdb

NOTE: Overriding is not supported on this maintenance screen. Active overrides show the associated point in alert (*). Only values with capital letter reference point names are variables available for read operation.

Table 2 — LID Display Data (cont)

EXAMPLE 11 — MAINTENANCE (MAINT02) DISPLAY SCREEN

To access this display from the **LID default** screen:

1. Press **MENU** .
2. Press **SERVICE** .
3. Scroll down to highlight **CONTROL ALGORITHM STATUS**.
4. Press **SELECT** .
5. Scroll down to highlight **MAINT02**.
6. Press **SELECT** .

DESCRIPTION	RANGE/STATUS	UNITS	REFERENCE POINT NAME
OVERRIDE STATUS			
MOTOR WINDING TEMP	–40-245 (–40-118)	DEG F (DEG C)	MTRW
Override Threshold	150-200 (66-93)	DEG F (DEG C)	mt_over
CONDENSER PRESSURE	–0-420 (–0-2896)	PSI (kPa)	CRP
Override Threshold	90-200 (621-1379)	PSI (kPa)	cp_over
EVAPORATOR REFRIG TEMP	–40-245 (–40-118)	DEG F (DEG C)	ERT
Override Threshold	2-45 (1-7.2)	DEG F (DEG C)	rt_over
DISCHARGE TEMPERATURE	–40-245 (–40-118)	DEG F (DEG C)	CMPD
Alert Threshold	125-200 (52-93)	DEG F (DEG C)	cd_alert
BEARING TEMPERATURE	–40-245 (–40-118)	DEG F (DEG C)	MTRB
Alert Threshold	165-185 (74-85)	DEG F (DEG C)	tb_alert
ACTUAL SUPERHEAT	–20-99 (–29-37)	DEG F (DEG C)	SUPERHEAT
Superheat Required	0-99 (–18-37)	DEG F (DEG C)	SUPR_REQ
System Alarm/Alert None = 0, Alert = 1, Alarm = 2	0-2		ALRMALRD

NOTE: Overriding is not supported on this maintenance screen. Active overrides show the associated point in alert (*). Only values with capital letter reference point names are variables available for read operation.

EXAMPLE 12 — MAINTENANCE (MAINT03) DISPLAY SCREEN

To access this display from the **LID default** screen:

1. Press **MENU** .
2. Press **SERVICE** .
3. Scroll down to highlight **CONTROL ALGORITHM STATUS**.
4. Press **SELECT** .
5. Scroll down to highlight **MAINT03**.
6. Press **SELECT** .

DESCRIPTION	RANGE/STATUS	UNITS	REFERENCE POINT NAME
SURGE/HGBP ACTIVE ?	NO/YES		
Active Delta P	0-200 (0-1379)	PSI (kPa)	dp_a
Active Delta T	0-200 (0-111)	DEG F (DEG C)	dt_a
Calculated Delta T	0-200 (0-111)	DEG F (DEG C)	dt_c
SURGE PROTECTION COUNTS	0-12		spc

NOTE: Override is not supported on this maintenance screen. Only values with capital letter reference point names are variables available for read operation.

EXAMPLE 13 — MAINTENANCE (MAINT04) DISPLAY SCREEN

To access this display from the **LID default** screen:

1. Press **MENU** .
2. Press **SERVICE** .
3. Scroll down to highlight **CONTROL ALGORITHM STATUS**.
4. Press **SELECT** .
5. Scroll down to highlight **MAINT04**.
6. Press **SELECT** .

DESCRIPTION	RANGE/STATUS	UNITS	REFERENCE POINT NAME
LEAD/LAG: Configuration	DISABLE,LEAD,LAG,STANDBY, INVALID		leadlag
Current Mode	DISABLE,LEAD,LAG,STANDBY, CONFIG		llmode
Load Balance Option	DISABLE/ENABLE		loadbal
LAG Start Time	20-60	MIN	lagstart
LAG Stop Time	20-60	MIN	lagstop
Prestart Fault Time	20-30	MIN	prefltt
Pulldown: Delta T/Min	x.xx	Δ DEG F (Δ DEG C)	pull_dt
Satisfied?	No/Yes		pull_sat
LEAD CHILLER in Control	No/Yes		leadctrl
LAG CHILLER: Mode	Reset,Off,Local,CCN		lagmode
Run Status	Timeout,Recycle,Startup,Ramping,Running Demand,Override,Shutdown,Abnormal,Pumpdown		lagstat
Start/Stop	Stop,Start,Retain		lag_s_s
Recovery Start Request	No/Yes		lag_rec
STANDBY CHILLER: Mode	Reset,Off,Local,CCN		stdmode
Run Status	Timeout,Recycle,Startup,Ramping,Running Demand,Override,Shutdown,Abnormal,Pumpdown		stdstat
Start/Stop	Stop,Start,Retain		std_s_s
Recovery Start Request	No/Yes		std_rec

NOTES:
1. Only values with capital letter reference point names are variables available for read operation. Forcing is not supported on this maintenance screen.
2. Δ = Difference or change.

PIC System Functions

NOTE: Words not part of paragraph headings and printed in all capital letters can be viewed on the LID (e.g., LOCAL, CCN, RUNNING, ALARM, etc.). Words printed in *both* in all capital letters and italics can also be viewed on the LID and are parameters (*CONTROL MODE*, *COOLING SET POINT*, *TARGET GUIDE VANE POS*, etc.) with associated values (e.g., modes, temperatures, pressures, percentages, on, off, enable, disable, etc.). Works printed in all capital letters and in a box represent softkeys on the LID control panel (e.g., ENTER and EXIT). See Table 2 for examples of the type of information that can appear on the LID screens. Figures 12-18 give an overview of LID operations and menus.

CAPACITY CONTROL — The PIC controls the chiller capacity by modulating the inlet guide vanes in response to chilled water temperature changes away from the *WATER/BRINE CONTROL POINT.* The *WATER/BRINE CONTROL POINT* may be changed by a CCN network device or is determined by the PIC adding any active chilled water reset to the *WATER/BRINE SET POINT.* The PIC uses the *PROPORTIONAL INC (Increase) BAND*, *PROPORTIONAL DEC (Decrease) BAND*, and the *PROPORTIONAL ECW (Entering Chilled Water) GAIN* to determine how fast or slow to respond. *WATER/BRINE CONTROL POINT* may be viewed or overridden from the STATUS01screen.

ECW CONTROL OPTION — If this option is enabled, the PIC uses the *ENTERING CHILLED WATER* temperature to modulate the vanes instead of the *LEAVING CHILLED WATER* temperature. The *ENTERING CHILLED WATER* control option may be viewed or modified on the CONFIG screen, which is accessed from the EQUIPMENT CONFIGURATION table.

CONTROL POINT DEADBAND — This is the tolerance range on the chilled water/brine temperature control point. If the water temperature goes outside the *WATER/BRINE DEADBAND*, the PIC opens or closes the guide vanes until the temperature is within tolerance. The PIC may be configured with a 0.5 to 2 F (0.3 to 1.1 C) deadband. *WATER/BRINE DEADBAND* may be viewed or modified on the SERVICE1 screen, which is accessed from the EQUIPMENT SERVICE table.

For example, a 1° F (0.6° C) deadband setting controls the water temperature within ±0.5° F (0.3° C) of the control point. This may cause frequent guide vane movement if the chilled water load fluctuates frequently. A value of 1° F (0.6° C) is the default setting.

DIFFUSER CONTROL — On 19XR compressors equipped with a variable discharge diffuser, the PIC adjusts the diffuser actuator position (*ACTUAL DIFFUSER POS* on the MAINT01 screen) to correspond to the actual guide vane position (*ACTUAL GUIDE VANE POS* on the STATUS01 screen).

The diffuser control can be enabled or disabled from the SERVICE03 screen. See Table 2, Example 9. In addition, the diffuser and guide vane load points may be viewed and modified from this screen. These points must be correct for the compressor size. The diffuser opening can be incremented from fully open to completely closed. A 0% setting is fully open; a 100% setting is completely closed. To obtain the proper settings for Diffuser Control, contact a Carrier Engineering representative.

PROPORTIONAL BANDS AND GAIN — Proportional band is the rate at which the guide vane position is corrected in proportion to how far the chilled water/brine temperature is from the control point. Proportional gain determines how quickly the guide vanes react to how quickly the temperature is moving from the *WATER/BRINE CONTROL POINT.* The proportional bands and gain may be viewed or modified from the SERVICE3 screen, which is accessed from the EQUIPMENT SERVICE table.

The Proportional Band — There are two response modes, one for temperature response above the control point, the other for the response below the control point.

The temperature response above the control point is called the *PROPORTIONAL INC BAND,* and it can slow or quicken guide vane response to chilled water/brine temperatures above the *DEADBAND*. The *PROPORTIONAL INC BAND* can be adjusted from a setting of 2 to 10; the default setting is 6.5.

The response below the control point is called the *PROPORTIONAL DEC BAND*, and it can slow or quicken the guide vane response to chilled water temperature below the deadband plus the control point. The *PROPORTIONAL DEC BAND* can be adjusted on the LID from a setting of 2 to 10. The default setting is 6.0.

NOTE: Increasing either of these settings causes the guide vanes to respond more slowly than they would at a lower setting.

The *PROPORTIONAL ECW GAIN* can be adjusted on the LID display for values of 1, 2, or 3; the default setting is 2. Increase this setting to increase guide vane response to a change in entering chilled water temperature.

DEMAND LIMITING — The PIC responds to the *ACTIVE DEMAND LIMIT* set point by limiting the opening of the guide vanes. It compares the *ACTIVE DEMAND LIMIT* set point to the *DEMAND LIMIT SOURCE* (either the *COMPRESSOR MOTOR LOAD* or the *COMPRESSOR MOTOR CURRENT*), depending on how the control is configured. *DEMAND LIMIT SOURCE* is on the SERVICE1 screen. The default source is the compressor motor current.

CHILLER TIMERS — The PIC maintains 2 runtime clocks, known as *COMPRESSOR ONTIME* and *SERVICE ONTIME*. *COMPRESSOR ONTIME* indicates the total lifetime compressor run hours. This timer can register up to 500,000 hours before the clock turns back to zero. The *SERVICE ONTIME* is a resettable timer that can be used to indicate the hours since the last service visit or any other event. The time can be changed from the LID to whatever value is desired. This timer can register up to 32,767 hours before it rolls over to zero.

The chiller also maintains a start-to-start timer and a stop-to-start timer. These timers limit how soon the chiller can be started. See the Start-Up/Shutdown/Recycle Sequence section, page 40, for more information on this topic.

OCCUPANCY SCHEDULE — The chiller schedule, described in the Time Schedule Operation section (page 19), determines when the chiller can run. Each schedule consists of from 1 to 8 occupied or unoccupied time periods, set by the operator. The chiller can be started and run during an occupied time period (when *OCCUPIED ?* is set to YES on the STATUS01 display screen). It cannot be started or run during an unoccupied time period (when *OCCUPIED ?* is set to NO on the STATUS01 display screen). These time periods can be set for each day of the week and for holidays. The day begins with 0000 hours and ends with 2400 hours. The default setting for *OCCUPIED ?* is YES, unless an unoccupied time period is in effect.

These schedules can be set up to follow a building's occupancy schedule, or the chiller can be set so to run 100% of the time, if the operator wishes. The schedules also can be bypassed by forcing the *CHILLER START/STOP* parameter on the STATUS01 screen to START. For more information on forced starts, see Local Start-Up, page 40.

The schedules also can be overridden to keep the chiller in an occupied state for up to 4 hours, on a one time basis. See the Time Schedule Operation section, page 19.

Figure 17 shows a schedule for a typical office building with a 3-hour, off-peak, cool-down period from midnight to 3 a.m., following a weekend shutdown. Holiday periods are in an unoccupied state 24 hours per day. The building operates Monday through Friday, 7:00 a.m. to 6:00 p.m., and Saturdays from 6:00 a.m. to 1:00 p.m. This schedule also includes the Monday midnight to 3:00 a.m. weekend cool-down schedule.

NOTE: This schedule is for illustration only and is not intended to be a recommended schedule for chiller operation.

Whenever the chiller is in the LOCAL mode, it uses Occupancy Schedule 01 (OCCPC01S). When the chiller is in the ICE BUILD mode, it uses Occupancy Schedule 02 (OCCPC02S). When the chiller is in CCN mode, it uses Occupancy Schedule 03 (OCCPC03S).

The CCN *SCHEDULE NUMBER* is configured on the CONFIG display screen, accessed from the EQUIPMENT CONFIGURATION table. See Table 2, Example 5. *SCHEDULE NUMBER* can be changed to any value from 03 to 99. If this number is changed on the CONFIG screen, the operator must go to the ATTACH TO NETWORK DEVICE screen to upload the new number into the SCHEDULE screen. See Fig. 16.

Safety Controls — The PIC monitors all safety control inputs and, if required, shuts down the chiller or limits the guide vanes to protect the chiller from possible damage from any of the following conditions:

- high bearing temperature
- high motor winding temperature
- high discharge temperature
- low discharge superheat*
- low oil pressure
- low cooler refrigerant temperature/pressure
- condenser high pressure or low pressure
- inadequate water/brine cooler and condenser flow
- high, low, or loss of voltage
- excessive motor acceleration time
- excessive starter transition time
- lack of motor current signal
- excessive motor amps
- excessive compressor surge
- temperature and transducer faults

*Superheat is the difference between saturation temperature and sensible temperature. The high discharge temperature safety measures only sensible temperature.

Starter faults or optional protective devices within the starter can shut down the chiller. The protective devices you have for your application depend on what options were purchased.

⚠ CAUTION

If compressor motor overload occurs, check the motor for grounded or open phases before attempting a restart.

If the PIC control initiates a safety shutdown, it displays the reason for the shutdown (the fault) on the LID display screen along with a primary and secondary message, energizes an alarm relay in the starter, and blinks the alarm light on the control center. The alarm is stored in memory and can be viewed on the ALARM HISTORY screen on the LID, along with a message for troubleshooting.

To give more precise information or warnings on the chiller's operating condition, the operator can define alert limits on various monitored inputs. Safety contact and alert limits are defined in Table 3. Alarm and alert messages are listed in the Troubleshooting Guide section, page 66.

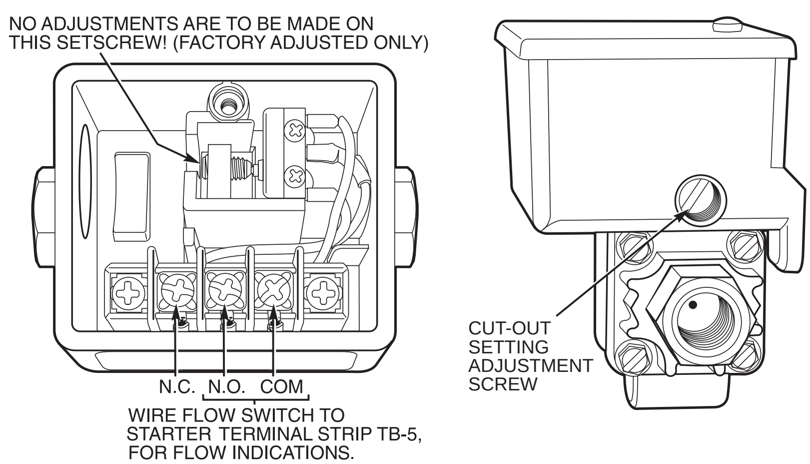
To define alert limits, access the appropriate screen and scroll to the desired variable using the $\boxed{\text{NEXT}}$ and $\boxed{\text{PREVIOUS}}$ softkeys. When the variable is highlighted, press the $\boxed{\text{SELECT}}$ softkey. Use the $\boxed{\text{INCREASE}}$ or $\boxed{\text{DECREASE}}$ softkey to change the value. When the desired value is reached, press $\boxed{\text{ENTER}}$ to define the new limit. To abort the change, press the $\boxed{\text{QUIT}}$ softkey.

Table 3 — Protective Safety Limits and Control Settings

MONITORED PARAMETER	LIMIT	APPLICABLE COMMENTS
TEMPERATURE SENSORS OUT OF RANGE	–40 to 245 F (–40 to 118.3 C)	Must be outside range for 2 seconds
PRESSURE TRANSDUCERS OUT OF RANGE	0.08 to 0.98 Voltage Ratio	Must be outside range for 2 seconds. Ratio = Input Voltage ÷ Voltage Reference
COMPRESSOR DISCHARGE TEMPERATURE	>220 F (104.4 C)	Preset, alert setting configurable
MOTOR WINDING TEMPERATURE	>220 F (104.4 C)	Preset, alert setting configurable
BEARING TEMPERATURE	>185 F (85 C)	Preset, alert setting configurable
EVAPORATOR REFRIGERANT TEMPERATURE	<33 F (for water chilling) (0.6° C)	Preset, configure chilled medium for water (Service1 table)
	<Brine Refrigerant Trippoint (set point adjustable from 0 to 40 F [–18 to 4 C] for brine chilling)	Configure chilled medium for brine (Service1 table). Adjust brine refrigerant trippoint for proper cutout
TRANSDUCER VOLTAGE	<4.5 vdc > 5.5 vdc	Preset
CONDENSER PRESSURE — SWITCH	>263 ± 7 psig (1813 ± 48 kPa), reset at 180 ± 10 (1241 ± 69 kPa)	Preset
— CONTROL	185 psig (1276 kPa)	Preset
OIL PRESSURE — SWITCH	Cutout <11 psid (76 kPad) ± 1.5 psid (10.3 kPad) Cut-in >16.5 psid (114 kPad) ± 4 psid (27.5 kPad)	Preset, no calibration needed
— CONTROL	Cutout <15 psid (103 kPad) Alert <18 psid (124 kPad)	Preset
LINE VOLTAGE — HIGH	>110% for one minute	Preset, based on transformed line voltage to 24 vac rated-input to the Starter Management Module. Also monitored at PSIO power input.
— LOW	<90% for one minute or ≤85% for 3 seconds	
— SINGLE-CYCLE	<50% for one cycle	
COMPRESSOR MOTOR LOAD	>110% for 30 seconds	Preset
	<10% with compressor running	Preset
	>10% with compressor off	Preset
STARTER ACCELERATION TIME (Determined by inrush current going below 100% compressor motor load)	>45 seconds	For chillers with reduced voltage mechanical and solid-state starters
	>10 seconds	For chillers with full voltage starters (Configured on Service1 table)
STARTER TRANSITION	>75 seconds	Reduced voltage starters only
CONDENSER FREEZE PROTECTION	Energizes condenser pump relay if condenser refrigerant temperature or condenser entering water temperature is below the configured condenser freeze point temperature. Deenergizes when the temperature is 5 F (3 C) above condenser freeze point temperature.	CONDENSER FREEZE POINT configured in Service01 table with a default setting of 34 F (1 C).
DISCHARGE SUPERHEAT	Minimum value calculated based on operating conditions and then compared to actual superheat.	Calculated minimum required superheat and actual superheat are shown on MAINT02 screen.
VARIABLE DIFFUSER OPERATION	Detects discharge pulses caused by incorrect diffuser position.	Preset, no calibration needed.

Flow Switches (Field Supplied)

Operate water pumps with chiller off. Manually reduce water flow and observe switch for proper cutout. Safety shutdown occurs when cutout time exceeds 3 seconds.



Shunt Trip (Option) — The function of the shunt trip option on the PIC is to act as a safety trip. The shunt trip is wired from an output on the SMM to a shunt trip equipped motor circuit breaker. If the PIC tries to shut down the compressor using a normal shutdown procedure but is unsuccessful for 30 seconds, the shunt trip output is energized and causes the circuit breaker to trip off. If ground fault protection has been applied to the starter, the ground fault trip also energizes the shunt trip to trip the circuit breaker. Protective devices in the starter can also energize the shunt trip. The shunt trip feature can be tested using the Control Test feature.

Default Screen Freeze — When the chiller is in an alarm state, the default LID display ″freezes,″ that is, it stops updating. The first line of the LID default screen displays a primary alarm message; the second line displays a secondary alarm message.

The LID default screen freezes to enable the operator to see the conditions of the chiller *at the time of the alarm.* If the value in alarm is one normally displayed on the default screen, it flashes between normal and reverse print. The LID default screen remains frozen until the condition that caused the alarm is remedied by the operator.

Knowledge of the operating state of the chiller at the time an alarm occurs is useful when troubleshooting. Additional chiller information can be viewed on the status screens. Troubleshooting information is recorded in the ALARM HISTORY table, which can be accessed from the SERVICE menu.

To determine what caused the alarm, the operator should read both the primary and secondary default screen messages, as well as the alarm history. The primary message indicates the most recent alarm condition, The secondary message gives more detail on the alarm condition. Since there may be more than one alarm condition, another alarm message may appear after the first condition is cleared. Check the ALARM HISTORY screen for additional help in determining the reasons for the alarms. Once all existing alarms are cleared (by pressing the RESET softkey), the default LID display returns to normal operation.

Ramp Loading Control — The ramp loading control slows down the rate at which the compressor loads up. This control can prevent the compressor from loading up during the short period of time when the chiller is started and the chilled water loop has to be brought down to normal design conditions. This helps reduce electrical demand charges by slowly bringing the chilled water to control point. The total power draw during this period remains almost unchanged.

There are 2 methods of ramp loading with the PIC. Ramp loading can be based on chilled water temperature or on motor load.

1. Temperature ramp loading (*TEMP PULLDOWN*) limits the rate at which either leaving chilled water or entering chilled water temperature decreases. The rate is configured by the operator. The lowest temperature ramp rate will be used the first time the chiller is started (at commissioning). The lowest temperature ramp rate will also be used if chiller power has been off for 3 hours or more (even if the motor ramp load is selected as the ramp loading method).
2. Motor load ramp loading (*LOAD PULLDOWN*) limits the rate at which the compressor motor current or compressor motor load (*COMPRESSOR MOTOR LOAD*) increases. The rate is configured by the operator.

The *TEMP PULLDOWN*, *LOAD PULLDOWN*, and *SELECT RAMP TYPE* parameters may be viewed or modified on the CONFIG screen, which is accessed from the EQUIPMENT CONFIGURATION table (see Table 2, Example 5). Motor load is the default ramp loading type.

Capacity Override (Table 4) — Capacity overrides can prevent some safety shutdowns caused by exceeding the motor amperage limit, refrigerant low temperature safety limit, motor high temperature safety limit, and condenser high pressure limit. In all cases there are 2 stages of compressor vane control.

1. The vanes are prevented from opening further, and the status line on the LID indicates the reason for the override.
2. The vanes are closed until the condition decreases to below the first step set point. Then the vanes are released to normal capacity control.

Whenever the motor current demand limit set point (*ACTIVE DEMAND LIMIT*) is reached, it activates a capacity override, again, with a 2-step process. Exceeding 110% of the rated load amps for more than 30 seconds will initiate a safety shutdown.

The compressor high lift (surge prevention) set point will cause a capacity override as well. When the surge prevention set point is reached, the controller normally will only prevent the guide vanes from opening. If so equipped, the hot gas bypass valve will open instead of holding the vanes. See the Surge Prevention Algorithm section, page 34.

High Discharge Temperature Control — If the discharge temperature increases above 160 F (71.1 C), the guide vanes are proportionally opened to increase gas flow through the compressor. If the leaving chilled water temperature is then brought 5° F (2.8° C) below the control set point temperature, the PIC will bring the chiller into the recycle mode.

Oil Sump Temperature Control — The oil sump temperature control is regulated by the PIC, which uses the oil heater relay when the chiller is shut down.

As part of the pre-start checks executed by the controls, the oil sump temperature (*OIL SUMP TEMPERATURE*) is compared to the cooler refrigerant temperature (*EVAPORATOR REFRIG TEMP*). If the difference between these 2 temperatures is 50 F (27.8 C) or less, the start-up will be delayed until the oil temperature is 50 F (27.8 C) or more. Once this temperature is confirmed, the start-up continues.

The oil heater relay is energized whenever the chiller compressor is off and the oil sump temperature is less than 150 F (65.6 C) or the oil sump temperature is less than the cooler refrigerant temperature plus 70° F (39° C). The oil heater is turned off when the oil sump temperature is either

- more than 160 F (71.1 C), or
- more than 155 F (68.3 C) and more than the cooler refrigerant temperature plus 75° F (41.6° C).

The oil heater is always off during start-up or when the compressor is running.

The oil pump is also energized during the time the oil is being heated (for 60 seconds at the end of every 30 minutes).

Oil Cooler — The oil must be cooled when the compressor is running. This is accomplished through a small, plate-type heat exchanger (also called the oil cooler) located behind the oil pump. The heat exchanger uses liquid condenser refrigerant as the cooling liquid. Refrigerant thermostatic expansion valves (TXVs) regulate refrigerant flow to control the oil temperature entering the bearings. The bulbs for the expansion valves are strapped to the oil supply line leaving the heat exchanger, and the valves are set to maintain 110 F (43 C).

NOTE: The TXVs are not adjustable. The oil sump temperature may be at a lower temperature during compressor operations.

Table 4 — Capacity Overrides

OVERRIDE CAPACITY CONTROL	FIRST STAGE SET POINT			SECOND STAGE SET POINT	OVERRIDE TERMINATION
	View/Modify on LID Screen	Default Value	Configurable Range	Value	Value
HIGH CONDENSER PRESSURE	SERVICE1	125 psig (862 kPa)	90 to 200 psig (620 to 1379 kPa)	>Override Set Point + 4 psid (28 kPad)	<Override Set Point
HIGH MOTOR TEMPERATURE	SERVICE1	>200 F (93.3 C)	150 to 200 F (66 to 93 C)	>Override Set Point +10° F (6° C)	<Override Set Point
LOW REFRIGERANT TEMPERATURE (Refrigerant Override Delta Temperature)	SERVICE1	<3° F (1.6° C) (Above Trippoint)	2° to 5° F (1° to 3° C)	≤Trippoint + Override ΔT –1° F (0.56° C)	>Trippoint + Override ΔT +2° F (1.2° C)
HIGH COMPRESSOR LIFT (Surge Prevention)	SERVICE1	Min: T1 — 1.5° F (0.8° C) P1 — 50 psid (345 kPad) Max: T2 — 10° F (5.6° C) P2 — 85 psid (586 kPad)	0.5° to 15° F (0.3° to 8.3° C) 30 to 170 psid (207 to1172 kPad) 0.5° to 15° F (0.3° to 8.3° C) 30 to 170 psid (207 to 1172 kPad)	None	Within Lift Limits Plus Surge/ HGBP Deadband Setting
MANUAL GUIDE VANE TARGET	MAINT01	Automatic	0 to 100%	None	Release of Manual Control
MOTOR LOAD — ACTIVE DEMAND LIMIT	STATUS01	100%	40 to 100%	≥5% of Set Point	2% Lower Than Set Point
LOW DISCHARGE SUPERHEAT	MAINT02	Calculated Minimum Superheat for Conditions	None	2° F (1.1° C) Below Calculated Minimum Superheat	1° F (0.56° C) Above Calculated Minimum Superheat

Remote Start/Stop Controls — A remote device, such as a timeclock that uses a set of contacts, may be used to start and stop the chiller. However, the device should not be programmed to start and stop the chiller in excess of 2 or 3 times every 12 hours. If more than 8 starts in 12 hours (the *STARTS IN 12 HOURS* parameter on the STATUS01 screen) occur, an excessive starts alarm displays, preventing the chiller from starting. The operator must press the $\boxed{\text{RESET}}$ softkey on the LID to override the starts counter and start the chiller. If the chiller records 12 starts (excluding recycle starts) in a sliding 12-hour period, it can be restarted only by pressing the $\boxed{\text{RESET}}$ softkey followed by the $\boxed{\text{LOCAL}}$ or $\boxed{\text{CCN}}$ softkey. This ensures that, if the automatic system is malfunctioning, the chiller will not repeatedly cycle on and off. If the automatic restart after a power failure option (*AUTO RESTART OPTION* on the CONFIG screen) is not activated when a power failure occurs, and if the remote contact is closed, the chiller will indicate an alarm because of the loss of voltage.

The contacts for remote start are wired into the starter at terminal strip TB5, terminals 8A and 8B. See the certified drawings for further details on contact ratings. The contacts must be dry (no power).

Spare Safety Inputs — Normally closed (NC) discrete inputs for additional field-supplied safeties may be wired to the spare protective limits input channel in place of the factory-installed jumper. (Wire multiple inputs in series.) The opening of any contact will result in a safety shutdown and a display on the LID. Refer to the certified drawings for safety contact ratings.

Analog temperature sensors may also be added to the options modules, if installed. The analog temperature sensors may be programmed to cause an alert on the CCN network. This alert will not shut the chiller down.

Spare Safety Alarm Contacts — Two spare sets of alarm contacts are provided in the starter. The contact ratings are provided in the certified drawings. The contacts are located on terminal strip TB6, terminals 5A and 5B, and terminals 5C and 5D.

Condenser Pump Control — The chiller will monitor the condenser pressure (*CONDENSER PRESSURE*) and may turn on the condenser pump if the condenser pressure becomes too high while the compressor is shut down. The condenser pressure override (*COND PRESS OVERRIDE*) parameter is used to determine this pressure point. *COND PRESS OVERRIDE* is found in the SERVICE1 display screen, which is accessed from the EQUIPMENT SERVICE table. The default value is 125 psig (862 kPa).

If the *CONDENSER PRESSURE* is greater than or equal to the *COND PRESS OVERRIDE,* and the entering condenser water temperature (*ENTERING CONDENSER WATER)* is less than 115 F (46 C), the condenser pump will energize to try to decrease the pressure. The pump will turn off when the condenser pressure is 5 psi (34 kPa) less than the pressure override or when the condenser refrigerant temperature (*CONDENSER REFRIG TEMP)* is within 3° F (1.7° C) of the entering condenser water temperature (*ENTERING CONDENSER WATER*).

Condenser Freeze Prevention — This control algorithm helps prevent condenser tube freeze-up by energizing the condenser pump relay. The PIC controls the pump and, by starting it, helps to prevent the water in the condenser from freezing. The PIC can perform this function whenever the chiller is not running *except* when it is either actively in pumpdown or in pumpdown/lockout with the freeze prevention disabled.

When the *CONDENSER REFRIG TEMP* is less than or equal to the *CONDENSER FREEZE POINT,* or the

ENTERING CONDENSER WATER temperature is less than or equal to the *CONDENSER FREEZE POINT*, the *CONDENSER WATER PUMP* is energized until the *CONDENSER REFRIG TEMP* is greater than the *CONDENSER FREEZE POINT* plus 5° F (2.7° C). An alarm is generated if the chiller is in PUMPDOWN mode and the pump is energized. An alert is generated if the chiller is not in PUMPDOWN mode and the pump is energized. If the chiller is in RECYCLE SHUTDOWN mode, the mode will transition to a non-recycle shutdown.

Tower Fan Relay — Low condenser water temperature can cause the chiller to shut down when refrigerant temperature is low. The tower fan relay, located in the starter, is controlled by the PIC to energize and deenergize as the pressure differential between cooler and condenser vessels changes. This prevents low condenser water temperature and maximizes chiller efficiency. The tower fan relay can only accomplish this if the relay has been added to the cooling tower temperature controller.

The tower fan relay is turned on whenever the condenser water pump is running, flow is verified, and the difference between cooler and condenser pressure is more than 30 psid (207 kPad) for entering condenser water temperature greater than 85 F (29 C).

The tower fan relay is turned off when the condenser pump is off, flow is stopped, and the cooler refrigerant temperature is less than the override temperature, or the differential pressure is less than 28 psid (193 kPad) for entering condenser water less than 80 F (27 C).

The *TOWER FAN RELAY* parameter is accessed from the STATUS02 screen.

> IMPORTANT: A field-supplied water temperature control system for condenser water should be installed. The system should maintain the leaving condenser water temperature at a temperature that is 20° F (11° C) above the leaving chilled water temperature.

> **⚠ CAUTION**
>
> The tower fan relay control is not a substitute for a condenser water temperature control. When used with a water temperature control system, the tower fan relay control can be used to help prevent low condenser water temperatures.

Auto. Restart After Power Failure — This option may be enabled or disabled and may be viewed or modified on the CONFIG screen, which is accessed from the EQUIPMENT CONFIGURATION table. If the Auto. Restart option is enabled, the chiller will start up automatically after a power failure has occurred (after a single cycle dropout; low, high, or loss of voltage; and the power is within ± 10% of normal). The 15- and 3-minute inhibit timers are ignored during this type of start-up.

When power is restored after the power failure and if the compressor had been running, the oil pump will energize for one minute before energizing the cooler pump. Auto. restart will then continue like a normal start-up.

If power to the PSIO module has been off for more than 3 hours or the timeclock has been set for the first time, start the compressor with the slowest temperature-based ramp load rate possible in order to minimize oil foaming.

The oil pump is energized occasionally during the time the oil is being brought up to proper temperature in order to eliminate refrigerant that has migrated to the oil sump during the power failure. The pump turns on for 60 seconds at the end of every 30-minute period until the chiller is started.

Water/Brine Reset — Three types of chilled water or brine reset are available and can be viewed or modified on the CONFIG screen, which is accessed from the EQUIPMENT CONFIGURATION table.

The LID default screen indicates when the chilled water reset is active. The control point (*WATER/BRINE CONTROL POINT*) temperature on the STATUS01 screen indicates the chiller's current reset temperature.

To activate a reset type, access the CONFIG screen and input all configuration information for that reset type. Then, input the reset type number (1, 2, or 3) in the *SELECT/ENABLE RESET TYPE* input line.

RESET TYPE 1: REQUIRES AN OPTIONAL 8-INPUT MODULE — Reset Type 1 is an automatic chilled water temperature reset based on a remote temperature sensor input signal (4 to 20 mA). Reset Type 1 permits up to ± 30° F (± 16° C) of automatic reset to the chilled water or brine temperature set point. The 4 to 20 mA signal is hardwired into the first 8-input module.

If the 4 to 20 mA signal to the 8-input module is externally powered, the signal is wired to terminals J1-5(+) and J1-6(–). If the signal is internally powered by the 8-input module (for example, when using variable resistance), the signal is wired to J1-7(+) and J1-6(–). The PIC must now be configured on the SERVICE2 screen to ensure that the appropriate power source is identified.

RESET TYPE 2: REQUIRES AN OPTIONAL 8-INPUT MODULE — Reset Type 2 is an automatic chilled water temperature reset based on a remote temperature sensor input signal. Reset Type 2 permits ± 30° F (± 16° C) of automatic reset to the set point based on a temperature sensor wired to the first 8-input module (see wiring diagrams or certified drawings). The temperature sensor must be wired to terminal J1-19 and J1-20.

To configure Reset Type 2, enter the temperature of the remote sensor at the point where no temperature reset will occur (*REMOTE TEMP [NO RESET]*). Next, enter the temperature at which the full amount of reset will occur (*REMOTE TEMP [FULL RESET]*). Then, enter the maximum amount of reset required to operate the chiller (*DEGREES RESET*). Reset Type 2 can now be activated.

RESET TYPE 3 — Reset Type 3 is an automatic chilled water temperature reset based on cooler temperature difference. Reset Type 3 adds ± 30° F (± 16° C) based on the temperature difference between the entering and leaving chilled water temperature. This is the only type of reset that does not require a first 8-input module. In addition, no wiring is required for this type of reset, since it already uses the cooler water sensors.

To configure Reset Type 3, enter the chilled water temperature difference (the difference between entering and leaving chilled water) at which no temperature reset occurs (*CHW DELTA T [NO RESET]*). This chilled water temperature difference is usually the full design load temperature difference. Next, enter the difference in chilled water temperature at which the full amount of reset occurs (*CHW DELTA T [FULL RESET]*). Finally, enter the amount of reset (*DEGREES RESET*). Reset Type 3 can now be activated.

Demand Limit Control Option (Requires Optional 8-Input Module) — The demand limit control option (*20 mA DEMAND LIMIT OPTION*) is externally controlled by a 4 to 20 mA signal from an energy management system (EMS). The option is set up on the CONFIG screen. When enabled, 4 mA is the 100% demand set point with an operator-configured minimum demand at a 20 mA set point (*DEMAND LIMIT AT 20 mA*).

The demand reset input from an energy management system is hard-wired into the first 8-input module. The signal may be internally powered by the module or externally powered. If the signal is externally powered, the signal is wired to terminals J1-1(+) and J1-2(–). If the signal is internally powered, the signal is wired to terminals J1-3(+) and J1-2(–).

Surge Prevention Algorithm — This is an operator-configurable feature that can determine if lift conditions are too high for the compressor and then take corrective action. Lift is defined as the difference between the pressure at the impeller eye and at the impeller discharge. The maximum lift a particular impeller wheel can perform varies with the gas flow across the impeller and the size of the wheel.

A surge condition occurs when the lift becomes so high the gas flow across the impeller reverses. This condition can eventually cause chiller damage. The surge prevention algorithm notifies the operator that chiller operating conditions are marginal and to take action to help prevent chiller damage such as lowering entering condenser water temperature.

The surge prevention algorithm first determines if corrective action is necessary. The algorithm checks 2 sets of operator-configured data points, the minimum load points (*MIN. LOAD POINTS [T1/P1]*) and the full load points (*FULL LOAD POINTS [T2/P2]*). These points have default settings as defined on the SERVICE1 screen or on Table 4.

The surge prevention algorithm function and settings are graphically displayed in Fig. 19 and 20. The two sets of load points on the graph (default settings are shown) describe a line the algorithm uses to determine the maximum lift of the compressor. When the actual differential pressure between the cooler and condenser and the temperature difference between the entering and leaving chilled water are above the line on the graph (as defined by the minimum and full load points), the algorithm goes into a corrective action mode. If the actual values are below the line, the algorithm takes no action. Information on modifying the default set points of the minimum and full load points may be found in the Input Service Configurations section, page 50.

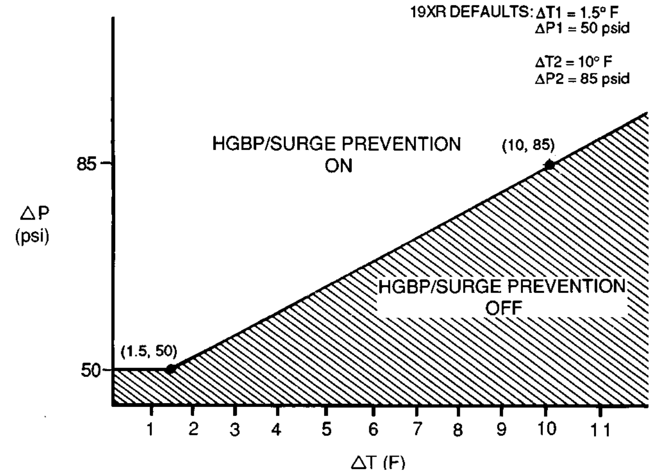
Corrective action can be taken by making one of 2 choices. If a hot gas bypass line is present and the hot gas option is selected on the SERVICE1 table (*SURGE LIMIT/HGBP OPTION* is set to 1), the hot gas bypass valve can be energized. If the hot gas bypass option is not selected (*SURGE LIMIT/HGBP OPTION* is set to 0), hold the guide vanes. See Table 4, Capacity Overrides. Both of these corrective actions try to reduce the lift experienced by the compressor and help prevent a surge condition.

Surge Protection — The PIC monitors surge, which is a fluctuation in compressor motor amperage. Each time the fluctuation exceeds an operator-specified limit (*SURGE DELTA PERCENT AMPS*), the PIC counts the surge. If more than 12 surges occur within an operator-specified time (*SURGE TIME PERIOD*), the PIC initiates a surge protection shutdown of the chiller.

The surge limit can be adjusted from the SERVICE1 screen. Scroll down to the *SURGE DELTA PERCENT AMPS* parameter, and use the INCREASE or DECREASE softkey to adjust the percent of surge. The default setting is 10% amps.

The surge time period can also be adjusted from the SERVICE1 screen. Scroll to the *SURGE TIME PERIOD* parameter, and use the INCREASE or DECREASE softkey to adjust the amount of time. The default setting is 6 minutes.

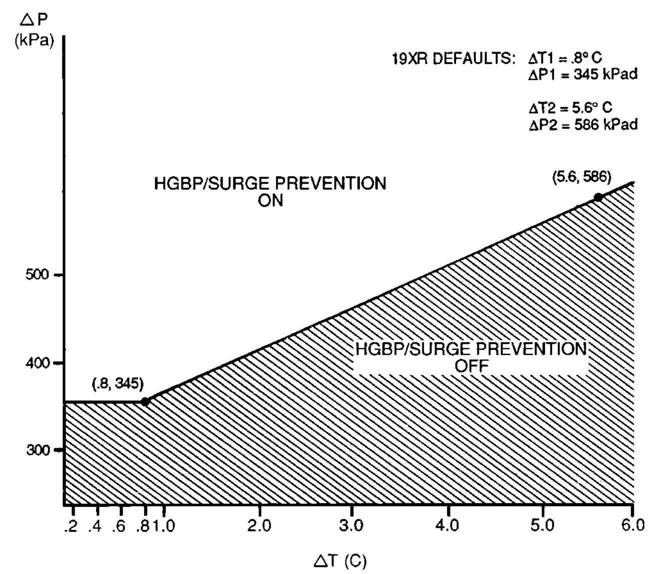
Access the maintenance screen (MAINT03) to monitor the surge count (*SURGE PROTECTION COUNTS*).



LEGEND

ECW — Entering Chilled Water
HGBP — Hot Gas Bypass
LCW — Leaving Chilled Water
ΔP = (Condenser Psi) − (Cooler Psi)
ΔT = (ECW) − (LCW)

Fig. 19 — 19XR Hot Gas Bypass/Surge Prevention with Default English Settings



LEGEND

ECW — Entering Chilled Water
HGBP — Hot Gas Bypass
LCW — Leaving Chilled Water
ΔP = (Condenser kPa) − (Cooler kPa)
ΔT = (ECW) − (LCW)

Fig. 20 — 19XR Hot Gas Bypass/Surge Prevention with Default Metric Settings

Lead/Lag Control — The lead/lag control system automatically starts and stops a lag or second chiller in a 2-chiller water system. A third chiller can be added to the lead/lag system as a standby chiller to start up in case the lead or lag chiller in the system has shut down during an alarm condition and additional cooling is required. Refer to Fig. 15 and 16 for menu, table, and screen selection information.

NOTE: The lead/lag function can be configured and monitored on the LEAD/LAG screen, which is accessed from the SERVICE menu and EQUIPMENT CONFIGURATION table. See Table 2, Example 6. Lead/lag status during chiller operation can be viewed on the MAINT04 display screen, which is accessed from the SERVICE menu and CONTROL ALGORITHM STATUS table. See Table 2, Example 13.

Lead/Lag System Requirements:

- all chillers in the system must have PSIO software capable of performing the lead/lag function
- water pumps MUST be energized from the PIC controls
- water flows should be constant
- the CCN time schedules for all chillers must be identical

Operation Features:

- 2 chiller lead/lag
- addition of a third chiller for backup
- manual rotation of lead chiller
- load balancing if configured
- staggered restart of the chillers after a power failure
- chillers may be piped in parallel or in series chilled water flow

COMMON POINT SENSOR INSTALLATION — Lead/lag operation does not require a common chilled water point sensor. Common point sensors can be added to the 8-input option module, if desired. Refer to the certified drawings for termination of sensor leads.

NOTE: If the common point sensor option is chosen on a chilled water system, each chiller should have its own 8-input option module and common point sensor installed. Each chiller uses its own common point sensor for control when that chiller is designated as the lead chiller. The PIC cannot read the value of common point sensors installed on other chillers in the chilled water system.

When installing chillers in series, a common point sensor should be used. If a common point sensor is not used, the leaving chilled water sensor of the upstream chiller must be moved into the leaving chilled water pipe of the downstream chiller.

If return chilled water control is required on chillers piped in series, the common point return chilled water sensor should be installed. If this sensor is not installed, the return chilled water sensor of the downstream chiller must be relocated to the return chilled water pipe of the upstream chiller.

To properly control the common supply point temperature sensor when chillers are piped in parallel, the water flow through the shutdown chillers must be isolated so no water bypass around the operating chiller occurs. The common point sensor option must not be used if water bypass around the operating chiller is occurring.

CHILLER COMMUNICATION WIRING — Refer to the chiller's Installation Instructions, Carrier Comfort Network Interface section for information on chiller communication wiring.

LEAD/LAG OPERATION — The PIC not only has the ability to operate 2 chillers in lead/lag, but it can also start a designated standby chiller when either the lead or lag chiller is faulted and capacity requirements are not met. The lead/lag option only operates when the chillers are in CCN mode. If any other chiller configured for lead/lag is set to the LOCAL or OFF modes, it will be unavailable for lead/lag operation.

NOTE: The lead/lag function can be configured and monitored on the LEADLAG screen, which is accessed from the SERVICE menu and EQUIPMENT CONFIGURATION table. See Table 2, Example 6. Lead/lag status during chiller operation can be viewed on the MAINT04 display screen, which is accessed from the SERVICE menu and CONTROL ALGORITHM STATUS table. See Table 2, Example 13.

<u>Lead/Lag Chiller Configuration and Operation</u>

- A chiller is designated the lead chiller when its *LEAD/LAG SELECT* value on the LEAD/LAG screen is set to ''1.''
- A chiller is designated the lag chiller when its *LEAD/LAG SELECT* value is set to ''2.''
- A chiller is designated as a standby chiller when its *LEAD/LAG SELECT* value is set to ''3.''
- A value of ''0'' disables the lead/lag designation of a chiller.

To configure the *LAG ADDRESS* value on the LEAD/LAG screen, always enter the address of the other chiller on the system. For example, if you are configuring chiller A, enter the address for chiller B as the lag address. If you are configuring chiller B, enter the address for chiller A as the lag address. This makes it easier to rotate the lead and lag chillers.

If the address assignments in the *LAG ADDRESS* and *STANDBY ADDRESS* parameters conflict, the lead/lag function is disabled and an alert (!) message displays. For example, if the *LAG ADDRESS* matches the lead chiller's address, the lead/lag will be disabled and an alert (!) message displayed. The lead/lag maintenance screen (MAINT04) displays the message 'INVALID' in the *LEAD/LAG CONFIGURATION* and *CURRENT MODE* fields.

The lead chiller responds to normal start/stop controls such as the occupancy schedule, a forced start or stop, and remote start contact inputs. After completing start-up and ramp loading, the PIC evaluates the need for additional capacity. If additional capacity is needed, the PIC initiates the start-up of the chiller configured at the *LAG ADDRESS*. If the lag chiller is faulted (in alarm) or is in the OFF or LOCAL modes, the chiller at the *STANDBY ADDRESS* (if configured) is requested to start. After the second chiller is started and is running, the lead chiller monitors conditions and evaluates whether the capacity has been reduced enough for the lead chiller to sustain the system alone. If the capacity is reduced enough for the lead chiller to sustain the *WATER/BRINE CONTROL POINT* temperatures alone, then the operating lag chiller is stopped.

If the lead chiller is stopped in CCN mode for any reason other than an alarm (*) condition, the lag and standby chillers are also stopped. If the configured lead chiller stops for an alarm condition, the configured lag chiller takes the lead chiller's place as the lead chiller, and the standby chiller serves as the lag chiller.

If the configured lead chiller does not complete the start-up before the *PRESTART FAULT TIMER* (a user-configured value) elapses, then the lag chiller starts and the lead chiller shuts down. The lead chiller then monitors the start request from the acting lead chiller. The *PRESTART FAULT TIMER* is initiated at the time of a start request. The *PRESTART FAULT TIMER* provides a timeout if there is a prestart alert condition that prevents the chiller from starting in a timely manner. The *PRESTART FAULT TIMER* parameter is on the LEAD/LAG screen, which is accessed from the EQUIPMENT CONFIGURATION table of the SERVICE menu.

If the lag chiller does not achieve start-up before the *PRESTART FAULT TIMER* elapses, the lag chiller stops, and the standby chiller is requested to start, if configured and ready.

Standby Chiller Configuration and Operation — A chiller is designated as a standby chiller when its *LEAD/LAG SELECT* value on the LEAD/LAG screen is set to ''3.'' The standby chiller can operate as a replacement for the lag chiller only if one of the other two chillers is in an alarm (*) condition (as shown on the LID panel). If both lead and lag chillers are in an alarm (*) condition, the standby chiller defaults to operate in CCN mode, based on its configured occupancy schedule and remote contacts input.

Lag Chiller Start-Up Requirements — Before the lag chiller can be started, the following conditions must be met:

1. Lead chiller ramp loading must be complete.
2. Lead chiller water temperature must be greater than the *WATER/BRINE CONTROL POINT* temperature (see the STATUS01 screen) plus 1/2 the *WATER/BRINE DEADBAND* temperature (see the SERVICE1 screen).

   NOTE: The chilled water temperature sensor may be the leaving chilled water sensor, the return water sensor, the common supply water sensor, or the common return water sensor, depending on which options are configured and enabled.
3. Lead chiller *ACTIVE DEMAND LIMIT* (see the STATUS01 screen) value must be greater than 95% of full load amps.
4. Lead chiller temperature pulldown rate (*TEMP PULLDOWN DEG/MIN* on the CONFIG screen) of the chilled water temperature is less than 0.5° F (0.27° C) per minute.
5. The lag chiller status indicates it is in CCN mode and is not in an alarm condition. If the current lag chiller is in an alarm condition, the standby chiller becomes the active lag chiller, if it is configured and available.
6. The configured *LAG START TIMER* entry has elapsed. The *LAG START TIMER* starts when the lead chiller ramp loading is completed. The *LAG START TIMER* entry is on the LEAD/LAG screen, which is accessed from the EQUIPMENT CONFIGURATION table of the SERVICE menu.

When all the above requirements have been met, the lag chiller is forced to a STARTUP mode. The PIC control then monitors the lag chiller for a successful start. If the lag chiller fails to start, the standby chiller, if configured, is started.

Lag Chiller Shutdown Requirements — The following conditions must be met in order for the lag chiller to be stopped.

1. Lead chiller compressor motor load value (*COMPRESSOR MOTOR LOAD* on the STATUS01 screen) is less than the lead chiller percent capacity plus 15%.

   NOTE: Lead chiller percent capacity = 100 – *LAG PERCENT CAPACITY.* The *LAG PERCENT CAPACITY* parameter is on the LEADLAG screen, which is accessed from the EQUIPMENT CONFIGURATION table on the SERVICE menu.
2. The lead chiller chilled water temperature is less than the *WATER/BRINE CONTROL POINT* temperature (see the STATUS01 screen) plus ½ the *WATER/BRINE DEADBAND* temperature (see the SERVICE1 screen).
3. The configured *LAG STOP TIME* entry has elapsed. The *LAG STOP TIME* starts when the chilled water temperature is less than the chilled water *WATER/BRINE CONTROL POINT* plus ½ of the *WATER/BRINE DEADBAND* and the lead chiller compressor motor load (*COMPRESSOR MOTOR LOAD* on the STATUS01 screen) is less than the lead chiller percent capacity plus 15% (see note below). The timer is ignored if the chilled water temperature reaches 3° F (1.67° C) below the *WATER/BRINE CONTROL POINT* and the lead chiller *COMPRESSOR MOTOR LOAD* value is less than the lead chiller percent capacity plus 15%.

NOTE: Lead chiller percent capacity = 100 – *LAG PERCENT CAPACITY.* The *LAG PERCENT CAPACITY* parameter is on the LEAD/LAG screen, which is accessed from the EQUIPMENT CONFIGURATION table on the SERVICE menu.

FAULTED CHILLER OPERATION — If the lead chiller shuts down because of an alarm (*) condition, it stops communicating to the lag and standby chillers. After 30 seconds, the lag chiller becomes the acting lead chiller and starts and stops the standby chiller, if necessary.

If the lag chiller goes into alarm when the lead chiller is also in alarm, the standby chiller reverts to a stand-alone CCN mode of operation.

If the lead chiller is in an alarm (*) condition (as shown on the LID panel), press the $\boxed{\text{RESET}}$ softkey to clear the alarm. The chiller is placed in CCN mode. The lead chiller communicates and monitors the RUN STATUS of the lag and standby chillers. If both the lag and standby chillers are running, the lead chiller does not attempt to start and does not assume the role of lead chiller until either the lag or standby chiller shuts down. If only one chiller is running, the lead chiller waits for a start request from the operating chiller. When the configured lead chiller starts, it assumes its role as lead chiller.

LOAD BALANCING — When the *LOAD BALANCE OPTION* (see LEAD/LAG screen) is enabled, the lead chiller sets the *ACTIVE DEMAND LIMIT* in the lag chiller to the lead chiller's compressor motor load value (*COMPRESSOR MOTOR LOAD* on the STATUS01 screen). This value has limits of 40% to 100%. When the lag chiller *ACTIVE DEMAND LIMIT* is set, the *WATER/BRINE CONTROL POINT* must be modified to a value of 3° F (1.67° C) less than the lead chiller's *WATER/BRINE CONTROL POINT* value. If the *LOAD BALANCE OPTION* is disabled, the *ACTIVE DEMAND LIMIT* and the *WATER/BRINE CONTROL POINT* are forced to the same value as the lead chiller.

AUTO. RESTART AFTER POWER FAILURE — When an auto. restart condition occurs, each chiller may have a delay added to the start-up sequence, depending on its lead/lag configuration. The lead chiller does not have a delay. The lag chiller has a 45-second delay. The standby chiller has a 90-second delay. The delay time is added after the chiller water flow is verified. The PIC ensures the guide vanes are closed. After the guide vane position is confirmed, the delay for lag and standby chillers occurs prior to energizing the oil pump. The normal start-up sequence then continues. The auto. restart delay sequence occurs whether the chiller is in CCN or LOCAL mode and is intended to stagger the compressor motor starts. Preventing the motors from starting simultaneously helps reduce the inrush demands on the building power system.

Ice Build Control — The ice build control option automatically sets the chilled *WATER/BRINE CONTROL POINT* of the chiller to a temperature that allows ice building for thermal storage.

NOTE: For ice build control to operate properly, the PIC must be in CCN mode.

NOTE: See Fig. 15 and 16 for more information on ice build-related menus.

The PIC can be configured for ice build operation.

- From the SERVICE menu, access the EQUIPMENT CONFIGURATION table. From there, select the CONFIG screen to enable or disable the *ICE BUILD OPTION*. See Table 2, Example 5.
- The *ICE BUILD SETPOINT* can be configured from the SETPOINT display, which is accessed from the PIC main menu. See Table 2, Example 4.
- The ice build schedule can be viewed or modified from the SCHEDULE table. From this table, select the ice build schedule (OCCPC02S) screen. See Fig. 17 and the section on Time Schedule Operation, page 19, for more information on modifying chiller schedules.

The ice build time schedule defines the period(s) during which ice build is active if the ice build option is enabled. If the ice build time schedule overlaps other schedules, the ice build time schedule takes priority. During the ice build period, the *WATER/BRINE CONTROL POINT* is set to the *ICE BUILD SETPOINT* for temperature control. The *ICE BUILD RECYCLE OPTION* and *ICE BUILD TERMINATION* parameters, accessed from the CONFIG screen, allow the chiller operator to recycle or terminate the ice build cycle. The ice build cycle can be configured to terminate if:

- the *ENTERING CHILLED WATER* temperature is less than the *ICE BUILD SETPOINT*. In this case, the operator sets the *ICE BUILD TERMINATION* parameter to 0 on the CONFIG screen.
- the REMOTE CONTACT inputs from an ice level indicator are opened. In this case, the operator sets the *ICE BUILD TERMINATION* parameter to 1 on the CONFIG screen.
- the chilled water temperature is less than the ice build set point *and* the remote contact inputs from an ice level indicator are open. In this case, the operator sets the *ICE BUILD TERMINATION* parameter to 2 on the CONFIG screen.
- the end of the ice build time schedule has been reached.

ICE BUILD INITIATION — The ice build time schedule (OCCPC02S) is the means for activating the ice build option. The ice build option is enabled if

- a day of the week and a time period on the ice build time schedule are enabled. The SCHEDULE screen shows an X in the day field and ON/OFF times are designated for the day(s),
- *and* the *ICE BUILD OPTION* is enabled.

The following events take place (unless overridden by a higher authority CCN device).

1. *CHILLER START/STOP* is forced to START.
2. The *WATER/BRINE CONTROL POINT* is forced to the *ICE BUILD SETPOINT*.
3. Any force (Auto) is removed from the *ACTIVE DEMAND LIMIT*.

NOTE: A parameter's value can be forced, that is, the value can be manually changed at the LID by an operator, changed from another CCN device, or changed by other algorithms in the PIC control system.

NOTE: Items 1-3 (shown above) do not occur if the chiller is configured and operating as a lag or standby chiller for lead/lag operation and is actively being controlled by a lead chiller. The lead chiller communicates the *ICE BUILD SET POINT*, the desired *CHILLER START/STOP* state, and the *ACTIVE DEMAND LIMIT* to the lag or standby chiller as required for ice build, if configured to do so.

START-UP/RECYCLE OPERATION — If the chiller is not running when ice build activates, the PIC checks the following conditions, based on the *ICE BUILD TERMINATION* value, to avoid starting the compressor unnecessarily:

- if *ICE BUILD TERMINATION* is set to the *TEMP* option and the *ENTERING CHILLED WATER* temperature is less than or equal to the *ICE BUILD SETPOINT*;
- if *ICE BUILD TERMINATION* is set to the *CONTACTS* option and the remote contacts are open;
- if the *ICE BUILD TERMINATION* is set to the *BOTH* (temperature and contacts) option and the *ENTERING CHILLED WATER* temperature is less than or equal to the *ICE BUILD SETPOINT and* the remote contacts are open.

The *ICE BUILD RECYCLE OPTION* on the CONFIG screen determines whether or not the chiller will go into an ice build RECYCLE mode.

- If the *ICE BUILD RECYCLE OPTION* is set to DSABLE (disable), the PIC reverts to normal temperature control when the ice build function terminates.
- If the *ICE BUILD RECYCLE OPTION* is set to ENABLE, the PIC goes into an *ICE BUILD RECYCLE* mode and the chilled water pump relay remains energized to keep the chilled water flowing when the ice build function terminates. If the temperature of the *ENTERING CHILLED WATER* increases above the *ICE BUILD SETPOINT* plus the *RECYCLE RESTART DELTA T* value, the compressor restarts and controls the chilled water/brine temperature to the *ICE BUILD SETPOINT*.

TEMPERATURE CONTROL DURING ICE BUILD — During ice build, the capacity control algorithm uses the *WATER/BRINE CONTROL POINT* minus 5 F (2.7 C) to control the *LEAVING CHILLED WATER* temperature. (See Table 2, Example 10, the *CAPACITY CONTROL* parameter on the MAINT01 screen.) The following options, which can be configured on the CONFIG screen, are ignored during ice build:

- *ECW CONTROL OPTION* and any temperature reset options
- *20 mA DEMAND LIMIT OPTION*

TERMINATION OF ICE BUILD — The ice build function terminates under the following conditions:

1. Time Schedule — When the current time on the ice build time schedule (OCCPC02S) is *not* set as an ice build time period.
2. Entering Chilled Water Temperature — Compressor operation terminates, based on temperature, if the *ICE BUILD TERMINATION* parameter is set to 0 (*TEMP*) and the *ENTERING CHILLED WATER* temperature is less than the *ICE BUILD SETPOINT*. If the *ICE BUILD RECYCLE OPTION* is set to ENABLE, a recycle shutdown occurs and recycle start-up depends on the *LEAVING CHILLED WATER* temperature being greater than the *WATER/BRINE CONTROL POINT* plus the *RECYCLE RESTART DELTA T* temperature.
3. Remote Contacts/Ice Level Input — Compressor operation terminates when the *ICE BUILD TERMINATION* parameter is set to 1 (*CONTACTS*) and the remote contacts are open. In this case, the contacts provide ice level termination control. The contacts are used to stop the ice build function when a time period on the ice build schedule (OCCPC02S) *is* set for ice build operation. The remote contacts can still be opened and closed to start and stop the chiller when a specific time period on the ice build schedule is *not* set for ice build.

4. Entering Chilled Water Temperature and Remote Contacts — Compressor operation terminates when the *ICE BUILD TERMINATION* parameter is set to 2 (*BOTH*) and the conditions described above in items 2 and 3 for entering chilled water temperature and remote contacts have occurred.

NOTE: It is not possible to override the *CHILLER START/STOP*, *WATER/BRINE CONTROL POINT*, and *ACTIVE DEMAND LIMIT* variables from CCN devices (with a priority less than 4) during the ice build period. However, a CCN device can override these settings during 2-chiller lead/lag operation.

RETURN TO NON-ICE BUILD OPERATIONS — The ice build function forces the chiller to start, even if all other schedules indicate that the chiller should stop. When the ice build function terminates, the chiller returns to normal temperature control and start/stop schedule operation. The *CHILLER START/STOP* and *WATER/BRINE CONTROL POINT* return to normal operation. If the *CHILLER START/STOP* or *WATER/BRINE CONTROL POINT* has been forced (with a device of less than 4 priority) before the ice build function started, when the ice build function ends, the previous forces (of less than 4 priority) are not automatically restored.

Attach to Network Device Control — The Service menu includes the ATTACH TO NETWORK DEVICE screen. From this screen, the operator can:

- enter the time schedule number (if changed) for OCCPC03S, as defined in the SERVICE1 screen
- attach the LID to any CCN device, if the chiller has been connected to a CCN network. This may include other PIC-controlled chillers.
- change to a new PSIO or LID module
- upgrade software

Figure 21 shows the ATTACH TO NETWORK DEVICE screen. The *LOCAL* parameter is always the PSIO module address of the chiller on which the LID is mounted. Whenever the controller identification of the PSIO changes, the change is reflected automatically in the BUS and ADDRESS columns for the local device. See Fig. 16.

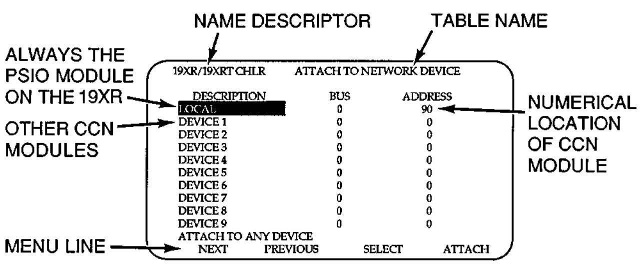


Fig. 21 — Example of Attach to Network Device Screen

When the ATTACH TO NETWORK DEVICE screen is accessed, information can not be read from the LID on any device until one of the devices listed on that screen is attached. The LID erases information about the module to which it was attached to make room for information on another device. Therefore, a CCN module must be attached when this screen is entered.

To attach a device, highlight it using the SELECT softkey and press the ATTACH softkey. The message ''UPLOADING TABLES, PLEASE WAIT'' displays. The LID then uploads the highlighted device or module. If the module address cannot be found, the message ''COMMUNICATION FAILURE'' appears. The LID then reverts back to the ATTACH TO DEVICE screen. Try another device or check the address of the device that would not attach. The upload process time for each CCN module is different. In general, the uploading process takes 3 to 5 minutes. Before leaving the ATTACH TO NETWORK DEVICE screen, select the local device. Otherwise, the LID will be unable to display information on the local chiller.

ATTACHING TO OTHER CCN MODULES — If the chiller PSIO has been connected to a CCN Network or other PIC controlled chillers through CCN wiring, the LID can be used to view or change parameters on the other controllers. Other PIC chillers can be viewed and set points changed (if the other unit is in CCN control), if desired, from this particular LID module.

To view the other devices, move to the ATTACH TO NETWORK DEVICE table. Move the highlight bar to any device number. Press SELECT softkey to change the bus number and address of the module to be viewed. Press EXIT softkey to move back to the ATTACH TO NETWORK DEVICE table. If the module number is not valid, the ''COMMUNICATION FAILURE'' message will show and a new address number must be entered or the wiring checked. If the module is communicating properly, the ''UPLOAD IN PROGRESS'' message will flash and the new module can now be viewed.

Whenever there is a question regarding which module on the LID is currently being shown, check the device name descriptor on the upper left hand corner of the LID screen. See Fig. 21.

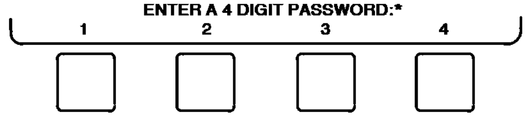
When the CCN device has been viewed, the ATTACH TO NETWORK DEVICE table should be used to attach to the PIC that is on the chiller. Move to the ATTACH TO NETWORK DEVICE table (LOCAL should be highlighted) and press the ATTACH softkey to upload the LOCAL device. The PSIO for the 19XR will be uploaded and the local LID default screen will display.

NOTE: The LID will not automatically reattach to the PSIO module on the chiller. Press the ATTACH softkey to attach to the LOCAL device and view the chiller PSIO.

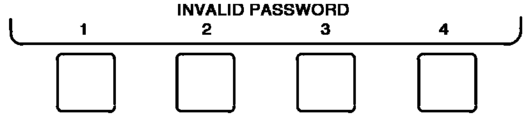
Service Operation — An overview of the tables and screens available for the SERVICE function is shown in Fig. 16.

TO ACCESS THE SERVICE SCREENS — When the SERVICE screens are accessed, a password must be entered.

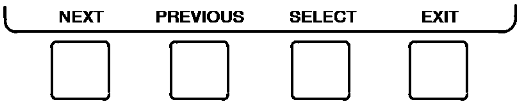
1. From the main MENU screen, press the SERVICE softkey. The softkeys now correspond to the numerals 1, 2, 3, 4.
2. Press the four digits of the password, one at a time. An asterisk (*) appears as each digit is entered.



NOTE: The initial factory-set password is 1 - 1- 1 - 1. If the password is incorrect, an error message is displayed.



If this occurs, return to Step 1 and try to access the SERVICE screens again. If the password is correct, the softkey labels change to:



and the LID screen displays the following list of available SERVICE screens:

- Alarm History
- Control Test
- Control Algorithm Status
- Equipment Configuration
- Equipment Service
- Time and Date
- Attach to Network Device
- Log Out of Device
- Controller Identification
- LID Configuration

See Fig. 16 for additional screens and tables available from the SERVICE screens listed above. Use the EXIT softkey to return to the main MENU screen.

NOTE: To prevent unauthorized persons from accessing the LID service screens, the LID automatically signs off and password-protects itself if a key has not been pressed for 15 minutes. The sequence is as follows. Fifteen minutes after the last key is pressed, the default screen displays, the LID screen light goes out (analogous to a screen saver), and the LID logs out of the password-protected SERVICE menu. Other screen and menus, such as the STATUS screen can be accessed without the password by pressing the appropriate softkey.

TO LOG OUT OF NETWORK DEVICE — To access this screen and log out of a network device, from the default LID screen, press the MENU and SERVICE softkeys. Enter the password and, from the SERVICE menu, highlight LOG OUT OF NETWORK DEVICE and press the SELECT softkey. The LID default screen will now be displayed.

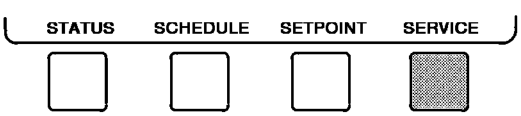
HOLIDAY SCHEDULING (Fig. 22) — The time schedules may be configured for special operation during a holiday period. When modifying a time period, the ''H'' at the end of the days of the week field signifies that the period is applicable to a holiday. (See Fig. 17.)

The broadcast function must be activated for the holidays configured on the HOLIDEF screen to work properly. Access the BRODEF screen from the EQUIPMENT CONFIGURATION table and answer ''Yes'' to the activated function. Note that when the chiller is connected to a CCN Network, only one chiller or CCN device can be configured as the broadcast device. The controller that is configured as the broadcaster is the device responsible for transmitting holiday, time, and daylight-savings dates throughout the network.

To access the BRODEF screen, see the SERVICE menu structure, Fig. 16.

To view or change the holiday periods for up to 18 different holidays, perform the following operation:

1. At the Menu screen, press SERVICE to access the Service menu.



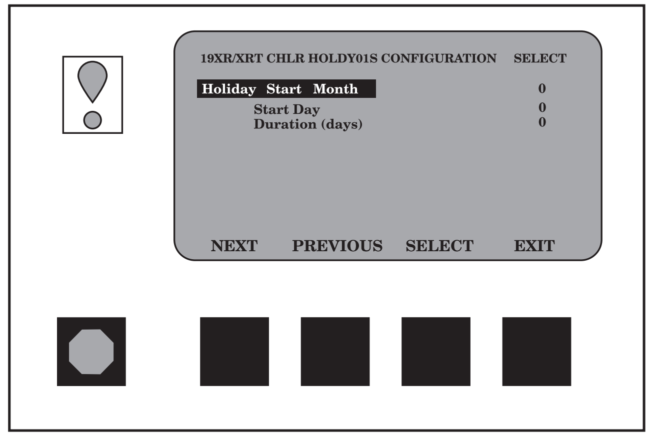
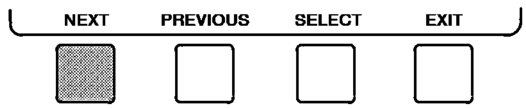
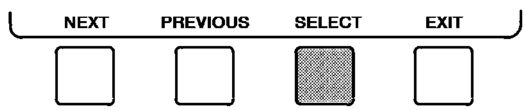


Fig. 22 — Example of Holiday Period Screen

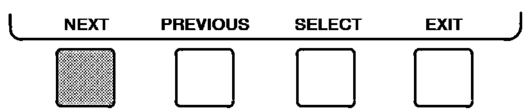
2. If not logged on, follow the instructions for To Log On or To Log Off. Once logged on, press NEXT until Equipment Configuration is highlighted.



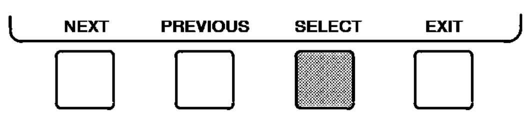
3. Once Equipment Configuration is highlighted, press SELECT to access.



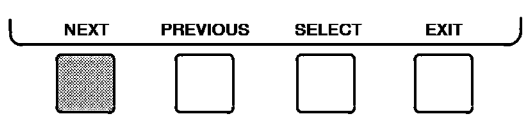
4. Press NEXT until HOLIDEF is highlighted. This is the Holiday Definition table.



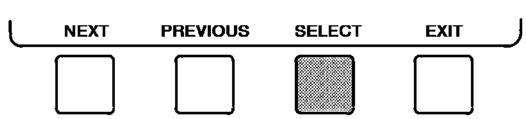
5. Press SELECT to enter the Data Table Select screen. This screen lists 18 holiday tables.



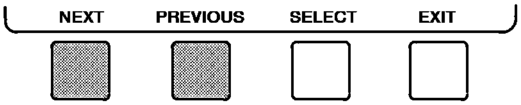
6. Press NEXT to highlight the holiday table that is to be viewed or changed. Each table is one holiday period, starting on a specific date, and lasting up to 99 days.



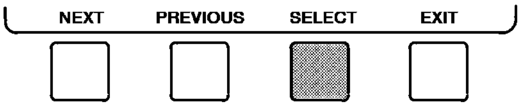
7. Press SELECT to access the holiday table. The Configuration Select table now shows the holiday start month and day, and how many days the holiday period will last.



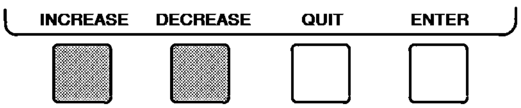
8. Press NEXT or PREVIOUS to highlight the month, day, or duration.



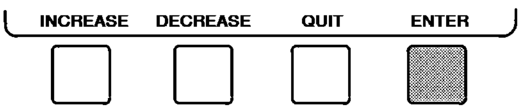
9. Press SELECT to modify the month, day, or duration.



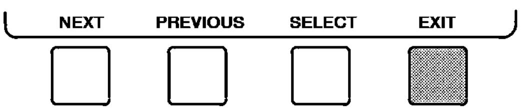
10. Press INCREASE or DECREASE to change the selected value.



11. Press ENTER to save the changes.



12. Press EXIT to return to the previous menu.



START-UP/SHUTDOWN/RECYCLE SEQUENCE (Fig. 23)

Local Start-Up — Local start-up (or a manual start-up) is initiated by pressing the LOCAL menu softkey on the default LID screen. Local start-up can proceed when the chiller schedule indicates that the current time and date have been established as a run time and date, and after the internal 15-minute start-to-start and the 1-minute stop-to-start inhibit timers have expired. If the timers have not expired the *RUN STATUS* parameter on the STATUS01 screen now reads *TIMEOUT*.

NOTE: The time schedule is said to be ''occupied.'' If the *OCCUPIED ?* parameter on the STATUS01 screen is set to YES. For more information on occupancy schedules, see the sections on Time Schedule Operation (page 19), Occupancy Schedule (page 28), and To Prevent Accidental Start-Up (page 55), and Fig. 17.

If the *OCCUPIED ?* parameter on the STATUS01 screen is set to NO, the chiller can be forced to start as follows. From the default LID screen, press the MENU and STATUS softkeys. Scroll to highlight STATUS01. Press the SELECT softkey. Scroll to highlight *CHILLER START/STOP*. Press the START softkey to override the schedule and start the chiller.

NOTE: The chiller will continue to run until this forced start is released, regardless of the programmed schedule. To release the forced start, highlight *CHILLER START/STOP* from the STATUS01 screen and press the RELEASE softkey. This action returns the chiller to the start and stop times established by the schedule.

The chiller may also be started by overriding the time schedule. From the default screen, press the MENU and SCHEDULE softkeys. Scroll down and select the current schedule. Select OVERRIDE, and set the desired override time.

Another condition for start-up must be met for chillers that have the *REMOTE CONTACTS OPTION* on the EQUIPMENT CONFIGURATION screen set to ENABLE. For these chillers, the *REMOTE CONTACTS INPUT* parameter on the STATUS01 screen must be ON. From the LID default screen, press the MENU and STATUS softkeys. Scroll to highlight STATUS01 and press the SELECT softkey. Scroll down the STATUS01 screen to highlight *REMOTE CONTACTS INPUT* and press the SELECT softkey. Then, press the ON softkey. To end the override, select *REMOTE CONTACTS INPUT* and press the RELEASE softkey.

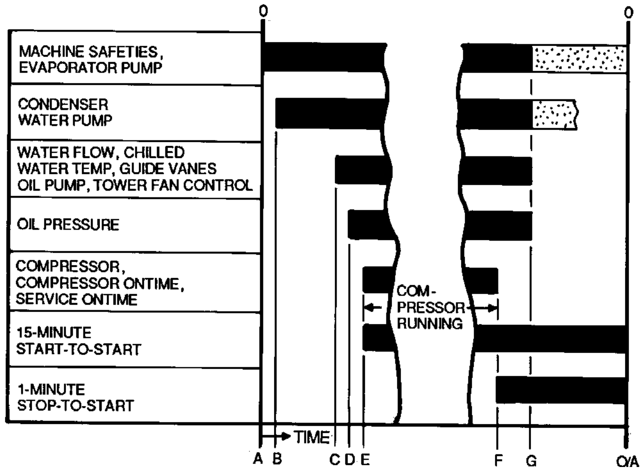
Once local start-up begins, the PIC performs a series of pre-start tests to verify that all pre-start alerts and safeties are within the limits shown in Table 3. The *RUN STATUS* parameter on the STATUS01 screen line now reads PRESTART. If a test is not successful, the start-up is delayed or aborted. If the tests are successful, the chilled water/brine pump relay energizes, and the STATUS01 screen line now reads STARTUP.

Five seconds later, the condenser pump relay energizes. Thirty seconds later the PIC monitors the chilled water and condenser water flow switches and waits until the *WATER FLOW VERIFY TIME* (operator-configured, default 5 minutes) expires to confirm flow. After flow is verified, the chilled water/brine temperature is compared to *WATER/BRINE CONTROL POINT* plus *WATER/BRINE DEADBAND*. If the temperature is less than or equal to this value, the PIC turns off the condenser pump relay and goes into a RECYCLE mode.

If the water/brine temperature is high enough, the start-up sequence continues and checks the guide vane position. If the guide vanes are more than 6% open, the start-up waits until the PIC closes the vanes. If the vanes are closed and the oil pump pressure is less than 4 psi (28 kPa), the oil pump relay energizes. The PIC then waits until the oil pressure *(OIL PRESS VERIFY TIME,* operator-configured, default of 15 seconds) reaches a maximum of 18 psi (124 kPa). After oil pressure is verified, the PIC waits 15 seconds, and the compressor start relay (1CR) energizes to start the compressor.

Compressor ontime and service ontime timers start, and the compressor starts in 12 hours counter and the number of starts over a 12-hour period counter advance by one.

Failure to verify any of the requirements up to this point will result in the PIC aborting the start and displaying the applicable pre-start mode of failure on the LID default screen. A pre-start failure does not advance the starts in 12 hours counter. Any failure after the 1CR relay has energized results in a safety shutdown, advances the starts in the 12 hours counter by one, and displays the applicable shutdown status on the LID display.



A — START INITIATED — Prestart checks made; evaporator pump started.
B — Condenser water pump started (5 seconds after A).
C — Water flows verified (30 seconds to 5 minutes maximum after B). Chilled water temperatures checked against control point. Guide vanes checked for closure. Oil pump started; tower fan control enabled.
D — Oil pressure verified (45 seconds minimum, 300 seconds maximum after C).
E — Compressor motor starts, compressor ontime and service ontime start, 15-minute inhibit timer starts (10 seconds after D), total compressor starts advances by one, and the number of starts over a 12-hour period advances by one.
F — SHUTDOWN INITIATED — Compressor motor stops, compressor ontime and service ontime stops, and 1-minute inhibit timer starts.
G — Oil pump and evaporator pumps deenergized (60 seconds after F). Condenser pump and tower fan control may continue to operate if condenser pressure is high. Evaporator pump may continue if in RECYCLE mode.
O/A — Restart permitted (both inhibit timers expired: minimum of 15 minutes after E; minimum of 1 minute after F).

Fig. 23 — Control Sequence

Shutdown Sequence — Chiller shutdown begins if any of the following occurs:

- the STOP button is pressed for at least one second (the alarm light blinks once to confirm the stop command)
- a recycle condition is present (see Chilled Water Recycle Mode section)
- the time schedule has gone into unoccupied mode
- the chiller protective limit has been reached and chiller is in alarm
- the start/stop status is overridden to stop from the CCN network or the LID

When a stop signal occurs, the shutdown sequence first stops the compressor by deactivating the start relay. A status message of ‘‘SHUTDOWN IN PROGRESS, COMPRESSOR DEENERGIZED’’ is displayed, and the compressor ontime and service ontime stop. The guide vanes are then brought to the closed position. The oil pump relay and the chilled water/brine pump relay shut down 60 seconds after the compressor stops. The condenser water pump shuts down when the *CONDENSER REFRIGERANT TEMP* is less than the *CONDENSER PRESSURE OVERRIDE* minus 5 psi (34 kPa), or is less than or equal to the *ENTERING CONDENSER WATER TEMP* plus 3° F(2° C). The stop-to-start timer now begins to count down. If the start-to-start timer value is still greater than the value of the start-to-stop timer, then this time displays on the LID.

Certain conditions that occur during shutdown can change this sequence.

- If the *COMPRESSOR MOTOR LOAD* is greater than 10% after shutdown, or the starter contacts remain energized, the oil pump and chilled water pump remain energized and the alarm is displayed.
- If the *ENTERING CONDENSER WATER* temperature is greater than 115 F (46 C) at shutdown, the condenser pump is deenergized after the 1CR compressor start relay.
- If the chiller shuts down due to low refrigerant temperature, the chilled water pump continues to run until the *LEAVING CHILLED WATER* temperature is greater than the *CONTROL POINT* temperature, plus 5° F (3° C).

Automatic Soft Stop Amps Threshold — The soft stop amps threshold feature closes the guide vanes of the compressor automatically if a non-recycle, non-alarm stop signal occurs before the compressor motor is deenergized.

If the STOP button is pressed, the guide vanes close to a preset amperage percent or until the guide vane is less than 2% open. The compressor then shuts off.

If the chiller enters an alarm state or if the compressor enters a RECYCLE mode, the compressor deenergizes immediately.

To activate the soft stop amps threshold feature, scroll to the bottom of SERVICE1 screen on the LID. Use the INCREASE or DECREASE softkey to set the *SOFT STOP AMPS THRESHOLD* parameter to the percent of amps at which the motor will shut down. The default setting is 100% amps (no soft stop). The range is 40 to 100%.

When the soft stop amps threshold feature is being applied, a status message, ‘‘SHUTDOWN IN PROGRESS, COMPRESSOR UNLOADING’’ displays on the LID.

The soft stop amps threshold function can be terminated and the compressor motor deenergized immediately by depressing the STOP button twice.

Chilled Water Recycle Mode — The chiller may cycle off and wait until the load increases to restart when the compressor is running in a lightly loaded condition. This cycling is normal and is known as ″recycle.″ A recycle shutdown is initiated when any of the following conditions are true:

- the chiller is in LCW control, the difference between the *LEAVING CHILLED WATER* temperature and *ENTERING CHILLED WATER* temperature is less than the *RECYCLE SHUTDOWN DELTA T* (found in the SERVICE1 table) *and* the *LEAVING CHILLED WATER* temperature is below the *CONTROL POINT and* the *CONTROL POINT* has not increased in the last 5 minutes.
- the *ECW CONTROL OPTION* is enabled, the difference between the *ENTERING CHILLED WATER* temperature and the *LEAVING CHILLED WATER* temperature is less than the *RECYCLE SHUTDOWN DELTA T* (found in the SERVICE1 table), *and* the *ENTERING CHILLED WATER* temperature is below the *CONTROL POINT and* the *CONTROL POINT* has not increased in the last 5 minutes.
- the *LEAVING CHILLED WATER* temperature is within 3° F (2° C) of the *BRINE REFRIG TRIPPOINT.*

When the chiller is in RECYCLE mode, the chilled water pump relay remains energized so the chilled water temperature can be monitored for increasing load. The recycle control uses *RECYCLE RESTART DELTA T* to check when the compressor should be restarted. This is an operator-configured function which defaults to 5° F (3° C). This value can be viewed or modified on the SERVICE1 table. The compressor will restart when the chiller is:

- in LCW CONTROL and the *LEAVING CHILLED WATER* temperature is greater than the *CONTROL POINT* plus the *RECYCLE RESTART DELTA T.*
- in ECW CONTROL and the *ENTERING CHILLED WATER* temperature is greater than the *CONTROL POINT* plus the *RECYCLE RESTART DELTA T.*

Once these conditions are met, the compressor initiates a start-up with a normal start-up sequence.

An alert condition may be generated if 5 or more recycle start-ups occur in less than 4 hours. Excessive recycling can reduce chiller life; therefore, compressor recycling due to extremely low loads should be reduced.

To reduce compressor recycling, use the time schedule to shut the chiller down during known low load operation period, or increase the chiller load by running the fan systems. If the hot gas bypass is installed, adjust the values to ensure that hot gas is energized during light load conditions. Increase the *RECYCLE RESTART DELTA T* on the SERVICE1 table to lengthen the time between restarts.

The chiller should not be operated below design minimum load without a hot gas bypass installed.

Safety Shutdown — A safety shutdown is identical to a manual shutdown with the exception that, during a safety shutdown, the LID displays the reason for the shutdown, the alarm light blinks continuously, and the spare alarm contacts are energized.

After a safety shutdown, the RESET softkey must be pressed to clear the alarm. If the alarm condition is still present, the alarm light continues to blink. Once the alarm is cleared, the operator must press the CCN or LOCAL softkeys to restart the chiller.

⚠ CAUTION

Do not reset starter loads or any other starter safety for 30 seconds after the compressor has stopped. Voltage output to the compressor start signal is maintained for 10 seconds to determine starter fault.

BEFORE INITIAL START-UP

Job Data Required

- list of applicable design temperatures and pressures (product data submittal)
- chiller certified prints
- starting equipment details and wiring diagrams
- diagrams and instructions for special controls or options
- 19XR Installation Instructions
- pumpout unit instructions

Equipment Required

- mechanic's tools (refrigeration)
- digital volt-ohmmeter (DVM)
- clamp-on ammeter
- electronic leak detector
- absolute pressure manometer or wet-bulb vacuum indicator (Fig. 24)
- 500-v insulation tester (megohmmeter) for compressor motors with nameplate voltage of 600 v or less, or a 5000-v insulation tester for compressor motor rated above 600 v

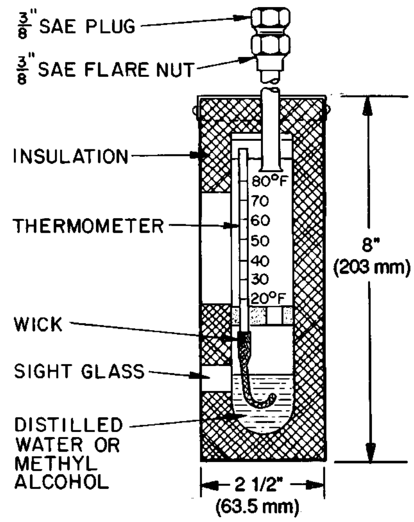


Fig. 24 — Typical Wet-Bulb Type Vacuum Indicator

Using the Optional Storage Tank and Pumpout System — Refer to Chillers with Storage Tanks section, page 59 for pumpout system preparation, refrigerant transfer, and chiller evacuation.

Remove Shipping Packaging — Remove any packaging material from the control center, power panel, guide vane actuator, motor cooling and oil reclaim solenoids, motor and bearing temperature sensor covers, and the factory-mounted starter.

Open Oil Circuit Valves — Check to ensure the oil filter isolation valves (Fig. 4) are open by removing the valve cap and checking the valve stem.

Tighten All Gasketed Joints and Guide Vane Shaft Packing — Gaskets and packing normally relax by the time the chiller arrives at the jobsite. Tighten all gasketed joints and the guide vane shaft packing to ensure a leak-tight chiller.

Check Chiller Tightness — Figure 25 outlines the proper sequence and procedures for leak testing.

19XR chillers are shipped with the refrigerant contained in the condenser shell and the oil charge in the compressor. The cooler is shipped with a 15 psig (103 kPa) refrigerant charge. Units may be ordered with the refrigerant shipped separately, along with a 15 psig (103 kPa) nitrogen-holding charge in each vessel.

To determine if there are any leaks, the chiller should be charged with refrigerant. Use an electronic leak detector to check all flanges and solder joints after the chiller is pressurized. If any leaks are detected, follow the leak test procedure.

If the chiller is spring isolated, keep all springs blocked in both directions to prevent possible piping stress and damage during the transfer of refrigerant from vessel to vessel during the leak test process, or any time refrigerant is being transferred. Adjust the springs when the refrigerant is in operating condition and the water circuits are full.

Refrigerant Tracer — Carrier recommends the use of an environmentally acceptable refrigerant tracer for leak testing with an electronic detector or halide torch.

Ultrasonic leak detectors can also be used if the chiller is under pressure.

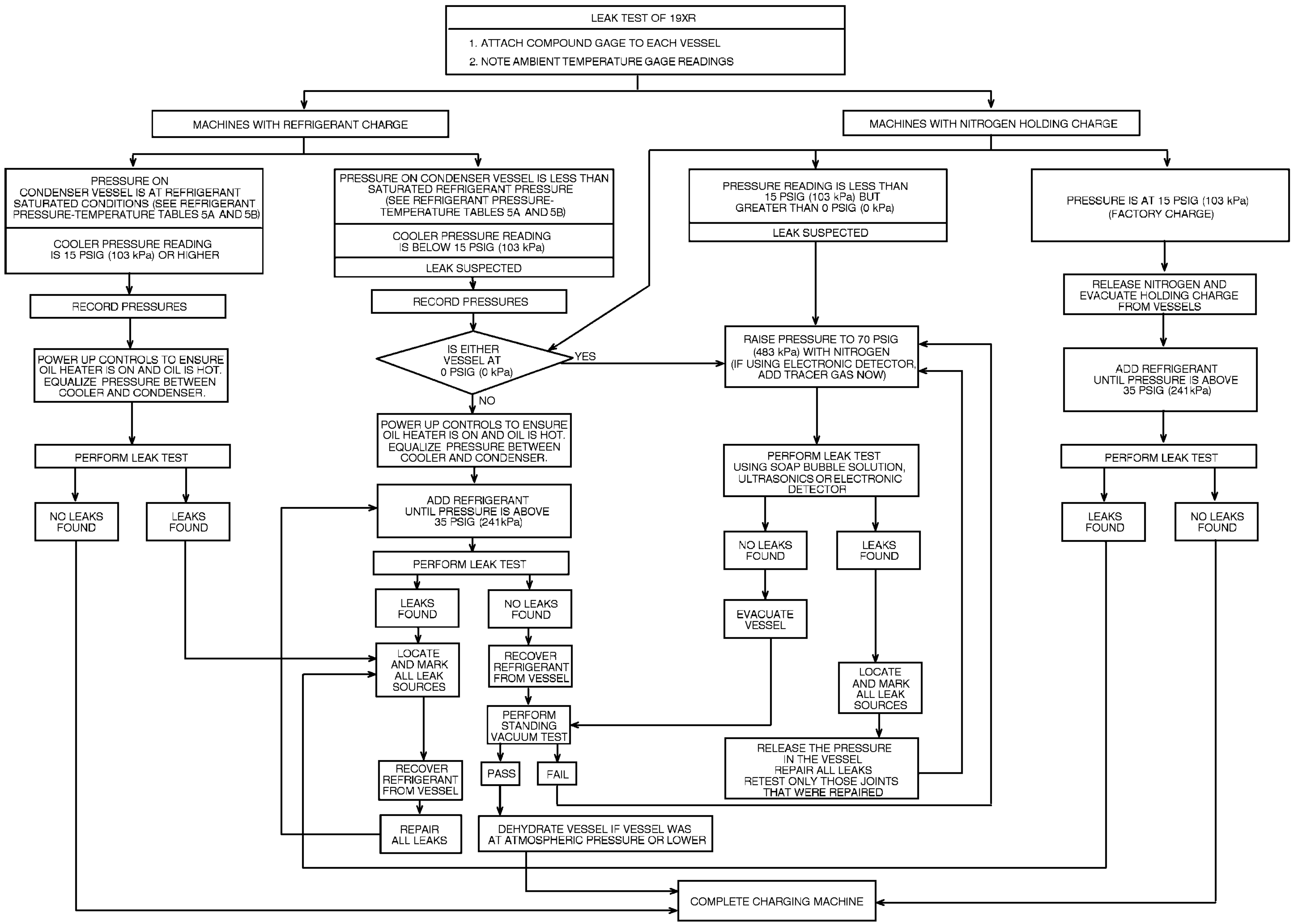


Fig. 25 — 19XR Leak Test Procedures

⚠ WARNING

Do not use air or oxygen as a means of pressurizing the chiller. Mixtures of HFC-134a and air can undergo combustion.

Leak Test Chiller — Due to regulations regarding refrigerant emissions and the difficulties associated with separating contaminants from the refrigerant, Carrier recommends the following leak test procedure. See Fig. 25 for an outline of the leak test procedure. Refer to Fig. 26 and 27 during pumpout procedures and Tables 5A and 5B for refrigerant pressure/temperature values.

1. If the pressure readings are normal for the chiller condition:
   a. Evacuate the holding charge from the vessels, if present.
   b. Raise the chiller pressure, if necessary, by adding refrigerant until pressure is at the equivalent saturated pressure for the surrounding temperature. Follow the pumpout procedures in the Transfer Refrigerant from Pumpout Storage Tank to Chiller section, Steps 1a - e, page 59.

   ⚠ WARNING

   Never charge liquid refrigerant into the chiller if the pressure in the chiller is less than 35 psig (241 kPa) for HFC-134a. Charge as a gas only, with the cooler and condenser pumps running, until this pressure is reached, using PUMPDOWN LOCKOUT and TERMINATE LOCKOUT mode on the PIC. Flashing of liquid refrigerant at low pressures can cause tube freeze-up and considerable damage.

   c. Leak test chiller as outlined in Steps 3 - 9.
2. If the pressure readings are abnormal for the chiller condition:
   a. Prepare to leak test chillers shipped with refrigerant (Step 2h).
   b. Check for large leaks by connecting a nitrogen bottle and raising the pressure to 30 psig (207 kPa). Soap test all joints. If the test pressure holds for 30 minutes, prepare the test for small leaks (Steps 2g - h).
   c. Plainly mark any leaks that are found.
   d. Release the pressure in the system.
   e. Repair all leaks.
   f. Retest the joints that were repaired.
   g. After successfully completing the test for large leaks, remove as much nitrogen, air, and moisture as possible, given the fact that small leaks may be present in the system. This can be accomplished by following the dehydration procedure, outlined in the Chiller Dehydration section, page 46.
   h. Slowly raise the system pressure to a maximum of 160 psig (1103 kPa) but no less than 35 psig (241 kPa) for HFC-134a by adding refrigerant. Proceed with the test for small leaks (Steps 3-9).
3. Check the chiller carefully with an electronic leak detector, halide torch, or soap bubble solution.
4. Leak Determination — If an electronic leak detector indicates a leak, use a soap bubble solution, if possible, to confirm. Total all leak rates for the entire chiller. Leakage at rates greater than 1 lb/year (0.45 kg/year) for the entire chiller must be repaired. Note the total chiller leak rate on the start-up report.
5. If no leak is found during the initial start-up procedures, complete the transfer of refrigerant gas from the pumpout storage tank to the chiller (see Transfer Refrigerant from Pumpout Storage Tank to Chiller section, page 59). Retest for leaks.
6. If no leak is found after a retest:
   a. Transfer the refrigerant to the pumpout storage tank and perform a standing vacuum test as outlined in the Standing Vacuum Test section, below.
   b. If the chiller fails the standing vacuum test, check for large leaks (Step 2b).
   c. If the chiller passes the standing vacuum test, dehydrate the chiller. Follow the procedure in the Chiller Dehydration section. Charge the chiller with refrigerant (see Transfer Refrigerant from Pumpout Storage Tank to Chiller section, page 59).
7. If a leak is found after a retest, pump the refrigerant back into the pumpout storage tank or, if isolation valves are present, pump the refrigerant into the non-leaking vessel (see Pumpout and Refrigerant Transfer procedures section).
8. Transfer the refrigerant until the chiller pressure is at 18 in. Hg (40 kPa absolute).
9. Repair the leak and repeat the procedure, beginning from Step 2h, to ensure a leak-tight repair. (If the chiller is opened to the atmosphere for an extended period, evacuate it before repeating the leak test.)

Standing Vacuum Test — When performing the standing vacuum test or chiller dehydration, use a manometer or a wet bulb indicator. Dial gages cannot indicate the small amount of acceptable leakage during a short period of time.

1. Attach an absolute pressure manometer or wet bulb indicator to the chiller.
2. Evacuate the vessel (see Pumpout and Refrigerant Transfer Procedures section, page 57) to at least 18 in. Hg vac, ref 30-in. bar (41 kPa), using a vacuum pump or the pumpout unit.
3. Valve off the pump to hold the vacuum and record the manometer or indicator reading.
4. a. If the leakage rate is less than 0.05 in. Hg (0.17 kPa) in 24 hours, the chiller is sufficiently tight.
   b. If the leakage rate exceeds 0.05 in. Hg (0.17 kPa) in 24 hours, repressurize the vessel and test for leaks. If refrigerant is available in the other vessel, pressurize by following Steps 2-10 of Return Chiller To Normal Operating Conditions section, page 61. If not, use nitrogen and a refrigerant tracer. Raise the vessel pressure in increments until the leak is detected. If refrigerant is used, the maximum gas pressure is approximately 70 psig (483 kPa) for HFC-134a at normal ambient temperature. If nitrogen is used, limit the leak test pressure to 230 psig (1585 kPa) maximum.
5. Repair the leak, retest, and proceed with dehydration.

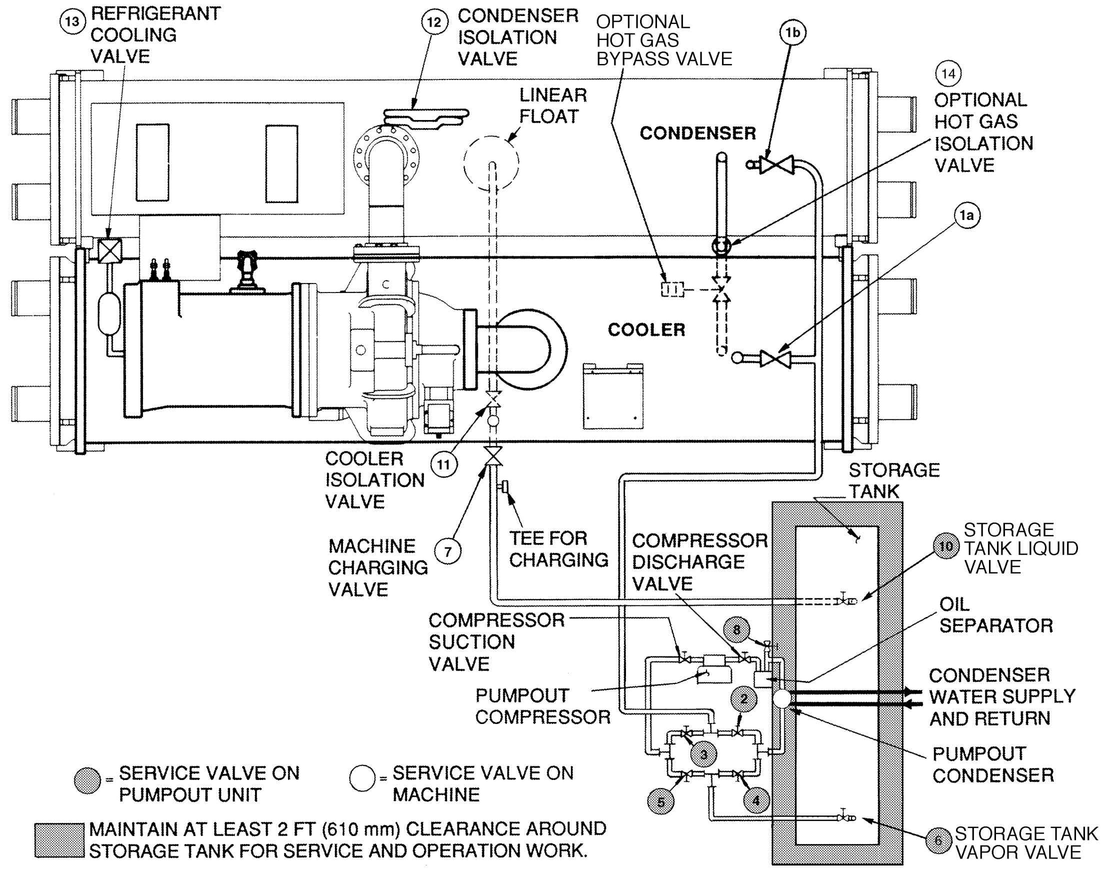


Fig. 26 — Typical Optional Pumpout System Piping Schematic with Storage Tank

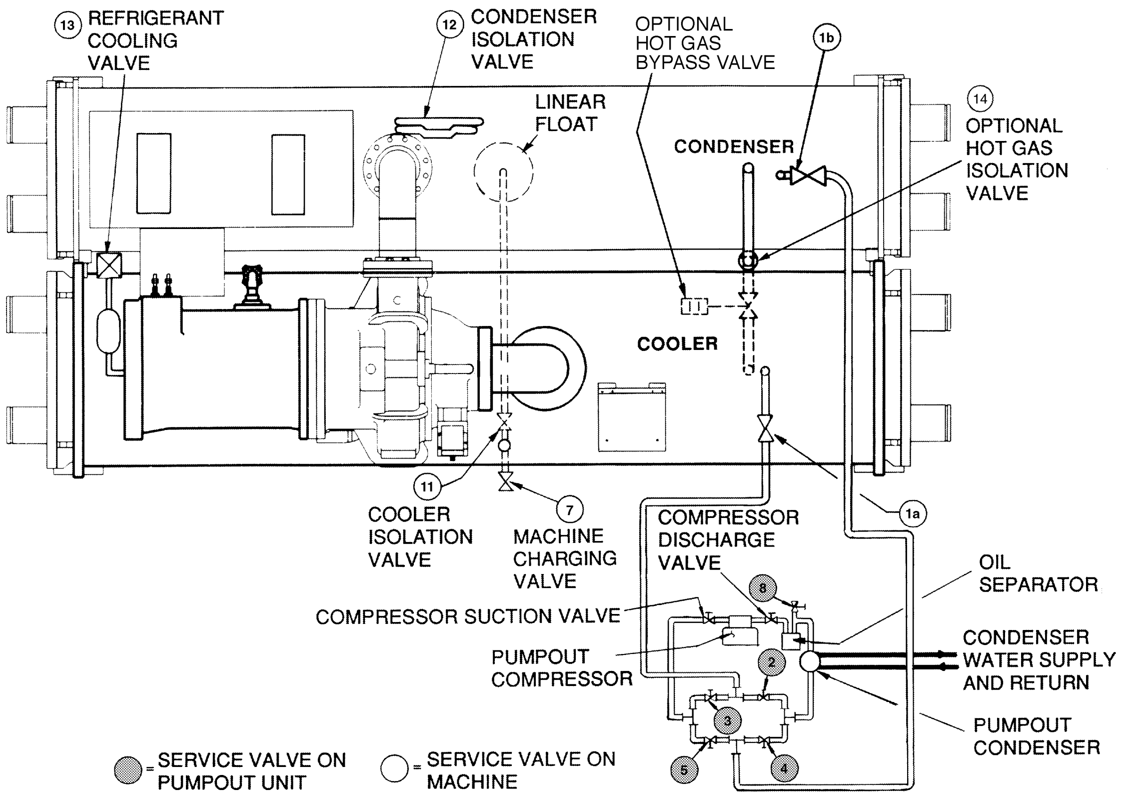


Fig. 27 — Typical Optional Pumpout System Piping Schematic without Storage Tank

Chiller Dehydration — Dehydration is recommended if the chiller has been open for a considerable period of time, if the chiller is known to contain moisture, or if there has been a complete loss of chiller holding charge or refrigerant pressure.

⚠ CAUTION

Do not start or megohm-test the compressor motor or oil pump motor, even for a rotation check, if the chiller is under dehydration vacuum. Insulation breakdown and severe damage may result.

⚠ WARNING

Inside-delta type starters must be disconnected by an isolation switch before placing the machine under a vacuum because one lead of each phase is live with respect to ground even though there is not a complete circuit to run the motor. To be safe, isolate any starter before evacuating the chiller if you are not sure if there are live leads to the hermetic motor.

Dehydration can be done at room temperatures. Using a cold trap (Fig. 28) may substantially reduce the time required to complete the dehydration. The higher the room temperature, the faster dehydration takes place. At low room temperatures, a very deep vacuum is required to boil off any moisture. If low ambient temperatures are involved, contact a qualified service representative for the dehydration techniques required.

Perform dehydration as follows:

1. Connect a high capacity vacuum pump (5 cfm [.002 m^3/s] or larger is recommended) to the refrigerant charging valve (Fig. 2). Tubing from the pump to the chiller should be as short in length and as large in diameter as possible to provide least resistance to gas flow.
2. Use an absolute pressure manometer or a wet bulb vacuum indicator to measure the vacuum. Open the shutoff valve to the vacuum indicator only when taking a reading. Leave the valve open for 3 minutes to allow the indicator vacuum to equalize with the chiller vacuum.
3. If the entire chiller is to be dehydrated, open all isolation valves (if present).
4. With the chiller ambient temperature at 60 F (15.6 C) or higher, operate the vacuum pump until the manometer reads 29.8 in. Hg vac, ref 30 in. bar. (0.1 psia)(–100.61 kPa) or a vacuum indicator reads 35 F (1.7 C). Operate the pump an additional 2 hours.

   Do not apply a greater vacuum than 29.82 in. Hg vac (757.4 mm Hg) or go below 33 F (.56 C) on the wet bulb vacuum indicator. At this temperature and pressure, isolated pockets of moisture can turn into ice. The slow rate of evaporation (sublimation) of ice at these low temperatures and pressures greatly increases dehydration time.
5. Valve off the vacuum pump, stop the pump, and record the instrument reading.
6. After a 2-hour wait, take another instrument reading. If the reading has not changed, dehydration is complete. If the reading indicates vacuum loss, repeat Steps 4 and 5.
7. If the reading continues to change after several attempts, perform a leak test up to the maximum 160 psig (1103 kPa) pressure. Locate and repair the leak, and repeat dehydration.

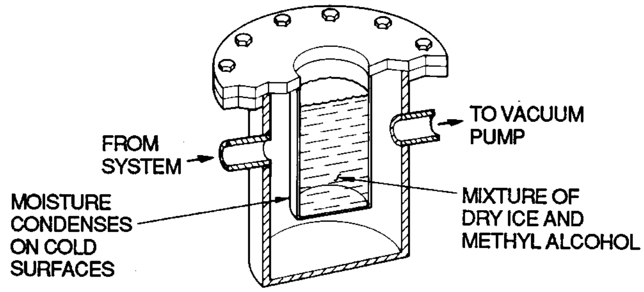


Fig. 28 — Dehydration Cold Trap

Inspect Water Piping — Refer to piping diagrams provided in the certified drawings and the piping instructions in the 19XR Installation Instructions manual. Inspect the piping to the cooler and condenser. Be sure that the flow directions are correct and that all piping specifications have been met.

Piping systems must be properly vented with no stress on waterbox nozzles and covers. Water flows through the cooler and condenser must meet job requirements. Measure the pressure drop across the cooler and the condenser.

⚠ CAUTION

Water must be within design limits, clean, and treated to ensure proper chiller performance and to reduce the potential of tube damage due to corrosion, scaling, or erosion. Carrier assumes no responsibility for chiller damage resulting from untreated or improperly treated water.

Check Optional Pumpout Compressor Water Piping — If the optional pumpout storage tank and/or pumpout system are installed, check to ensure the pumpout condenser water has been piped in. Check for field-supplied shutoff valves and controls as specified in the job data. Check for refrigerant leaks on field-installed piping. See Fig. 26 and 27.

Check Relief Devices — Be sure the relief devices have been piped to the outdoors in compliance with the latest edition of ANSI/ASHRAE Standard 15 and applicable local safety codes. Piping connections must allow for access to the valve mechanism for periodic inspection and leak testing.

The 19XR relief valves are set to relieve at the 185 psig (1275 kPa) chiller design pressure.

Inspect Wiring

⚠ WARNING

Do not check the voltage supply without proper equipment and precautions. Serious injury may result. Follow power company recommendations.

⚠ CAUTION

Do not apply any kind of test voltage, even for a rotation check, if the chiller is under a dehydration vacuum. Insulation breakdown and serious damage may result.

1. Examine the wiring for conformance to the job wiring diagrams and all applicable electrical codes.

Table 5A — HFC-134a Pressure — Temperature (F)

TEMPERATURE, F	PRESSURE (psig)
0	6.50
2	7.52
4	8.60
6	9.66
8	10.79
10	11.96
12	13.17
14	14.42
16	15.72
18	17.06
20	18.45
22	19.88
24	21.37
26	22.90
28	24.48
30	26.11
32	27.80
34	29.53
36	31.32
38	33.17
40	35.08
42	37.04
44	39.06
46	41.14
48	43.28
50	45.48
52	47.74
54	50.07
56	52.47
58	54.93
60	57.46
62	60.06
64	62.73
66	65.47
68	68.29
70	71.18
72	74.14
74	77.18
76	80.30
78	83.49
80	86.17
82	90.13
84	93.57
86	97.09
88	100.70
90	104.40
92	108.18
94	112.06
96	116.02
98	120.08
100	124.23
102	128.47
104	132.81
106	137.25
108	141.79
110	146.43
112	151.17
114	156.01
116	160.96
118	166.01
120	171.17
122	176.45
124	181.83
126	187.32
128	192.93
130	198.66
132	204.50
134	210.47
136	216.55
138	222.76
140	229.09

Table 5B — HFC-134a Pressure — Temperature (C)

TEMPERATURE, C	PRESSURE GAGE (kPa)
-18.0	44.8
-16.7	51.9
-15.6	59.3
-14.4	66.6
-13.3	74.4
-12.2	82.5
-11.1	90.8
-10.0	99.4
-8.9	108.0
-7.8	118.0
-6.7	127.0
-5.6	137.0
-4.4	147.0
-3.3	158.0
-2.2	169.0
-1.1	180.0
0.0	192.0
1.1	204.0
2.2	216.0
3.3	229.0
4.4	242.0
5.0	248.0
5.6	255.0
6.1	261.0
6.7	269.0
7.2	276.0
7.8	284.0
8.3	290.0
8.9	298.0
9.4	305.0
10.0	314.0
11.1	329.0
12.2	345.0
13.3	362.0
14.4	379.0
15.6	396.0
16.7	414.0
17.8	433.0
18.9	451.0
20.0	471.0
21.1	491.0
22.2	511.0
23.3	532.0
24.4	554.0
25.6	576.0
26.7	598.0
27.8	621.0
28.9	645.0
30.0	669.0
31.1	694.0
32.2	720.0
33.3	746.0
34.4	773.0
35.6	800.0
36.7	828.0
37.8	857.0
38.9	886.0
40.0	916.0
41.1	946.0
42.2	978.0
43.3	1010.0
44.4	1042.0
45.6	1076.0
46.7	1110.0
47.8	1145.0
48.9	1180.0
50.0	1217.0
51.1	1254.0
52.2	1292.0
53.3	1330.0
54.4	1370.0
55.6	1410.0
56.7	1451.0
57.8	1493.0
58.9	1536.0
60.0	1580.0

2. On low-voltage compressors (600 v or less) connect a voltmeter across the power wires to the compressor starter and measure the voltage. Compare this reading to the voltage rating on the compressor and starter nameplates.
3. Compare the ampere rating on the starter nameplate to rating on the compressor nameplate. The overload trip amps must be 108% to 120% of the rated load amps.
4. The starter for a centrifugal compressor motor must contain the components and terminals required for PIC refrigeration control. Check the certified drawings.
5. Check the voltage to the following components and compare it to the nameplate values: oil pump contact, pumpout compressor starter, and power panel.
6. Ensure that fused disconnects or circuit breakers have been supplied for the oil pump, power panel, and pumpout unit.
7. Ensure all electrical equipment and controls are properly grounded in accordance with job drawings, certified drawings, and all applicable electrical codes.
8. Ensure the customer's contractor has verified proper operation of the pumps, cooling tower fans, and associated auxiliary equipment. This includes ensuring motors are properly lubricated and have proper electrical supply and proper rotation.
9. For field-installed starters only, test the chiller compressor motor and its power lead insulation resistance with a 500-v insulation tester such as a megohmmeter. (Use a 5000-v tester for motors rated over 600 v.) Factory-mounted starters do not require a megohm test.
   a. Open the starter main disconnect switch and follow lockout/tagout rules.

   ⚠ CAUTION

   If the motor starter is a solid-state starter, the motor leads must be disconnected from the starter before an insulation test is performed. The voltage generated from the tester can damage the starter solid-state components.

   b. With the tester connected to the motor leads, take 10-second and 60-second megohm readings as follows:

   6-Lead Motor — Tie all 6 leads together and test between the lead group and ground. Next tie the leads in pairs: 1 and 4, 2 and 5, and 3 and 6. Test between each pair while grounding the third pair.

   3-Lead Motor — Tie terminals 1, 2, and 3 together and test between the group and ground.

   c. Divide the 60-second resistance reading by the 10-second reading. The ratio, or polarization index, must be one or higher. Both the 10- and 60-second readings must be at least 50 megohms.

   If the readings on a field-installed starter are unsatisfactory, repeat the test at the motor with the power leads disconnected. Satisfactory readings in this second test indicate the fault is in the power leads.

   NOTE: Unit-mounted starters do not have to be megohm tested.
10. Tighten all wiring connections to the plugs on the SMM, 8-input, and PSIO modules.
11. Ensure the voltage selector switch inside the power panel is switched to the incoming voltage rating.
12. On chillers with free-standing starters, inspect the power panel to ensure that the contractor has fed the wires into the bottom of the panel. Wiring into the top of the panel can cause debris to fall into the contactors. Clean and inspect the contactors if this has occurred.

Carrier Comfort Network Interface — The Carrier Comfort Network (CCN) communication bus wiring is supplied and installed by the electrical contractor. It consists of shielded, 3-conductor cable with drain wire.

The system elements are connected to the communication bus in a daisy chain arrangement. The positive pin of each system element communication connector must be wired to the positive pins of the system element on either side of it; the negative pins must be wired to the negative pins; the signal ground pins must be wired to signal ground pins.

To attach the CCN communication bus wiring, refer to the certified drawings and wiring diagrams. The wire is inserted into the CCN communications plug (COMM1) on the PSIO module. This plug is also referred to as J5.

NOTE: Conductors and drain wire must be 20 AWG (American Wire Gage) minimum stranded, tinned copper. Individual conductors must be insulated with PVC, PVC/nylon, vinyl, Teflon, or polyethylene. An aluminum/polyester 100% foil shield and an outer jacket of PVC, PVC/nylon, chrome vinyl, or Teflon with a minimum operating temperature range of –20 C to 60 C is required. See table below for cables that meet the requirements.

MANUFACTURER	CABLE NO.
Alpha	2413 or 5463
American	A22503
Belden	8772
Columbia	02525

When connecting the CCN communication bus to a system element, a color code system for the entire network is recommended to simplify installation and checkout. The following color code is recommended:

SIGNAL TYPE	CCN BUS CONDUCTOR INSULATION COLOR	PSIO MODULE COMM 1 PLUG (J5) PIN NO.
+	RED	1
Ground	WHITE	2
–	BLACK	3

Check Starter

⚠ CAUTION

BE AWARE that certain automatic start arrangements *can engage the starter*. Open the disconnect *ahead* of the starter in addition to shutting off the chiller or pump.

Use the instruction and service manual supplied by the starter manufacturer to verify the starter has been installed correctly, to set up and calibrate the starter, and for complete troubleshooting information.

⚠ CAUTION

The main disconnect on the starter front panel may not deenergize all internal circuits. Open all internal and remote disconnects before servicing the starter.

Whenever a starter safety trip device activates, wait at least 30 seconds before resetting the safety. The microprocessor maintains its output to the 1CR relay for 10 seconds to determine the fault mode of failure.

MECHANICAL STARTER

1. Check all field wiring connections for tightness, clearance from moving parts, and correct connection.
2. Check the contactor(s) to ensure they move freely. Check the mechanical interlock between contactors to ensure that 1S and 2M contactors cannot be closed at the same time. Check all other electro-mechanical devices, such as relays and timers, for free movement. If the devices do not move freely, contact the starter manufacturer for replacement components.
3. Some dashpot-type magnetic overload relays must be filled with oil on the jobsite. If the starter is equipped with devices of this type, remove the fluid cups from these magnetic overload relays. Add the dashpot oil to the cups according to the instructions supplied with the starter. The oil is usually shipped in a small container attached to the starter frame near the relays. Use only the dashpot oil supplied with the starter. Do not substitute.

   Factory-filled dashpot overload relays need no oil at start-up and solid-state overload relays do not have oil.
4. Reapply starter control power *(not main chiller power*) to check the electrical functions. When using a reduced-voltage starter (such as a wye-delta type), check the transition timer for proper setting. The factory setting is 30 seconds (± 5 seconds), timed closing. The timer is adjustable in a range between 0 and 60 seconds, and settings other than the nominal 30 seconds may be chosen as needed (typically 20 to 30 seconds are used).

   After the timer has been set, ensure the starter (with relay 1CR closed) goes through a complete and proper start cycle.

BENSHAW, INC. REDISTART MICRO SOLID-STATE STARTER

⚠ WARNING

This equipment is at line voltage when AC power is connected. Pressing the STOP button does not remove voltage. Use caution when adjusting the potentiometers on the equipment.

1. Ensure all wiring connections are properly terminated to the starter.
2. Verify the ground wire to the starter is installed properly and is sufficient size.
3. Verify the motors are properly grounded to the starter.
4. Ensure all of the relays are properly seated in their sockets.
5. Verify the proper ac input voltage is brought into the starter according to the certified drawings.
6. Apply power to the starter.

Oil Charge — The oil charge for the 19XR compressor depends on the compressor Frame size:

- Frame 2 compressor — 5 gal (18.9 L)
- Frame 3 compressor — 8 gal (30 L)
- Frame 4 compressor — 10 gal (37.8 L)
- Frame 5 compressor — 18 gal (67.8 L)

The chiller is shipped with oil in the compressor. When the sump is full, the oil level should be no higher than the middle of the upper sight glass, and minimum level is the bottom of the lower sight glass (Fig. 2). If oil is added, it must meet Carrier's specification for centrifugal compressor use as described in the Oil Specification section. Charge the oil through the oil charging valve located near the bottom of the transmission housing (Fig. 2). The oil must be pumped from the oil container through the charging valve due to higher refrigerant pressure. The pumping device must be able to lift from 0 to 200 psig (0 to 1380 kPa) or above unit pressure. Oil should only be charged or removed when the chiller is shut down.

Power Up the Controls and Check the Oil Heater — Ensure that an oil level is visible in the compressor before energizing the controls. A circuit breaker in the starter energizes the oil heater and the control circuit. When first powered, the LID should display the default screen within a short period of time.

The oil heater is energized by powering the control circuit. This should be done several hours before start-up to minimize oil-refrigerant migration. The oil heater is controlled by the PIC and is powered through a contactor in the power panel. Starters contain a separate circuit breaker to power the heater and the control circuit. This arrangement allows the heater to energize when the main motor circuit breaker is off for service work or extended shutdowns. The oil heater relay status (*OIL HEATER RELAY*) can be viewed on the STATUS02 table on the LID. Oil sump temperature can be viewed on the LID default screen.

SOFTWARE VERSION — The software version is labeled on the PSIO module and on the back-side of the LID module. The software version also appears on both the controller ID and LID ID displays on the LID CONFIGURATION screen.

Set Up Chiller Control Configuration

⚠ WARNING

Do not operate the chiller before the control configurations have been checked and a Control Test has been satisfactorily completed. Protection by safety controls cannot be assumed until all control configurations have been confirmed.

As the 19XR unit is configured, all configuration settings should be written down. A log, such as the one shown on pages CL-1 to CL-12, provides a convenient list for configuration values.

Input the Design Set Points — Access the LID set point screen and view/modify the base demand limit set point, and *either* the LCW set point *or* the ECW set point. The PIC can control a set point to either the leaving or entering chilled water. This control method is set in the EQUIPMENT CONFIGURATION (CONFIG) table.

Input the Local Occupied Schedule (OCCPC01S) — Access the schedule OCCPC01S screen on the LID and set up the occupied time schedule according to the customer's requirements. If no schedule is available, the default is factory set for 24 hours occupied, 7 days per week including holidays.

For more information about how to set up a time schedule, see the Controls section, page 10.

The CCN Occupied Schedule (OCCPC03-99S) should be configured if a CCN system is being installed or if a secondary time schedule is needed.

NOTE: The default CCN Occupied Schedule is OCCPC03S.

Input Service Configurations — The following configurations require the LID screen to be in the SERVICE portion of the menu.

- password
- input time and date
- LID configuration
- controller identification
- service parameters
- equipment configuration
- automated control test

PASSWORD — When accessing the SERVICE tables, a password must be entered. All LIDs are initially set for a password of 1-1-1-1.

INPUT TIME AND DATE — Access the TIME AND DATE table on the SERVICE menu. Input the present time of day, date, and day of the week. The *HOLIDAY TODAY* parameter should only be configured to YES if the present day is a holiday.

NOTE: Because a schedule is integral to the chiller control sequence, the chiller will not start until the time and date have been set.

CHANGE LID CONFIGURATION IF NECESSARY — From the SERVICE table, access the LID CONFIGURATION screen. From there, view or modify the LID CCN address, change to English or SI units, and change the password. If there is more than one chiller at the jobsite, change the LID address on each chiller so that each chiller has its own address. Note and record the new address. Change the screen to SI units as required, and change the password if desired.

TO CHANGE THE PASSWORD — The password may be changed from the LID CONFIGURATION screen.

1. Press the MENU and SERVICE softkeys. Enter the current password and highlight LID CONFIGURATION. Press the SELECT softkey. Only the last 5 entries on the LID CONFIG screen can be changed: *BUS #*, *ADDRESS #*, *BAUD RATE*, *US IMP/METRIC*, and *PASSWORD*.
2. Use the ENTER softkey to scroll to *PASSWORD*. The first digit of the password is highlighted on the screen.
3. To change the digit, press the INCREASE or DECREASE softkey. When the desired digit is seen, press the ENTER softkey.
4. The next digit is highlighted. Change it, and the third and fourth digits in the same way the first was changed.
5. After the last digit is changed, the LID goes to the *BUS* parameter. Press the EXIT softkey to leave that screen and return to the SERVICE menu.

⚠ CAUTION

Be sure to remember the password. Retain a copy for future reference. Without the password, access to the SERVICE menu will not be possible unless a new PSIO Module is installed and downloaded.

TO CHANGE THE LID DISPLAY FROM ENGLISH TO METRIC UNITS — By default, the LID displays information in English units. To change to metric units, access the LID CONFIGURATION screen:

1. Press the MENU and SERVICE softkeys. Enter the password and highlight LID CONFIGURATION. Press the SELECT softkey.
2. Use the ENTER softkey to scroll to *US IMP/METRIC*.
3. Press the softkey that corresponds to the units desired for display on the LID (e.g., US or METRIC).

MODIFY CONTROLLER IDENTIFICATION IF NECESSARY — The PSIO module address can be changed from the CONTROLLER IDENTIFICATION screen. Change this address for each chiller if there is more than one chiller at the jobsite. Write the new address on the PSIO module for future reference.

Change the LID CCN address if there is more than one chiller on the job site. Access the LID CONFIGURATION screen to view or modify this address.

INPUT EQUIPMENT SERVICE PARAMETERS IF NECESSARY — The EQUIPMENT SERVICE table has three service tables: SERVICE1, SERVICE2, and SERVICE3.

Configure SERVICE1 Table — Access the SERVICE1 table to modify or view the following to job site parameters:

Chilled Medium	Water or Brine?
Brine Refrigerant Trippoint	Usually 3° F (1.7° C) below design refrigerant temperature
Surge Limiting or Hot Gas Bypass Option	Is HGBP installed?
Minimum Load Points (T1/P1)	Per job data — See Modify Load Points section
Full (Maximum) Load Points (T2/P2)	Per job data — See Modify Load Points section
Motor Rated Load Amps	Per job data
Motor Rated Line Voltage	Per job data
Motor Rated Line kW	Per job data (if kW meter installed)
Line Frequency	50 or 60 Hz
Compressor Starter Type	Reduced voltage or full?

NOTE: Other parameters on this screen are normally left at the default settings; however, they may be changed by the operator as required. The SERVICE2 and SERVICE3 screens can also be modified by the owner/operator as required.

Modify Minimum and Maximum Load Points (ΔT1/P1; Δ T2/P2) If Necessary — These pairs of chiller load points, located on the SERVICE1 screen, determine when to limit guide vane travel or open the hot gas bypass valve when surge prevention is needed. These points should be set based on individual chiller operating conditions.

If after configuring a value for these points, surge prevention is operating too soon or too late for conditions, these parameters should be changed by the operator.

An example of such a configuration is shown below.

Refrigerant: HCFC-134a

Estimated Minimum Load Conditions:

44 F (6.7 C) LCW
45.5 F (7.5 C) ECW
43 F (6.1 C) Suction Temperature
70 F (21.1 C) Condensing Temperature

Estimated Maximum Load Conditions:

44 F (6.7 C) LCW
54 F (12.2 C) ECW
42 F (5.6 C) Suction Temperature
98 F (36.7 C) Condensing Temperature

Calculate Maximum Load — To calculate the maximum load points, use the design load condition data. If the chiller full load cooler temperature difference is more than 15 F (8.3 C), estimate the refrigerant suction and condensing temperatures at this difference. Use the proper saturated pressure and temperature for the particular refrigerant used.

Suction Temperature:
42 F (5.6 C) = 37 psig (255 kPa) saturated refrigerant pressure (HFC-134a)

Condensing Temperature:
98 F (36.7 C) = 120 psig (1827 kPa) saturated refrigerant pressure (HFC-134a)

Maximum Load $\Delta T2$:
54 – 44 = 10° F (12.2 – 6.7 = 5.5° C)

Maximum Load $\Delta P2$:
120 – 37 = 83 psid (827 – 255 = 572 kPad)

To avoid unnecessary surge prevention, add about 10 psid (70 kPad) to $\Delta P2$ from these conditions:
$\Delta T2$ = 10° F (5.5° C)
$\Delta P2$ = 93 psid (642 kPad)

Calculate Minimum Load — To calculate the minimum load conditions, estimate the temperature difference the cooler will have at 10% load, then estimate what the suction and condensing temperatures will be at this point. Use the proper saturated pressure and temperature for the particular refrigerant used.

Suction Temperature:
43 F (6.1 C) = 38 psig (262 kPa) saturated refrigerant pressure (HFC-134a)

Condensing Temperature:
70 F (21.1 C) = 71 psig (490 kPa) saturated refrigerant pressure (HFC-134a)

Minimum Load $\Delta T1$ (at 20% Load): 2 F (1.1 C)

Minimum Load $\Delta P1$:
71 – 38 = 33 psid (490 – 262 = 228 kPad)

Again, to avoid unnecessary surge prevention, add 20 psid (140 kPad) at $\Delta P1$ from these conditions:
$\Delta T1$ = 2 F (1.1 C)
$\Delta P1$ = 53 psid (368 kPad)

If surge prevention occurs too soon or too late:

LOAD	SURGE PREVENTION OCCURS TOO SOON	SURGE PREVENTION OCCURS TOO LATE
At low loads (<50%)	Increase P1 by 10 psid (70 kPad)	Decrease P1 by 10 psid (70 kPad)
At high loads (>50%)	Increase P2 by 10 psid (70 kPad)	Decrease P2 by 10 psid (70 kPad)

Amp Correction Factors — The amp correction factor should be kept at its default value unless a Carrier technical representative advises a change. To modify the amp correction factor, enter the appropriate amp correction factor on the SERVICE1 screen of the EQUIPMENT SERVICE table.

CONFIGURE DIFFUSER CONTROL IF NECESSARY — If the compressor is equipped with a variable diffuser, access the SERVICE3 screen. Scroll to *DIFFUSER CONTROL* and press the ENABLE softkey. Enter the correct diffuser and guide vane values (*GUIDE VANE 25% LOAD PT, GUIDE VANE 50 % LOAD PT, GUIDE VANE 75% LOAD PT, DIFFUSER 25% LOAD PT, DIFFUSER 50% LOAD PT, DIFFUSER 75% LOAD PT*). To obtain proper settings for diffuser control, contact a Carrier Engineering representative.

Compressors with variable diffuser control have actuators tested and stamped with the milliamp (mA) value that results in 100% actuator rotation. These values are configured on the SERVICE3 screen. They are labeled *GUIDE VANE FULL SPAN* mA and *DIFFUSER FULL SPAN mA*.

MODIFY EQUIPMENT CONFIGURATION IF NECESSARY — The EQUIPMENT CONFIGURATION table has screens to select, view, or modify parameters. Carrier's certified drawings have the configuration values required for the jobsite. Modify these values only if requested.

CONFIG Screen Modifications — Access this screen from the EQUIPMENT CONFIGURATION table. Change the values on this screen according to specific job data. See the certified drawings for the correct values. Modifications can include:

- chilled water reset
- entering chilled water control (Enable/Disable)
- 4 to 20 mA demand limit
- auto restart option (Enable/Disable)
- remote contact option (Enable/Disable)

Owner-Modified CCN Tables— The following EQUIPMENT CONFIGURATION screens are described for reference only.

OCCDEFS — The OCCDEFS screen contains the Local and CCN time schedules, which can be modified here or on the SCHEDULE screen as described previously.

HOLIDEF — From the HOLIDEF screen, the days of the year that holidays are in effect can be configured. See the holiday paragraphs in the Controls section for more details.

BRODEF — The BRODEF screen defines the outside-air temperature and humidity sensor if one is installed. It also defines the start and end of daylight savings time. Enter the dates for the start and end of daylight savings if required for your location. BRODEF also activates the Broadcast function which enables the holiday periods that are defined on the LID to take effect.

Other Tables — The ALARMDEF, CONS-DEF, and RUNT-DEF screens contain parameters used with a CCN system. See the applicable CCN manual for more information on these screens. These tables can only be defined from a CCN Building Supervisor.

Check Voltage Supply — Access the STATUS01 screen and read the actual line voltage. This reading should be equal to the incoming power to the starter. Use a voltmeter to check incoming power at the starter power leads. If the readings are not equal, calibrate the voltage reading on the LID screen by highlighting the *LINE VOLTAGE: PERCENT* parameter. Press SELECT, then press INCREASE or DECREASE until the value equals the percent of rated voltage that is actually on the voltmeter. Press ENTER to complete the calibration.

Perform a Controls Test — Check the safety controls status by performing an automated control test. Access the CONTROL TEST table and select the Automated Tests function (Table 6).

The Automated Control Test checks all outputs and inputs for a function. The compressor must be in the OFF mode to operate the controls test, and the 24-v input to the SMM must be in range (according to the LINE VOLTAGE PERCENT on STATUS01 table). The compressor can be put in OFF mode by pressing the STOP push-button on the LID. Each test asks the operator to confirm the operation is occurring and whether or not to continue. If an error occurs, the operator can try to address the problem as the test is being done or note the problem and proceed to the next test.

NOTE: If during the control test the guide vanes do not open, verify the low pressure alarm is not active. (An active low pressure alarm causes the guide vanes to close.)

NOTE: The oil pump test will not energize the oil pump if cooler pressure is below –5 psig (–35 kPa).

When the control test is finished or the EXIT softkey is pressed, the test stops, and the CONTROL TEST menu displays. If a specific automated test procedure is not completed, access the particular control test to test the function when ready. The CONTROL TEST menu is described in the table below.

Automated Tests	As described above, a complete control test.
PSIO Thermistors	Check of all PSIO thermistors only.
Options Thermistors	Check of all options boards thermistors.
Transducers	Check of all transducers.
Guide Vane/Diffuser Actuator*	Check of the guide vane operation.
Pumps	Checks operation of pump outputs; either all pumps can be activated or individual pumps. Also tests associated inputs such as flow or pressure.
Discrete Outputs	Activation of all on/off outputs or each output individually.
Pumpdown/Lockout	Pumpdown prevents the low refrigerant alarm during evacuation so refrigerant can be removed from the unit. Also locks the compressor off and starts the water pumps.
Terminate Lockout	To charge refrigerant and enable the chiller to run after pumpdown lockout.

*Diffuser test will function only if diffuser control is enabled.

Check Optional Pumpout System Controls and Compressor — Controls include an on/off switch, a 3-amp fuse, the compressor overloads, an internal thermostat, a compressor contactor, and a refrigerant high pressure cutout. The high pressure cutout is factory set to open at 161 psig (1110 kPa) and reset at 130 psig (896 kPa). Ensure the water-cooled condenser has been connected. Loosen the compressor holddown bolts to allow free spring travel. Open the compressor suction and discharge the service valves. Ensure oil is visible in the compressor sight glass. Add oil if necessary.

See the Pumpout and Refrigerant Transfer Procedures and Optional Pumpout System Maintenance sections, pages 57 and 65, for details on the transfer of refrigerant, oil specifications, etc.

Table 6 — Control Test Menu Functions

TESTS TO BE PERFORMED	DEVICES TESTED
1. Automated Tests*	Operates the second through seventh tests
2. PSIO Thermistors	Entering chilled water Leaving chilled water Entering condenser water Leaving condenser water Discharge temperature Bearing temperature Motor winding temperature Oil sump temperature
3. Options Thermistors	Common chilled water supply sensor Common chilled water return sensor Remote reset sensor Temperature sensor Spare 1 Spare 2 Spare 3 Spare 4 Spare 5 Spare 6 Spare 7 Spare 8 Spare 9
4. Transducers	Evaporator pressure Condenser pressure Oil pressure differential Oil pump pressure
5. Guide Vane Actuator	Open Close
6. Pumps	All pumps or individual pumps may be activated: Oil pump — Confirm pressure Chilled water pump — Confirm flow Condenser water pump — Confirm flow
7. Discrete Outputs	All outputs or individual outputs may be energized: Hot gas bypass relay Oil heater relay Motor cooling relay Tower fan relay Alarm relay Shunt trip relay
8. Pumpdown/Lockout	When using pumpdown/lockout, observe freeze up precautions when removing charge: Instructs operator which valves to close and when Starts chilled water and condenser water pumps and confirms flows Monitors Evaporator pressure Condenser pressure Evaporator temperature during pumpout procedures Turns pumps off after pumpdown Locks out compressor
9. Terminate Lockout	Starts pumps and monitors flows Instructs operator which valves to open and when Monitors Evaporator pressure Condenser pressure Evaporator temperature during charging process Terminates compressor lockout

*During any of the tests that are not automated, an out-of-range reading will have an asterisk (*) next to the reading and a message will be displayed.

High Altitude Locations — Because the chiller is initially calibrated at sea level, it is necessary to recalibrate the pressure transducers if the chiller has been moved to a high altitude location. See the calibration procedure in the Troubleshooting Guide section.

Charge Refrigerant into Chiller

⚠ CAUTION

The transfer, addition, or removal of refrigerant in spring isolated chillers may place severe stress on external piping if springs have not been blocked in both up and down directions.

⚠ WARNING

Always operate the condenser and chilled water pumps during charging operations to prevent freeze-ups.

The standard 19XR chiller is shipped with the refrigerant already charged in the vessels. However, the 19XR may be ordered with a nitrogen holding charge of 15 psig (103 kPa). Evacuate the nitrogen from the entire chiller, and charge the chiller from refrigerant cylinders.

CHILLER EQUALIZATION WITHOUT A PUMPOUT UNIT

⚠ WARNING

When equalizing refrigerant pressure on the 19XR chiller after service work or during the initial chiller start-up, *do not use the discharge isolation valve to equalize.* Either the motor cooling isolation valve or the charging hose (connected between the pumpout valves on top of the cooler and condenser) should be used as the equalization valve.

To equalize the pressure differential on a refrigerant isolated 19XR chiller, use the terminate lockout function of the CONTROL TEST on the SERVICE menu. This helps to turn on pumps and advises the operator on proper procedures.

The following steps describe how to equalize refrigerant pressure in an isolated 19XR chiller without a pumpout unit.

1. Access terminate lockout function on the CONTROL TEST screen.
2. IMPORTANT: Turn on the chilled water and condenser water pumps to prevent freezing.
3. Slowly open the refrigerant cooling isolation valve. The chiller cooler and condenser pressures will gradually equalize. This process takes approximately 15 minutes.
4. Once the pressures have equalized, the cooler isolation valve, the condenser isolation valve, and the hot gas isolation valve may now be opened. Refer to Fig. 26 and 27, Valves 11, 12, and 14.

⚠ WARNING

Whenever turning the discharge isolation valve, be sure to reattach the valve locking device. This prevents the valve from opening or closing during service work or during chiller operation.

CHILLER EQUALIZATION WITH PUMPOUT UNIT — The following steps describe how to equalize refrigerant pressure on an isolated 19XR chiller using the pumpout unit.

1. Access the terminate lockout function on the CONTROL TEST screen.
2. IMPORTANT: Turn on the chilled water and condenser water pumps to prevent possible freezing.
3. Open valve 4 on the pumpout unit and open valves 1a and 1b on the chiller cooler and condenser, Fig. 26 and 27. Slowly open valve 2 on the pumpout unit to equalize the pressure. This process takes approximately 15 minutes.
4. Once the pressures have equalized, the discharge isolation valve, cooler isolation valve, optional hot gas bypass isolation valve, and the refrigerant isolation valve can be opened. Close valves 1a and 1b, and all pumpout unit valves.

⚠ WARNING

Whenever turning the discharge isolation valve, be sure to reattach the valve locking device. This prevents the valve from opening or closing during service work or during chiller operation.

The full refrigerant charge on the 19XR will vary with chiller components and design conditions, as indicated on the job data specifications. An approximate charge may be determined by adding the condenser charge to the cooler charge as listed in Table 7.

Table 7 — Refrigerant (HFC-134a) Charge

COOLER CODE	REFRIGERANT CHARGE		CONDENSER CODE	REFRIGERANT CHARGE	
	lb	kg		lb	kg
10	290	132	**10**	200	91
11	310	141	**11**	200	91
12	330	150	**12**	200	91
15	320	145	**15**	250	113
16	340	154	**16**	250	113
17	370	168	**17**	250	113
20	345	157	**20**	225	102
21	385	175	**21**	225	102
22	435	197	**22**	225	102
30	350	159	**30**	260	118
31	420	190	**31**	260	118
32	490	222	**32**	260	118
35	400	181	**35**	310	141
36	480	218	**36**	310	141
37	550	250	**37**	310	141
40	560	254	**40**	280	127
41	630	286	**41**	280	127
42	690	313	**42**	280	127
45	640	290	**45**	330	150
46	720	327	**46**	330	150
47	790	358	**47**	330	150
50	750	340	**50**	400	181
51	840	381	**51**	400	181
52	900	408	**52**	400	181
55	870	395	**55**	490	222
56	940	426	**56**	490	222
57	980	445	**57**	490	222
60	940	426	**60**	420	190
61	980	445	**61**	420	190
62	1020	463	**62**	420	190
65	1020	463	**65**	510	231
66	1060	481	**66**	510	231
67	1090	494	**67**	510	231
70	1220	553	**70**	780	354
71	1340	608	**71**	780	354
72	1440	653	**72**	780	354
75	1365	619	**75**	925	420
76	1505	683	**76**	925	420
77	1625	737	**77**	925	420
80	1500	680	**80**	720	327
81	1620	735	**81**	720	327
82	1730	785	**82**	720	327
85	1690	766	**85**	860	390
86	1820	825	**86**	860	390
87	1940	880	**87**	860	390

⚠ WARNING

Always operate the condenser and chilled water pumps whenever charging, transferring, or removing refrigrant from the chiller.

Use the CONTROL TEST terminate lockout function to monitor conditions and start the pumps.

If the chiller has been shipped with a holding charge, the refrigerant is added through the refrigerant charging valve (Fig. 26 and 27, valve 7) or to the pumpout charging connection. First evacuate the nitrogen holding charge from the chiller vessels. Charge the refrigerant as a gas until the system pressure exceeds 35 psig (141 kPa) for HFC-134a. After the chiller is beyond this pressure the refrigerant should be charged as a liquid until all the recommended refrigerant charge has been added.

TRIMMING REFRIGERANT CHARGE — The 19XR is shipped with the correct charge for the design duty of the chiller. Trimming the charge can best be accomplished when the design load is available. To trim the charge, check the temperature difference between the leaving chilled water temperature and cooler refrigerant temperature at full load design conditions. If necessary, add or remove refrigerant to bring the temperature difference to design conditions or minimum differential.

Table 7 lists the 19XR chiller refrigerant charges for each cooler and condenser code. Total refrigerant charge is the sum of the cooler and condenser charge.

INITIAL START-UP

Preparation — Before starting the chiller, verify:

1. Power is on to the main starter, oil pump relay, tower fan starter, oil heater relay, and the chiller control center.
2. Cooling tower water is at proper level and at-or-below design entering temperature.
3. Chiller is charged with refrigerant and all refrigerant and oil valves are in their proper operating positions.
4. Oil is at the proper level in the reservoir sight glasses.
5. Oil reservoir temperature is above 140 F (60 C) or refrigerant temperature plus 50° F (28° C).
6. Valves in the evaporator and condenser water circuits are open.

   NOTE: If the pumps are not automatic, ensure water is circulating properly.
7. Solid-state starter checks: The Power +15 and the Phase Correct LEDs must be lit before the starter will energize. If the Power +15 LED is not on, incoming voltage is not present or is incorrect. If the Phase Correct LED is not lit, rotate any 2 incoming phases to correct the phasing.

⚠ WARNING

Do not permit water or brine that is warmer than 110 F (43 C) to flow through the cooler or condenser. Refrigerant overpressure may discharge through the relief devices and result in the loss of refrigerant charge.

8. Access the STATUS01 screen. Scroll down to the *CHILLER START/STOP* parameter and press the [RELEASE] softkey to enable the chiller to start. This parameter is set to STOP at the factory in order to prevent accidental start-up.

Manual Operation of the Guide Vanes — Manual operation of the guide vanes helps to establish a steady motor current for calibrating the motor amps value.

In order to manually operate the guide vanes, it is necessary to override the target guide vane position value (*TARGET GUIDE VANE POS* parameter on the STATUS01 screen). Manual control is indicated by the word ''SUPVSR!'' flashing after the target guide vane parameter. Manual control is also indicated on the default screen on the run status line.

1. Access the STATUS01 screen and scroll to *TARGET GUIDE VANE POS*. See Table 2, Example 1. If the compressor is off, the value will read zero.
2. Highlight the *TARGET GUIDE VANE POS* line and press the [SELECT] softkey.
3. Press the [ENTER] softkey to override the automatic target. The screen will now read a value of zero, and the word ''SUPVSR!'' will flash.
4. Press the [SELECT] softkey, and then press the [RELEASE] softkey to release the vanes to AUTOMATIC mode. After a few seconds the word ''SUPVSR!'' will disappear.

Dry Run to Test Start-Up Sequence

1. Disengage the main motor disconnect on the starter front panel. This should only disconnect the motor power. Power to the controls, oil pump, and starter control circuit should still be energized.
2. Observe the default screen on the LID: the Status message in the upper left-hand corner reads, ''Manually Stopped.'' Press the [CCN] or [LOCAL] softkey to start. If the chiller controls do not go into a start mode, go to the Schedule screen and override the schedule or change the occupied time. Press the [LOCAL] softkey to begin the start-up sequences.
3. Verify the chilled water and condenser water pumps have energized.
4. Verify the oil pump has started and is pressurizing the lubrication system. After the oil pump has run about 11 seconds, the starter energizes and goes through its start-up sequence.
5. Check the main contactor for proper operation.
6. The PIC eventually shows an alarm for motor amps not sensed. Reset this alarm and continue with the initial start-up.

Check Motor Rotation

1. Engage the main motor disconnect on the front of the starter panel. The motor is now ready for a rotation check.
2. After the default screen status message states ''Ready for Start'' press the [LOCAL] softkey. The PIC control performs start-up checks.
3. When the starter is energized and the motor begins to turn, check for clockwise motor rotation (Fig. 29).

IF THE MOTOR ROTATION IS CLOCKWISE, allow the compressor to come up to speed.

IF THE MOTOR ROTATION IS NOT CLOCKWISE (as viewed through the sight glass), reverse any 2 of the 3 incoming power leads to the starter and recheck the rotation.

NOTE: Solid-state starters have phase protection and do not permit a start if the phase is not correct. If this happens, a Starter Fault message appears on the LID.

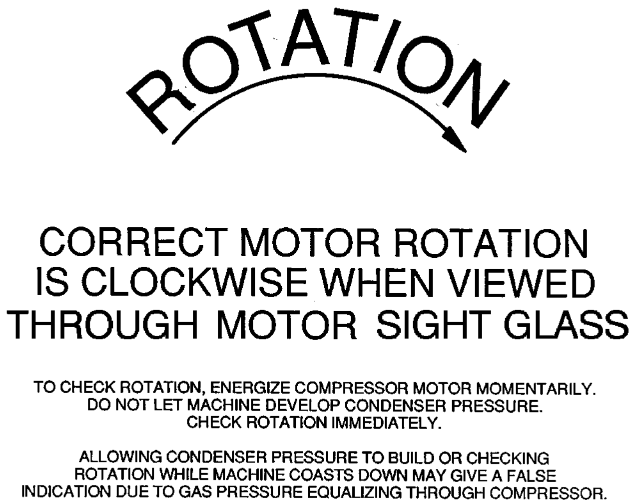


Fig. 29 − Correct Motor Rotation

⚠ CAUTION

Do not check motor rotation during coastdown. Rotation may have reversed during equalization of vessel pressures.

Check Oil Pressure and Compressor Stop

1. When the motor is at full speed, note the differential oil pressure reading on the LID default screen. It should be between 18 and 30 psid (124 to 206 kPad).
2. Press the Stop button and listen for any unusual sounds from the compressor as it coasts to a stop.

Calibrate Motor Current Demand Setting

1. Ensure that the compressor *MOTOR RATED LOAD AMPS* value in the SERVICE1 table has been configured. Place an ammeter on the line that passes through the motor load current transformer on the motor side of the power factor correction capacitors (if provided).
2. Start the compressor and establish a steady motor current value between 70% and 100% RLA by manually overriding the guide vane target value on the LID and setting the chilled water set point to a low value. Do not exceed 105% of the nameplate RLA.
3. When a steady motor current value in the desired range is achieved, compare the compressor *MOTOR RATED LOAD AMPS* value on the STATUS01 screen to the actual amps shown on the ammeter on the starter. Adjust the amps value on the LID to match the actual value seen at the starter if there is a difference. Highlight the amps value. Then press the SELECT softkey. Press the INCREASE or DECREASE softkey to bring the value to that indicated on the ammeter. Press the ENTER softkey when the values are equal.
4. Ensure that the target guide vane position is released into automatic mode. See Steps 1-4 in the Manual Operation of the Guide Vanes section for information on how to put the guide vane position into automatic mode.

To Prevent Accidental Start-Up — A chiller STOP override setting may be entered to prevent accidental start-up during service or whenever necessary. Access the STATUS01 screen and using the NEXT or PREVIOUS softkeys, highlight the *CHILLER START/STOP* parameter. Override the current START value by pressing the SELECT softkey. Press the STOP softkey followed by the ENTER softkey. The word SUPVSR! displays on the LID indicating the override is in place.

To restart the chiller the STOP override setting must be removed. Access the STATUS01 screen and using NEXT or PREVIOUS softkeys highlight *CHILLER START/STOP*. The 3 softkeys that appear represent 3 choices:

- START — forces the chiller ON
- STOP — forces the chiller OFF
- RELEASE — puts the chiller under remote or schedule control.

To return the chiller to normal control, press the RELEASE softkey followed by the ENTER softkey. For more information, see Local Start-Up, page 40.

The default LID screen message line indicates which command is in effect.

Check Chiller Operating Condition — Check to be sure that chiller temperatures, pressures, water flows, and oil and refrigerant levels indicate the system is functioning properly.

Instruct the Customer Operator — Ensure the operator(s) understand all operating and maintenance procedures. Point out the various chiller parts and explain their function as part of the complete system.

COOLER-CONDENSER — Float chamber, relief devices, refrigerant charging valve, temperature sensor locations, pressure transducer locations, Schrader fittings, waterboxes and tubes, and vents and drains.

OPTIONAL PUMPOUT STORAGE TANK AND PUMPOUT SYSTEM — Transfer valves and pumpout system, refrigerant charging and pumpdown procedure, and relief devices.

MOTOR COMPRESSOR ASSEMBLY — Guide vane actuator, transmission, motor cooling system, oil cooling system, temperature and pressure sensors, oil sight glasses, integral oil pump, isolatable oil filter, extra oil and motor temperature sensors, synthetic oil, and compressor serviceability.

MOTOR COMPRESSOR LUBRICATION SYSTEM — Oil pump, cooler filter, oil heater, oil charge and specification, operating and shutdown oil level, temperature and pressure, and oil charging connections.

CONTROL SYSTEM — CCN and LOCAL start, reset, menu, softkey functions, LID operation, occupancy schedule, set points, safety controls, and auxiliary and optional controls.

AUXILIARY EQUIPMENT — Starters and disconnects, separate electrical sources, pumps, and cooling tower.

DESCRIBE CHILLER CYCLES — Refrigerant, motor cooling, lubrication, and oil reclaim.

REVIEW MAINTENANCE — Scheduled, routine, and extended shutdowns, importance of a log sheet, importance of water treatment and tube cleaning, and importance of maintaining a leak-free chiller.

SAFETY DEVICES AND PROCEDURES — Electrical disconnects, relief device inspection, and handling refrigerant.

CHECK OPERATOR KNOWLEDGE — Start, stop, and shutdown procedures, safety and operating controls, refrigerant and oil charging, and job safety.

REVIEW THE START-UP, OPERATION, AND MAINTENANCE MANUAL.

OPERATING INSTRUCTIONS

Operator Duties

1. Become familiar with the chiller and related equipment before operating the chiller.
2. Prepare the system for start-up, start and stop the chiller, and place the system in a shutdown condition.
3. Maintain a log of operating conditions and document any abnormal readings.
4. Inspect the equipment, make routine adjustments, and perform a Control Test. Maintain the proper oil and refrigerant levels.
5. Protect the system from damage during shutdown periods.
6. Maintain the set point, time schedules, and other PIC functions.

Prepare the Chiller for Start-Up — Follow the steps described in the Initial Start-Up section, page 54.

To Start the Chiller

1. Start the water pumps, if they are not automatic.
2. On the LID default screen, press the $\boxed{\text{LOCAL}}$ or $\boxed{\text{CCN}}$ softkey to start the system. If the chiller is in the OCCUPIED mode and the start timers have expired, the start sequence will start. Follow the procedure described in the Start-Up/Shutdown/Recycle section, page 40.

Check the Running System — After the compressor starts, the operator should monitor the LID display and observe the parameters for normal operating conditions:

1. The oil reservoir temperature should be above 120 F (49 C) during shutdown and above 125 F (58 C) during compressor operation.
2. The bearing oil temperature accessed on the STATUS01 table should be 120 to 165 F (49 to 74 C). If the bearing temperature reads more than 180 F (83 C) with the oil pump running, stop the chiller and determine the cause of the high temperature. *Do not restart* the chiller until corrected.
3. The oil level should be visible anywhere in one of the two sight glasses. Foaming oil is acceptable as long as the oil pressure and temperature are within limits.
4. The oil pressure should be between 18 and 30 psid (124 to 207 kPad) differential, as seen on the LID default screen. Typically the reading will be 18 to 25 psid (124 to 172 kPad) at initial start-up.
5. The moisture indicator sight glass on the refrigerant motor cooling line should indicate refrigerant flow and a dry condition.
6. The condenser pressure and temperature varies with the chiller design conditions. Typically the pressure will range between 100 and 210 psig (690 to 1450 kPa) with a corresponding temperature range of 60 to 105 F (15 to 41 C). The condenser entering water temperature should be controlled below the specified design entering water temperature to save on compressor kilowatt requirements.
7. Cooler pressure and temperature also will vary with the design conditions. Typical pressure range will be between 60 and 80 psig (410 and 550 kPa), with temperature ranging between 34 and 45 F (1 and 8 C).
8. The compressor may operate at full capacity for a short time after the pulldown ramping has ended, even though the building load is small. The active electrical demand setting can be overridden to limit the compressor IkW, or the pulldown rate can be decreased to avoid a high demand charge for the short period of high demand operation. Pulldown rate can be based on load rate or temperature rate. It is accessed on the Equipment Configuration screen, CONFIG table (Table 2, Example 5).

To Stop the Chiller

1. The occupancy schedule starts and stops the chiller automatically once the time schedule is configured.
2. By pressing the STOP button for one second, the alarm light blinks once to confirm the button has been pressed. The compressor will then follow the normal shutdown sequence as described in the Controls section. The chiller will not restart until the $\boxed{\text{CCN}}$ or $\boxed{\text{LOCAL}}$ softkey is pressed. The chiller is now in the OFF control mode.

   If the chiller fails to stop, in addition to action the PIC initiates, the operator should close the guide vanes by overriding the guide vane target to zero to reduce chiller load; then, open the main disconnect. (See the Manual Operation of the Guide Vanes section on page 54.)

   > IMPORTANT: Do not attempt to stop the chiller by opening an isolating knife switch. High intensity arcing may occur.

   Do *not restart the chiller* until the problem is diagnosed and corrected.

After Limited Shutdown — No special preparations should be necessary. Follow the regular preliminary checks and starting procedures.

Preparation for Extended Shutdown — The refrigerant should be transferred into the pumpout storage tank (if supplied; see Pumpout and Refrigerant Transfer Procedures) to reduce chiller pressure and the possibility of leaks. Maintain a holding charge of 5 to 10 lbs (2.27 to 4.5 kg) of refrigerant or nitrogen to prevent air from leaking into the chiller.

If freezing temperatures are likely to occur in the chiller area, drain the chilled water, condenser water, and the pumpout condenser water circuits to avoid freeze-up. Keep the waterbox drains open.

Leave the oil charge in the chiller with the oil heater and controls energized to maintain the minimum oil reservoir temperature.

After Extended Shutdown — Ensure the water system drains are closed. It may be advisable to flush the water circuits to remove any soft rust which may have formed. This is a good time to brush the tubes if necessary.

Check the cooler pressure on the LID default screen and compare it to the original holding charge that was left in the chiller. If (after adjusting for ambient temperature changes) any loss in pressure is indicated, check for refrigerant leaks. See Check Chiller Tightness section, page 42.

Recharge the chiller by transferring refrigerant from the pumpout storage tank (if supplied). Follow the Pumpout and Refrigerant Transfer Procedures section, this page. Observe freeze-up precautions.

Carefully make all regular preliminary and running system checks. Perform a Control Test before start-up. If the compressor oil level appears abnormally high, the oil may have absorbed refrigerant. Ensure that the oil temperature is above 140 F (60 C) or above the cooler refrigerant temperature plus 50° F (27° C).

Cold Weather Operation — When the entering condenser water temperature drops very low, the operator should automatically cycle the cooling tower fans off to keep the temperature up. Piping may also be arranged to bypass the cooling tower. The PIC controls have a low limit tower fan relay (PR3) that can be used to assist in this control.

Manual Guide Vane Operation — It is possible to manually operate the guide vanes in order to check control operation or to control the guide vanes in an emergency. Manual operation is possible by overriding the target guide vane position. Access the STATUS01 screen on the LID and scroll down to highlight *TARGET GUIDE VANE POSITION*. To control the position, use the INCREASE or DECREASE softkey to adjust to the percentage of guide vane opening that is desired. Zero percent is fully closed; 100% is fully open. To release the guide vanes to automatic control, press the RELEASE softkey.

NOTE: Manual control increases the guide vane openings and overrides the pulldown rate during start-up. Motor current above the electrical demand setting, capacity overrides, and chilled water temperature below the control point override the manual target and close the guide vanes. For descriptions of capacity overrides and set points, see the Controls section.

Refrigeration Log — A refrigeration log (as shown in Fig. 30), is a convenient checklist for routine inspection and maintenance and provides a continuous record of chiller performance. It is also an aid when scheduling routine maintenance and diagnosing chiller problems.

Keep a record of the chiller pressures, temperatures, and liquid levels on a sheet similar to the one in Fig. 31. Automatic recording of PIC data is possible by using CCN devices such as the Data Collection module and a Building Supervisor. Contact a Carrier representative for more information.

PUMPOUT AND REFRIGERANT TRANSFER PROCEDURES

Preparation — The 19XR may come equipped with an optional pumpout storage tank, pumpout system, or pumpout compressor. The refrigerant can be pumped for service work to either the chiller compressor vessel or chiller condenser vessel by using the optional pumpout system. If a pumpout storage tank is supplied, the refrigerant can be isolated in the storage tank. The following procedures describe how to transfer refrigerant from vessel to vessel and perform chiller evacuations.

⚠ WARNING

Always run the chiller cooler and condenser water pumps and always charge or transfer refrigerant as a gas when the chiller pressure is less than 30 psig (207 kPa). Below these pressures, liquid refrigerant flashes into gas, resulting in extremely low temperatures in the cooler/condenser tubes and possibly causing tube freeze-up.

⚠ WARNING

During transfer of refrigerant into and out of the optional storage tank, carefully monitor the storage tank level gage. Do not fill the tank more than 90% of capacity to allow for refrigerant expansion. Overfilling may result in damage to the tank or personal injury.

⚠ CAUTION

Do not mix refrigerants from chillers that use different compressor oils. Compressor damage can result.

Operating the Optional Pumpout Unit

1. Be sure that the suction and the discharge service valves on the optional pumpout compressor are open (backseated) during operation. Rotate the valve stem fully counterclockwise to open. Frontseating the valve closes the refrigerant line and opens the gage port to compressor pressure.
2. Ensure that the compressor holddown bolts have been loosened to allow free spring travel.
3. Open the refrigerant inlet valve on the pumpout compressor.
4. Oil should be visible in the pumpout unit compressor sight glass under all operating conditions and during shutdown. If oil is low, add oil as described under Optional Pumpout System Maintenance section, page 65. The pumpout unit control wiring schematic is detailed in Fig. 31.

TO READ REFRIGERANT PRESSURES during pumpout or leak testing:

1. The LID display on the chiller control center is suitable for determining refrigerant-side pressures and low (soft) vacuum. To assure the desired range and accuracy when measuring evacuation and dehydration, use a quality vacuum indicator or manometer. This can be placed on the Schrader connections on each vessel (Fig. 7) by removing the pressure transducer.

Carrier®

Date________________

REFRIGERATION LOG CARRIER 19XR HERMETIC CENTRIFUGAL REFRIGERATION MACHINE

Plant___________ MACHINE MODEL NO.____________ MACHINE SERIAL NO.____________ REFRIGERANT TYPE___________

DATE__________	COOLER							CONDENSER							COMPRESSOR					OPER-ATOR INITIALS	REMARKS
TIME	Refrigerant		Water					Refrigerant		Water					BEARING TEMP	Oil			Motor		
			Pressure			Temp				Pressure			Temp						FLA________		
	Press.	Temp	In	Out	GPM	In	Out	Press.	Temp	In	Out	GPM	In	Out		Press. Diff.	Temp (reser-voir)	Level	Amperage (or vane position)		

REMARKS: Indicate shutdowns on safety controls, repairs made, oil or refrigerant added or removed, air exhausted and water drained from dehydrator. Include amounts.

Fig. 30 — Refrigeration Log

2. To determine pumpout storage tank pressure, a 30 in. -0-400 psi (-101-0-2769 kPa) gage is attached to the storage tank.
3. Refer to Fig. 26, 27, and 32 for valve locations and numbers.

⚠ CAUTION

Transfer, addition, or removal of refrigerant in spring-isolated chillers may place severe stress on external piping if springs have not been blocked in both up and down directions.

Chillers with Storage Tanks — If the chiller has isolation valves, leave them open for the following procedures. The letter ‘‘C’’ describes a closed valve. See Fig. 15, 16, 26, and 27.

TRANSFER REFRIGERANT FROM PUMPOUT STORAGE TANK TO CHILLER

1. Equalize refrigerant pressure.
   a. Use the PIC terminate lockout function on the PUMPDOWN LOCKOUT screen, accessed from the CONTROL TEST table to turn on the water pumps and monitor pressures.

⚠ WARNING

If the chilled water and condenser water pumps are not controlled by the PIC, these pumps must be started and stopped manually at the appropriate times during the refrigerant transfer procedure.

   b. Close pumpout unit valves 2, 4, 5, 8, and 10, and close chiller charging valve 7; open chiller isolation valves 11, 12, 13, and 14 (if present).
   c. Open pumpout unit/storage tank valves 3 and 6, open chiller valves 1a and 1b.

VALVE	1a	1b	2	3	4	5	6	7	8	10	11	12	13	14
CONDITION			C		C	C		C	C	C				

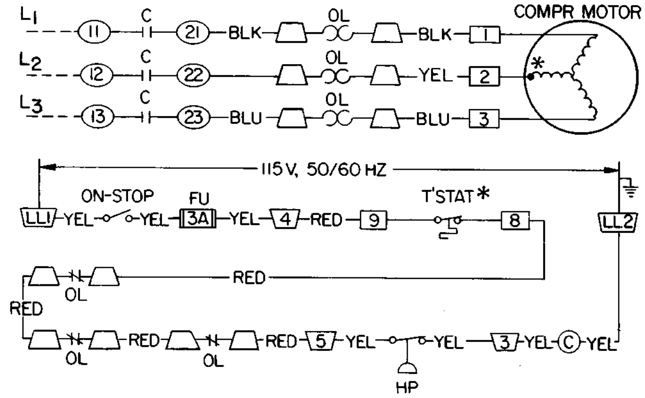
   d. Slowly open valve 5 to increase chiller pressure to 68 psig 35 psig (141 kPa) for HFC-134a. Feed refrigerant slowly to prevent freeze up.
   e. Open valve 5 fully after the pressure rises above the freeze point of the refrigerant. Open liquid line valves 7 and 10 until refrigerant pressure equalizes.

VALVE	1a	1b	2	3	4	5	6	7	8	10	11	12	13	14
CONDITION			C		C				C					

2. Transfer the remaining refrigerant.
   a. Close valve 5 and open valve 4.

VALVE	1a	1b	2	3	4	5	6	7	8	10	11	12	13	14
CONDITION			C			C			C					

   b. Turn off the chiller water pumps using the LID (or manually, if necessary).
   c. Turn off the pumpout condenser water, and turn on the pumpout compressor to push liquid out of the storage tank.
   d. Close liquid line valve 7.
   e. Turn off the pumpout compressor.
   f. Close valves 3 and 4.



LEGEND

C	—	Contactor
FU	—	Fuse, 3 Amps
HP	—	High-Pressure Cutout
OL	—	Compressor Overload
T'STAT	—	Internal Thermostat
		Compressor Terminal
		Contactor Terminal
		Overload Terminal
		Pumpout Unit Terminal

*Bimetal thermal protector imbedded in motor winding.

Fig. 31 — 19XR Pumpout Unit Wiring Schematic

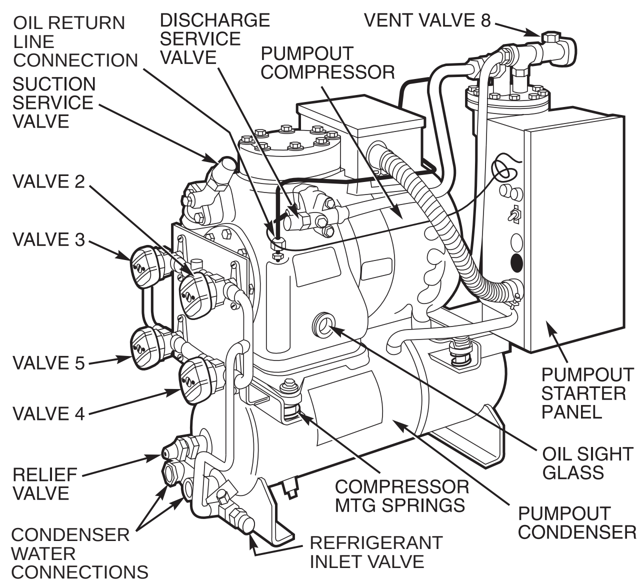


Fig. 32 — Optional Pumpout Unit

   g. Open valves 2 and 5.

VALVE	1a	1b	2	3	4	5	6	7	8	10	11	12	13	14
CONDITION				C	C			C	C					

   h. Turn on the pumpout condenser water.
   i. Run the pumpout compressor until the pumpout storage tank pressure reaches 5 psig (34 kPa) (18 in. Hg [40 kPa absolute] if repairing the tank).

j. Turn off the pumpout compressor.

k. Close valves 1a, 1b, 2, 5, 6, and 10.

VALVE	1a	1b	2	3	4	5	6	7	8	10	11	12	13	14
CONDITION	C	C	C	C	C	C	C	C	C	C				

l. Turn off pumpout condenser water.

TRANSFER REFRIGERANT FROM CHILLER TO PUMPOUT STORAGE TANK

1. Equalize refrigerant pressure.

a. Valve positions:

VALVE	1a	1b	2	3	4	5	6	7	8	10	11	12	13	14
CONDITION			C		C	C		C	C					

b. Slowly open valve 5. When the pressures are equalized, open liquid line valve 7 to allow liquid refrigerant to drain by gravity into the pumpout storage tank.

VALVE	1a	1b	2	3	4	5	6	7	8	10	11	12	13	14
CONDITION			C		C				C					

2. Transfer the remaining liquid.

a. Turn off the pumpout condenser water. Place the valves in the following positions:

VALVE	1a	1b	2	3	4	5	6	7	8	10	11	12	13	14
CONDITION				C	C				C					

b. Run the pumpout compressor for approximately 30 minutes; then close valve 10.

VALVE	1a	1b	2	3	4	5	6	7	8	10	11	12	13	14
CONDITION				C	C				C	C				

c. Turn off the pumpout compressor.

3. Remove any remaining refrigerant.

a. Turn on the chiller water pumps using the PIC PUMPDOWN LOCKOUT screen, accessed from the CONTROL TEST table. Turn on the pumps manually, if they are not controlled by the PIC.

b. Turn on the pumpout condenser water.

c. Place valves in the following positions:

VALVE	1a	1b	2	3	4	5	6	7	8	10	11	12	13	14
CONDITION			C			C			C	C				

d. Run the pumpout compressor until the chiller pressure reaches 30 psig (207 kPa) for HFC-134a. Then, shut off the pumpout compressor. Warm condenser water will boil off any entrapped liquid refrigerant and the chiller pressure will rise.

e. When the pressure rises to 40 psig (276 kPa) for HFC-134a, turn on the pumpout compressor until the pressure again reaches 30 psig (207 kPa), and then turn off the pumpout compressor. Repeat this process until the pressure no longer rises. Then, turn on the pumpout compressor and pump until the pressure reaches 18 in. Hg. (40 kPa absolute).

f. Close valves 1a, 1b, 3, 4, 6, and 10.

VALVE	1a	1b	2	3	4	5	6	7	8	10	11	12	13	14
CONDITION	C	C	C	C	C	C	C	C	C	C				

g. Turn off the pumpout condenser water and continue to use the PIC PUMPDOWN LOCKOUT screen functions, which lock out the chiller compressor for operation.

4. Establish a vacuum for service.

To conserve refrigerant, operate the pumpout compressor until the chiller pressure is reduced to 18 in. Hg vac., ref 30 in. bar. (40 kPa abs.) following Step 3e.

Chillers with Isolation Valves

TRANSFER ALL REFRIGERANT TO CHILLER CONDENSER VESSEL — For chillers with isolation valves, refrigerant can be stored in one chiller vessel or the other without the need for an external storage tank.

1. Push refrigerant into the chiller condenser.

a. Valve positions:

VALVE	1a	1b	2	3	4	5	8	11	12	13	14
CONDITION				C	C		C		C	C	C

b. Using the PIC controls, turn off the chiller water pumps and pumpout condenser water. If the chiller water pumps are not controlled through the PIC, turn them off manually.

c. Turn on the pumpout compressor to push the liquid refrigerant out of the chiller cooler vessel.

d. When all liquid refrigerant has been pushed into the chiller condenser vessel, close chiller isolation valve 11.

e. Access the PUMPDOWN LOCKOUT screen on the PIC CONTROL TEST table to turn on the chiller water pumps. If the chiller water pumps are not controlled by the PIC, turn them on manually.

f. Turn off the pumpout compressor.

2. Evacuate the refrigerant gas from chiller cooler vessel.

a. Close pumpout compressor valves 2 and 5, and open valves 3 and 4.

VALVE	1a	1b	2	3	4	5	8	11	12	13	14
CONDITION			C			C	C	C	C	C	C

b. Turn on the pumpout condenser water.

c. Run the pumpout compressor until the chiller cooler vessel pressure reaches 18 in. Hg vac (40 kPa abs.). Monitor pressures on the LID and on refrigerant gages.

d. Close valve 1a.

e. Turn off the pumpout compressor.

f. Close valves 1b, 3, and 4.

VALVE	1a	1b	2	3	4	5	8	11	12	13	14
CONDITION	C	C	C	C	C	C	C	C	C	C	C

g. Turn off the pumpout condenser water.

h. Proceed to the PUMPDOWN LOCKOUT function accessed from the LID CONTROL TEST table to turn off the chiller water pumps and lock out the chiller compressor. Turn off the chiller water pumps manually if they are not controlled by the PIC.

TRANSFER ALL REFRIGERANT TO CHILLER COOLER VESSEL

1. Push the refrigerant into the chiller cooler vessel.
   a. Valve positions:

VALVE	1a	1b	2	3	4	5	8	11	12	13	14
CONDITION			C			C	C		C	C	C

   b. Turn off the chiller water pumps (either through the PIC controls or manually, if necessary) and the pumpout condenser water.
   c. Turn on the pumpout compressor to push the refrigerant out of the chiller condenser.
   d. When all liquid refrigerant is out of the chiller condenser, close the cooler isolation valve 11.
   e. Turn off the pumpout compressor.
2. Evacuate the refrigerant gas from the chiller condenser vessel.
   a. Access the PUMPDOWN LOCKOUT function accessed from the LID CONTROL TEST table to turn on the chiller water pumps. Turn the chiller water pumps on manually if they are not controlled by the PIC.
   b. Close pumpout unit valves 3 and 4; open valves 2 and 5.

VALVE	1a	1b	2	3	4	5	8	11	12	13	14
CONDITION				C	C		C	C	C	C	C

   c. Turn on the pumpout condenser water.
   d. Run the pumpout compressor until the chiller condenser pressure reaches 18 in. Hg vac (40 kPa abs.). Monitor pressure at the LID and at refrigerant gages.
   e. Close valve 1b.
   f. Turn off the pumpout compressor.
   g. Close valves 1a, 2, and 5.

VALVE	1a	1b	2	3	4	5	8	11	12	13	14
CONDITION	C	C	C	C	C	C	C	C	C	C	C

   h. Turn off the pumpout condenser water.
   i. Proceed to the PUMPDOWN LOCKOUT test from the LID CONTROL TEST table to turn off the chiller water pumps and lock out the chiller compressor. Turn off the chiller water pumps manually if they are not controlled by the PIC.

RETURN CHILLER TO NORMAL OPERATING CONDITIONS

1. Ensure vessel that was opened has been evacuated.
2. Access the TERMINATE LOCKOUT function LID from the LID CONTROL TEST table to view vessel pressures and turn on chiller water pumps. If the chiller water pumps are not controlled by the PIC, turn them on manually.
3. Open valves 1a, 1b, and 3.

VALVE	1a	1b	2	3	4	5	8	11	12	13	14
CONDITION			C		C	C	C	C	C	C	C

4. Slowly open valve 5, gradually increasing pressure in the evacuated vessel to 35 psig (141 kPa). Feed refrigerant slowly to prevent tube freeze up.
5. Leak test to ensure vessel integrity.
6. Open valve 5 fully.

VALVE	1a	1b	2	3	4	5	8	11	12	13	14
CONDITION			C		C		C	C	C	C	C

7. Open valve 11 to equalize the liquid refrigerant level between the vessels.
8. Close valves 1a, 1b, 3, and 5.
9. Open isolation valves 12, 13, and 14 (if present).

VALVE	1a	1b	2	3	4	5	8	11	12	13	14
CONDITION	C	C	C	C	C	C	C				

10. Proceed to the TERMINATE LOCKOUT screen (accessed from the CONTROL TEST table) to turn off the water pumps and enable the chiller compressor for start-up. If the chiller water pumps are not controlled by the PIC, turn them off manually.

GENERAL MAINTENANCE

Refrigerant Properties — The standard refrigerant for the 19XR chiller is HFC-134a. At normal atmospheric pressure, HFC-134a will boil at –14 F (–25 C) and must, therefore, be kept in pressurized containers or storage tanks. The refrigerant is practically odorless when mixed with air and is noncombustible at atmospheric pressure. Read the Material Safety Data Sheet and the latest ASHRAE Safety Guide for Mechanical Refrigeration to learn more about safe handling of this refrigerant.

⚠ DANGER

HFC-134a will dissolve oil and some nonmetallic materials, dry the skin, and, in heavy concentrations, may displace enough oxygen to cause asphyxiation. When handling this refrigerant, protect the hands and eyes and avoid breathing fumes.

Adding Refrigerant — Follow the procedures described in Trim Refrigerant Charge section, page 62.

⚠ WARNING

Always use the compressor pumpdown function in the Control Test table to turn on the cooler pump and lock out the compressor when transferring refrigerant. Liquid refrigerant may flash into a gas and cause possible freeze-up when the chiller pressure is below 30 psig (207 kPa) for HFC-134a.

Removing Refrigerant — If the optional pumpout system is used, the 19XR refrigerant charge may be transferred to a pumpout storage tank or to the chiller condenser or cooler vessels. Follow the procedures in the Pumpout and Refrigerant Transfer Procedures section when transferring refrigerant from one vessel to another.

Adjusting the Refrigerant Charge — If the addition or removal of refrigerant is required to improve chiller performance, follow the procedures given under the Trim Refrigerant Charge section, this page.

Refrigerant Leak Testing — Because HFC-134a is above atmospheric pressure at room temperature, leak testing can be performed with refrigerant in the chiller. Use an electronic halide leak detector, soap bubble solution, or ultrasonic leak detector. Ensure that the room is well ventilated and free from concentration of refrigerant to keep false readings to a minimum. Before making any necessary repairs to a leak, transfer all refrigerant from the leaking vessel.

Leak Rate — It is recommended byASHRAE that chillers be taken off line immediately and repaired if the refrigerant leak rate for the entire chiller is more than 10% of the operating refrigerant charge per year.

In addition, Carrier recommends that leaks totalling less than the above rate but more than a rate of 1 lb (0.5 kg) per year should be repaired during annual maintenance or whenever the refrigerant is transferred for other service work.

Test After Service, Repair, or Major Leak — If all the refrigerant has been lost or if the chiller has been opened for service, the chiller or the affected vessels must be pressure tested and leak tested. Refer to the Leak Test Chiller section to perform a leak test.

⚠ WARNING

HFC-134a should not be mixed with air or oxygen and pressurized for leak testing. In general, this refrigerant should not be present with high concentrations of air or oxygen above atmospheric pressures, because the mixture can undergo combustion.

REFRIGERANT TRACER — Use an environmentally acceptable refrigerant as a tracer for leak test procedures.

TO PRESSURIZE WITH DRY NITROGEN — Another method of leak testing is to pressurize with nitrogen only and to use a soap bubble solution or an ultrasonic leak detector to determine if leaks are present.

NOTE: Pressurizing with dry nitrogen for leak testing should only be done if *all* refrigerant has been evacuated from the vessel.

1. Connect a copper tube from the pressure regulator on the cylinder to the refrigerant charging valve. Never apply full cylinder pressure to the pressurizing line. Follow the listed sequence.
2. Open the charging valve fully.
3. Slowly open the cylinder regulating valve.
4. Observe the pressure gage on the chiller and close the regulating valve when the pressure reaches test level. *Do not exceed* 140 psig (965 kPa).
5. Close the charging valve on the chiller. Remove the copper tube if it is no longer required.

Repair the Leak, Retest, and Apply Standing Vacuum Test — After pressurizing the chiller, test for leaks with an electronic halide leak detector, soap bubble solution, or an ultrasonic leak detector. Bring the chiller back to atmospheric pressure, repair any leaks found, and retest.

After retesting and finding no leaks, apply a standing vacuum test. Then dehydrate the chiller. Refer to the Standing Vacuum Test and Chiller Dehydration section (pages 44 and 46) in the Before Initial Start-Up section.

Checking Guide Vane Linkage — When the chiller is off, the guide vanes are closed and the actuator mechanism is in the position shown in Fig. 33. If slack develops in the drive chain, do the following to eliminate backlash:

1. With the chiller shut down and the actuator fully closed, remove the chain guard and loosen the actuator bracket holddown bolts.
2. Loosen guide vane sprocket adjusting bolts.
3. Pry bracket upwards to remove slack, then retighten the bracket holddown bolts.
4. Retighten the guide vane sprocket adjusting bolts. Ensure that the guide vane shaft is rotated fully in the clockwise direction in order close it fully.

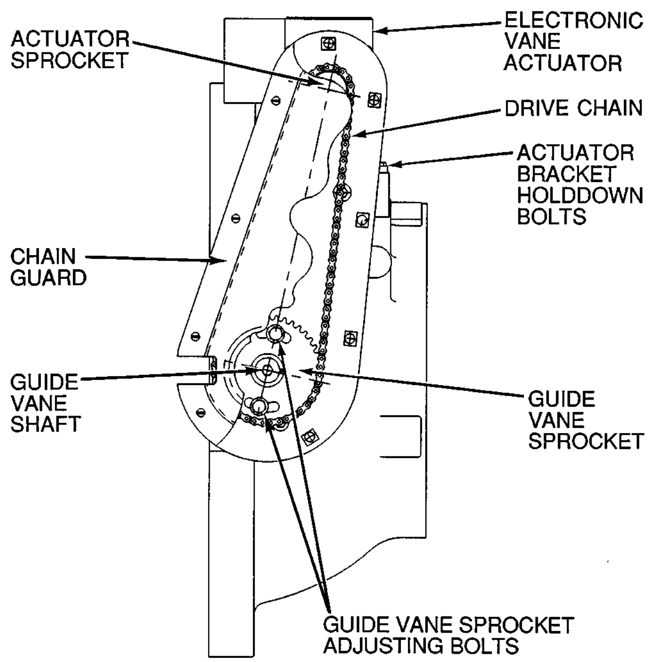


Fig. 33 — Guide Vane Actuator Linkage

Trim Refrigerant Charge — If, to obtain optimal chiller performance, it becomes necessary to adjust the refrigerant charge, operate the chiller at design load and then add or remove refrigerant slowly until the difference between the leaving chilled water temperature and the cooler refrigerant temperature reaches design conditions or becomes a minimum. *Do not overcharge.*

Refrigerant may be added either through the storage tank or directly into the chiller as described in the Charge Refrigerant into Chiller section.

To remove any excess refrigerant, follow the procedure in Transfer Refrigerant from Chiller to Pumpout Storage Tank section, Steps 1a and b, page 60.

WEEKLY MAINTENANCE

Check the Lubrication System — Mark the oil level on the reservoir sight glass, and observe the level each week while the chiller is shut down.

If the level goes below the lower sight glass, check the oil reclaim system for proper operation. If additional oil is required, add it through the oil drain charging valve (Fig. 2). A pump is required when adding oil against refrigerant pressure. The oil charge for the 19XR compressor depends on the compressor Frame size:

- Frame 2 compressor — 5 gal (18.9 L)
- Frame 3 compressor — 8 gal (30 L)
- Frame 4 compressor — 10 gal (37.8 L)
- Frame 5 compressor — 18 gal (67.8 L)

The added oil *must* meet Carrier specifications for the 19XR. Refer to Changing Oil Filter and Oil Changes section on this page. Any additional oil that is added should be logged by noting the amount and date. Any oil that is added due to oil loss that is not related to service will eventually return to the sump. It must be removed when the level is high.

An oil heater is controlled by the PIC to maintain oil temperature (see the Controls section) when the compressor is off. The LID STATUS02 screen displays whether the heater is energized or not. The heater is energized if the *OIL HEATER RELAY* parameter reads ON. If the PIC shows that the heater is energized and if the sump is still not heating up, the power to the oil heater may be off or the oil level may be too low. Check the oil level, the oil heater contactor voltage, and oil heater resistance.

The PIC does not permit compressor start-up if the oil temperature is too low. The PIC continues with start-up only after the temperature is within allowable limits.

SCHEDULED MAINTENANCE

Establish a regular maintenance schedule based on your actual chiller requirements such as chiller load, run hours, and water quality. *The time intervals listed in this section are offered as guides to service only*.

Service Ontime — The LID will display a *SERVICE ONTIME* value on the STATUS01 screen. This value should be reset to zero by the service person or the operator each time major service work is completed so that the time between service can be viewed and tracked.

Inspect the Control Center — Maintenance consists of general cleaning and tightening of connections. Vacuum the cabinet to eliminate dust build-up. If the chiller control malfunctions, refer to the Troubleshooting Guide section for control checks and adjustments.

⚠ CAUTION

Ensure power to the control center is off when cleaning and tightening connections inside the control center.

Check Safety and Operating Controls Monthly — To ensure chiller protection, the automated Control Test should be performed at least once per month. See Table 3 for safety control settings. See Table 6 for Control Test functions.

Changing Oil Filter — Change the oil filter on a yearly basis or when the chiller is opened for repairs. The 19XR has an isolatable oil filter so that the filter may be changed with the refrigerant remaining in the chiller. Use the following procedure:

1. Ensure the compressor is off and the disconnect for the compressor is open.
2. Disconnect the power to the oil pump.
3. Close the oil filter isolation valves (Fig. 2).
4. Connect an oil charging hose from the oil charging valve (Fig. 2) and place the other end in a clean container suitable for used oil. The oil drained from the filter housing should be used as an oil sample and sent to a laboratory for proper analysis. *Do not contaminate this sample.*
5. Slowly open the charging valve to drain the oil from the housing.

⚠ CAUTION

The oil filter housing is at a high pressure. Relieve this pressure slowly.

6. Once all oil has been drained, place some rags or absorbent material under the oil filter housing to catch any drips once the filter is opened. Remove the 4 bolts from the end of the filter housing and remove the filter cover.
7. Remove the filter retainer by unscrewing the retainer nut. The filter may now be removed and disposed of properly.
8. Replace the old filter with a new filter. Install the filter retainer and tighten down the retainer nut. Install the filter cover and tighten the 4 bolts.
9. Evacuate the filter housing by placing a vacuum pump on the charging valve. Follow the normal evacuation procedures. Shut the charging valve when done and reconnect the valve so that new oil can be pumped into the filter housing. Fill with the same amount that was removed; then close the charging valve.
10. Remove the hose from the charging valve, open the isolation valves to the filter housing, and turn on the power to the pump and the motor.

Oil Specification — If oil is added, it must meet the following Carrier specifications:

Oil Type for units using R-134a . . . . Inhibited polyolester-based synthetic compressor oil formatted for use with HFC, gear-driven, hermetic compressors.

ISO Viscosity Grade 68

The polyolester-based oil (P/N: PP23BZ103) may be ordered from your local Carrier representative.

Oil Changes — Carrier recommends changing the oil after the first year of operation and every five years thereafter as a minimum in addition to a yearly oil analysis. However, if a continuous oil monitoring system is functioning and a yearly oil analysis is performed, the time between oil changes can be extended.

TO CHANGE THE OIL

1. Transfer the refrigerant into the chiller condenser vessel (for isolatable vessels) or to a pumpout storage tank.
2. Mark the existing oil level.
3. Open the control and oil heater circuit breaker.
4. When the chiller pressure is 5 psig (34 kPa) or less, drain the oil reservoir by opening the oil charging valve (Fig. 2). Slowly open the valve against refrigerant pressure.
5. Change the oil filter at this time. See Changing Oil Filter section.
6. Change the refrigerant filter at this time, see the next section, Refrigerant Filter.
7. Charge the chiller with oil. Charge until the oil level is equal to the oil level marked in Step 2. Turn on the power to the oil heater and let the PIC warm it up to at least 140 F (60 C). Operate the oil pump manually, using the Control Test function, for 2 minutes. For shutdown conditions, the oil level should be full in the lower sight glass. If the oil level is above ½ full in the upper sight glass, remove the excess oil. The oil level should now be equal to the amount shown in Step 2.

Refrigerant Filter — A refrigerant filter/drier, located on the refrigerant cooling line to the motor, should be changed once a year or more often if filter condition indicates a need for more frequent replacement. Change the filter by closing the filter isolation valves (Fig. 4) and slowly opening the flare fittings with a wrench and back-up wrench to relieve the pressure. A moisture indicator sight glass is located beyond this filter to indicate the volume and moisture in the refrigerant. If the moisture indicator indicates moisture, locate the source of water immediately by performing a thorough leak check.

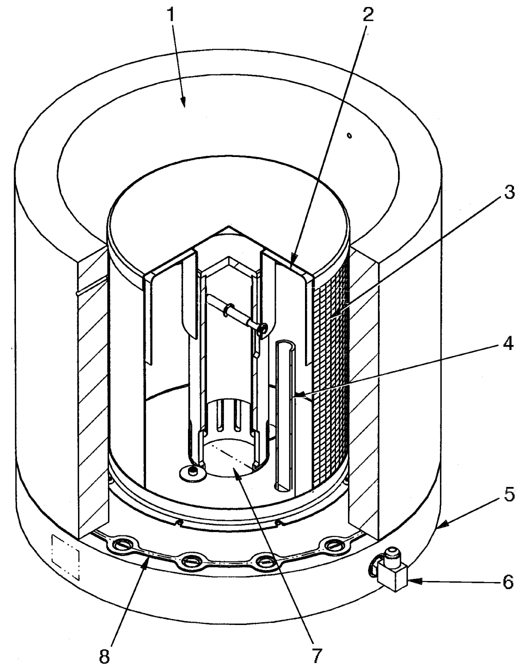
Oil Reclaim Filter — The oil reclaim system has a strainer on the eductor suction line, a strainer on the discharge pressure line, and a filter on the cooler scavenging line. Replace the filter once per year or more often if filter condition indicates a need for more frequent replacement. Change the filter by closing the filter isolation valves and slowly opening the flare fitting with a wrench and back-up wrench to relieve the pressure. Change the strainers once every 5 years or whenever refrigerant is evacuated from the cooler.

Inspect Refrigerant Float System — Perform this inspection every 5 years or when the condenser is opened for service.

1. Transfer the refrigerant into the cooler vessel or into a pumpout storage tank.
2. Remove the float access cover.
3. Clean the chamber and valve assembly thoroughly. Be sure the valve moves freely. Ensure that all openings are free of obstructions.
4. Examine the cover gasket and replace if necessary.

See Fig. 34 for a view of the float valve design. For linear float valve designs, inspect the orientation of the float slide pin. It must be pointed toward the bubbler tube for proper operation.

Inspect Relief Valves and Piping — The relief valves on this chiller protect the system against the potentially dangerous effects of overpressure. To ensure against damage to the equipment and possible injury to personnel, these devices must be kept in peak operating condition.



LEGEND

1 — Refrigerant Inlet from FLASC Chamber
2 — Linear Float Assembly
3 — Float Screen
4 — Bubble Line
5 — Float Cover
6 — Bubble Line Connection
7 — Refrigerant Outlet to Cooler
8 — Gasket

Fig. 34 — 19XR Float Valve Design

As a minimum, the following maintenance is required.

1. At least once a year, disconnect the vent piping at the valve outlet and carefully inspect the valve body and mechanism for any evidence of internal corrosion or rust, dirt, scale, leakage, etc.
2. If corrosion or foreign material is found, do not attempt to repair or recondition. *Replace the valve.*
3. If the chiller is installed in a corrosive atmosphere or the relief valves are vented into a corrosive atmosphere, inspect the relief valves at more frequent intervals.

Compressor Bearing and Gear Maintenance — The key to good bearing and gear maintenance is proper lubrication. Use the proper grade of oil, maintained at recommended level, temperature, and pressure. Inspect the lubrication system regularly and thoroughly.

To inspect the bearings, a complete compressor teardown is required. Only a trained service technician should remove and examine the bearings. The cover plate on older compressor bases was used for factory-test purposes and is not usable for bearing or gear inspection. The bearings and gears should be examined on a scheduled basis for signs of wear. The frequency of examination is determined by the hours of chiller operation, load conditions during operation, and the condition of the oil and the lubrication system. Excessive bearing wear can sometimes be detected through increased vibration or increased bearing temperature. If either symptom appears, contact an experienced and responsible service organization for assistance.

Inspect the Heat Exchanger Tubes

COOLER — Inspect and clean the cooler tubes at the end of the first operating season. Because these tubes have internal ridges, a rotary-type tube cleaning system is needed to fully clean the tubes. Inspect the tubes' condition to determine the scheduled frequency for future cleaning and to determine whether water treatment in the chilled water/brine circuit is adequate. Inspect the entering and leaving chilled water temperature sensors for signs of corrosion or scale. Replace a sensor if corroded or remove any scale if found.

CONDENSER — Since this water circuit is usually an open-type system, the tubes may be subject to contamination and scale. Clean the condenser tubes with a rotary tube cleaning system at least once per year and more often if the water is contaminated. Inspect the entering and leaving condenser water sensors for signs of corrosion or scale. Replace the sensor if corroded or remove any scale if found.

Higher than normal condenser pressures, together with the inability to reach full refrigeration load, usually indicate dirty tubes or air in the chiller. If the refrigeration log indicates a rise above normal condenser pressures, check the condenser refrigerant temperature against the leaving condenser water temperature. If this reading is more than what the design difference is supposed to be, the condenser tubes may be dirty or water flow may be incorrect. Because HFC-134a is a high-pressure refrigerant, air usually does not enter the chiller.

During the tube cleaning process, use brushes specially designed to avoid scraping and scratching the tube wall. Contact your Carrier representative to obtain these brushes. *Do not* use wire brushes.

⚠ CAUTION

Hard scale may require chemical treatment for its prevention or removal. Consult a water treatment specialist for proper treatment.

Water Leaks — The refrigerant moisture indicator on the refrigerant motor cooling line (Fig. 2) indicates whether there is water leakage during chiller operation. Water leaks should be repaired immediately.

⚠ CAUTION

The chiller must be dehydrated after repair of water leaks. See Chiller Dehydration section, page 46.

Water Treatment — Untreated or improperly treated water may result in corrosion, scaling, erosion, or algae. The services of a qualified water treatment specialist should be obtained to develop and monitor a treatment program.

⚠ CAUTION

Water must be within design flow limits, clean, and treated to ensure proper chiller performance and reduce the potential of tube damage due to corrosion, scaling, erosion, and algae. Carrier assumes no responsibility for chiller damage resulting from untreated or improperly treated water.

Inspect the Starting Equipment — Before working on any starter, shut off the chiller and open all disconnects supplying power to the starter.

⚠ WARNING

The disconnect on the starter front panel does not de-energize all internal circuits. Open all internal and remote disconnects before servicing the starter.

⚠ WARNING

Never open isolating knife switches while equipment is operating. Electrical arcing can cause serious injury.

Inspect starter contact surfaces for wear or pitting on mechanical-type starters. Do not sandpaper or file silverplated contacts. Follow the starter manufacturer's instructions for contact replacement, lubrication, spare parts ordering, and other maintenance requirements.

Periodically vacuum or blow off accumulated debris on the internal parts with a high-velocity, low-pressure blower.

Power connections on newly installed starters may relax and loosen after a month of operation. Turn power off and retighten. Recheck annually thereafter.

⚠ CAUTION

Loose power connections can cause voltage spikes, overheating, malfunctioning, or failures.

Check Pressure Transducers — Once a year, the pressure transducers should be checked against a pressure gage reading. Check all four transducers: the 2 oil differential pressure transducers, the condenser pressure transducer, and the cooler pressure transducer.

Note the evaporator and condenser pressure readings on the STATUS01 screen on the LID (*EVAPORATOR PRESSURE* and *CONDENSER PRESSURE*). Attach an accurate set of refrigeration gages to the cooler and condenser Schrader fittings. Compare the two readings. If there is a difference in readings, the transducer can be calibrated as described in the Troubleshooting Guide section. Oil differential pressure (*OIL PRESSURE* on the STATUS01 screen) should be zero whenever the compressor is off.

Optional Pumpout System Maintenance — For pumpout unit compressor maintenance details, refer to the 06D, 07D Installation, Start-Up, and Service Instructions.

OPTIONAL PUMPOUT COMPRESSOR OIL CHARGE — Use oil conforming to Carrier specifications for reciprocating compressor usage. Oil requirements are as follows:

ISO Viscosity 68
Carrier Part Number PP23BZ103

The total oil charge, 4.5 pints (2.6 L), consists of 3.5 pints (2.0 L) for the compressor and one additional pint (0.6 L) for the oil separator.

Oil should be visible in one of the compressor sight glasses during both operation and at shutdown. Always check the oil level before operating the compressor. Before adding or changing oil, relieve the refrigerant pressure as follows:

1. Attach a pressure gage to the gage port of either compressor service valve (Fig. 33).

2. Close the suction service valve and open the discharge line to the storage tank or the chiller.
3. Operate the compressor until the crankcase pressure drops to 2 psig (13 kPa).
4. Stop the compressor and isolate the system by closing the discharge service valve.
5. Slowly remove the oil return line connection (Fig. 32). Add oil as required.
6. Replace the connection and reopen the compressor service valves.

OPTIONAL PUMPOUT SAFETY CONTROL SETTINGS (Fig. 35) — The optional pumpout system high-pressure switch opens at 161 psig (1110 kPa) and closes at 130 psig (896 kPa). Check the switch setting by operating the pumpout compressor and slowly throttling the pumpout condenser water.

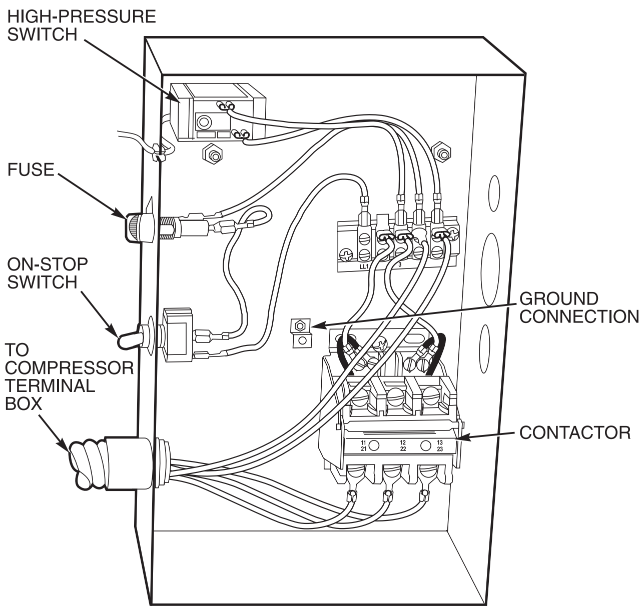


Fig. 35 — Optional Pumpout System Controls

Ordering Replacement Chiller Parts — When ordering Carrier specified parts, the following information must accompany an order:

- chiller model number and serial number
- name, quantity, and part number of the part required
- delivery address and method of shipment.

TROUBLESHOOTING GUIDE

Overview — The PIC has many features to help the operator and technician troubleshoot a 19XR chiller.

- The LID display shows the chiller's actual operating conditions and can be viewed while the unit is running.
- The LID default screen freezes when an alarm occurs. The freeze enables the operator to view the chiller conditions at the time of alarm. The STATUS screens continue to show current information. Once all alarms have been cleared (by correcting the problems and pressing the $\boxed{\text{RESET}}$ softkey), the LID default screen returns to normal operation.
- The CONTROL ALGORITHM STATUS screens (which include the MAINT01, MAINT02, MAINT03, and MAINT04 screens) display information that helps to diagnose problems with chilled water temperature control, chilled water temperature control overrides, hot gas bypass, surge algorithm status, and time schedule operation.
- The control test feature facilitates the proper operation and test of temperature sensors, pressure transducers, the guide vane actuator, oil pump, water pumps, tower control, and other on/off outputs while the compressor is stopped. It also has the ability to lock off the compressor and turn on water pumps for pumpout operation. The LID display shows the temperatures and pressures required during these operations.
- From other SERVICE tables, the operator/technician can access configured items, such as chilled water resets, override set points, etc.
- If an operating fault is detected, an alarm message is generated and displayed on the LID default screen. A more detailed message — along with a diagnostic message — is also stored into the ALARM HISTORY table.

Checking the Display Messages — The first area to check when troubleshooting the 19XR is the LID display. If the alarm light is flashing, check the primary and secondary message lines on the LID default screen (Fig. 12). These messages indicate where the fault is occurring. The ALARM HISTORY table on the LID SERVICE menu also contains an alarm message to further expand on this alarm. For a complete list of possible alarm messages, see Table 8. If the alarm light starts to flash while accessing a menu screen, press the $\boxed{\text{EXIT}}$ softkey to return to the default screen to read the alarm message. The compressor will not run while an alarm condition exists unless the alarm is caused by an unauthorized start or a failure to shut down.

Checking Temperature Sensors — All temperature sensors are thermistor-type sensors. This means that the resistance of the sensor varies with temperature. All sensors have the same resistance characteristics. If the controls are on, determine sensor temperature by measuring voltage drop; if the controls are powered off, determine sensor temperature by measuring resistance. Compare the readings to the values listed in Table 9A or 9B.

RESISTANCE CHECK — Turn off the control power and, from the module, disconnect the terminal plug of the sensor in question. With a digital ohmmeter, measure sensor resistance between receptacles as designated by the wiring diagram. The resistance and corresponding temperature are listed in Table 9A or 9B. Check the resistance of both wires to ground. This resistance should be infinite.

VOLTAGE DROP — The voltage drop across any energized sensor can be measured with a digital voltmeter while the control is energized. Table 9A or 9B lists the relationship between temperature and sensor voltage drop (volts dc measured across the energized sensor). Exercise care when measuring voltage to prevent damage to the sensor leads, connector plugs, and modules. Sensors should also be checked at the sensor plugs. Check the sensor wire at the sensor for 5 vdc if the control is powered on.

⚠ CAUTION

Relieve all refrigerant pressure or drain the water before replacing the temperature sensors.

CHECK SENSOR ACCURACY — Place the sensor in a medium of known temperature and compare that temperature to the measured reading. The thermometer used to determine the temperature of the medium should be of laboratory quality with 0.5° F (.25° C) graduations. The sensor in question should be accurate to within 2° F (1.2° C).

See Fig. 7 for sensor locations. The sensors are immersed directly in the refrigerant or water circuits. The wiring at each sensor is easily disconnected by unlatching the connector. These connectors allow only one-way connection to the sensor. When installing a new sensor, apply a pipe sealant or thread sealant to the sensor threads.

DUAL TEMPERATURE SENSORS — For servicing convenience, there are 2 sensors each on the bearing and motor temperature sensors. If one of the sensors is damaged, the other can be used by simply moving a wire.The number 2 terminal in the sensor terminal box is the common line. To use the second sensor, move the wire from the number 1 position to the number 3 position.

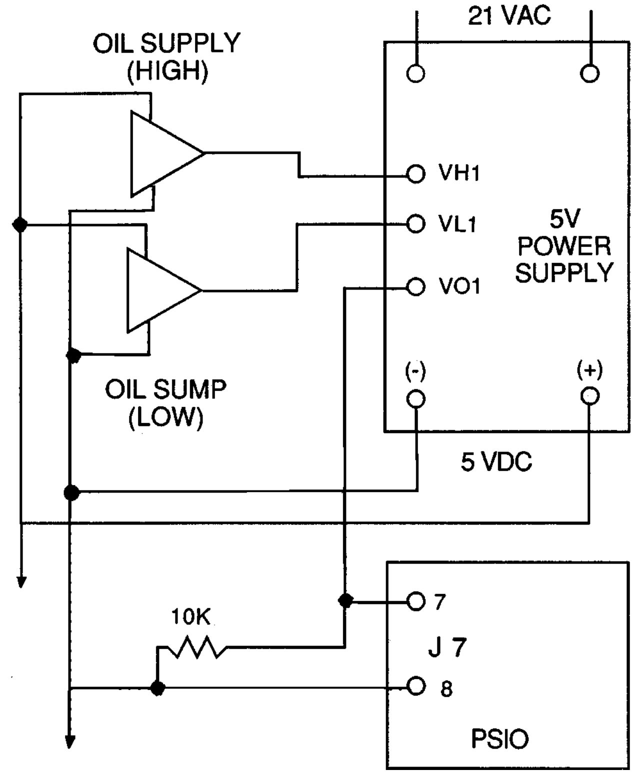
Checking Pressure Transducers — There are 4 pressure transducers on 19XR chillers. They determine cooler, condenser, and oil pressure. The cooler and condenser transducers are also used by the PIC to determine the refrigerant temperatures. The oil supply pressure transducer value and the oil transmission sump pressure transducer value difference is calculated by a differential pressure/power supply module located in the control center. See Fig. 36. The PSIO module then reads the differential pressure. In effect, the PSIO reads only one input for oil pressure for a total of 3 pressure inputs: cooler pressure, condenser pressure, and oil differential pressure. See the Check Pressure Transducers section (page 65) under Scheduled Maintenance.

All 3 pressure transducers can be calibrated if necessary. It is not usually necessary to calibrate at initial start-up. However, at high altitude locations, it is necessary to calibrate the transducers to ensure the proper refrigerant temperature/pressure relationship. Each transducer is supplied with 5 vdc power from a power supply. If the power supply fails, a transducer voltage reference alarm occurs. If the transducer reading is suspected of being faulty, check the supply voltage. It should be 5 vdc ± .5 v. If the supply voltage is correct, the transducer should be recalibrated or replaced.

COOLER AND CONDENSER PRESSURE CALIBRATION — Calibration can be checked by comparing the pressure readings from the transducer to an accurate refrigeration gage reading. These readings can be viewed or calibrated from the STATUS01 screen on the LID. The transducer can be checked and calibrated at 2 pressure points. These calibration points are 0 psig (0 kPa) and between 240 and 260 psig (1655 and 1793 kPa). To calibrate these transducers:

1. Shut down the compressor.
2. Disconnect the transducer in question from its Schrader fitting.

   NOTE: If the cooler or condenser vessels are at 0 psig (0 kPa) or are open to atmospheric pressure, the transducers can be calibrated for zero without removing the transducer from the vessel.
3. Access the STATUS01 screen and view the particular transducer reading (the *EVAPORATOR PRESSURE* or *CONDENSER PRESSURE* parameter on the STATUS01 screen). It should read 0 psi (0 kPa). If the reading is not 0 psi (0 kPa), but within ± 5 psi (35 kPa), the value may be set to zero by pressing the $\boxed{\text{SELECT}}$ softkey while the appropriate transducer parameter is highlighted on the LID screen. Then press the $\boxed{\text{ENTER}}$ softkey. The value will now go to zero.



19XR OIL PRESSURE INPUT

LEGEND

J — Connector
PSIO — Processor Input/Output Module
VH1 — High-Pressure Transducer Output
VL1 — Low-Pressure Transducer Output
VO1 — (VH1-VL1) + 0.467 ± 0.1V
● Connection
○ Terminal
▷ Transducer

Fig. 36 — Oil Differential Pressure/Power Supply Module

If the transducer value is not within the calibration range, the transducer returns to the original reading. If the LID pressure value is within the allowed range (noted above), check the voltage ratio of the transducer. To obtain the voltage ratio, divide the voltage (dc) input from the transducer by the supply voltage signal measured at the PSIO terminals J7-J34 and J7-J35. For example, the condenser transducer voltage input is measured at PSIO terminals J7-1 and J7-2. The voltage ratio must be between 0.80 vdc and 0.11 vdc for the software to allow calibration. Pressurize the transducer until the ratio is within range. Then attempt calibration again.

4. A high pressure point can also be calibrated between 240 and 260 psig (1655 and 1793 kPa) by attaching a regulated 250 psig (1724 kPa) pressure (usually from a nitrogen cylinder). The high pressure point can be calibrated by accessing the appropriate transducer parameter on the STATUS01 screen, highlighting the parameter, pressing the $\boxed{\text{SELECT}}$ softkey, and then using the $\boxed{\text{INCREASE}}$ or $\boxed{\text{DECREASE}}$ softkeys to adjust the value to the exact pressure on the refrigerant gage. Press the $\boxed{\text{ENTER}}$ softkey to finish the calibration. Pressures at high altitude locations must be compensated for, so the chiller temperature/pressure relationship is correct.

If the transducer reading returns to the previous value and the pressure is within the allowed range, check the voltage ratio of the transducer. Refer to Step 3 above. The voltage ratio for this high pressure calibration must be between 0.585 and 0.634 vdc to allow calibration. Change the pressure at the transducer until the ratio is within the acceptable range. Then attempt calibrate to the new pressure input.

The PIC does not allow calibration if the transducer is too far out of calibration. In this case, a new transducer must be installed and re-calibrated.

OIL DIFFERENTIAL PRESSURE/POWER SUPPLY MODULE CALIBRATION — (See Fig. 37.) The oil reservoir in the 19XR chiller is not common to cooler pressure. Differential oil pressure cannot be determined by comparing pump output to cooler pressure. Instead, use the method described below.

Oil transmission sump pressure and oil supply pressure are fed to a comparator circuit on a 5V power supply board. The output of this circuit, which represents differential oil pressure, is fed to the PSIO. Calibrate the oil differential pressure to 0 psid (0 kPad) by selecting the oil pressure input on the STATUS01 screen. Turn off the oil pump and make sure the transducers are connected. Press the ENTER softkey to set the point to zero. No high-end calibration is needed or possible.

TRANSDUCER REPLACEMENT — Since the transducers are mounted on Schrader-type fittings, there is no need to remove refrigerant from the vessel when replacing the transducers. Disconnect the transducer wiring by pulling up on the locking tab while pulling up on the weather-tight connecting plug from the end of the transducer. *Do not pull on the transducer wires.* Unscrew the transducer from the Schrader fitting. When installing a new transducer, do not use pipe sealer (which can plug the sensor). Put the plug connector back on the sensor and snap into place. Check for refrigerant leaks.

⚠ WARNING

Be sure to use a back-up wrench on the Schrader fitting whenever removing a transducer, since the Schrader fitting may back out with the transducer, causing a large leak and possible injury to personnel.

Control Algorithms Checkout Procedure — One of the tables on the LID SERVICE menu is CONTROL ALGORITHM STATUS. The maintenance screens may be viewed from the CONTROL ALGORITHM STATUS table to see how a particular control algorithm is operating.

These maintenance screens are very useful in helping to determine how the control temperature is calculated and guide vane positioned and for observing the reactions from load changes, control point overrides, hot gas bypass, surge prevention, etc. The tables are:

MAINT01	Capacity Control	This table shows all values used to calculate the chilled water/brinecontrol point.
MAINT02	Override Status	Details of all chilled water control override values.
MAINT03	Surge/ HGBP Status	The surge and hot gas bypass control algorithm status is viewed from this screen. All values dealing with this control are displayed.
MAINT04	LEAD/LAG Status	Indicates LEAD/LAG operation status.
OCCDEFM	Time Schedules Status	The Local and CCN occupied schedules are displayed here to help the operator quickly determine whether the schedule is in the ″occupied″ mode or not.
WSMDEFME	Water System Manager Status	The water system manager is a CCN module that can turn on the chiller and change the chilled water control point. This screen indicates the status of this system.

Control Test — The Control Test feature can check all the thermistor temperature sensors, including those on the options modules, pressure transducers, pumps and their associated flow switches, the guide vane actuator, and other control outputs such as hot gas bypass. The tests can help to determine whether a switch is defective or a pump relay is not operating, as well as other useful troubleshooting issues. During pumpdown operations, the pumps are energized to prevent freeze-up and the vessel pressures and temperatures are displayed. The Pumpdown/Lockout feature prevents compressor start-up when there is no refrigerant in the chiller or if the vessels are isolated. The Terminate Lockout feature ends the Pumpdown/Lockout after the pumpdown procedure is reversed and refrigerant is added.

LEGEND TO TABLES 8A - 8N

1CR_AUX	— Compressor Start Contact
CA_P	— Compressor Current
CCN	— Carrier Comfort Network
CDFL	— Condenser Water Flow
CHIL_S_S	— Chiller Start/Stop
CHW	— Chilled Water
CHWS	— Chilled Water Supply
CHWR	— Chilled Water Return
CMPD	— Discharge Temperature
CRP	— Condenser Refrigerant Pressure
CRT	— Condenser Refrigerant Temperature
ERT	— Evaporator Refrigerant Temperature
EVFL	— Chilled Water Flow
GV_TRG	— Target Guide Vane Position
LID	— Local Interface Device
MTRB	— Bearing Temperature
MTRW	— Motor Winding Temperature
OILPD	— Oil Pressure
OILT	— Oil Sump Temperature
PIC	— Product Integrated Control
PSIO	— Processor/Sensor Input/Output Module
RLA	— Rated Load Amps
RUN_AUX	— Compressor Run Contact
SPR_PL	— Spare Protective Limit Input
SMM	— Starter Management Module
STR_FLT	— Starter Fault
TXV	— Thermostatic Expansion Valve
V_P	— Line Voltage: Percent
V_REF	— Voltage Reference

Table 8 — LID Primary and Secondary Messages and Custom Alarm/Alert Messages with Troubleshooting Guides

A. SHUTDOWN WITH ON/OFF/RESET-OFF

PRIMARY MESSAGE	SECONDARY MESSAGE	PROBABLE CAUSE/REMEDY
MANUALLY STOPPED — PRESS	CCN OR LOCAL TO START	PIC in OFF mode, press the CCN or local softkey to start unit.
TERMINATE PUMPDOWN MODE	TO SELECT CCN OR LOCAL	Enter the Control Test screen and select Terminate Lockout to unlock compressor.
SHUTDOWN IN PROGRESS	COMPRESSOR UNLOADING	Chiller unloading before shutdown due to Soft Stop feature.
SHUTDOWN IN PROGRESS	COMPRESSOR DEENERGIZED	Chiller compressor is being commanded to stop. Water pumps are deenergized within one minute.
ICE BUILD	OPERATION COMPLETE	Chiller shutdown from Ice Build operation.

B. TIMING OUT OR TIMED OUT

PRIMARY MESSAGE	SECONDARY MESSAGE	PROBABLE CAUSE/REMEDY
READY TO START IN XX MIN	UNOCCUPIED MODE	Time schedule for PIC is unoccupied (*OCCUPIED ?* parameter on STATUS01 screen is set to NO).Chiller starts only when *OCCUPIED ?* is set to YES.
READY TO START IN XX MIN	REMOTE CONTACTS OPEN	Remote contacts have stopped chiller. Close contacts to start.
READY TO START IN XX MIN	STOP COMMAND IN EFFECT	*CHILLER START/STOP* on STATUS01 screen manually forced to stop. Press RELEASE softkey to start.
READY TO START IN XX MIN	RECYCLE RESTART PENDING	Chiller in recycle mode.
READY TO START	UNOCCUPIED MODE	Time schedule for PIC is in an ″unoccupied″ state. Chiller starts only when in an ″occupied″ state. Check the time and date on the SERVICE menu TIME AND DATE screen.
READY TO START	REMOTE CONTACTS OPEN	Remote contacts have stopped chiller. Close contacts to start.
READY TO START	STOP COMMAND IN EFFECT	*CHILLER START/STOP* on STATUS01 manually forced to stop. Press RELEASE softkey to start.
READY TO START IN XX MIN	REMOTE CONTACTS CLOSED	Chiller timer counting down unit. Ready for start.
READY TO START IN XX MIN	OCCUPIED MODE	Chiller timer counting down unit. Ready for start.
READY TO START	REMOTE CONTACTS CLOSED	Chiller timers complete, unit start will commence.
READY TO START	OCCUPIED MODE	Chiller timers complete, unit start will commence.
STARTUP INHIBITED	LOADSHED IN EFFECT	CCN loadshed module commanding chiller to stop.
READY TO START IN XX MIN	START COMMAND IN EFFECT	*CHILLER START/STOP* on STATUS 01 screen has been manually forced to start. Chiller will start regardless of time schedule or remote contact status.

C. IN RECYCLE SHUTDOWN

PRIMARY MESSAGE	SECONDARY MESSAGE	PROBABLE CAUSE/REMEDY
RECYCLE RESTART PENDING	OCCUPIED MODE	Unit in RECYCLE mode; chilled water temperature is not high enough to start.
RECYCLE RESTART PENDING	REMOTE CONTACT CLOSED	Unit in RECYCLE mode; chilled water temperature is not high enough to start.
RECYCLE RESTART PENDING	START COMMAND IN EFFECT	*CHILLER START/STOP* on STATUS01 screen manually forced to start Chilled water temperature is not high enough to start.
RECYCLE RESTART PENDING	ICE BUILD MODE	Chiller in ICE BUILD mode. Chilled water temperature is satisfied for ice build set point temperature requirements.

Table 8 — LID Primary and Secondary Messages and Custom Alarm/Alert Messages with Troubleshooting Guides (cont)

D. PRE-START ALERTS: These alerts only delay start-up. When alert is corrected, the start-up will continue. No reset is necessary.

PRIMARY MESSAGE	SECONDARY MESSAGE	ALARM MESSAGE/PRIMARY CAUSE	ADDITIONAL CAUSE/REMEDY
PRESTART ALERT	STARTS LIMIT EXCEEDED	STARTS EXCESSIVE Compressor Starts (8 in 12 hours)	Press the RESET softkey if additional start is required. Reassess start-up requirements.
PRESTART ALERT	HIGH MOTOR TEMPERATURE	MTRW [VALUE] exceeded limit of [LIMIT]*. Check motor temperature.	Check motor cooling line for proper operation. Check for excessive starts within a short time span.
PRESTART ALERT	HIGH BEARING TEMPERATURE	MTRB [VALUE] exceeded limit of [LIMIT]*. Check thrust bearing temperature.	Check oil heater for proper operation. Check for low oil level, partially closed oil supply valves, etc. Check sensor accuracy.
PRESTART ALERT	HIGH DISCHARGE TEMP	CMPD [VALUE] exceeded limit of [LIMIT]*. Check discharge temperature.	Check sensor accuracy. Allow discharge temperature to cool. Check for excessive starts.
PRESTART ALERT	LOW REFRIGERANT TEMP	ERT [VALUE] exceeded limit of [LIMIT]*. Check refrigerant temperature.	Check transducer accuracy. Check for low chilled water/brine supply temperature.
PRESTART ALERT	LOW OIL TEMPERATURE	OILT [VALUE] exceeded limit of [LIMIT]*. Check oil temperature.	Check oil heater power, oil heater relay. and oil level.
PRESTART ALERT	LOW LINE VOLTAGE	V__ P [VALUE] exceeded limit of [LIMIT]*. Check voltage suppy.	Check voltage supply. Check voltage transformers. Consult power utility if voltage is low. Check voltage calibration on *LINE VOLTAGE* parameter on STATUS01 screen.
PRESTART ALERT	HIGH LINE VOLTAGE	V__ P [VALUE] exceeded limit of [LIMIT]*. Check voltage supply.	Check voltage supply. Check voltage transformers. Consult power utility if voltage is low. Check voltage calibration on *LINE VOLTAGE* parameter on STATUS01 screen
PRESTART ALERT	HIGH CONDENSER PRESSURE	CRP [VALUE] exceeded limit of [LIMIT]*. Check condenser water and transducer.	Check for high condenser water temperature. Check transducer accuracy.

*[LIMIT] is shown on the LID as temperature, pressure, voltage, etc., predefined or selected by the operator as an override or an alert. [VALUE] is the actual pressure, temperature, voltage, etc., at which the control tripped.

E. NORMAL OR AUTO.-RESTART

PRIMARY MESSAGE	SECONDARY MESSAGE	PROBABLE CAUSE/REMEDY
STARTUP IN PROGRESS	OCCUPIED MODE	Chiller starting. Chiller time schedule is in ″occupied″ state (*OCCUPIED ?* parameter on STATUS01 screen is set to YES).
STARTUP IN PROGRESS	REMOTE CONTACT CLOSED	Chiller starting. Remote contacts are closed.
STARTUP IN PROGRESS	START COMMAND IN EFFECT	Chiller starting. *CHILLER START/STOP* on STATUS01 manually forced to start.
AUTORESTART IN PROGRESS	OCCUPIED MODE	Chiller starting. Time schedule is in ″occupied″ state (*OCCUPIED ?* parameter on STATUS01 screen is set to YES).
AUTORESTART IN PROGRESS	REMOTE CONTACT CLOSED	Chiller starting. Remote contacts are closed.
AUTORESTART IN PROGRESS	START COMMAND IN EFFECT	Chiller starting. *CHILLER START/STOP* on STATUS01 manually forced to start.

Table 8 — LID Primary and Secondary Messages and Custom Alarm/Alert Messages with Troubleshooting Guides (cont)

F. START-UP FAILURES: This is an alarm condition. A manual reset is required to clear.

PRIMARY MESSAGE	SECONDARY MESSAGE	ALARM MESSAGE/PRIMARY CAUSE	ADDITIONAL CAUSE/REMEDY
FAILURE TO START	LOW OIL PRESSURE	OILPD [VALUE] exceeded limit of [LIMIT]*. Check oil pump system.	Check for closed oil supply valves. Check oil filter. Check for low oil temperature. Check transducer accuracy.
FAILURE TO START	OIL PRESS SENSOR FAULT	OILPD [VALUE] exceeded limit of [LIMIT]*. Check oil pressure sensor.	Check for excessive refrigerant in oil sump. Run oil pump manually for 5 minutes. Check transducer calibration. Check wiring. Replace transducer if necessary.
FAILURE TO START	LOW CHILLED WATER FLOW	EVFL Evap Flow Fault: Check water pump/flow switch.	Check wiring to flow switch. Check using CONTROL TEST feature for proper switch operation.
FAILURE TO START	LOW CONDENSER WATER FLOW	CDFL Cond. Flow Fault: Check water pump/flow switch.	Check wiring to flow switch. Check using CONTROL TEST feature for proper switch operation.
FAILURE TO START	STARTER FAULT	STR_FLT Starter Fault: Check Starter for Fault Source.	A starter protective device has faulted. Check starter for ground fault, voltage trip, temperature trip, etc.
FAILURE TO START	STARTER OVERLOAD TRIP	STR_FLT Starter Overload Trip: Check amps calibration/reset overload.	Reset overloads before restart. Fault on starter reset or motor protection system.
FAILURE TO START	LINE VOLTAGE DROPOUT	V_P Single-Cycle Dropout Detected: Check voltage supply. Check power monitor in starter.	Check voltage supply. Check transformers for supply. Check with utility if voltage supply is erratic. Monitor must be installed to confirm consistent, single-cycle dropouts. Check low oil pressure switch.
FAILURE TO START	HIGH CONDENSER PRESSURE	High Condenser Pressure [LIMIT]*. Check switch and water temperature/flow.	Check for proper design condenser flow and temperature. Check condenser approach. Check high pressure switch.
FAILURE TO START	EXCESS ACCELERATION TIME	CA_P Excess Acceleration: Check guide vane closure at start-up.	Check that guide vanes are closed at start-up. Check starter for proper operation. Reduce unit pressure if possible.
FAILURE TO START	STARTER TRANSITION FAULT	RUN_AUX Starter Transition Fault: Check 1CR/1M/Interlock mechanism.	Check starter for proper operation. Run contact has failed to close.
FAILURE TO START	1CR AUX CONTACT FAULT	1CR_AUX Starter Contact Fault: Check 1CR/1M aux. contacts.	Check starter for proper operation. Start contact failed to close.
FAILURE TO START	MOTOR AMPS NOT SENSED	CA_P Motor Amps Not Sensed: Check motor load signal; close disconnect switch.	Check for proper motor amps signal to SMM. Check wiring from SMM to current transformer or starter control or motor protection system. Check main motor circuit breaker for trip.
FAILURE TO START	LOW OIL PRESSURE	Low Oil Pressure [LIMIT]*. Check oil pressure switch/pump. Check SMM voltage output to ICR relay.	The oil pressure differential switch is open when the compressor tried to start. Check the switch for proper operation. Also, check the oil pump interlock (2C aux) in the power panel and the high condenser pressure switch.

*[LIMIT] is shown on the LID as the temperature, pressure, voltage, etc., set point predefined or selected by the operator as an override, alert, or alarm condition. [VALUE] is the actual pressure, temperature, voltage, etc., at which the control tripped.

Table 8 — LID Primary and Secondary Messages and Custom Alarm/Alert Messages with Troubleshooting Guides (cont)

G. COMPRESSOR JUMPSTART AND REFRIGERANT PROTECTION

PRIMARY MESSAGE	SECONDARY MESSAGE	ALARM MESSAGE/PRIMARY CAUSE	ADDITIONAL CAUSE/REMEDY
UNAUTHORIZED OPERATION	UNIT SHOULD BE STOPPED	CA__P Emergency: Compressor running without control authorization.	Compressor is running with more than 10% RLA and the PIC control is trying to shut it down. Turn power off to compressor if unable to stop. Determine cause before repowering.
POTENTIAL FREEZE-UP	EVAP PRESS/TEMP TOO LOW	ERT Emergency: Freeze-up prevention.	Determine cause. If pumping refrigerant out of chiller, stop operation and go over pumpout procedures.
FAILURE TO STOP	DISCONNECT POWER	RUN__AUX Emergency: DISCONNECT POWER.	Starter and run and start contacts are energized while control tried to shut down. Disconnect power to starter.
LOSS OF COMMUNCIATION	WITH STARTER	Loss of Communication with Starter: Check chiller.	Check wiring from PSIO to SMM. Check SMM module troubleshooting procedures.
STARTER CONTACT FAULT	ABNORMAL 1CR OR RUN AUX	1CR__AUX Starter Contact Fault: Check 1CR/1M aux. contacts.	Starter run and start contacts energized while chiller was off. Disconnect power.
POTENTIAL FREEZE UP	COND PRESS/TEMP TOO LOW	CRT Emergency: Freeze-up prevention.	The condenser pressure transducer is reading a pressure that could freeze the water in the condenser tubes. Check for condenser refrigerant leaks, bad transducers, or transferred refrigerant. From the CONTROL TEST screen, place the unit in Pumpdown/Lockout mode to eliminate ALARM if vessel is evacuated.

*[LIMIT] is shown on the LID as the temperature, pressure, voltage, etc., set point predefined or selected by the operator as an override, alert, or alarm condition. [VALUE] is the actual pressure, temperature, voltage, etc., at which the control tripped.

H. NORMAL RUN WITH RESET, TEMPERATURE, OR DEMAND

PRIMARY MESSAGE	SECONDARY MESSAGE	PROBABLE CAUSE/REMEDY
RUNNING — RESET ACTIVE	4-20MA SIGNAL	Reset program active based upon CONFIG screen setup.
RUNNING — RESET ACTIVE	REMOTE SENSOR CONTROL	
RUNNING — RESET ACTIVE	CHW TEMP DIFFERENCE	
RUNNING — TEMP CONTROL	LEAVING CHILLED WATER	Default method of temperature control.
RUNNING — TEMP CONTROL	ENTERING CHILLED WATER	ECW control activated on CONFIG screen.
RUNNING — TEMP CONTROL	TEMPERATURE RAMP LOADING	Ramp loading in effect. Use SERVICE1 table to modify.
RUNNING — DEMAND LIMITED	BY DEMAND RAMP LOADING	Ramp loading in effect. Use SERVICE1 table to modify.
RUNNING — DEMAND LIMITED	BY LOCAL DEMAND SET POINT	Demand limit set point is $<$ actual demand.
RUNNING — DEMAND LIMITED	BY 4-20MA SIGNAL	Demand limit is active based on CONFIG screen setup.
RUNNING — DEMAND LIMITED	BY CCN SIGNAL	
RUNNING — DEMAND LIMITED	BY LOADSHED/REDLINE	
RUNNING — TEMP CONTROL	HOT GAS BYPASS	Hot gas bypass is energized. See Surge Prevention Algorithm in the control section.
RUNNING — DEMAND LIMITED	BY LOCAL SIGNAL	Active demand limit manually overridden or STATUS01 table.
RUNNING — TEMP CONTROL	ICE BUILD MODE	Chiller is running under Ice Build temperature control.

Table 8 — LID Primary and Secondary Messages and Custom Alarm/Alert Messages with Troubleshooting Guides (cont)

I. NORMAL RUN OVERRIDES ACTIVE (ALERTS)

PRIMARY MESSAGE	SECONDARY MESSAGE	ALARM MESSAGE/PRIMARY CAUSE	ADDITIONAL CAUSE/REMEDY
RUN CAPACITY LIMITED	HIGH CONDENSER PRESSURE	CRP [VALUE]* exceeded limit of [LIMIT]*. Condenser pressure override.	See Capacity Overrides, Table 4. Correct operating condition, modify set point, or release override.
RUN CAPACITY LIMITED	HIGH MOTOR TEMPERATURE	MTRW [VALUE]* exceeded limit of [LIMIT]*. Motor temperature override.	
RUN CAPACITY LIMITED	LOW EVAP REFRIG TEMP	ERT [VALUE]* exceeded limit of [LIMIT]*. Check refrigerant charge level.	
RUN CAPACITY LIMITED	HIGH COMPRESSOR LIFT	Surge Prevention Override; lift too high for compressor.	
RUN CAPACITY LIMITED	MANUAL GUIDE VANE TARGET	GV_TRG Run Capacity Limited: Manual guide vane target.	
RUN CAPACITY LIMITED	LOW DISCHARGE SUPERHEAT	No message.	Check for chiller refrigerant overcharge.

*[LIMIT] is shown on the LID as the temperature, pressure, voltage, etc., set point predefined or selected by the operator as an override, alert, or alarm condition. [VALUE] is the actual temperature, pressure, voltage, etc., at which the control tripped.

J. OUT-OF-RANGE SENSOR FAILURES

PRIMARY MESSAGE	SECONDARY MESSAGE	ALARM MESSAGE/PRIMARY CAUSE	ADDITIONAL CAUSE/REMEDY
SENSOR FAULT	LEAVING CHW TEMPERATURE	Sensor Fault: Check leaving CHW sensor.	See sensor test procedure and check sensors for proper operation and wiring.
SENSOR FAULT	ENTERING CHW TEMPERATURE	Sensor Fault: Check entering CHW sensor.	
SENSOR FAULT	CONDENSER PRESSURE	Sensor Fault: Check condenser pressure transducer.	
SENSOR FAULT	EVAPORATOR PRESSURE	Sensor Fault: Check evaporator pressure transducer.	
SENSOR FAULT	BEARING TEMP	Bearing temperature fault.	
SENSOR FAULT	MOTOR WINDING TEMP	Sensor Fault: Check motor temperature sensor.	
SENSOR FAULT	DISCHARGE TEMPERATURE	Sensor Fault: Check discharge temperature sensor.	
SENSOR FAULT	OIL SUMP TEMPERATURE	Sensor Fault: Check oil sump temperature sensor.	
SENSOR FAULT	OIL PRESSURE	Sensor Fault: Check oil pressure transducer.	

K. CHILLER PROTECT LIMIT FAULTS

⚠ WARNING

Excessive numbers of the same fault can lead to severe chiller damage. Seek service expertise.

PRIMARY MESSAGE	SECONDARY MESSAGE	ALARM MESSAGE/PRIMARY CAUSE	ADDITIONAL CAUSE/REMEDY
PROTECTIVE LIMIT	HIGH DISCHARGE TEMP	CMPD [VALUE] exceeded limit of [LIMIT]*. Check discharge temperature.	Check discharge temperature immediately. Check sensor for accuracy; check for proper condenser flow and temperature; check oil reservoir temperature. Check condenser for fouled tubes or air in chiller. Check for proper guide vane actuator operation.
PROTECTIVE LIMIT	LOW REFRIGERANT TEMP	ERT [VALUE] exceeed limit of [LIMIT]*. Check evap pump and flow switch.	Check for proper amount of refrigerant charge; check for proper water flow and temperaures. Check for proper guide vane actuator operation.
PROTECTIVE LIMIT	HIGH MOTOR TEMPERATURE	MTRW [VALUE] exceeded limit of [LIMIT]*. Check motor cooling and solenoid.	Check motor temperature immediately. Check sensor for accuracy. Check for proper condenser flow and temperature. Check motor cooling system for restrictions. Check motor cooling solenoid for proper operation. Check refrigerant filter.
PROTECTIVE LIMIT	HIGH BEARING TEMPERATURE	MTRB [VALUE] exceeded limit of [LIMIT]*. Check oil cooling control.	Check for throttled oil supply isolation valves. Valves should be wide open. Check oil cooler thermal expansion valve. Check sensor accuracy. Check journal and thrust bearings. Check refrigerant filter. Check for excessive oil sump level.

*[LIMIT] is shown on the LID as the temperature, pressure, voltage, etc., set point predefined or selected by the operator as an override, alert, or alarm condition. [VALUE] is the actual temperature, pressure, voltage, etc., at which the control tripped. [OPEN] indicates that an input circuit is open.

Table 8 — LID Primary and Secondary Messages and Custom Alarm/Alert Messages with Troubleshooting Guides (cont)

K. CHILLER PROTECT LIMIT FAULTS (cont)

> **⚠ WARNING**
>
> Excessive numbers of the same fault can lead to severe chiller damage. Seek service expertise.

PRIMARY MESSAGE	SECONDARY MESSAGE	ALARM MESSAGE/PRIMARY CAUSE	ADDITIONAL CAUSE/REMEDY
PROTECTIVE LIMIT	LOW OIL PRESSURE	OILPD [VALUE] exceeded limit of [LIMIT]*. Check oil pump and transducer.	Check power to oil pump and oil level. Check for dirty filters or oil foaming at start-up. Check for thermal overload cutout. Reduce ramp load rate if foaming noted. NOTE: This alarm is not related to pressure switch problems.
		Low Oil Pressure [OPEN]*. Check oil pressure switch/pump and 2C aux.	Check the oil pressure switch for proper operation. Check oil pump for proper pressure. Check for excessive refrigerant in oil system.
PROTECTIVE LIMIT	NO MOTOR CURRENT	CA__P Loss of Motor Current: Check sensor.	Check wiring: Check torque setting on solid-state starter. Check for main circuit breaker trip. Check power supply to PSIO module.
PROTECTIVE LIMIT	POWER LOSS	V__P Power Loss: Check voltage supply.	Check 24-VAC input on the SMM. Check transformers to SMM. Check power to PSIO module. Check distribution bus. Consult power company.
PROTECTIVE LIMIT	POWER LOSS	Line voltage dropout single cycle.	
PROTECTIVE LIMIT	LOW LINE VOLTAGE	V__P [VALUE] exceeded limit of [LIMIT]*. Check voltage supply.	
PROTECTIVE LIMIT	HIGH LINE VOLTAGE	V__P [VALUE] exceeded limit of [LIMIT]*. Check voltage supply.	
PROTECTIVE LIMIT	LOW CHILLED WATER FLOW	EVFL Flow Fault: Check evap pump/flow switch.	Perform pumps test on CONTROL TEST screen and verify proper switch operation. Check all water valves and pump operation.
PROTECTIVE LIMIT	LOW CONDENSER WATER FLOW	CDFL Flow Fault: Check cond pump/flow switch.	
PROTECTIVE LIMIT	HIGH CONDENSER PRESSURE	High Cond Pressure [OPEN]*. Check switch, oil pressure contact, and water temp/flow.	Check the high-pressure switch. Check for proper condenser pressures and condenser waterflow. Check for fouled tubes. Check the 2C aux. contact and the oil pressure switch in the power panel. This alarm is not caused by the transducer.
		High Cond Pressure [VALUE]: Check switch, water flow, and transducer.	Check water flow in condenser. Check for fouled tubes. Transducer should be checked for accuracy. This alarm is not caused by the high pressure switch.
PROTECTIVE LIMIT	1CR AUX CONTACT FAULT	1CR__AUX Starter Contact Fault: Check 1CR/1M aux contacts.	1CR auxiliary contact opened while chiller was running. Check starter for proper operation.
PROTECTIVE LIMIT	RUN AUX CONTACT FAULT	RUN__AUX Starter Contact Fault: Check 1CR/1M aux contacts.	Run auxiliary contact was opened while chiller was running. Check starter for proper operation.
PROTECTIVE LIMIT	CCN OVERRIDE STOP	CHIL__S__S CCN Override Stop while in LOCAL run mode.	CCN has signaled chiller to stop. Reset and restart when ready. If the signal was sent by the LID, select RELEASE softkey for *CHILLER START/STOP* parameter on STATUS01 screen.
PROTECTIVE LIMIT	SPARE SAFETY DEVICE	SRP__PL Spare Safety Fault: Check contacts.	Spare safety input has tripped or factory-installed jumper is not present.
PROTECTIVE LIMIT	EXCESSIVE MOTOR AMPS	CA__P [VALUE] exceeded limit of [LIMIT]*. High Amps; Check guide vane drive.	Check motor current for proper calibration. Check guide vane drive and actuator for proper operation.
PROTECTIVE LIMIT	EXCESSIVE COMPR SURGE	Compressor Surge: Check condenser water temp and flow.	Check condenser flow and temperatures. Check configuration of surge protection.
PROTECTIVE LIMIT	STARTER FAULT	STR__FLT Starter Fault: Check starter for fault source.	Check starter for possible ground fault, reverse rotation, voltage trip, etc.
PROTECTIVE LIMIT	STARTER OVERLOAD TRIP	STR__FLT Starter Overload Trip: Check amps calibration/reset overload.	Reset overloads and reset alarm. Check motor current calibration or overload calibration (do not field-calibrate overloads).
PROTECTIVE LIMIT	TRANSDUCER VOLTAGE FAULT	V__REF [VALUE] exceeded limit of [LIMIT]*. Check transducer power supply.	Check transformer power (5 vdc) supply to transducers. Power must be 4.5 to 5.5 vdc.
PROTECTIVE LIMIT	LOW DISCHARGE SUPERHEAT	Check for oil in refrigerant or check for refrigerant overcharge.	Required and actual superheat can be viewed on the MAINT02 screen during operation. Check discharge sensor, condenser pressure transducer, condenser flow, guide vane operation, refrigerant charge (amount and oil content).
PROTECTIVE LIMIT	DIFFUSER POSITION FAULT	Check guide vane and diffuser actuator	Either the guide vane or diffuser actuator may be malfunctioning.

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Table 8 — LID Primary and Secondary Messages and Custom Alarm/Alert Messages with Troubleshooting Guides (cont)

L. CHILLER ALERTS

PRIMARY MESSAGE	SECONDARY MESSAGE	ALARM MESSAGE/PRIMARY CAUSE	ADDITIONAL CAUSE/REMEDY
RECYCLE ALERT	HIGH AMPS AT SHUTDOWN	High Amps at Recycle: Check guide vane drive.	Check that guide vanes are closing. Check that motor amps correction calibration is correct. Check actuator for proper operation.
SENSOR FAULT ALERT	LEAVING COND WATER TEMP	Sensor Fault: Check leaving condenser water sensor.	Check sensor. See sensor test procedure.
SENSOR FAULT ALERT	ENTERING COND WATER TEMP	Sensor Fault: Check entering condenser water sensor.	
LOW OIL PRESSURE ALERT	CHECK OIL FILTER	Low Oil Pressure Alert: Check oil	Check oil filter. Check for improper oil level or temperature.
AUTORESTART PENDING	POWER LOSS	V_P Power Loss: Check voltage supply.	Check power supply if there are excessive compressor starts occurring.
AUTORESTART PENDING	LOW LINE VOLTAGE	V_P [VALUE] exceeded limit of [LIMIT]*. Check voltage supply.	
AUTORESTART PENDING	HIGH LINE VOLTAGE	V_P [VALUE] exceeded limit of [LIMIT*]. Check voltage supply.	
SENSOR ALERT	HIGH DISCHARGE TEMP	CMPD [VALUE] exceeded limit of [LIMIT]*. Check discharge temperature.	Discharge temperature exceeded the alert threshold. Check entering condenser water temperature.
SENSOR ALERT	HIGH BEARING TEMPERATURE	MTRB [VALUE] exceeded limit of [LIMIT]*. Check thrust bearing temperature.	Thrust bearing temperature exceeded the alert threshold. Check for closed valves, improper oil level or temperatures.
CONDENSER PRESSURE ALERT	PUMP RELAY ENERGIZED	CRP High Condenser Pressure [LIMIT]*. Pump energized to reduce pressure.	Check ambient conditions. Check condenser pressure for accuracy.
RECYCLE ALERT	EXCESSIVE RECYCLE STARTS	Excessive recycle starts.	The chiller load is too small to keep the chiller on line and there have been more than 5 restarts in 4 hours. Increase chiller load, adjust hot gas bypass, increase *RECYCLE RESTART DELTA T* on SERVICE1 screen.

*[LIMIT] is shown on the LID as the temperature, pressure, voltage, etc., set point predefined or selected by the operator as an override, alert, or alarm condition. [VALUE] is the actual temperature, pressure, voltage, etc., at which the control tripped.

M. SPARE SENSOR ALERT MESSAGES

PRIMARY MESSAGE	SECONDARY MESSAGE	ALARM MESSAGE/PRIMARY CAUSE	ADDITIONAL CAUSE/REMEDY
SPARE SENSOR ALERT	COMMON CHWS SENSOR	Sensor Fault: Check common CHWS sensor.	Check alert temperature set points on SERVICE2 LID screen. Check sensor for accuracy if reading is not accurate.
SPARE SENSOR ALERT	COMMON CHWR SENSOR	Sensor Fault: Check common CHWR sensor.	
SPARE SENSOR ALERT	REMOTE RESET SENSOR	Sensor Fault: Check remote reset temperature sensor.	
SPARE SENSOR ALERT	TEMP SENSOR — SPARE 1	Sensor Fault: Check temperature sensor — Spare 1.	
SPARE SENSOR ALERT	TEMP SENSOR — SPARE 2	Sensor Fault: Check temperature sensor — Spare 2.	
SPARE SENSOR ALERT	TEMP SENSOR — SPARE 3	Sensor Fault: Check temperature sensor — Spare 3.	
SPARE SENSOR ALERT	TEMP SENSOR — SPARE 4	Sensor Fault: Check temperature sensor — Spare 4.	
SPARE SENSOR ALERT	TEMP SENSOR — SPARE 5	Sensor Fault: Check temperature sensor — Spare 5.	
SPARE SENSOR ALERT	TEMP SENSOR — SPARE 6	Sensor Fault: Check temperature sensor — Spare 6.	
SPARE SENSOR ALERT	TEMP SENSOR — SPARE 7	Sensor Fault: Check temperature sensor — Spare 7.	
SPARE SENSOR ALERT	TEMP SENSOR — SPARE 8	Sensor Fault: Check temperature sensor — Spare 8.	
SPARE SENSOR ALERT	TEMP SENSOR — SPARE 9	Sensor Fault: Check temperature sensor — Spare 9.	

Table 8 — LID Primary and Secondary Messages and Custom Alarm/Alert Messages with Troubleshooting Guides (cont)

N. OTHER PROBLEMS/MALFUNCTIONS

DESCRIPTION/MALFUNCTION	PROBABLE CAUSE/REMEDY
Chilled Water/Brine Temperature Too High (Chiller Running)	Chilled water set point set too high. Access set point on LID and verify. Capacity override or excessive cooling load (chiller at design capacity). Check LID status messages. Check for outside air infiltration into conditioned space. Condenser temperature too high. Check for proper flow, examine cooling tower operation, check for air or water leaks, check for fouled tubes. Refrigerant level low. Check for leaks, add refrigerant, and trim charge. Liquid bypass in waterbox. Examine division plates and gaskets for leaks. Guide vanes fail to open. Use Control Test feature to check operation. Chilled water control point too high. Access CONTROL ALGORITHM STATUS screens and check chilled water control operation. Guide vanes fail to open fully. Be sure that the guide vane target is released. Check guide vane linkage. Check limit switch in actuator. Check that sensor is in the proper terminals.
Chilled Water/Brine Temperature Too Low (Chiller Running)	Chilled water set point set too low. Access set point on LID and verify. Chilled water control point too low. Access control algorithm status and check chilled water control for proper resets. High discharge temperature keeps guide vanes open. Guide vanes fail to close. Be sure that guide vane target is released. Check chilled water sensor accuracy. Check guide vane linkage. Check actuator operation.
Chilled Water Temperature Fluctuates. Vanes Hunt	Deadband too narrow. Configure LID for a larger deadband. See SERVICE1 screen. Proportional bands too narrow. Either *PROPORTIONAL INC BAND* or *PROPORTIONAL DEC BAND* should be increased (SERVICE3 screen). Loose guide vane drive. Adjust chain drive. Defective vane actuator. Check through Control Test. Defective temperature sensor. Check sensor accuracy.
Low Oil Sump Temperature While Running (Less than 100 F [38 C])	Check for proper oil level (not enough oil). Check for proper refrigerant level (too much refrigerant).
At Power Up, Default Screen Does Not Appear, "Tables Loading" Message Continually Appears	Check for proper communications wiring on PSIO module. Check that the COMM1 communications wires from the LID are terminated to the COMM1 PSIO connection.
SMM Communications Failure	Check that PSIO communication plugs are connected correctly. Check SMM communication plug. Check for proper SMM power supply. See Control Modules section on page 79.
High Oil Temperature While Running	Check for proper oil level (too much oil). Check that TXV valve is operating properly.
Blank LID Screen	Increase contrast potentiometer. See Fig. 39. Check red LED on LID for proper operation, (power supply).
"Communications Failure" Highlighted Message At Bottom of LID Screen	LID is not properly addressed to the PSIO. Check ATTACH TO NETWORK DEVICE screen to see if *LOCAL DEVICE* parameter is set to local PSIO address. Check LEDs on PSIO. Is red LED operating properly? Are green LEDs blinking? See control module troubleshooting section.
Controls Test Disabled	Press the "Stop" pushbutton. The PIC must be in the OFF mode for the controls test to operate. Clear all alarms. Check line voltage percent on STATUS01 screen. The percent must be within 90% to 110%. Check voltage input to SMM.
Vanes Will Not Open In Control Test	Low pressure alarm is active. Put chiller into pumpdown mode or equalize pressure. Check guide vane actuator wiring.
Oil Pump Does Not Run	Check oil pump voltage supply. Cooler vessel pressure is under vacuum. Pressurize the vessel. Check temperature overload cutout switch.
LID Default Screen Does Not Update	This is normal operation when an alarm condition exists. The LID screen freezes the moment the alarm occurs to aid in troubleshooting. Go to the STATUS01 screen to view current information.
Chiller Does Not Stop When STOP Button is Pressed	The STOP button wiring connector at the LID module is not properly connected, or the chiller is in the Soft Stop mode and the guide vanes are closing.
LID Screen is Dark	The backlight turns off after 15 minutes of non-use. Press any softkey to restore light.

Table 9A — Thermistor Temperature (F) vs Resistance/Voltage Drop

TEMPERATURE (F)	VOLTAGE DROP (V)	RESISTANCE (Ohms)
−25	4.821	98,010
−24	4.818	94,707
−23	4.814	91,522
−22	4.806	88,449
−21	4.800	85,486
−20	4.793	82,627
−19	4.786	79,871
−18	4.779	77,212
−17	4.772	74,648
−16	4.764	72,175
−15	4.757	69,790
−14	4.749	67,490
−13	4.740	65,272
−12	4.734	63,133
−11	4.724	61,070
−10	4.715	59,081
−9	4.705	57,162
−8	4.696	55,311
−7	4.688	53,526
−6	4.676	51,804
−5	4.666	50,143
−4	4.657	48,541
−3	4.648	46,996
−2	4.636	45,505
−1	4.624	44,066
0	4.613	42,679
1	4.602	41,339
2	4.592	40,047
3	4.579	38,800
4	4.567	37,596
5	4.554	36,435
6	4.540	35,313
7	4.527	34,231
8	4.514	33,185
9	4.501	32,176
10	4.487	31,202
11	4.472	30,260
12	4.457	29,351
13	4.442	28,473
14	4.427	27,624
15	4.413	26,804
16	4.397	26,011
17	4.381	25,245
18	4.366	24,505
19	4.348	23,789
20	4.330	23,096
21	4.313	22,427
22	4.295	21,779
23	4.278	21,153
24	4.258	20,547
25	4.241	19,960
26	4.223	19,393
27	4.202	18,843
28	4.184	18,311
29	4.165	17,796
30	4.145	17,297
31	4.125	16,814
32	4.103	16,346
33	4.082	15,892
34	4.059	15,453
35	4.037	15,027
36	4.017	14,614
37	3.994	14,214
38	3.968	13,826
39	3.948	13,449
40	3.927	13,084
41	3.902	12,730
42	3.878	12,387
43	3.854	12,053
44	3.828	11,730
45	3.805	11,416
46	3.781	11,112
47	3.757	10,816
48	3.729	10,529
49	3.705	10,250
50	3.679	9,979
51	3.653	9,717
52	3.627	9,461
53	3.600	9,213
54	3.575	8,973
55	3.547	8,739
56	3.520	8,511
57	3.493	8,291
58	3.464	8,076
59	3.437	7,868
60	3.409	7,665
61	3.382	7,468
62	3.353	7,277
63	3.323	7,091
64	3.295	6,911
65	3.267	6,735
66	3.238	6,564
67	3.210	6,399
68	3.181	6,238
69	3.152	6,081
70	3.123	5,929
71	3.093	5,781
72	3.064	5,637
73	3.034	5,497
74	3.005	5,361
75	2.977	5,229
76	2.947	5,101
77	2.917	4,976
78	2.884	4,855
79	2.857	4,737
80	2.827	4,622
81	2.797	4,511
82	2.766	4,403
83	2.738	4,298
84	2.708	4,196
85	2.679	4,096
86	2.650	4,000
87	2.622	3,906
88	2.593	3,814
89	2.563	3,726
90	2.533	3,640
91	2.505	3,556
92	2.476	3,474
93	2.447	3,395
94	2.417	3,318
95	2.388	3,243
96	2.360	3,170
97	2.332	3,099
98	2.305	3,031
99	2.277	2,964
100	2.251	2,898
101	2.217	2,835
102	2.189	2,773
103	2.162	2,713
104	2.136	2,655
105	2.107	2,597
106	2.080	2,542
107	2.053	2,488
108	2.028	2,436
109	2.001	2,385
110	1.973	2,335
111	1.946	2,286
112	1.919	2,239
113	1.897	2,192
114	1.870	2,147
115	1.846	2,103
116	1.822	2,060
117	1.792	2,018
118	1.771	1,977
119	1.748	1,937
120	1.724	1,898
121	1.702	1,860
122	1.676	1,822
123	1.653	1,786
124	1.630	1,750
125	1.607	1,715
126	1.585	1,680
127	1.562	1,647
128	1.538	1,614
129	1.517	1,582
130	1.496	1,550
131	1.474	1,519
132	1.453	1,489
133	1.431	1,459
134	1.408	1,430
135	1.389	1,401
136	1.369	1,373
137	1.348	1,345
138	1.327	1,318
139	1.308	1,291
140	1.291	1,265
141	1.289	1,240
142	1.269	1,214
143	1.250	1,190
144	1.230	1,165
145	1.211	1,141
146	1.192	1,118
147	1.173	1,095
148	1.155	1,072
149	1.136	1,050
150	1.118	1,029
151	1.100	1,007
152	1.082	986
153	1.064	965
154	1.047	945
155	1.029	925
156	1.012	906
157	0.995	887
158	0.978	868
159	0.962	850
160	0.945	832
161	0.929	815
162	0.914	798
163	0.898	782
164	0.883	765
165	0.868	750
166	0.853	734
167	0.838	719
168	0.824	705
169	0.810	690
170	0.797	677
171	0.783	663
172	0.770	650
173	0.758	638
174	0.745	626
175	0.734	614
176	0.722	602
177	0.710	591
178	0.700	581
179	0.689	570
180	0.678	561
181	0.668	551
182	0.659	542
183	0.649	533
184	0.640	524
185	0.632	516
186	0.623	508
187	0.615	501
188	0.607	494
189	0.600	487
190	0.592	480
191	0.585	473
192	0.579	467
193	0.572	461
194	0.566	456
195	0.560	450
196	0.554	445
197	0.548	439
198	0.542	434
199	0.537	429
200	0.531	424
201	0.526	419
202	0.520	415
203	0.515	410
204	0.510	405
205	0.505	401
206	0.499	396
207	0.494	391
208	0.488	386
209	0.483	382
210	0.477	377
211	0.471	372
212	0.465	367
213	0.459	361
214	0.453	356
215	0.446	350
216	0.439	344
217	0.432	338
218	0.425	332
219	0.417	325
220	0.409	318
221	0.401	311
222	0.393	304
223	0.384	297
224	0.375	289
225	0.366	282

Table 9B — Thermistor Temperature (C) vs Resistance/Voltage Drop

TEMPERATURE (C)	VOLTAGE DROP (V)	RESISTANCE (Ohms)
−40	4.896	168 230
−39	4.889	157 440
−38	4.882	147 410
−37	4.874	138 090
−36	4.866	129 410
−35	4.857	121 330
−34	4.848	113 810
−33	4.838	106 880
−32	4.828	100 260
−31	4.817	94 165
−30	4.806	88 480
−29	4.794	83 170
−28	4.782	78 125
−27	4.769	73 580
−26	4.755	69 250
−25	4.740	65 205
−24	4.725	61 420
−23	4.710	57 875
−22	4.693	54 555
−21	4.676	51 450
−20	4.657	48 536
−19	4.639	45 807
−18	4.619	43 247
−17	4.598	40 845
−16	4.577	38 592
−15	4.554	38 476
−14	4.531	34 489
−13	4.507	32 621
−12	4.482	30 866
−11	4.456	29 216
−10	4.428	27 633
−9	4.400	26 202
−8	4.371	24 827
−7	4.341	23 532
−6	4.310	22 313
−5	4.278	21 163
−4	4.245	20 079
−3	4.211	19 058
−2	4.176	18 094
−1	4.140	17 184
0	4.103	16 325
1	4.065	15 515
2	4.026	14 749
3	3.986	14 026
4	3.945	13 342
5	3.903	12 696
6	3.860	12 085
7	3.816	11 506
8	3.771	10 959
9	3.726	10 441
10	3.680	9 949
11	3.633	9 485
12	3.585	9 044
13	3.537	8 627
14	3.487	8 231
15	3.438	7 855
16	3.387	7 499
17	3.337	7 161
18	3.285	6 840
19	3.234	6 536
20	3.181	6 246
21	3.129	5 971
22	3.076	5 710
23	3.023	5 461
24	2.970	5 225
25	2.917	5 000
26	2.864	4 786
27	2.810	4 583
28	2.757	4 389
29	2.704	4 204
30	2.651	4 028
31	2.598	3 861
32	2.545	3 701
33	2.493	3 549
34	2.441	3 404
35	2.389	3 266
36	2.337	3 134
37	2.286	3 008
38	2.236	2 888
39	2.186	2 773
40	2.137	2 663
41	2.087	2 559
42	2.039	2 459
43	1.991	2 363
44	1.944	2 272
45	1.898	2 184
46	1.852	2 101
47	1.807	2 021
48	1.763	1 944
49	1.719	1 871
50	1.677	1 801
51	1.635	1 734
52	1.594	1 670
53	1.553	1 609
54	1.513	1 550
55	1.474	1 493
56	1.436	1 439
57	1.399	1 387
58	1.363	1 337
59	1.327	1 290
60	1.291	1 244
61	1.258	1 200
62	1.225	1 158
63	1.192	1 118
64	1.160	1 079
65	1.129	1 041
66	1.099	1 006
67	1.069	971
68	1.040	938
69	1.012	906
70	0.984	876
71	0.949	836
72	0.920	805
73	0.892	775
74	0.865	747
75	0.838	719
76	0.813	693
77	0.789	669
78	0.765	645
79	0.743	623
80	0.722	602
81	0.702	583
82	0.683	564
83	0.665	547
84	0.648	531
85	0.632	516
86	0.617	502
87	0.603	489
88	0.590	477
89	0.577	466
90	0.566	456
91	0.555	446
92	0.545	436
93	0.535	427
94	0.525	419
95	0.515	410
96	0.506	402
97	0.496	393
98	0.486	385
99	0.476	376
100	0.466	367
101	0.454	357
102	0.442	346
103	0.429	335
104	0.416	324
105	0.401	312
106	0.386	299
107	0.370	285

Control Modules

⚠ CAUTION

Turn controller power off before servicing controls. This ensures safety and prevents damage to the controller.

The Processor/Sensor Input/Output module (PSIO), 8-input (Options) modules, Starter Management Module (SMM), and the Local Interface Device (LID) module perform continuous diagnostic evaluations of the hardware to determine its condition. Proper operation of all modules is indicated by LEDs (light-emitting diodes) located on the side of the LID and on the top horizontal surface of the PSIO, SMM, and 8-input modules.

RED LED — If the red LED:

- blinks continuously at a 2-second interval, the module is operating properly
- is lit continuously, there is a problem that requires replacing the module
- is off continuously, the power should be checked
- blinks 3 times per second, a software error has been discovered and the module must be replaced

If there is no input power, check the fuses and circuit breaker. If the fuse is good, check for a shorted secondary of the ransformer or, if power is present to the module, replace the module.

GREEN LEDs — There are one or 2 green LEDs on the PSIO. These LEDs indicate the communication status between different parts of the controller and the network modules as follows:

Green LED closest to communications connection — Communication with SMM and 8-input module; must blink continuously (COMM3).

Other Green LED — Communication with LID; must blink every 3 to 5 seconds (COMM3).

8-Input Modules and SMM

Green LED — Communication with PSIO module; blinks continuously.

Notes on Module Operation

1. The chiller operator monitors and modifies configurations in the microprocessor by using the 4 softkeys and the LID. Communication between the LID and the PSIO is accomplished through the CCN bus. The communication between the PSIO, SMM, and both 8-input modules is accomplished through the sensor bus, which is a 3-wire cable.

   On sensor bus terminal strips, Terminal 1 of PSIO module is connected to Terminal 1 of each of the other modules. Terminals 2 and 3 are connected in the same manner. See Fig. 37-41. If a Terminal 2 wire is connected to Terminal 1, the system does not work.

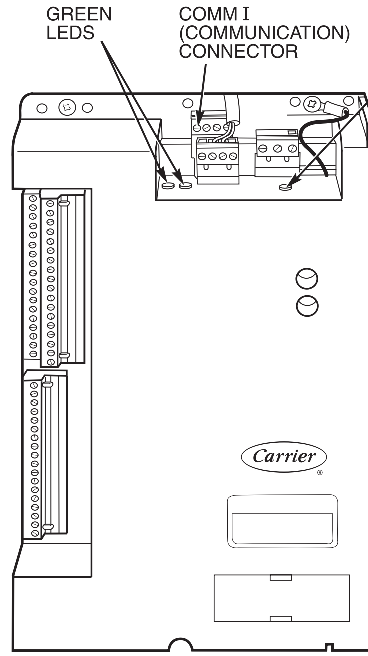
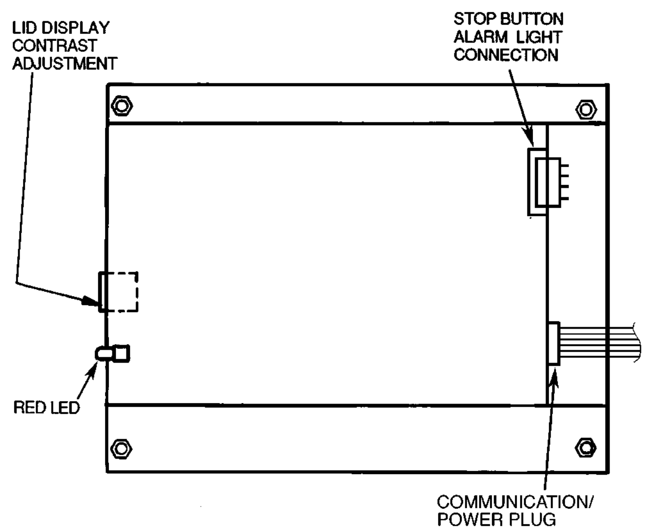


Fig. 37 — PSIO Module Address Selector Switch Locations and LED Locations



NOTE: Addresses are only changed through the LID screen or CCN.

Fig. 38 — LID Module (Rear View) and LED Locations

2. If a green LED is on continuously, check the communication wiring. If a green LED is off, check the red LED operation. If the red LED is normal, check the module address switches (Fig. 37-41). Proper addresses are:

MODULE	ADDRESS	
	S1	S2
SMM (Starter Management Module)	3	2
8-input Options Module 1	6	4
8-input Options Module 2	7	2

   If all modules indicate a communications failure, check the communications plug on the PSIO module for proper seating. Also check the wiring (CCN bus — 1:red, 2:wht, 3:blk; Sensor bus — 1:red, 2:blk, 3:clr/wht). If the connections are good and the condition persists, replace the PSIO module.

   If only one 8-input module or SMM indicates a communication failure, check the communications plug on that module. If the connection is good and the condition persists, replace the module.

   All system operating intelligence resides in the PSIO module. Some safety shutdown logic resides in the SMM in case communications are lost between the 2 modules. The PSIO monitors conditions using input ports on the PSIO, the SMM, and the 8-input modules. Outputs are controlled by the PSIO and SMM as well.

3. Power is supplied to the modules within the control panel via 21-vac power sources.

   The transformers are located within the power panel, with the exception of the SMM, which operates from a 24-vac power source and has its own 24-vac transformer located in the starter.

   In the power panel, T1 supplies power to the LID, the PSIO, and the 5-vac power supply for the transducers. The other 21-vac transformer, T4, supplies power to both 8-input modules (if present). T4 can supply power to two modules. If additional modules are added, another power supply will be required.

   Power is connected to Terminals 1 and 2 of the power input connection on each module.

Processor/Sensor Input/Output Module (PSIO) (Fig. 39)

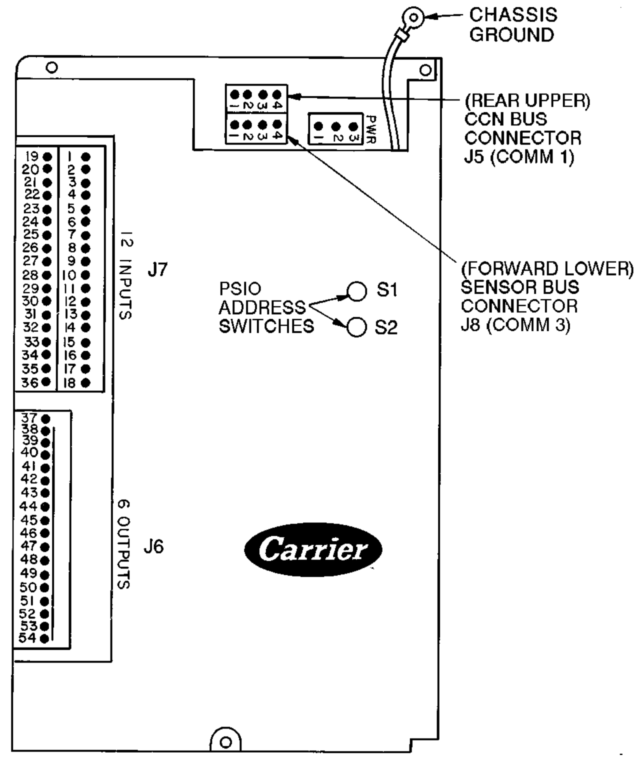
INPUTS — Each input channel has 3 terminals, only 2 of which are normally used depending on the application of the chiller. Refer to individual chiller wiring diagrams for the correct terminal numbers for your application.

OUTPUTS — Output is 20 vdc. There are 3 terminals per output, only 2 of which are used depending on the specific application. Refer to the chiller wiring diagram. for your specific application for the correct terminal numbers.

Starter Management Module (SMM) (Fig. 40)

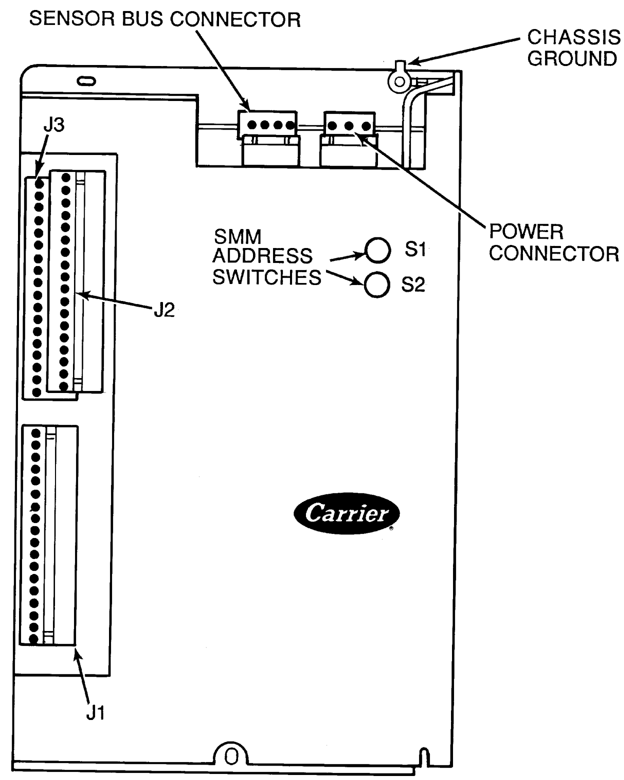
INPUTS — Inputs on strips J2 and J3 are a mix of analog and discrete (on/off) inputs. The specific application of the chiller determines which terminals are used. Refer to the individual chiller wiring diagram for the correct terminal numbers for your application.

OUTPUTS — Outputs are 24 vdc and wired to strip J1. There are 2 terminals per output.



NOTE: Address switches on this module can be at any position. Addresses can only be changed through the LID or CCN.

Fig. 39 — Processor/Sensor Input/Output (PSIO) Module



NOTE: SSM address switches should be set as follows: S1 set at 3; S2 set at 2.

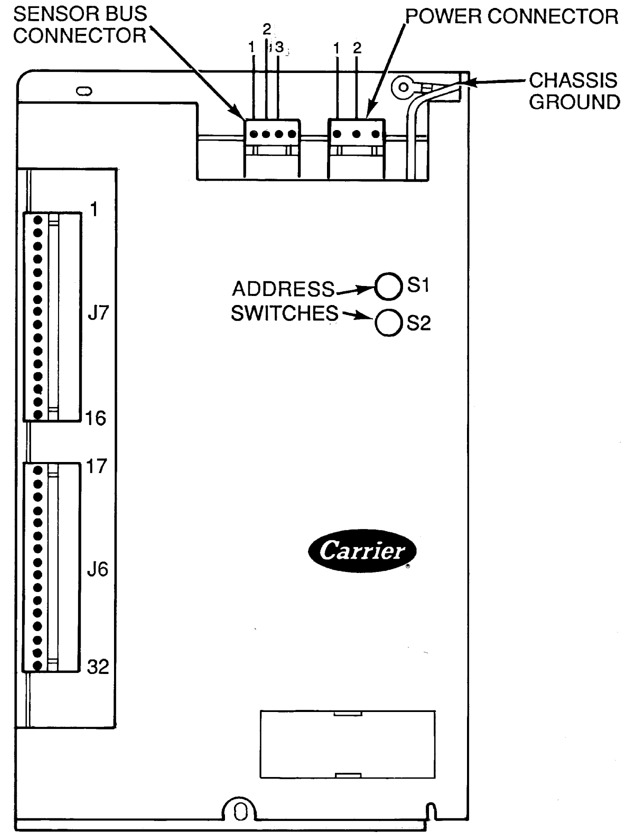
Fig. 40 — Starter Management Module (SSM)

Options Modules (8-Input) — The options modules are optional additions to the PIC and are used to add temperature reset inputs, spare sensor inputs, and demand limit inputs. Each option module contains 8 inputs; each input is meant for a specific duty. See the wiring diagram for your application for exact module wire terminations. Inputs for each of the options modules available include the following:

OPTION MODULE 1
4 to 20 mA Auto. Demand Reset
4 to 20 mA Auto. Chilled Water Reset
Common Chilled Water Supply Temperature
Common Chilled Water Return Temperature
Remote Temperature Reset Sensor
Spare Temperature 1
Spare Temperature 2
Spare Temperature 3

OPTION MODULE 2
4 to 20 mA Spare 1
4 to 20 mA Spare 2
Spare Temperature 4
Spare Temperature 5
Spare Temperature 6
Spare Temperature 7
Spare Temperature 8
Spare Temperature 9

Terminal block connections are provided on the options modules. All sensor inputs are field wired and installed. Options module number 1 can be factory or field installed. Options module 2 is shipped separately and must be field installed. For installation, refer to your specific unit or field wiring diagrams. Be sure to address the module with the proper module number (Fig. 41) and to configure the chiller for each feature that is used.



SWITCH SETTING	OPTION MODULE 1	OPTION MODULE 2
S1	6	7
S2	4	2

Fig. 41 — Options Module

Replacing Defective Processor Modules — The module replacement part number is printed on a small label on the front of the PSIO module. The chiller model and serial numbers are printed on the chiller nameplate located on an exterior corner post. The proper software is factory-installed by Carrier in the replacement module. When ordering a replacement processor module (PSIO), specify the complete replacement part number, full chiller model number, and chiller serial number. The installer must configure the new module to the original chiller data. Follow the procedures described in the Set Up Chiller Control Configuration section on page 49.

> **⚠ CAUTION**
>
> Electrical shock can cause personal injury. Disconnect all electrical power before servicing.

INSTALLATION

1. Verify the existing PSIO module is defective by using the procedure described in the Troubleshooting Guide section, page 66, and the Control Modules section, page 79. Do not select the ATTACH TO NETWORK DEVICE table if the LID indicates a communication failure.
2. Data regarding the PSIO configuration should have been recorded and saved. This data must be reconfigured into the new PSIO using the LID. If this data is not available, follow the procedures described in the Set Up Chiller Control Configuration section.

   If a CCN Building Supervisor or Service Tool is available, the module configuration should have already been uploaded into memory. When the new module is installed, the configuration can be downloaded from the computer.

   Any communication wires from other chillers or CCN modules should be disconnected to prevent the new PSIO module from uploading incorrect run hours into memory.
3. To install this module, record values for the *TOTAL COMPRESSOR STARTS* and the *COMPRESSOR ONTIME* from the STATUS01 screen on the LID.
4. Power off the controls.
5. Remove the old PSIO. DO NOT install the new PSIO at this time.
6. Turn on the control power. When the LID screen displays, press the MENU softkey, then press the SERVICE softkey. Enter your password, if applicable. Highlight the ATTACH TO NETWORK DEVICE line. Press the SELECT softkey. Now, press the ATTACH softkey. The LID displays “UPLOADING TABLES, PLEASE WAIT” and then displays “COMMUNICATIONS FAILURE.” Press the EXIT softkey.

7. Turn the control power off.
8. Install the new PSIO module. Turn the control power back on.
9. The LID now automatically uploads the new PSIO module.
10. Access the STATUS01 table and highlight the *TOTAL COMPRESSOR STARTS* parameter. Press the SELECT softkey. Increase or decrease the value to match the starts value recorded in Step 3. Press the ENTER softkey when you reach the correct value. Now, move the highlight bar to the *COMPRESSOR ONTIME* parameter. Press the SELECT softkey. Increase or decrease the run hours value to match the value recorded in Step 2. Press the ENTER softkey when the correct value is reached.
11. Complete the PSIO installation. Following the instructions in the Input Service Configurations section, page 50, input all the proper configurations such as the time, date, etc. Recalibrate the motor amps and check the pressure transducer calibrations. PSIO installation is now complete.

Solid-State Starters — For troubleshooting information pertaining to the Wye-Delta starter, refer to the IQ-1000 II Motor Protection System User's Manual supplied by the starter vendor.

Troubleshooting informtion pertaining to the Benshaw, Inc., solid-state starter may be found in the following paragraphs and in the Carrier REDISTART Micro Instruction Manual supplied by the starter vendor.

Attempt to solve the problem by using the following preliminary checks before consulting the troubleshooting tables found in the Benshaw manual.

⚠ WARNING

1. Motor terminals or starter output lugs or wire should not be touched without disconnecting the incoming power supply. The silicon control rectifiers (SCRs) although technically turned off still have AC mains potential on the output of the starter.
2. Power is present on all yellow wiring throughout the system even though the main circuit breaker in the unit is off.

With power off:

- Inspect for physical damage and signs of arcing, overheating, etc.
- Verify the wiring to the starter is correct.
- Verify all connections in the starter are tight.
- Check the control transformer fuses.

TESTING SILICON CONTROL RECTIFIERS IN THE BENSHAW, INC., SOLID-STATE STARTERS — If an SCR is suspected of being defective, use the following procedure as part of a general troubleshooting guide.

1. Verify power is applied.
2. Verify the state of each SCR light-emitting diode (LED) on the micropower card.

   NOTE: All LEDs should be lit. If any red or green side of these LEDs is not lit, the line voltage is not present or one or more SCRs has failed.
3. Check incoming power. If voltage is not present check the incoming line. If voltage is present, proceed to Steps 4 through 11.

   NOTE: If after completing Steps 4 - 11 all measurements are within specified limits, the SCRs are functioning normally. If after completing Steps 4 - 11 resistance measurements are outside the specified limits, the motor leads on the starter power lugs T1 through T6 should be removed and the steps repeated. This will identify if abnormal resistance measurements are being influenced by the motor windings.
4. Remove power from the starter unit.
5. Using an ohmmeter, perform the following resistance measurements and record the results:

MEASURE BETWEEN	SCR PAIRS BEING CHECKED	RECORDED VALUE
T1 and T6	3 and 6	
T2 and T4	2 and 5	
T3 and T5	1 and 4	

   If all measured values are greater than 5K ohms, proceed to Step 10. If any values are less than 5K ohms, one or more of the SCRs in that pair is shorted.
6. Remove both SCRs in the pair (See SCR Removal/Installation).
7. Using an ohmmeter, measure the resistance (anode to cathode) of each SCR to determine which device has failed.

   NOTE: Both SCRs may be defective, but typically, only one is shorted. If both SCRs provide acceptable resistance measurements, proceed to Step 10.
8. Replace the defective SCR(s).
9. Retest the ‘‘pair’’ for resistance values indicated above.
10. On the right side of the firing card, measure the resistance between the red and white gate/cathode leads for each SCR (1 through 6). A measurement between 5 and 50 ohms is normal. Abnormally high values may indicate a failed gate for that SCR.

⚠ CAUTION

If any red or white SCR gate leads are removed from the firing card or an SCR, care must be taken to ensure the leads are replaced EXACTLY as they were (white wires to gates, and red wires to cathodes on both the firing card and SCR), or damage to the starter and/or motor may result.

11. Replace the SCRs and retest the pair.

SCR REMOVAL/INSTALLATION — Refer to Fig. 42.

1. Remove the SCR by loosening the clamping bolts on each side of the SCR,
2. After the SCR has been removed and the bus work is loose, apply a thin coat of either silicon based thermal joint compound or a joint compound for aluminum or copper wire connections to the contact surfaces of the replacement SCR. This allows for improved heat dissipation and electrical conductivity.

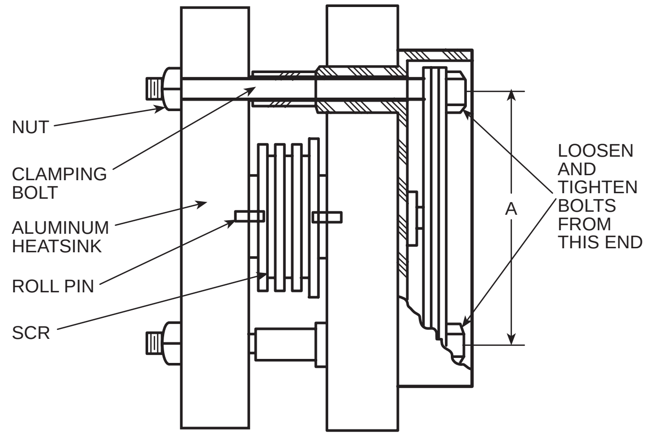
3. Place the SCR between the roll pins on the heatsink assemblies so the roll pins fit into the small holes in each side of the SCR.

   NOTE: Ensure the SCR is installed so the cathode side is the side from which the red wire extends. The heatsink is labeled to show the correct orientation.
4. Hand tighten the bolts until the SCR contacts the heatsink.
5. Using quarter-turn increments, alternating between clamping bolts, apply the appropriate number of whole turns referencing the table in Fig. 42.

⚠ CAUTION

Care must be taken to prevent nut rotation while tightening the bolts. If the nut rotates while tighting the bolt, SCR replacement must be started over.

6. Reconnect the red (cathode) wire from the SCR and the white (anode-gate) wire to the appropriate location on the firing card (i.e., SCR1 wires to firing card terminal G1-white wire, and K1-red wire).
7. Reconnect all other wiring and bus work.
8. Return starter to normal operation.



SCR PART NUMBER BISCR	CLAMP SIZE	A DIMENSION (in.)	NO OF TURNS	BOLT LENGTH (in.)
6601218	1030	2.75 (70 mm)	1½	3.0 (76 mm)
6601818	1030	2.75 (70 mm)	1½	3.0 (76 mm)
8801230	1035	2.75 (70 mm)	1¾	3.5 (89 mm)
8801830	1035	2.75 (70 mm)	1¾	3.0 (89 mm)
15001850	2040	4.00 (102 mm)	2¾	4.0 (102 mm)
15001850	2050	4.00 (102 mm)	2¾	5.0 (127 mm)
220012100	Consult Benshaw Representative			
330018500	Consult Benshaw Representative			

Fig. 42 — SCR Installation

Physical Data — Tables 10A-17 and Fig. 43-50 provide additional information on component weights, compressor fits and clearances, physical and electrical data, and wiring schematics for the operator's convenience during troubleshooting.

Table 10A — Heat Exchanger Data (English)

CODE	NUMBER OF TUBES		ENGLISH					
			Dry (Rigging) Weight (lb)		Chiller Charge			
					Refrigerant Weight		Water Volume (gal)	
	Cooler	Condenser	Cooler Only	Condenser Only	Cooler	Condenser	Cooler	Condenser
10	142	180	2,742	2,704	290	200	34	42
11	161	200	2,812	2,772	310	200	37	45
12	180	225	2,883	2,857	330	200	40	49
15	142	180	3,003	2,984	320	250	39	48
16	161	200	3,089	3,068	340	250	43	52
17	180	225	3,176	3,173	370	250	47	57
20	200	218	3,442	3,523	345	225	48	48
21	240	266	3,590	3,690	385	225	55	55
22	282	315	3,746	3,854	435	225	62	63
30	200	218	4,137	3,694	350	260	55	55
31	240	267	4,319	3,899	420	260	64	65
32	280	315	4,511	4,100	490	260	72	74
35	200	218	4,409	4,606	400	310	61	62
36	240	267	4,617	4,840	480	310	70	72
37	280	315	4,835	5,069	550	310	80	83
40	324	370	5,898	6,054	560	280	89	96
41	364	417	6,080	6,259	630	280	97	106
42	400	463	6,244	6,465	690	280	105	114
45	324	370	6,353	6,617	640	330	98	106
46	364	417	6,561	6,851	720	330	108	117
47	400	463	6,748	7,085	790	330	116	127
50	431	509	7,015	7,285	750	400	115	128
51	485	556	7,262	7,490	840	400	126	137
52	519	602	7,417	7,683	900	400	133	136
55	431	509	7,559	7,990	870	490	127	142
56	485	556	7,839	8,214	940	490	139	152
57	519	602	8,016	8,434	980	490	147	162
60	557	648	8,270	8,286	940	420	144	159
61	599	695	8,462	8,483	980	420	153	168
62	633	741	8,617	8,676	1020	420	160	177
65	557	648	8,943	9,204	1020	510	160	176
66	599	695	9,161	9,428	1060	510	169	187
67	633	741	9,338	9,648	1090	510	177	197
70	644	781	12,395	13,139	1220	780	224	209
71	726	870	12,821	13,568	1340	780	243	229
72	790	956	13,153	13,969	1440	780	257	248
75	644	781	13,293	14,211	1365	925	245	234
76	726	870	13,780	14,702	1505	925	266	257
77	790	956	14,159	15,160	1625	925	283	278
80	829	990	16,156	15,746	1500	720	285	264
81	901	1080	16,530	16,176	1620	720	302	284
82	976	1170	16,919	16,606	1730	720	319	304
85	829	990	17,296	17,001	1690	860	313	295
86	901	1080	17,723	17,492	1820	860	331	318
87	976	1170	18,169	17,984	1940	860	351	341

NOTES:
1. Cooler data: based on a cooler with standard wall tubing, 2-pass, 150 psig, nozzle-in-head waterbox with victaulic grooves. Weight includes suction elbow, control panel, and distribution piping. Weight does not include compressor.
2. Condenser data: based on a condenser with standard wall tubing, 2-pass, 150 psig, nozzle-in-head waterbox with victaulic grooves. Weight includes the float valve, discharge elbow, and distribution piping. Weight does not include unit-mounted starter, isolation valves, and pumpout unit.

Table 10B — Heat Exchanger Data (SI)

CODE	NUMBER OF TUBES		SI					
			Dry (Rigging) Weight (kg)		Chiller Charge			
	Cooler	Condenser	Cooler Only	Condenser Only	Refrigerant Weight		Water Volume (L)	
					Cooler	Condenser	Cooler	Condenser
10	142	180	1244	1227	132	91	129	158
11	161	200	1276	1257	141	91	140	170
12	180	225	1308	1296	150	91	152	185
15	142	180	1362	1354	145	113	149	183
16	161	200	1401	1392	154	113	163	198
17	180	225	1441	1439	168	113	178	216
20	200	218	1561	1598	156	102	183	181
21	240	266	1628	1674	175	102	207	210
22	282	315	1699	1748	197	102	234	239
30	200	218	1877	1676	159	118	208	210
31	240	267	1959	1769	190	118	242	246
32	280	315	2046	1860	222	118	271	282
35	200	218	2000	2089	181	141	232	233
36	240	267	2094	2195	218	141	266	273
37	280	315	2193	2299	249	141	301	314
40	324	370	2675	2746	254	127	338	365
41	364	417	2757	2839	286	127	368	400
42	400	463	2832	2933	313	127	396	433
45	324	370	2881	3001	290	150	372	403
46	364	417	2976	3108	327	150	407	442
47	400	463	3060	3214	358	150	438	481
50	431	509	3181	3304	340	181	435	483
51	485	556	3293	3397	381	181	477	518
52	519	602	3364	3484	408	181	502	552
55	431	509	3428	3624	395	222	481	536
56	485	556	3555	3725	426	222	527	575
57	519	602	3635	3825	445	222	557	613
60	557	648	3751	3758	426	190	546	601
61	599	695	3838	3847	445	190	578	636
62	633	741	3908	3935	463	190	604	669
65	557	648	4056	4174	463	231	605	668
66	599	695	4155	4276	481	231	641	707
67	633	741	4235	4376	494	231	671	745
70	644	781	5622	5960	553	354	848	791
71	726	870	5816	6154	608	354	919	867
72	790	956	5966	6336	653	354	974	937
75	644	781	6030	6446	619	420	927	885
76	726	870	6251	6669	683	420	1009	971
77	790	956	6423	6877	737	420	1072	1052
80	829	990	7328	7142	680	327	1080	1000
81	901	1080	7498	7337	735	327	1143	1075
82	976	1170	7674	7532	785	327	1208	1150
85	829	990	7845	7712	767	390	1183	1118
86	901	1080	8029	7934	826	390	1254	1205
87	976	1170	8241	8158	880	390	1329	1291

NOTES:
1. Cooler data: based on a cooler with standard wall tubing, 2-pass, 150 psig, nozzle-in-head waterbox with victaulic grooves. Weight includes suction elbow, control panel, and distribution piping. Weight does not include compressor.
2. Condenser data: based on a condenser with standard wall tubing, 2-pass, 150 psig, nozzle-in-head waterbox with victaulic grooves. Weight includes the float valve, discharge elbow, and distribution piping. Weight does not include unit-mounted starter, isolation valves, and pumpout unit.

Table 11 — 19XR Additional Data for Marine Waterboxes*

HEAT EXCHANGER FRAME, PASS	ENGLISH			SI		
	Psig	Rigging Weight (lb)	Water Volume (gal)	kPa	Rigging Weight (kg)	Water Volume (L)
FRAME 2, 1 AND 2 PASS	150	730	84	1034	331	318
FRAME 2, 2 PASS	150	365	42	1034	166	159
FRAME 3, 1 AND 2 PASS	150	730	84	1034	331	317
FRAME 3, 2 PASS	150	365	42	1034	166	159
FRAME 4, 1 AND 3 PASS	150	1060	123	1034	481	465
FRAME 4, 2 PASS	150	530	61	1034	240	231
FRAME 5, 1 AND 3 PASS	150	1240	139	1034	562	526
FRAME 5, 2 PASS	150	620	69	1034	281	263
FRAME 6, 1 AND 3 PASS	150	1500	162	1034	680	612
FRAME 6, 2 PASS	150	750	81	1034	340	306
FRAME 7, 1 AND 3 PASS	150	2010	326	1034	912	1234
FRAME 7, 2 PASS	150	740	163	1034	336	617
FRAME 8, 1 AND 3 PASS	150	1855	406	1034	841	1537
FRAME 8, 2 PASS	150	585	203	1034	265	768
FRAME 2, 1 AND 3 PASS	300	860	84	2068	390	318
FRAME 2, 2 PASS	300	430	42	2068	195	159
FRAME 3, 1 AND 3 PASS	300	860	84	2068	390	317
FRAME 3, 2 PASS	300	430	42	2068	195	159
FRAME 4, 1 AND 3 PASS	300	1210	123	2068	549	465
FRAME 4, 2 PASS	300	600	61	2068	272	231
FRAME 5, 1 AND 3 PASS	300	1380	139	2068	626	526
FRAME 5, 2 PASS	300	690	69	2068	313	263
FRAME 6, 1 AND 3 PASS	300	1650	162	2068	748	612
FRAME 6, 2 PASS	300	825	81	2068	374	306
FRAME 7, 1 AND 3 PASS	300	3100	326	2068	1406	1234
FRAME 7, 2 PASS	300	2295	163	2068	830	617
FRAME 8, 1 AND 3 PASS	300	2745	405	2068	1245	1533
FRAME 8, 2 PASS	300	1630	203	2068	739	768

*Add to heat exchanger data for total weights or volumes.

NOTES:
1. Weight adder shown is the same for cooler and condenser of equal frame size.
2. For the total weight of a vessel with a marine waterbox, add these values to the heat exchanger weights (or volumes).

Table 12 — Compressor Weights

COMPONENT	FRAME 2 COMPRESSOR WEIGHT		FRAME 3 COMPRESSOR WEIGHT		FRAME 4 COMPRESSOR WEIGHT		FRAME 5 COMPRESSOR WEIGHT	
	lb	kg	lb	kg	lb	kg	lb	kg
SUCTION ELBOW	50	23	54	24	175	79	210	95
DISCHARGE ELBOW	60	27	46	21	157	71	140	63
TRANSMISSION*	320	145	730	331	656	298	1000	454
SUCTION HOUSING	300	136	350	159	446	202	1200	544
IMPELLER SHROUD	35	16	80	36	126	57	250	113
COMPRESSOR BASE	1260	571	1050	476	1589	721	3695	1676
DIFFUSER	35	16	70	32	130	59	300	136
OIL PUMP	125	57	150	68	150	68	185	84
MISCELLANEOUS	100	45	135	61	144	65	220	100
TOTAL WEIGHT (Less Motor and Elbows)	2300	1043	2660	1207	3712	1684	6850	3107

*Transmission weight does not include rotor, shaft, and gear. See Table 15.

Table 13 — 19XR Motor Weights Standard and High Efficiency Motors

MOTOR SIZE	ENGLISH					SI				
	Stator Weight* (lb)		Rotor Weight† (lb)		End Bell Cover (lb)	Stator Weight* (kg)		Rotor Weight† (kg)		End Bell Cover (kg)
	60 Hz	50 Hz	60 Hz	50 Hz		60 Hz	50 Hz	60 Hz	50 Hz	
BD	1030	1030	240	240	185	467	467	109	109	84
BE	1070	1070	250	250	185	485	485	113	113	84
BF	1120	1120	265	265	185	508	508	120	120	84
BG	1175	1175	290	290	185	533	533	132	132	84
BH	1175	1175	290	290	185	533	533	132	132	84
CB	1235	1290	239	254	274	560	585	108	115	124
CC	1260	1295	249	259	274	572	587	113	117	124
CD	1286	1358	258	273	274	583	616	117	124	124
CE	1305	1377	265	279	274	592	625	120	127	124
CL	1324	1435	280	296	274	601	651	127	134	124
CM	1347	1455	303	296	274	611	660	137	134	124
CN	1369	1467	316	316	274	621	665	143	143	124
CP	1411	1479	329	316	274	640	671	149	143	124
CQ	1455	1522	329	336	274	660	690	149	152	124
DB	1665	1725	361	391	236	755	782	164	177	107
DC	1681	1737	391	404	236	763	788	177	183	107
DD	1977	1997	536	546	318	897	906	243	248	144
DE	2018	2089	550	550	318	915	948	249	249	144
DF	2100	2139	575	567	318	953	972	261	257	144
DG	2187	2153	599	599	318	992	977	272	272	144
DH	2203	2195	604	604	318	999	996	274	274	144
DJ	2228	2305	614	614	318	1011	1046	279	279	144
EH	3060	3120	701	751	414	1388	1415	318	341	188
EJ	3105	3250	716	751	414	1408	1474	325	341	188
EK	3180	3250	716	768	414	1442	1474	325	348	188
EL	3180	3370	737	801	414	1442	1529	334	363	188
EM	3270	3370	737	801	414	1483	1529	334	363	188
EN	3270	3520	801	851	414	1483	1597	363	386	188
EP	3340	3520	830	851	414	1515	1597	376	386	188

*Stator weight includes stator and shell.
†Rotor weight includes rotor and shaft.

NOTE: When different voltage motors have different weights the largest weight is given.

Table 14A — 19XR Waterbox Cover Weights — English (lb)

HEAT EXCHANGER	WATERBOX DESCRIPTION	FRAME 1		FRAME 2		FRAME 3	
		Standard Nozzles	Flanged	Standard Nozzles	Flanged	Standard Nozzles	Flanged
COOLER/ CONDENSER	NIH, 1 Pass Cover, 150 psig	177	204	320	350	320	350
	NIH, 2 Pass Cover, 150 psig	185	218	320	350	320	350
	NIH, 3 Pass Cover, 150 psig	180	196	300	340	300	340
	NIH/MWB End Cover, 150 psig	136	136	300	300	300	300
	NIH, 1 Pass Cover, 300 psig	248	301	411	486	411	486
	NIH, 2 Pass Cover, 300 psig	255	324	411	518	411	518
	NIH, 3 Pass Cover, 300 psig	253	288	433	468	433	468
	NIH/MWB End Cover, 300 psig	175	175	400	400	400	400

HEAT EXCHANGER	WATERBOX DESCRIPTION	FRAME 4		FRAME 5		FRAME 6	
		Standard Nozzles	Flanged	Standard Nozzles	Flanged	Standard Nozzles	Flanged
COOLER/ CONDENSER	NIH, 1 Pass Cover, 150 psig	485	521	616	652	802	838
	NIH, 2 Pass Cover, 150 psig	487	540	590	663	770	843
	NIH, 3 Pass Cover, 150 psig	504	520	629	655	817	843
	NIH/MWB End Cover, 150 psig	379	379	428	428	583	583
	NIH, 1 Pass Cover, 300 psig	593	668	764	839	880	956
	NIH, 2 Pass Cover, 300 psig	594	700	761	878	844	995
	NIH, 3 Pass Cover, 300 psig	621	656	795	838	901	952
	NIH/MWB End Cover, 300 psig	569	569	713	713	833	833

HEAT EXCHANGER	WATERBOX DESCRIPTION	FRAME 7 COOLER		FRAME 7 CONDENSER	
		Standard Nozzles	Flanged	Standard Nozzles	Flanged
COOLER/ CONDENSER	NIH, 1 Pass Cover, 150 psig	1392	1469	1205	1282
	NIH, 2 Pass Cover, 150 psig	1345	1461	1163	1279
	NIH, 3 Pass Cover, 150 psig	1434	1471	1222	1280
	NIH/MWB End Cover, 150 psig	1022	1022	920	920
	NIH, 1 Pass Cover, 300 psig	1985	2150	1690	1851
	NIH, 2 Pass Cover, 300 psig	1934	2174	1628	1862
	NIH, 3 Pass Cover, 300 psig	2009	2090	1714	1831
	NIH/MWB End Cover, 300 psig	1567	1567	1923	1923

HEAT EXCHANGER	WATERBOX DESCRIPTION	FRAME 8 COOLER		FRAME 8 CONDENSER	
		Standard Nozzles	Flanged	Standard Nozzles	Flanged
COOLER/ CONDENSER	NIH, 1 Pass Cover, 150 psig	1831	1909	1682	1760
	NIH, 2 Pass Cover, 150 psig	1739	1893	1589	1744
	NIH, 3 Pass Cover, 150 psig	1851	1909	1702	1761
	NIH/MWB End Cover, 150 psig	1480	1480	1228	1228
	NIH, 1 Pass Cover, 300 psig	2690	2854	2394	2549
	NIH, 2 Pass Cover, 300 psig	2595	2924	2269	2578
	NIH, 3 Pass Cover, 300 psig	2698	2861	2417	2529
	NIH/MWB End Cover, 300 psig	1440	1440	1767	1767

LEGEND

NIH — Nozzle-in-Head
MWB — Marine Waterbox

NOTE: Weight for NIH 2-Pass Cover, 150 psig is included in the heat exchanger weights shown in Tables 10A and 10B.

Table 14B — 19XR Waterbox Cover Weights — SI (kg)

HEAT EXCHANGER	WATERBOX DESCRIPTION	FRAME 1		FRAME 2		FRAME 3	
		Standard Nozzles	Flanged	Standard Nozzles	Flanged	Standard Nozzles	Flanged
COOLER/ CONDENSER	NIH, 1 Pass Cover, 150 psig	80	93	145	159	145	159
	NIH, 2 Pass Cover, 150 psig	84	99	145	159	145	159
	NIH, 3 Pass Cover, 150 psig	82	89	136	154	140	154
	NIH/MWB End Cover, 150 psig	62	62	136	136	136	136
	NIH, 1 Pass Cover, 300 psig	112	137	186	220	186	220
	NIH, 2 Pass Cover, 300 psig	116	147	186	235	186	235
	NIH, 3 Pass Cover, 300 psig	115	131	196	212	196	212
	NIH/MWB End Cover, 300 psig	79	79	181	181	181	181

HEAT EXCHANGER	WATERBOX DESCRIPTION	FRAME 4		FRAME 5		FRAME 6	
		Standard Nozzles	Flanged	Standard Nozzles	Flanged	Standard Nozzles	Flanged
COOLER/ CONDENSER	NIH, 1 Pass Cover, 150 psig	220	236	279	296	364	380
	NIH, 2 Pass Cover, 150 psig	221	245	268	301	349	382
	NIH, 3 Pass Cover, 150 psig	229	236	285	297	371	381
	NIH/MWB End Cover, 150 psig	172	172	194	194	265	265
	NIH, 1 Pass Cover, 300 psig	269	303	347	381	399	434
	NIH, 2 Pass Cover, 300 psig	269	318	345	398	383	451
	NIH, 3 Pass Cover, 300 psig	282	298	361	380	409	432
	NIH/MWB End Cover, 300 psig	258	258	323	323	378	378

HEAT EXCHANGER	WATERBOX DESCRIPTION	FRAME 7 COOLER		FRAME 7 CONDENSER	
		Standard Nozzles	Flanged	Standard Nozzles	Flanged
COOLER/ CONDENSER	NIH, 1 Pass Cover, 150 psig	631	666	547	582
	NIH, 2 Pass Cover, 150 psig	610	663	528	580
	NIH, 3 Pass Cover, 150 psig	650	667	554	581
	NIH/MWB End Cover, 150 psig	464	464	417	417
	NIH, 1 Pass Cover, 300 psig	900	975	767	840
	NIH, 2 Pass Cover, 300 psig	877	986	738	845
	NIH, 3 Pass Cover, 300 psig	911	948	777	831
	NIH/MWB End Cover, 300 psig	711	711	872	872

HEAT EXCHANGER	WATERBOX DESCRIPTION	FRAME 8 COOLER		FRAME 8 CONDENSER	
		Standard Nozzles	Flanged	Standard Nozzles	Flanged
COOLER/ CONDENSER	NIH, 1 Pass Cover, 150 psig	831	866	763	798
	NIH, 2 Pass Cover, 150 psig	789	859	721	791
	NIH, 3 Pass Cover, 150 psig	840	866	772	799
	NIH/MWB End Cover, 150 psig	671	671	557	557
	NIH, 1 Pass Cover, 300 psig	1220	1295	1086	1156
	NIH, 2 Pass Cover, 300 psig	1177	1326	1029	1169
	NIH, 3 Pass Cover, 300 psig	1224	1298	1096	1147
	NIH/MWB End Cover, 300 psig	653	653	802	802

LEGEND

NIH — Nozzle-in-Head
MWB — Marine Waterbox

NOTE: Weight for NIH 2-Pass Cover, 150 psig is included in the heat exchanger weights shown in Tables 10A and 10B.

Table 15 — Optional Pumpout System Electrical Data

MOTOR CODE	CONDENSER UNIT	VOLTS-PH-Hz	MAX RLA	LRA
1	19EA47-748	575-3-60	3.8	23.0
4	19EA42-748	200/208-3-60	10.9	63.5
5	19EA44-748	230-3-60	9.5	57.5
6	19EA46-748	400/460-3-50/60	4.7	28.8

LEGEND

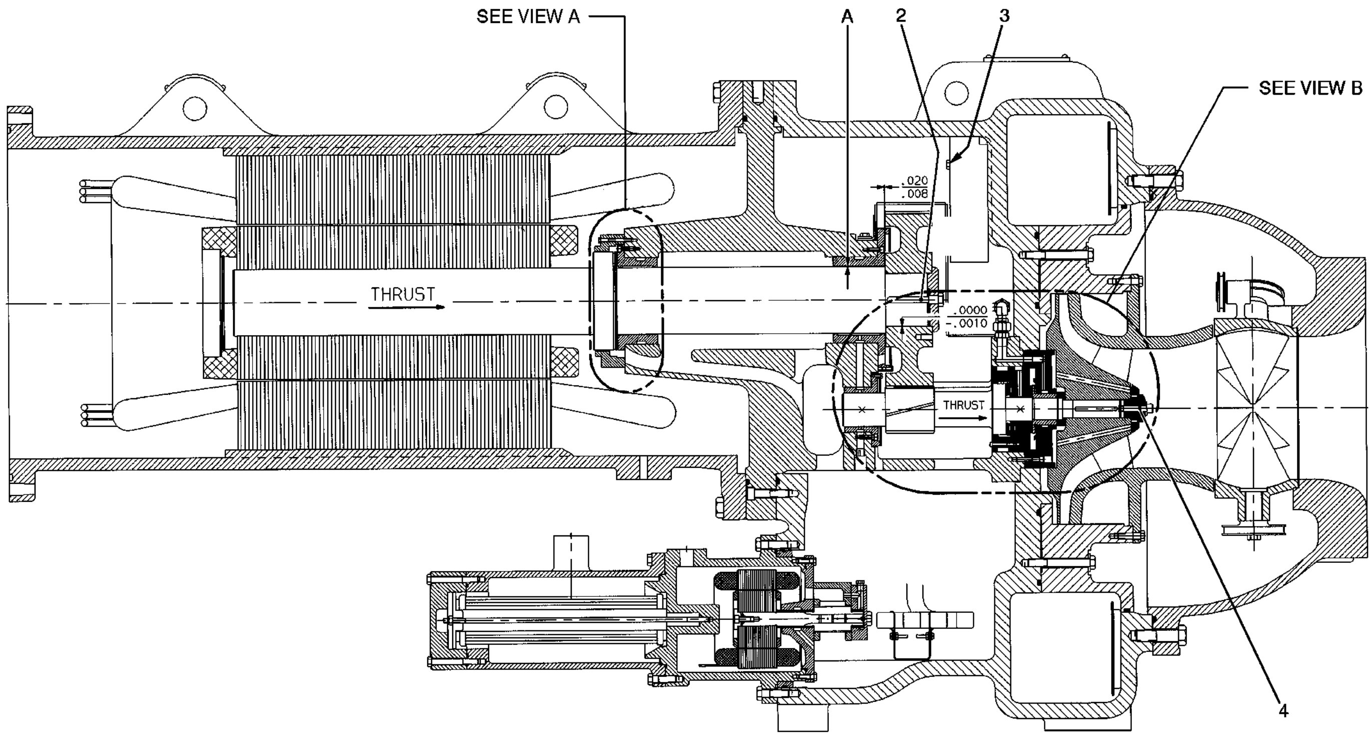
LRA — Locked Rotor Amps
RLA — Rated Load Amps

Table 16 — Additional Miscellaneous Weights

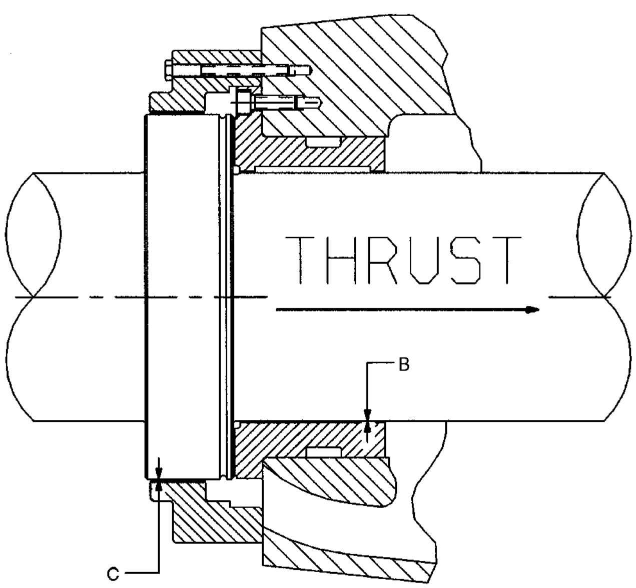
ITEM	Lb	Kg
CONTROL CABINET	30	14
UNIT-MOUNTED STARTER	500	227
OPTIONAL ISOLATION VALVES	115	52

Table 17 — Motor Voltage Code

MOTOR VOLTAGE CODE		
Code	**Volts**	**Frequency**
60	200	60
61	230	60
62	380	60
63	416	60
64	460	60
65	575	60
66	2400	60
67	3300	60
68	4160	60
69	6900	60
50	230	50
51	346	50
52	400	50
53	3000	50
54	3300	50
55	6300	50



COMPRESSOR, TRANSMISSION AREA



VIEW A
LOW SPEED SHAFT THRUST DISK

Compressor Assembly Torques

ITEM	DESCRIPTION	TORQUE	
		ft-lb	N·m
1*	Oil Heater Retaining Nut	20	28
2	Bull Gear Retaining Bolt	80-85	108-115
3	Demister Bolts	15-19	20-26
4	Impeller Retaining Bolt	44-46	60-62
5*	Motor Terminals (Low Voltage)	50	68
6*	Guide Vane Shaft Seal Nut	25	34
7*	Motor Terminals (High Voltage)		
	— Insulator	2-4	2.7-5.4
	— Packing Nut	5	6.8
	— Brass Jam Nut	10	13.6

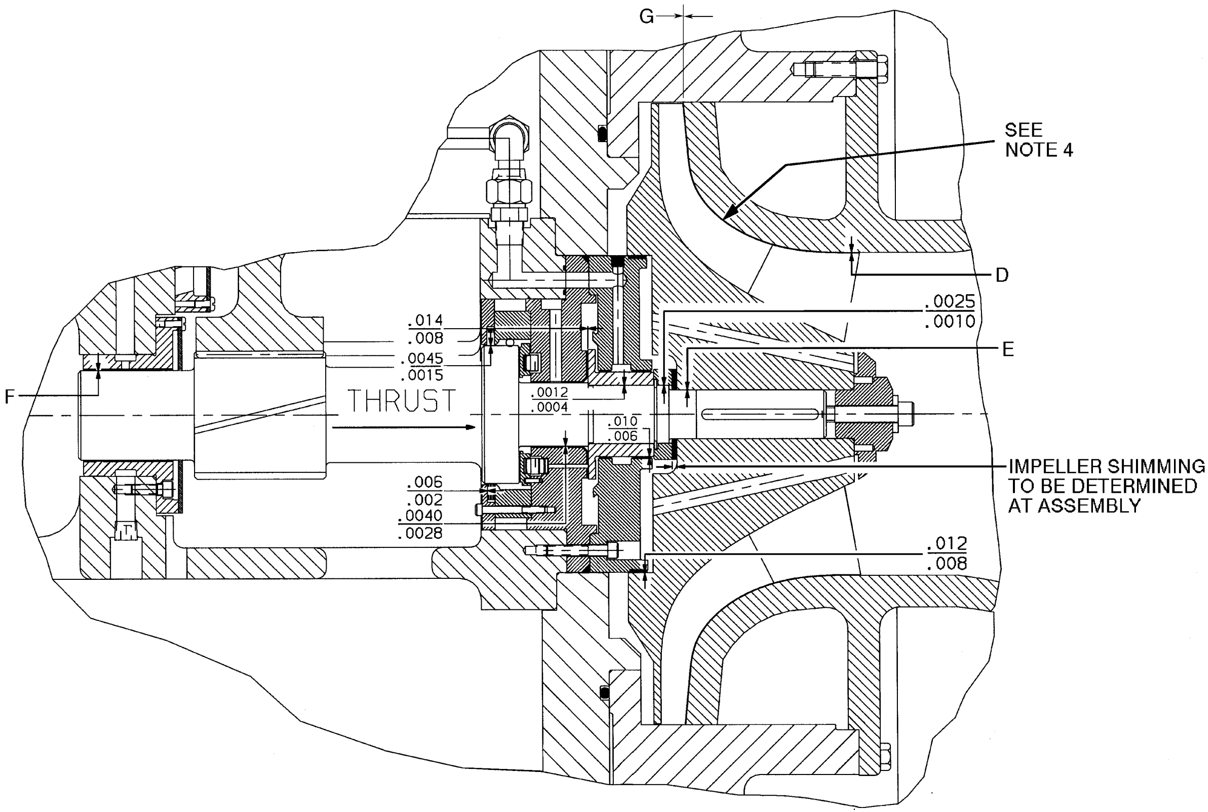
LEGEND

N·m — Newton meters

*Not shown.

NOTES:
1. All clearances for cylindrical surfaces are diametrical.
2. Dimensions are with rotor in thrust position.
3. Dimensions shown are in inches.
4. Impeller spacing should be performed in accordance with most recent Carrier Service Bulletin on impeller spacing.

Fig. 43 — Compressor Fits and Clearances

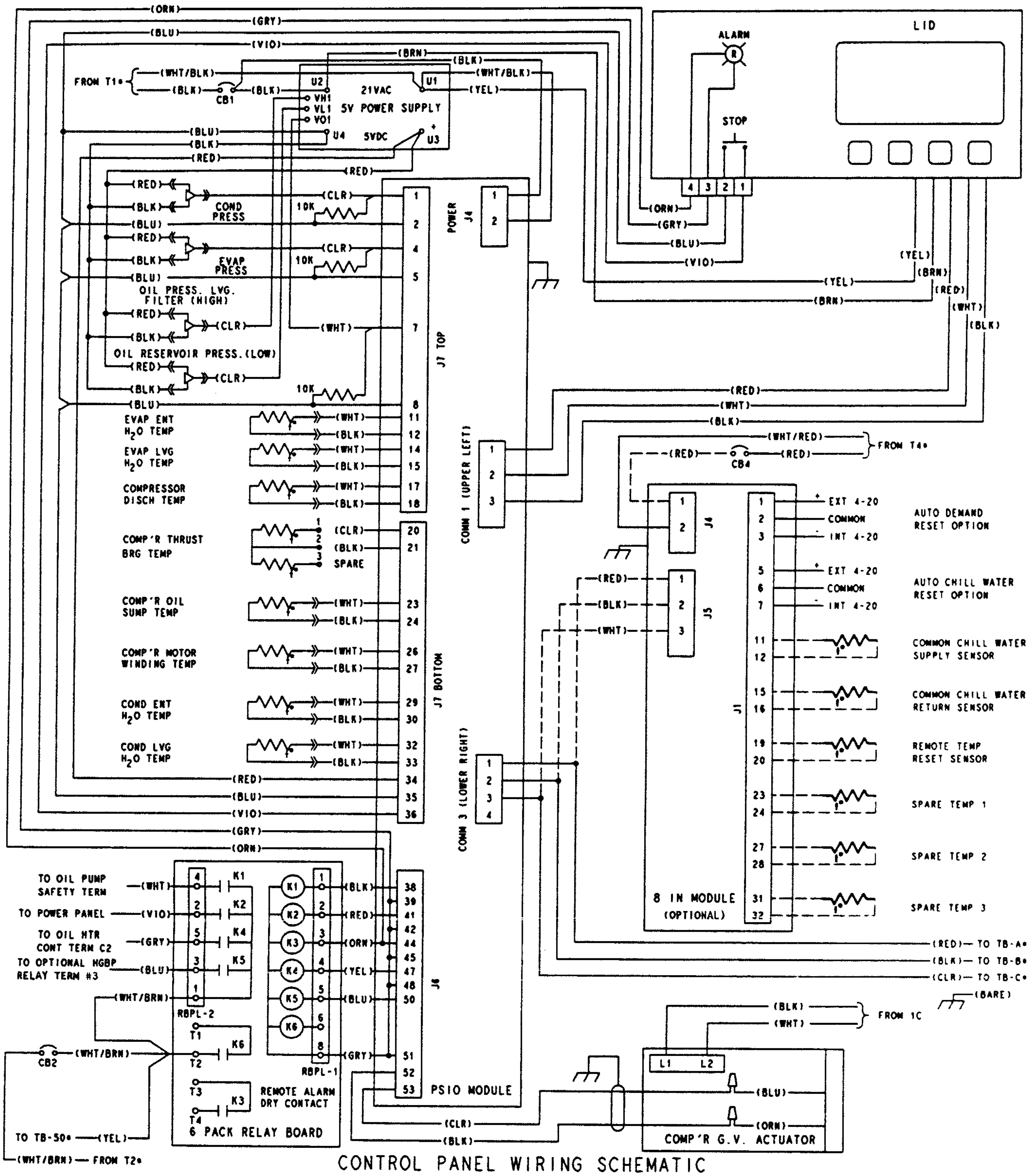


VIEW B − HIGH SPEED SHAFT

19XR Compressor Clearances

ITEM	COMPRESSOR CODE			
	221-299	321-389	421-489	521-599
A	.0050 / .0040	.0050 / .0040	.0055 / .0043	.0069 / .0059
B	.0050 / .0040	.0050 / .0040	.0053 / .0043	.0065 / .0055
C	.0115 / .0055	.0115 / .0080	.0100 / .0050	.0010 / .0060
D	.0190 / .0040	.022 / .012	.027 / .017	.0350 / .0250
E	.002 / .0005	− .0020 / − .0005	− .0029 / − .0014	.0031 / .0017
F	.0050 / .0040	.0050 / .0040	.0048 / .0038	.0062 / .0052
G	.157 / .0257	.157 / .0257	.0340 / .0240	.0530 / .0430

Fig. 43 — Compressor Fits and Clearances (cont)

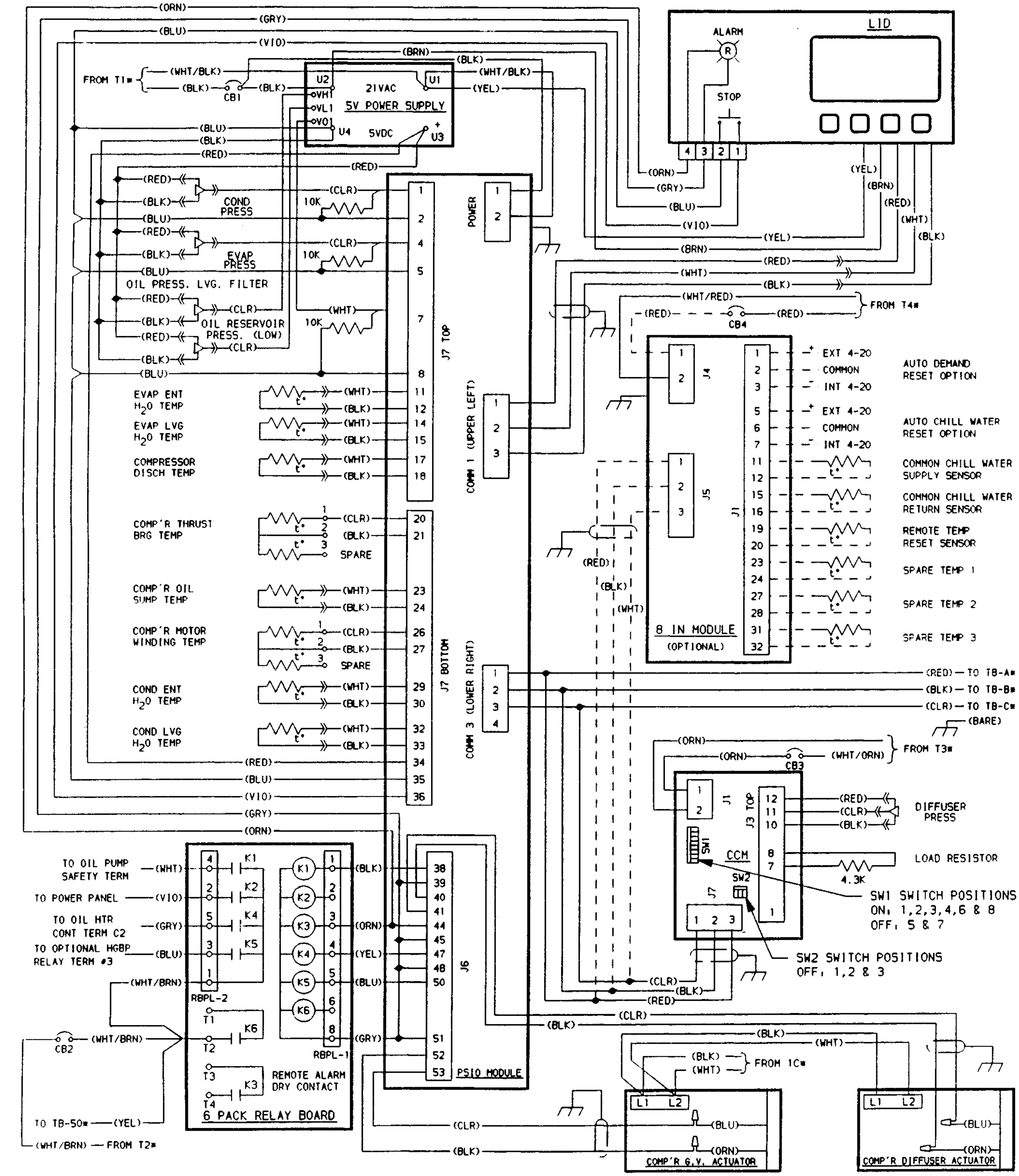


LEGEND

2C-C1	—	Oil Pump Contactor Coil Terminal
CB	—	Circuit Breaker
COMM	—	Communication
G.V.	—	Guide Vane
HGBP	—	Hot Gas Bypass
HTR	—	Heater
J	—	Module Connector
LID	—	Local Interface Device
L1, L2	—	Line Terminals
PSIO	—	Processor/Sensor Input/Output Module
RBPL	—	Relay Board Plug
T	—	Terminal
T1*,T2*,T3*,T4*	—	Power Panel Transformers
TB	—	Terminal Board
*		Denotes Power Panel Connection
○		Denotes Component Terminal
		Denotes Wire Crimp Joint
—››—		Denotes Conductor Male/Female Connector
– – – –		Option Wiring

IMPORTANT: Wiring shown is typical and **not** intended to show detail for a specific installation. Refer to certified field wiring diagrams.

Fig. 44 — Electronic PIC Controls Wiring Schematic (Frame 2, 3, 4 Compressor)



LEGEND

1C	—	Power Panel Transformers
CB	—	Circuit Breaker
CCM	—	Chiller Control Module
COMM	—	Communication
COMP'R	—	Compressor
COND	—	Condenser
DISCH	—	Discharge
ENT	—	Entering
EVAP	—	Evaporator
G.V.	—	Guide Vane
HGBP	—	Hot Gas Bypass
HTR	—	Heater
J	—	Jack
LID	—	Local Interface Device
L1,L2	—	3-Phase Line Terminals
LVG	—	Leaving
PSIO	—	Processor/Sensor Input/Output Module
RBPL	—	Relay Board Plug
T1,T2,T3,T4	—	Power Panel Transformers
TB	—	Terminal Board
*		Denotes Power Panel Connection
○		Denotes Component Terminal
		Denotes Wire Crimp Joint
—›)—		Denotes Conductor Male/Female Connector
– – – –		Option Wiring

IMPORTANT: Wiring shown is typical and **not** intended to show detail for a specific installation. Refer to certified field wiring diagrams.

Fig. 45 — Electronic PIC Controls Wiring Schematic (Frame 5 Compressor)

LEGEND

C	—	Contactor
CB	—	Circuit Breaker
COM	—	Common
FW	—	Field Wiring
HGBP	—	Hot Gas Bypass
HTR	—	Heater
J	—	Module Connector
M	—	Oil Pump Motor
PSIO	—	Processor/Sensor Input/Output Module
Q.C.	—	Quick-Connect
RB	—	Relay Board
RBPL	—	Relay Board Plug
SMM	—	Starter Management Module
SOL	—	Solenoid
T	—	Transformer
TB	—	Terminal B

- △ Denotes Oil Pump Terminal
- □ Denotes Power Panel Terminal
- ○ Denotes Component Terminal
- ● Wire Splice
- ▬ ▬ ▬ Option Wiring
- ⟶⟩— Denotes Male/Female Q.C.
- * Denotes Chiller Control Panel
- ** Denotes Motor Starter Panel Connection
- Denotes (3) Conductor Connector

IMPORTANT: Wiring shown is typical and **not** intended to show detail for a specific installation. Refer to certified field wiring diagrams.

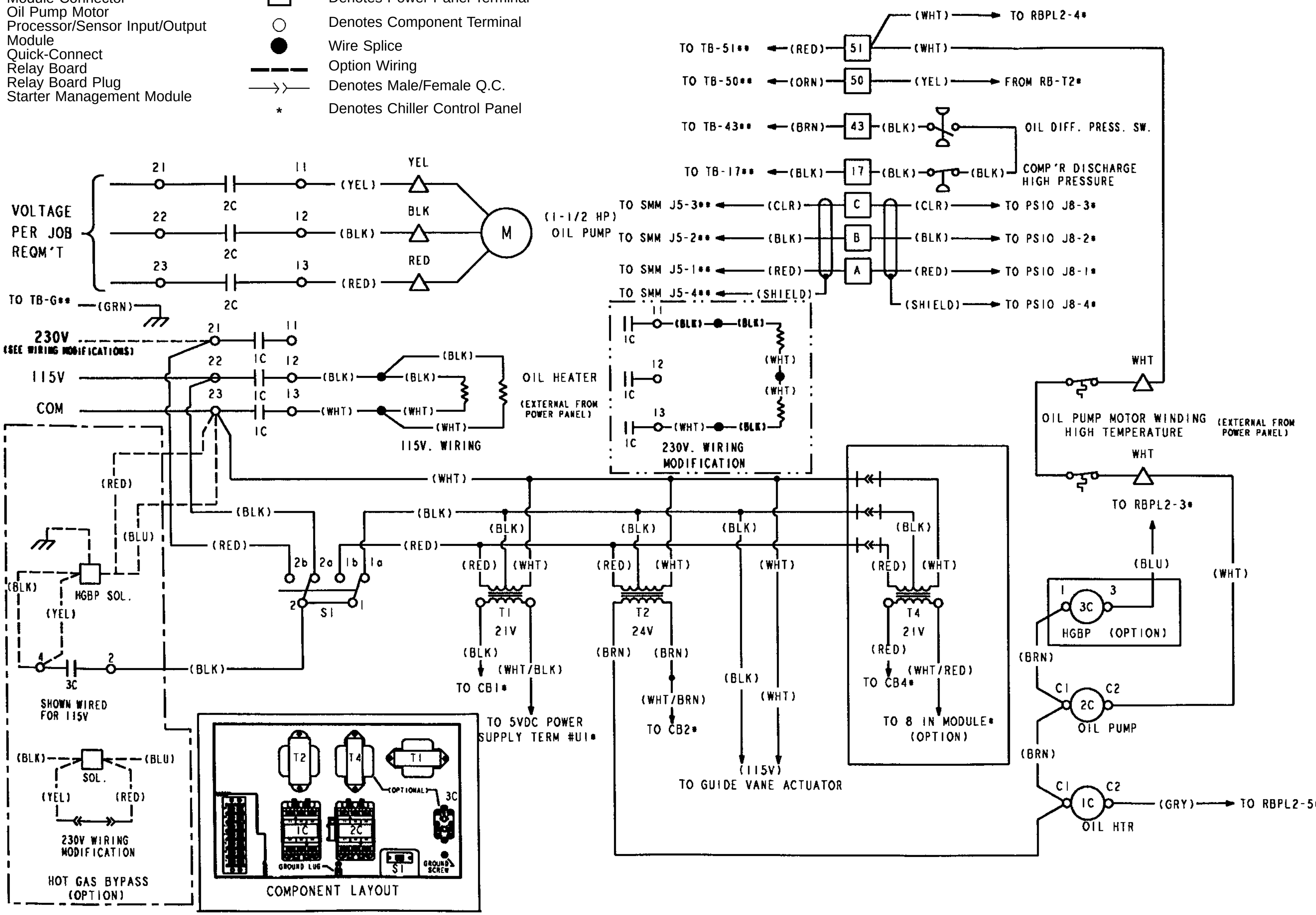


Fig. 46 — Power Panel Wiring (Chillers with Frame 1-5 Heat Exchangers)

LEGEND

C	—	Contactor
CB	—	Circuit Breaker
CCM	—	Chiller Control Module
COM	—	Common
COMM	—	Communication
FW	—	Field Wiring
HGBP	—	Hot Gas Bypass
HTR	—	Heater
J	—	Module Connector
LID	—	Local Interface Device
M	—	Oil Pump Motor
PSIO	—	Processor/Sensor Input/ Output Module
Q.C.	—	Quick-Connect
RB	—	Relay Board
RBPL	—	Relay Board Plug
SMM	—	Starter Management Module
SOL	—	Solenoid
T	—	Transformer
TB	—	Terminal Board

△	Denotes Oil Pump Terminal
□	Denotes Power Panel Terminal
○	Denotes Component Terminal
●	Wire Splice
— — —	Option Wiring
(FW)	Field Wiring (by Others)
→)—	Denotes Male/Female Q.C.
*	Denotes Chiller Control Panel Connection
**	Denotes Motor Starter Panel Connection
⊣≪⊢	Denotes (3) Conductor Connector

IMPORTANT: Wiring shown is typical and **not** intended to show detail for a specific installation. Refer to certified field wiring diagrams.

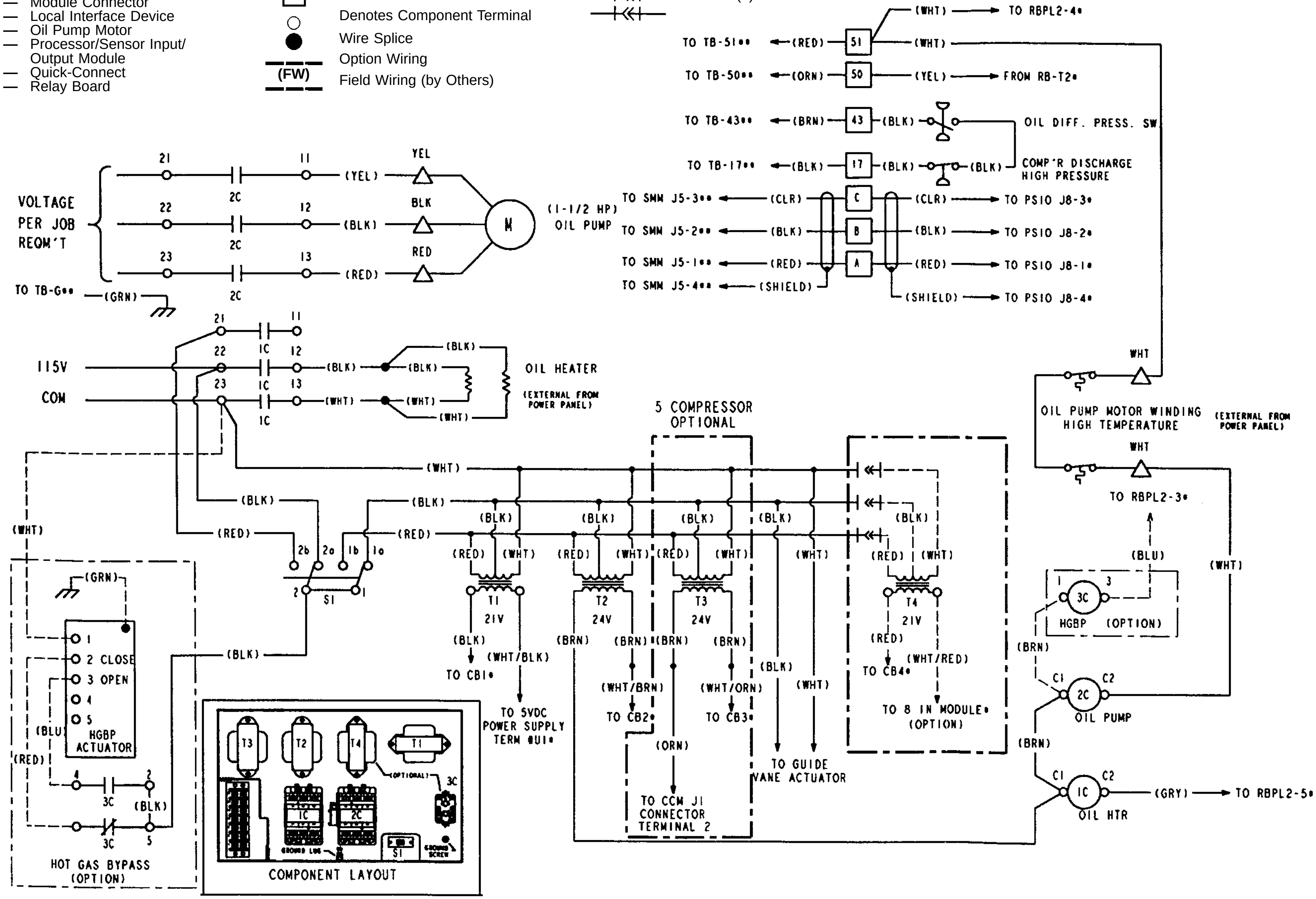


Fig. 47 — Power Panel Wiring (Chillers with Frame 6-8 Heat Exchangers)

LEGEND

HGBP — Hot Gas Bypass
TB — Terminal Board

△ Oil Pump Terminal

□ Power Panel Terminal

○ Component Terminal

● Wire Splice

– – – – Option Equipment

——— Option Wiring

→≻— Male/Female Q.C.

* Machine Control Panel Connection

** Motor Starter Panel Connection

IMPORTANT: Wiring shown is typical and **not** intended to show detail for a specific installation. Refer to certified field wiring diagrams.

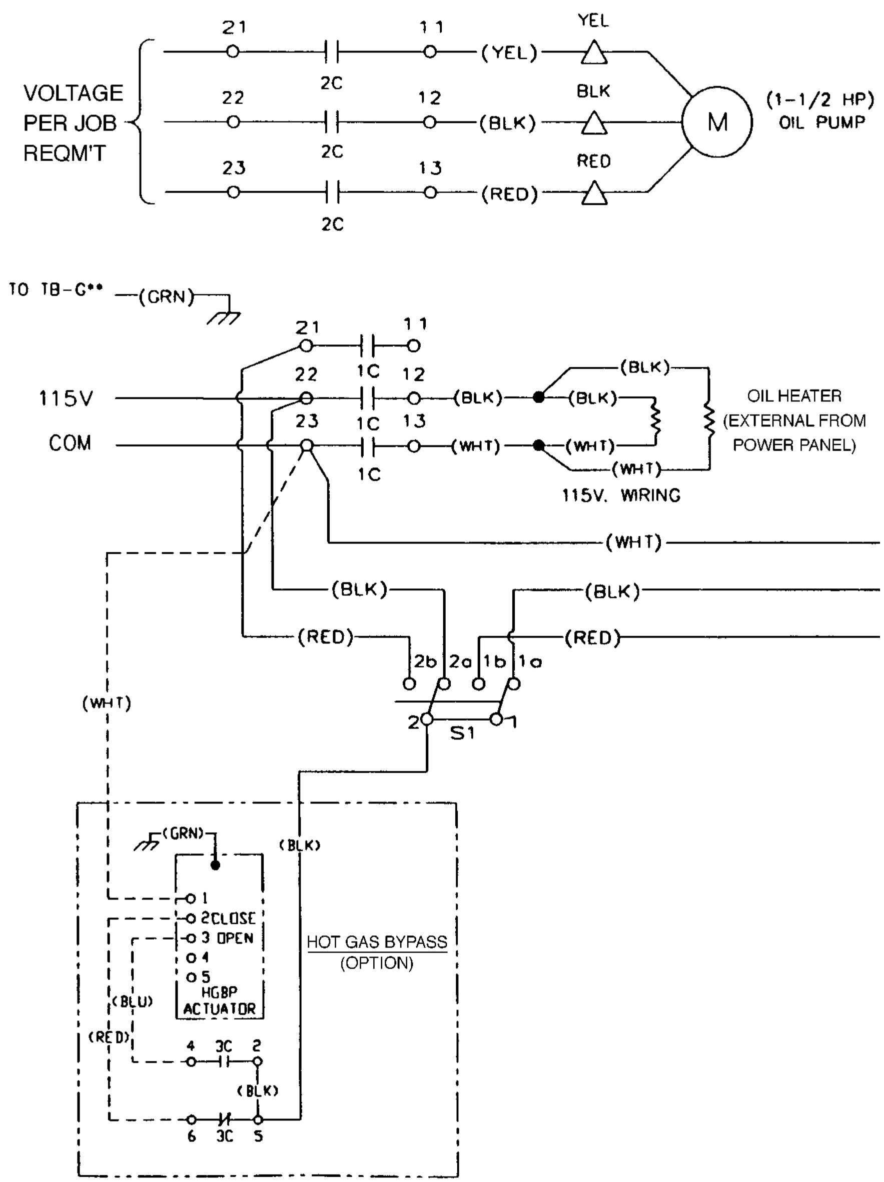
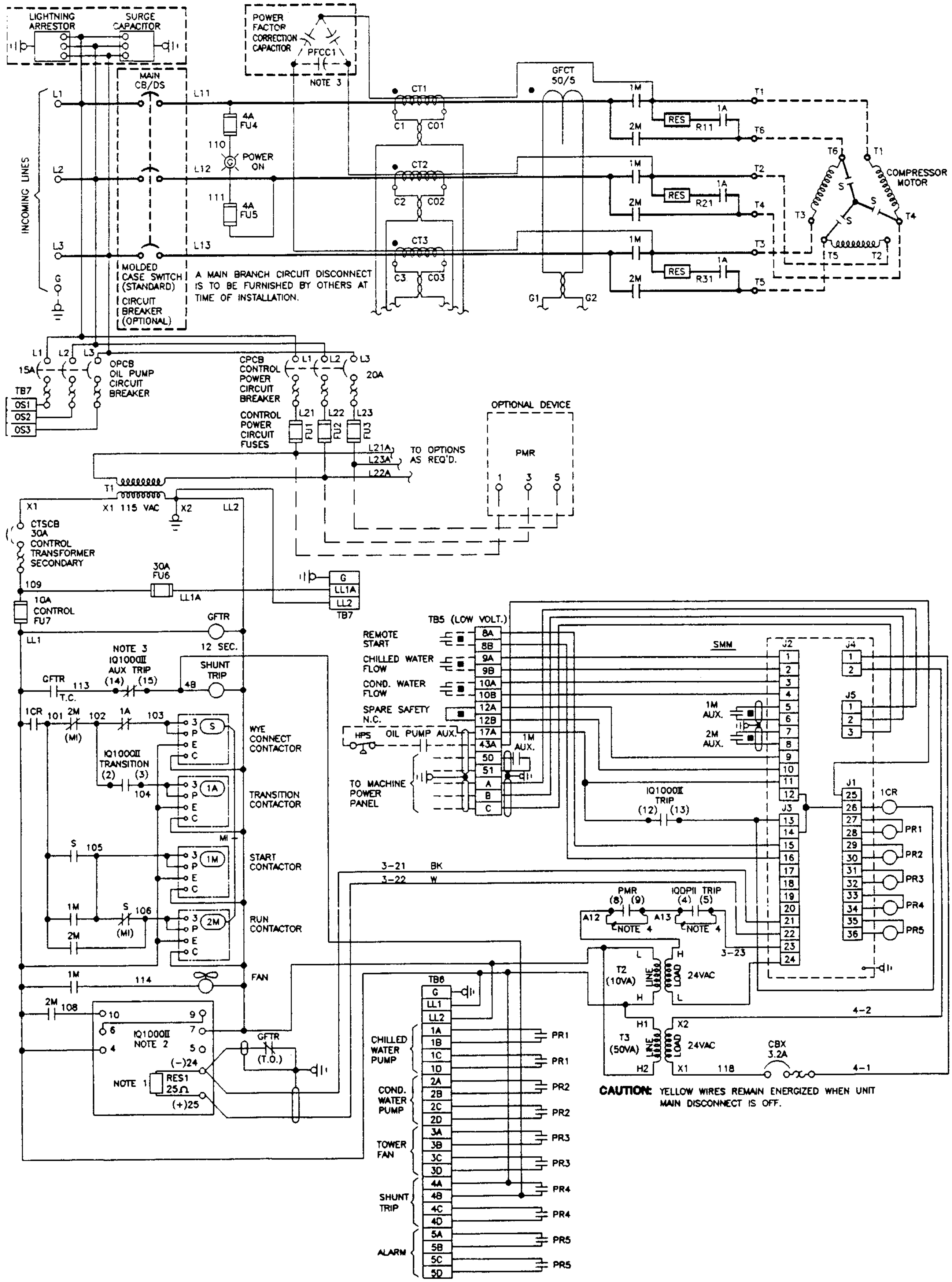


Fig. 48 — Hot Gas Bypass Valve Wiring Schematic (Frame 5 Compressor)



IMPORTANT: Wiring shown is typical and **not** intended to show detail for a specific installation. Refer to certified field wiring diagrams.

Fig. 49 — Cutler-Hammer (Wye-Delta) Unit Mounted Starter Wiring Schematic

LEGEND AND NOTES FOR FIG. 49

LEGEND

A	—	Ampere
AUX	—	Auxiliary
C	—	Connector
CB	—	Circuit Breaker
CPCB	—	Control Power Circuit Breaker
CR	—	Control Relay
CT	—	Current Transformer
CTS	—	Control Transformer Secondary
CTSCB	—	Control Transformer Circuit Breaker
DS	—	Disconnect
FU	—	Fuse
G	—	Ground
GFCT	—	Ground Fault Current Transformer
GFTR	—	Ground Fault Timer Relay
H	—	Power Transformer Primary Terminal
HPS	—	High-Pressure Switch
IQ 1000 II	—	Trade Name (Motor Protection System)
IQDP II	—	Trade Name: IQ Data Plus II (Line Metering and Protection System)
J	—	Module Connector
L1, L2, L3	—	3-Phase Line Terminals
LL1	—	Control Power (Feed)
LL2	—	Control Power (Neutral)
LLGF	—	Low Level Ground Fault
L.O.	—	Late Opening
M	—	Main Compressor Contactor
N.C.	—	Normally Closed
PFCC	—	Power Factor Correction Capacitor
PMR	—	Phase Loss Reversal Relay
PR	—	Pilot Relay
RES	—	Resistor
S	—	Compressor Motor Start Contactor
SMM	—	Starter Management Module
SW	—	Switch
T	—	Motor Terminal
TB	—	Terminal Board
T.C.	—	Timed Closed
VA	—	Volt Ampere
VAC	—	Volts Alternating Current
X	—	Variable

Dry Contact

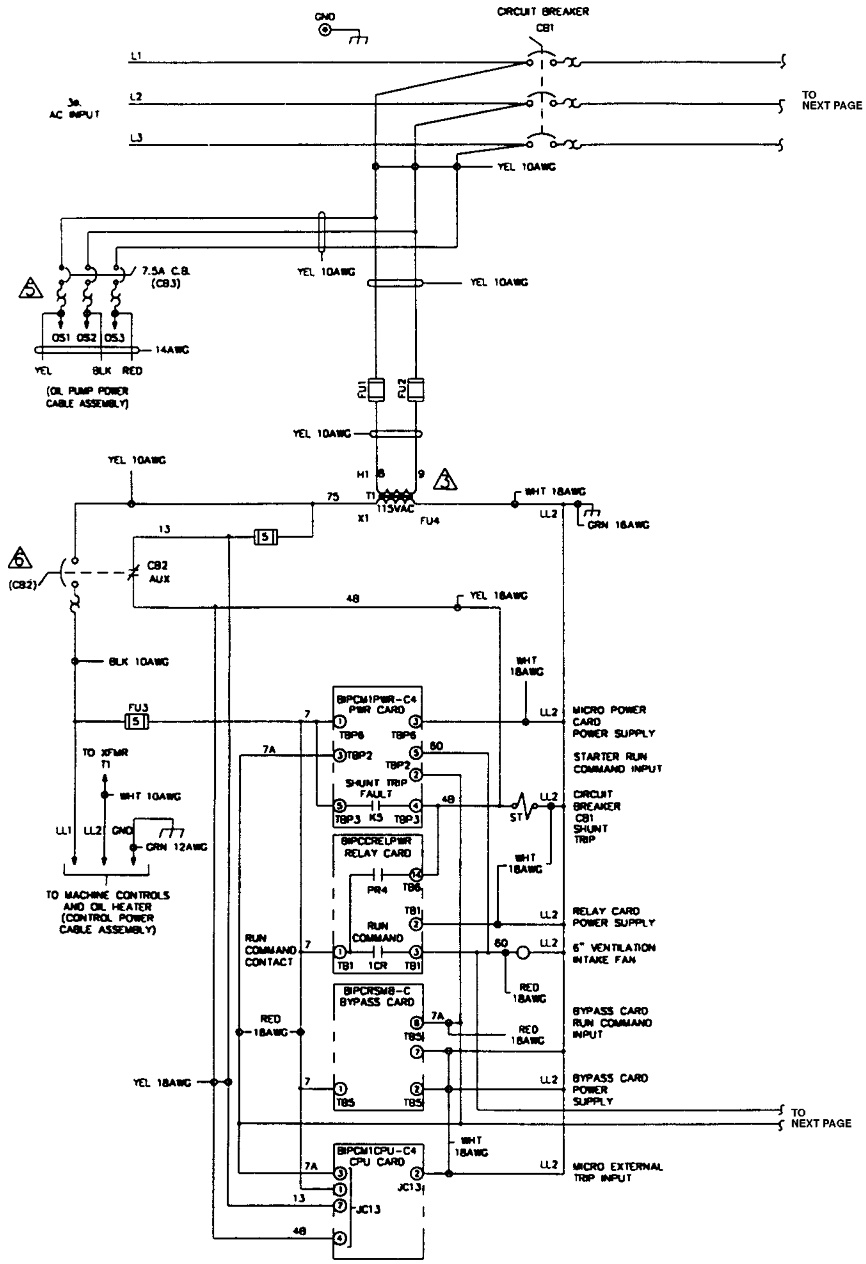
Field Wiring

Carrier Wiring

Carrier Specified Device

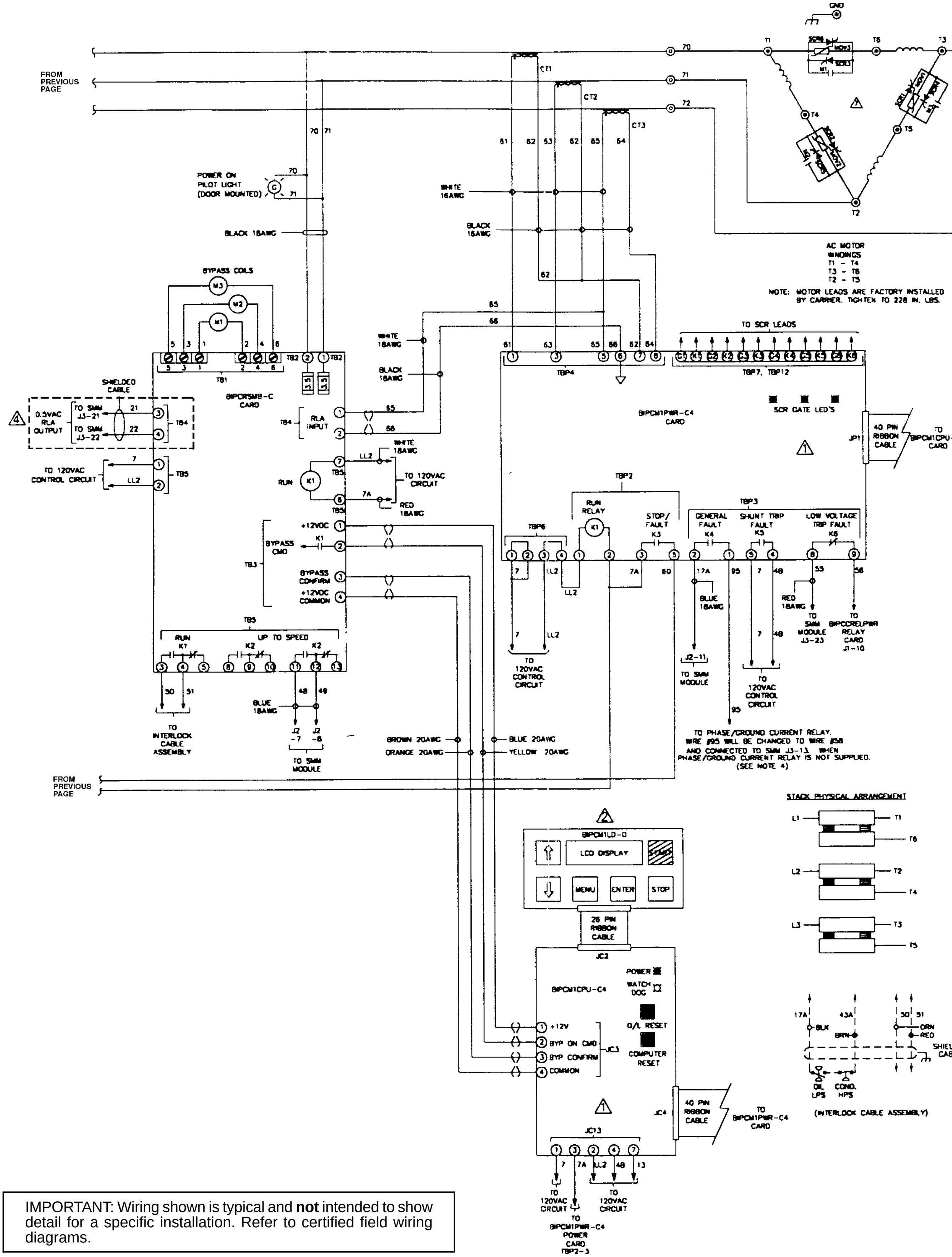
Optional Feature

NOTES:
1. Load sensing resistor to provide 0.5 volts at rated load amps.
2. IQ-1000-II Features: (refer to IQ manual for details)
   IQ AUX TRIP CONTACT set for ground fault only.
   IQ TRIP CONTACT set for remaining protection features.
   IQ TRIP MODE 2.
   IQ TRANSITION CONTACT TR (Start to Run).
   IQ Incomplete Sequence Feedback contact 2M.
3. Power factor correction capacitors (when required) are connected ahead of all current transformers for proper calibration and sensing by the IQ devices.
4. Provide jumper at terminal block if option is not required.



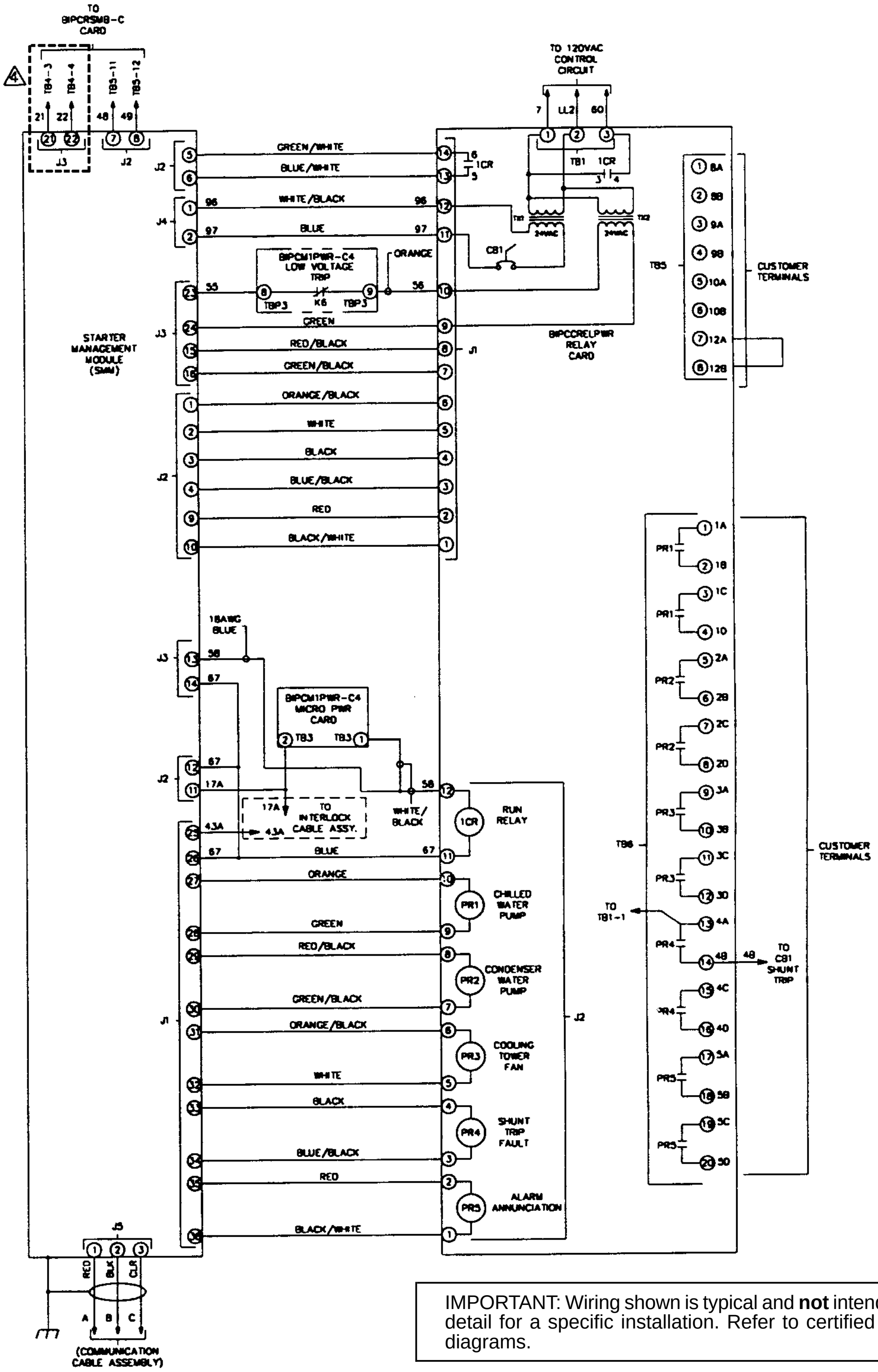
IMPORTANT: Wiring shown is typical and **not** intended to show detail for a specific installation. Refer to certified field wiring diagrams.

Fig. 50 — Benshaw (Inside the Delta) Unit Mounted Starter Wiring Schematic



IMPORTANT: Wiring shown is typical and **not** intended to show detail for a specific installation. Refer to certified field wiring diagrams.

Fig. 50 — Benshaw (Inside the Delta) Unit Mounted Starter Wiring Schematic (cont)



IMPORTANT: Wiring shown is typical and **not** intended to show detail for a specific installation. Refer to certified field wiring diagrams.

Fig. 50 — Benshaw (Inside the Delta) Unit Mounted Starter Wiring Schematic (cont)

LEGEND FOR FIG. 50

AC	—	Alternating Current
AUX	—	Auxiliary
AWG	—	American Wire Gage
BYP	—	Bypass
CB	—	Circuit Breaker
CMD	—	Command
COM	—	Common
COND	—	Condenser
CPU	—	Central Processing Unit
CR	—	Control Relay
CT	—	Current Transformer
FU	—	Fuse
GND	—	Ground
GRN	—	Ground
HPS	—	High-Pressure Switch
J	—	Jack
JC	—	Plug Designator
K	—	Relay
L1, L2	—	Line Terminals
LCD	—	Liquid Crystal Display
LED	—	Light-Emitting Diode
LL	—	Control Power
LPS	—	Low-Pressure Switch
M1, M2, M3	—	Bypass Coils
MOV	—	Metal Oxide Varistor
NC	—	Normally Closed
O/L	—	Overload Reset
OS	—	3-Phase Current Power Source to Oil Pump
PR	—	Pilot Relay
PWR	—	Power
RLA	—	Rated Load Amps
SCR	—	Silicon Control Rectifier
SMM	—	Starter Management Module
ST	—	Shunt Trip
T	—	Transformer
T1, T2, T3	—	AC Motor Winding Leads
TB	—	Terminal Board
TBP	—	Terminal Board Plug
TEMP	—	Temperature
VAC	—	Volts Alternating Current
XFMR	—	Transformer

 Wire Node Symbol — May Have Terminal Block

 Power Terminal

 PC Board Terminal

NOTES FOR FIG. 50

1. LED status with power applied and prior to run command:

   ● "ON"

   ○ "OFF"

2. Local Start pushbutton is disabled.

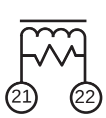
3. Transformer T1 primary and secondary connection and fuses FU1/FU2 value dependent on system voltage and model connect per the following table:

MODEL	**32STS**		
Transformer Size	**1 KVA**		
System Voltage	**Primary Connection**	**Secondary Connection**	**FU1-2 Amps**
200	H1, H2	X1, X2	12
230 60 Hz	H1, H4 (Jumper H1-H3 and H2-H4)	X1, X2	10
230 50 Hz	H1, H4 (Jumper H1-H3 and H2-H4)	X1, X2	10
346	H1, H2	X1, X4 (Jumper X1-X3 and X2-X4)	7
380	H1, H2	X1, X4 (Jumper X1-X3 and X2-X4)	6
400	H1, H3	X1, X4 (Jumper X1-X3 and X2-X4)	6
416	H1, H4	X1, X4 (Jumper X1-X3 and X2-X4)	6
460	H1, H4 (Jumper H2-H3)	X1, X2	5
575	H1, H2	X1, X2	4

MODEL	**32STT**		
Transformer Size	**2 KVA**		
System Voltage	**Primary Connection**	**Secondary Connection**	**FU1-2 Amps**
200	H1, H2	X1, X3	20
230 60 Hz	H1, H2	X1, X3	20
230 50 Hz	H1, H2	X1, X3	20
346	H1, H4	X1, X3	12
380	H1, H2	X1, X4 (Jumper H1-H3 and H2-H4)	12
400	H1, H3	X1, X4 (Jumper H1-H3 and H2-H4)	12
416	H1, H4	X1, X4 (Jumper H1-H3 and H2-H4)	12
460	H1, H3	X1, X3	10
575	H1, H4	X1, X3	8

MODEL	**32STC, 32STJ**		
Transformer Size	**3 KVA**		
System Voltage	**Primary Connection**	**Secondary Connection**	**FU1-2 Amps**
200	H1, H2	X1, X3	30
230 60 Hz	H1, H2	X1, X3	30
230 50 Hz	H1, H4	X1, X4 (Jumper X1-X3 and X2-X4)	30
346	H1, H2 (Jumper H1-H3 and H2-H4)	X1, X4 (Jumper X1-X3 and X2-X4)	20
380	H1, H2	X1, X4 (Jumper X1-X3 and X2-X4)	20
400	H1, H3	X1, X4 (Jumper X1-X3 and X2-X4)	15
416	H1, H4	X1, X4 (Jumper X1-X3 and X2-X4)	15
460	H1, H3	X1, X3	15
575	H1, H4	X1, X3	12

4. On some units, an independent motor current transformer (CT) and load resistor is used to provide signal to the SMM:



5. Circuit Breaker CB3 provided on models 32STG, 32STJ and 32STT. CB3 provided on model 32STS only when Option 1 (Ammeter/Voltmeter) is specified.

6. Circuit Breaker CB2 is rated at 20 A for models 32STG, 32STJ and 32STT. CB2 is rated at 15 A for model 32STS.

7. MOVs are used on power stack assemblies for system voltages of 200 through 480 VAC (as shown). Resistor/Capacitor networks are used on power stack assemblies in place of MOVs for a system voltage of 575 VAC (not shown).

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